Introduction to Systematic Review and Meta-analysis

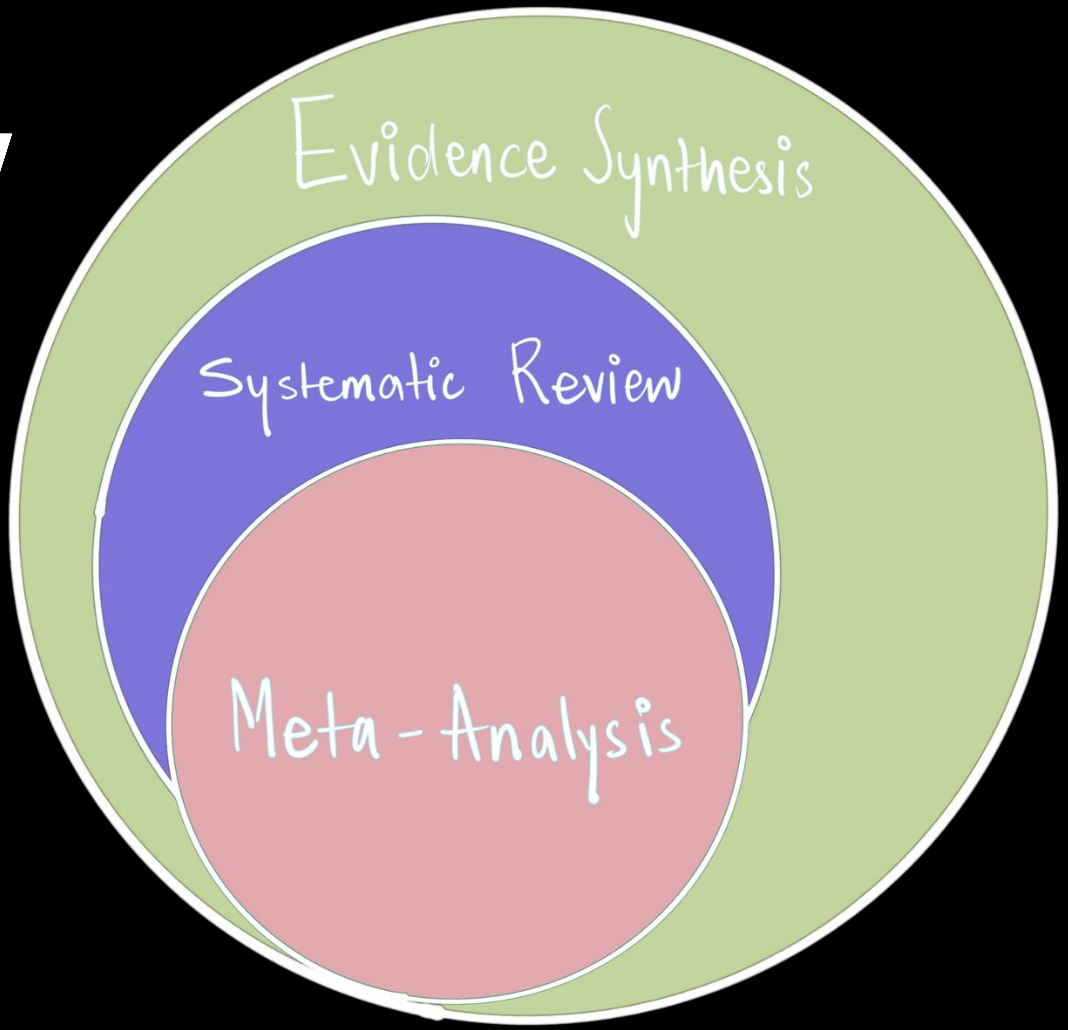
Pre-conference Workshop: Behaviour 2025

Instructor:

Shreya Dimri

Department of Evolutionary Biology

Bielefeld University (Germany)



- Shreya Dimri (she/her)
- 4th year PhD Student
- Doing a lot of meta-analysis
- Behaviour ecology and evolution
- Open Science practitioner
- Meta-research enthusiast

https://github.com/shreyadimri

https://bsky.app/profile/shreyadimri.bsky.social

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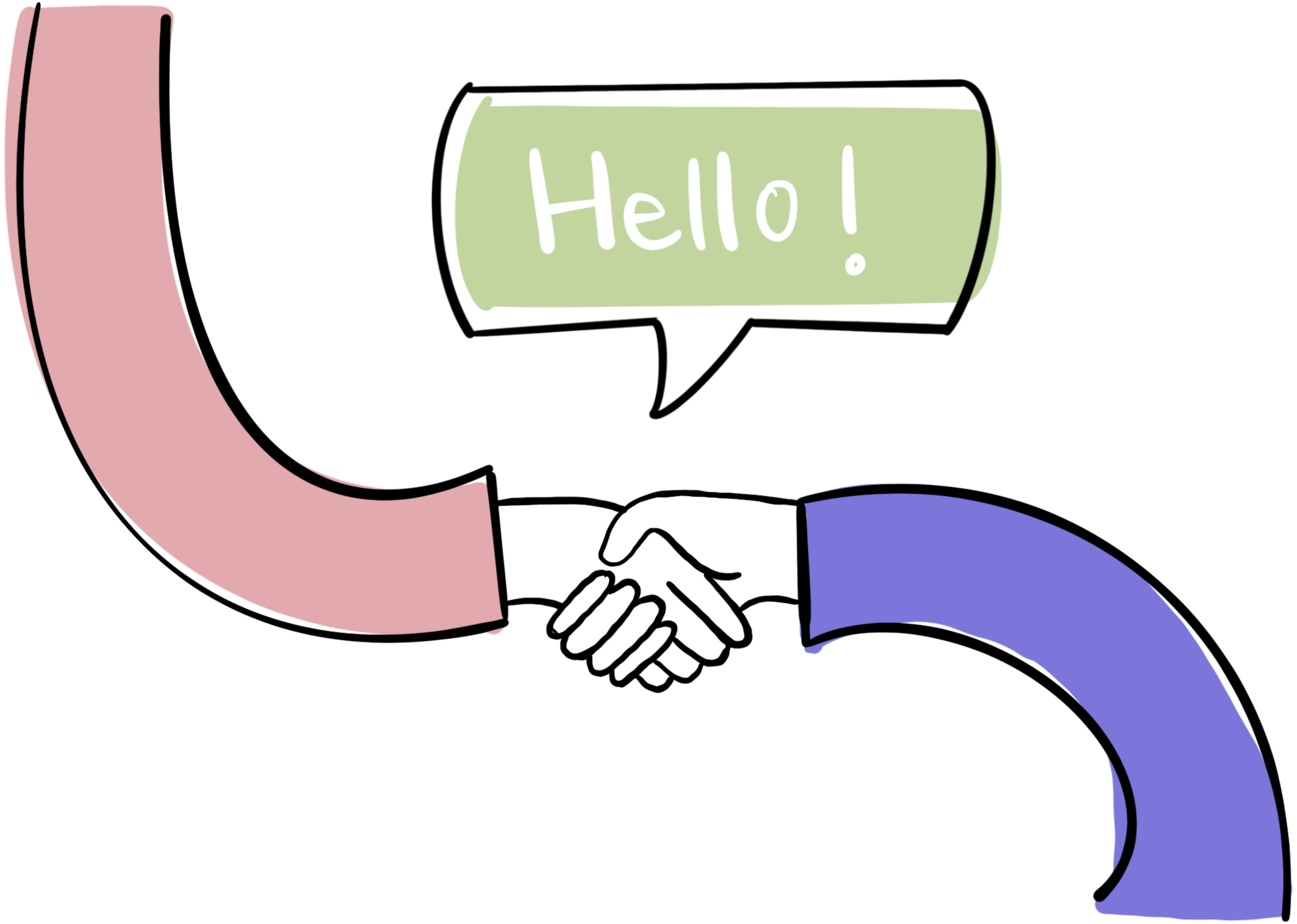
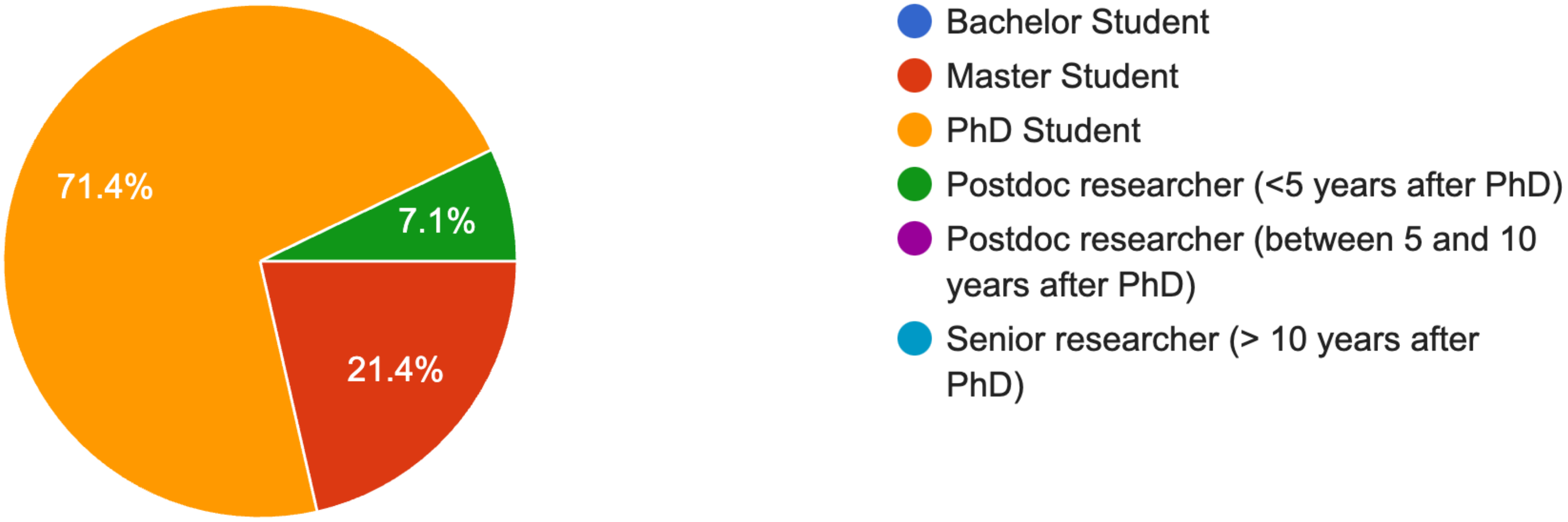

Hello !

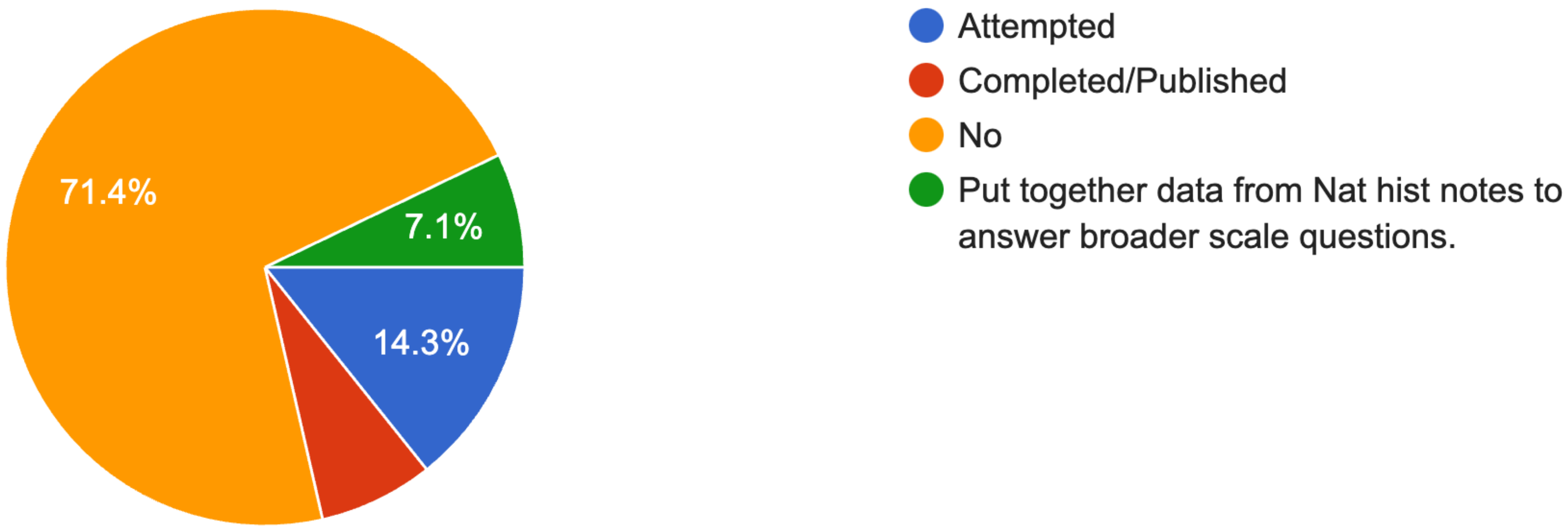
What is your research career stage?

14 responses



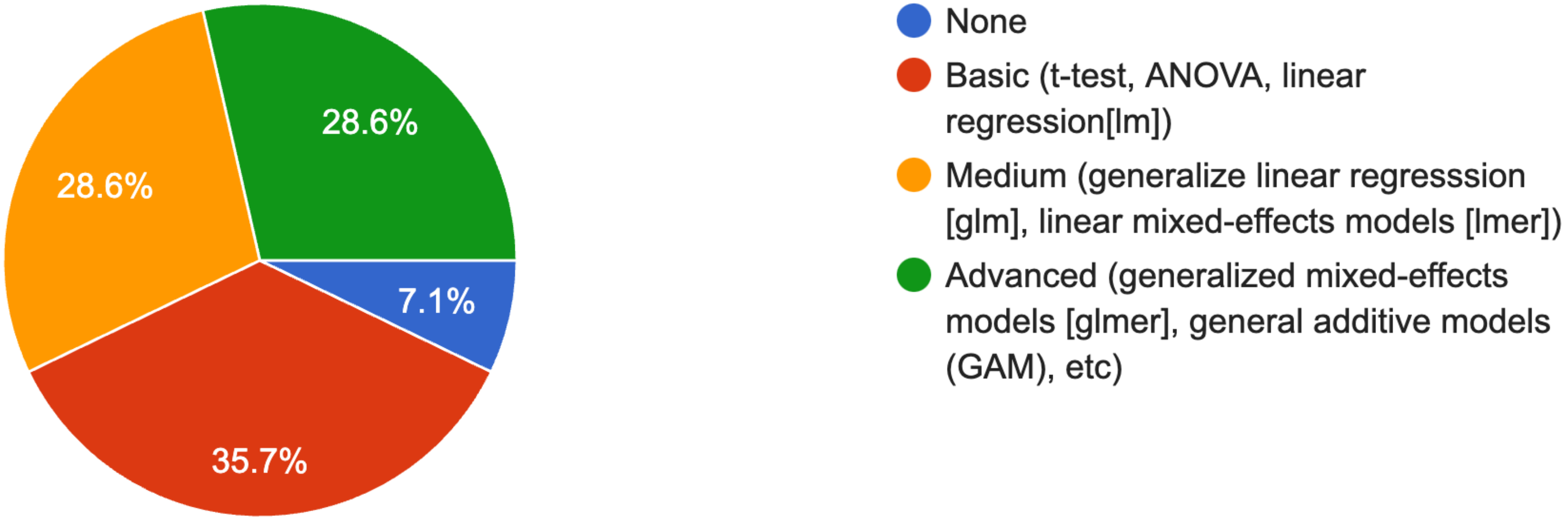
Have you previously attempted/completed/published a meta-analysis?

14 responses



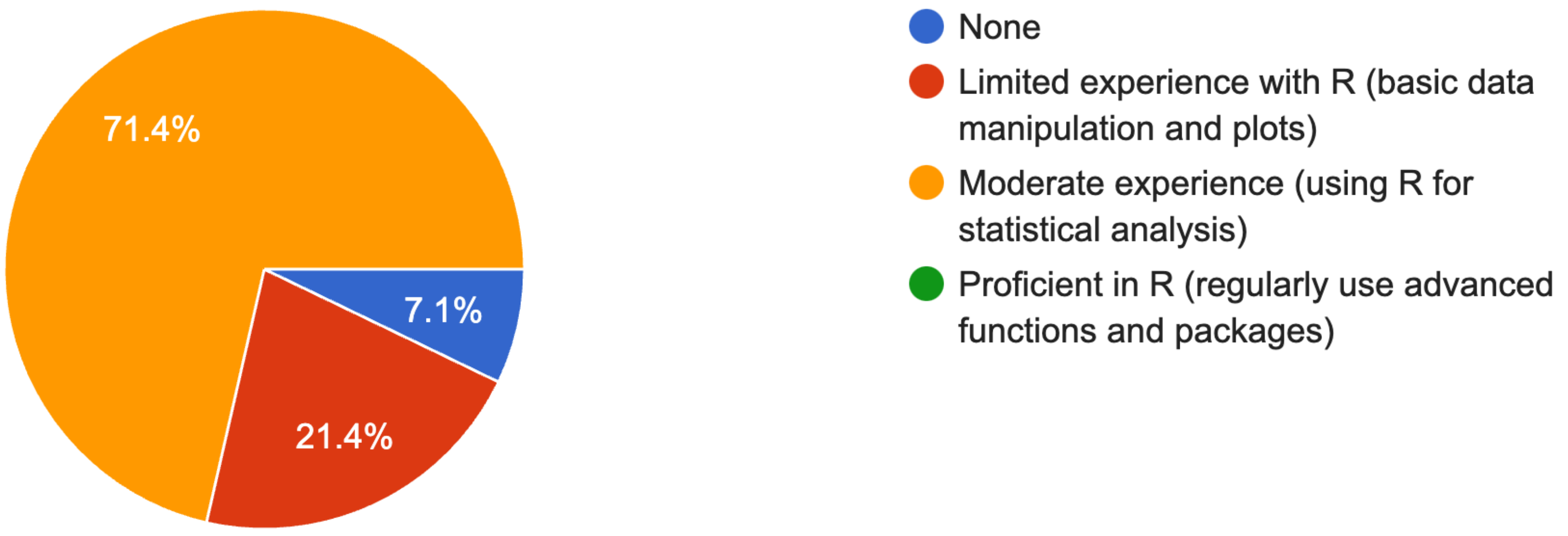
What is your statistical expertise?

14 responses



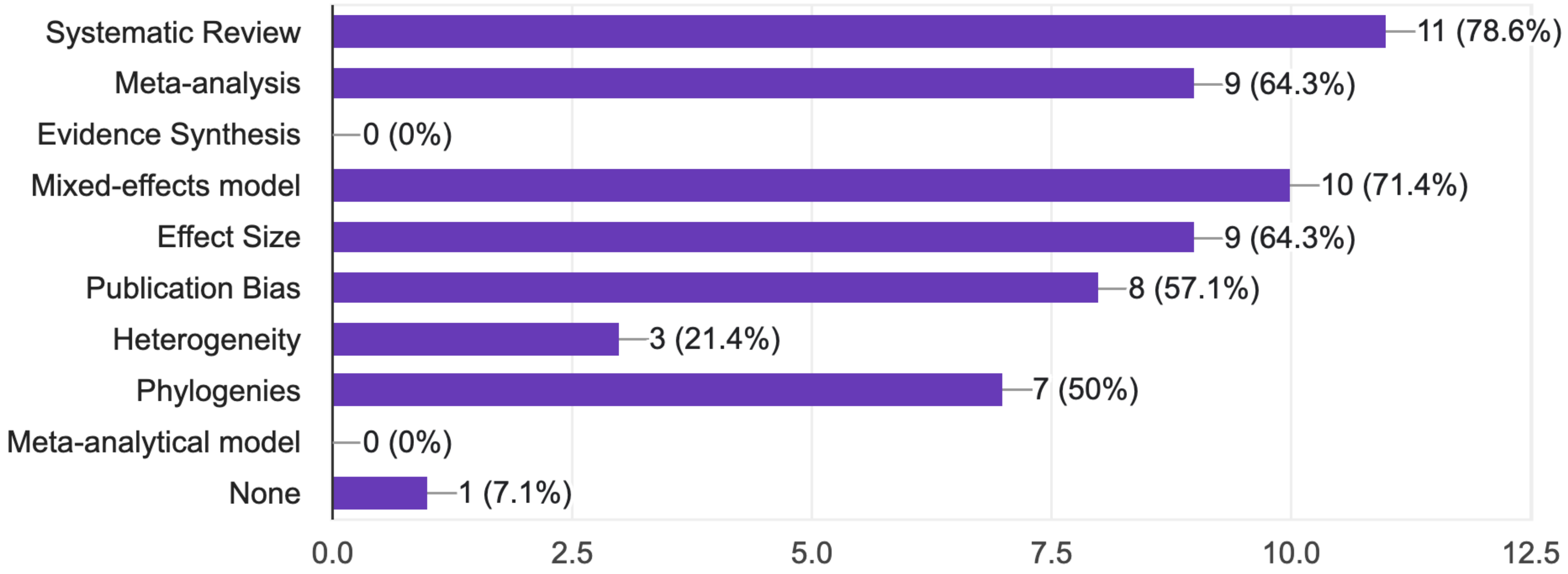
Do you have any experience working with R?

14 responses



Which of these terms are you familiar with

14 responses



What I plan to cover today

- Overview of the field: What is a Systematic Review and a Meta-Analysis?
- Foundations of a high-quality Systematic Review, Common Mistakes to avoid
- Demonstrate the process of searching and screening
- What effect sizes are out there for a meta-analysis?
- How do we extract data for a meta-analysis?
- What is heterogeneity, and what to do with it?
- What is publication bias?
- How to evaluate a meta-analysis? How to interpret some common graphs?

What I cannot promise today

- To cover all the statistical information/details out there
- To give you step-by-step code for all the possible analyses
- Run an “How to do a meta-analysis in R” workshop
- To go into great detail about the effect sizes and models
  (very important, but I am limited by time here!)
- To turn you into an expert on meta-analysis

How this workshop is designed

- Don’t let this become a monologue, you are the key component of this “Workshop”
- You do not need to take notes! All the resources will be provided to you!
- I am happy to chat afterwards, in the breaks, over emails or even collaborate!

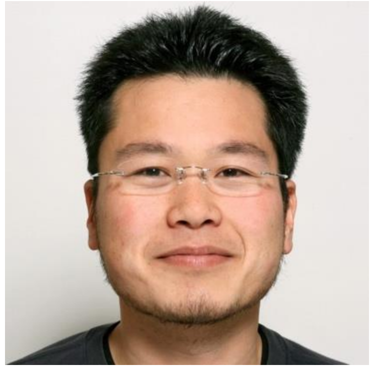
Outline of the workshop

- Time: 1:00 - 5:30 PM
- Introductions
- Systematic Review (70% of the time)
- Meta-analysis (30% of the time)

I stand upon the shoulder of the giants!

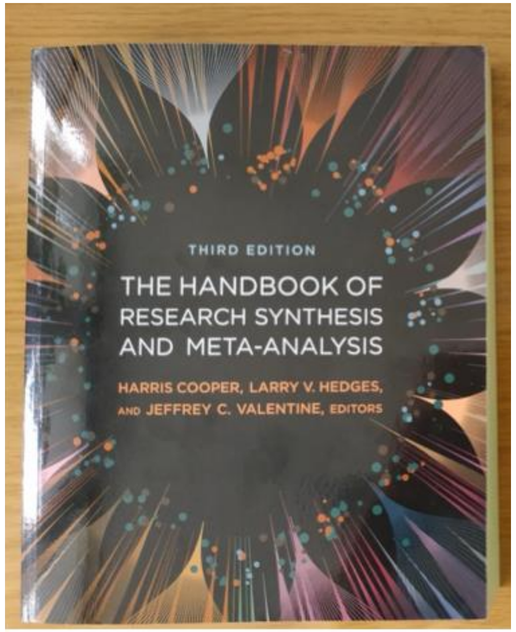
Alfredo Sánchez-Tójar

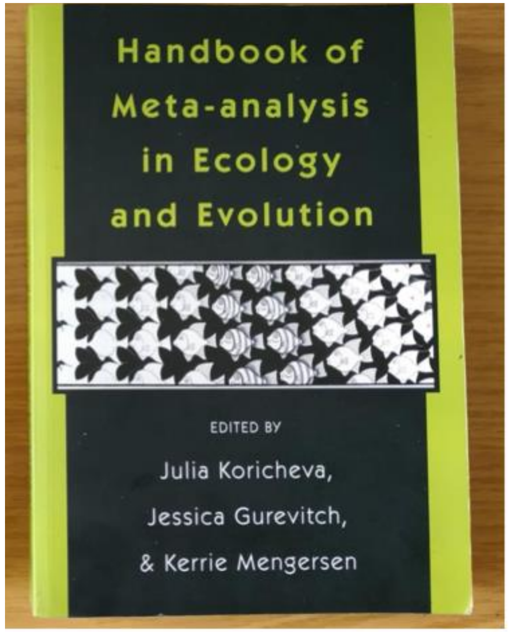
Wolfgang Viechtbauer



Shinichi Nakagawa

Let's get the resources out of the way!

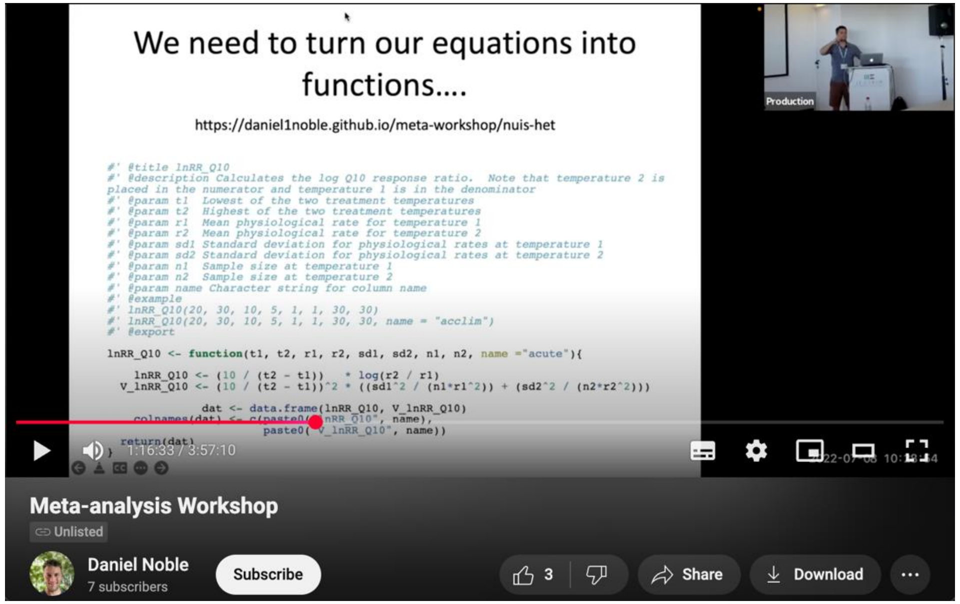




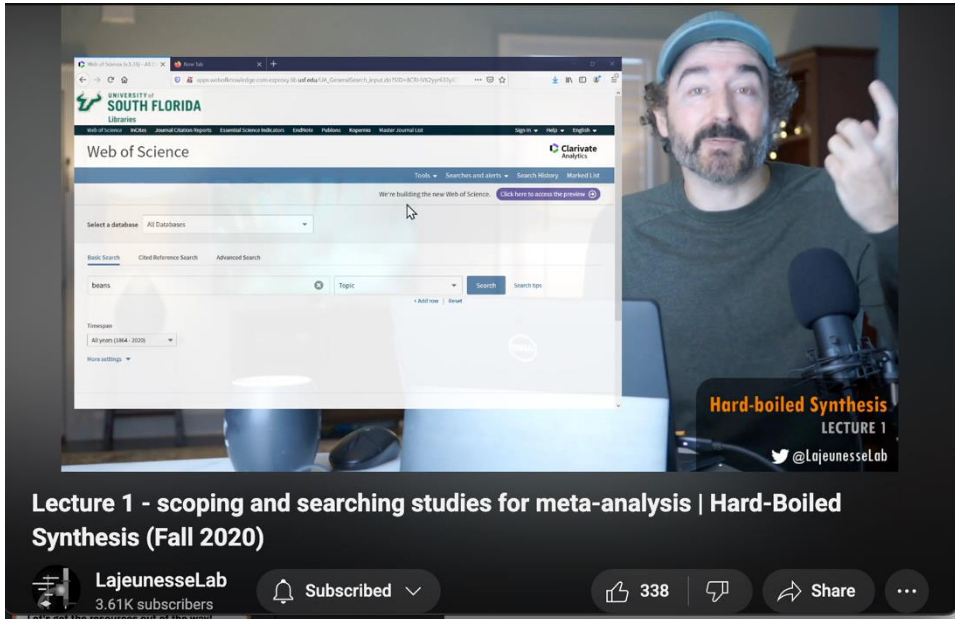
The free book: "Cochrane Handbook for Systematic Reviews of Interventions" (https://training.cochrane.org/handbook/current) is a tremendous resource.
We particularly like Chapter 6: Choosing effect measures and computing estimates of effect.

Another book that you can easily access online for Doing Meta-analysis in R book.

Let’s get the resources out of the way!

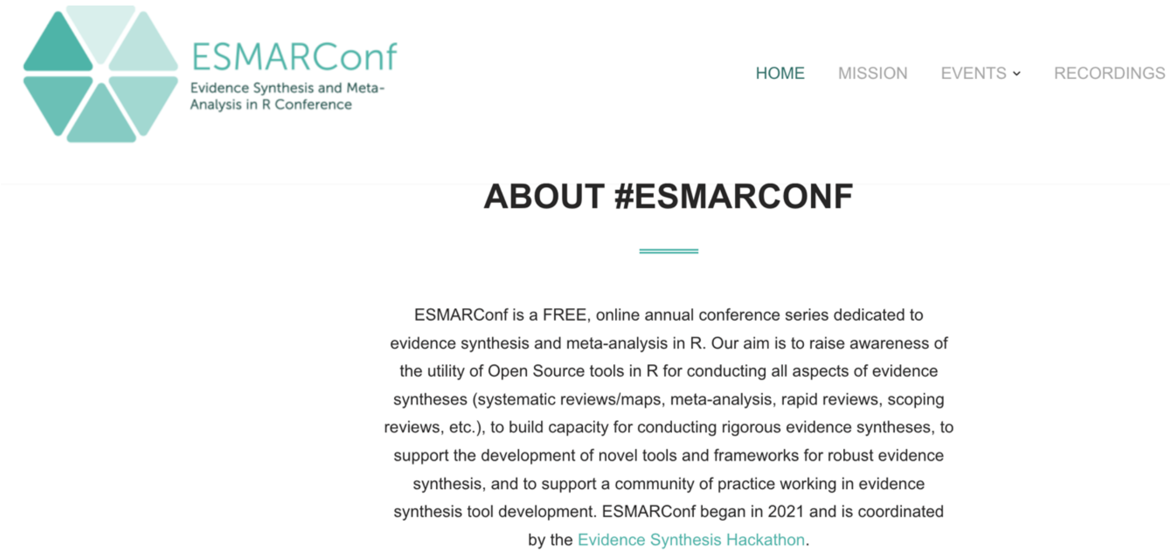


Check out *Daniel Noble*’s [4-hour workshop]



The [YouTube channel] of the [Lajeunesse Lab] is fantastic, featuring a video-by-video tutorial on how to perform systematic reviews and meta-analyses.
Don’t miss out [channel].

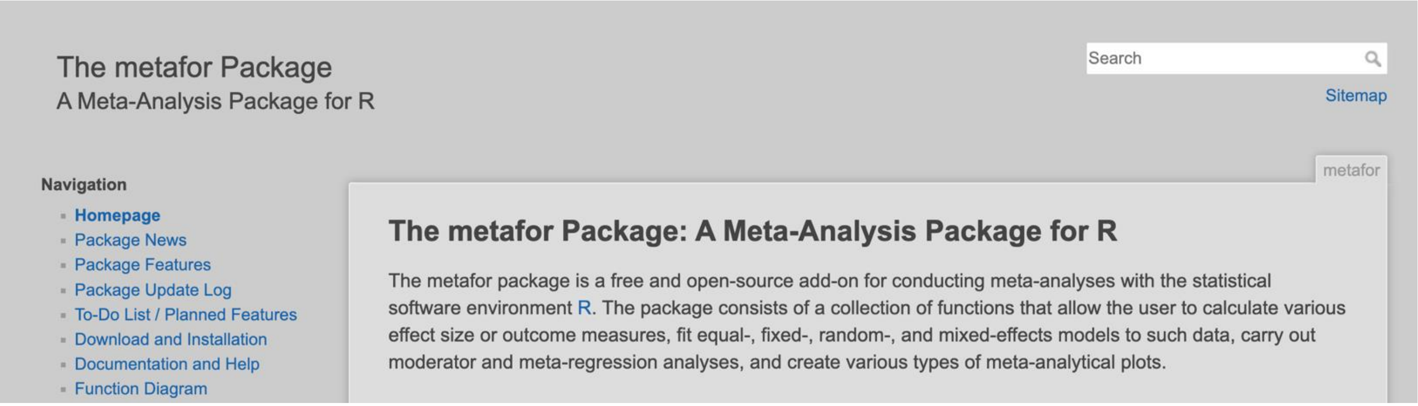
Let's get the resources out of the way!

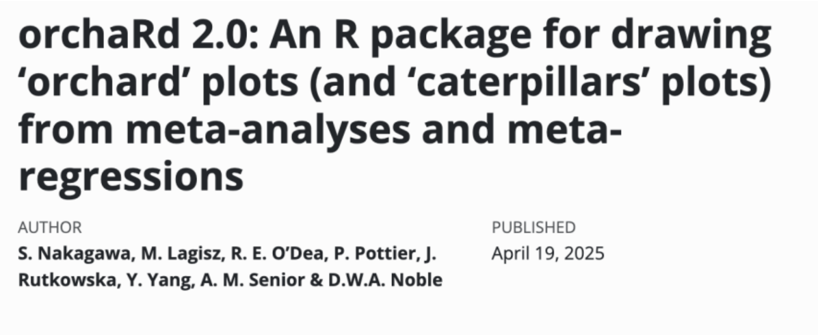


ESMARConf: Keep up to date with events and software created as part of the Evidence Synthesis Hackathon, and get involved if you can.

SORTEE: Not just for meta-analysis, for all science (ecology, evolution and even behaviour!)
They have an online conference in October!

Let’s get the resources out of the way!





synthesisr

rmetaverse/synthesisr

revtools

Let's get the resources out of the way!

Essential reading

- O'Dea et al. ([2021](#))
- Foo et al. (2021)
- Gurevitch et al. (2018)
- Nakagawa et al. (2017)
- Noble et al. (2017)
- Noble et al. (2022)
- Senior et al. (2016)
- Yang, Noble, et al. (2023)

Methods

Basics

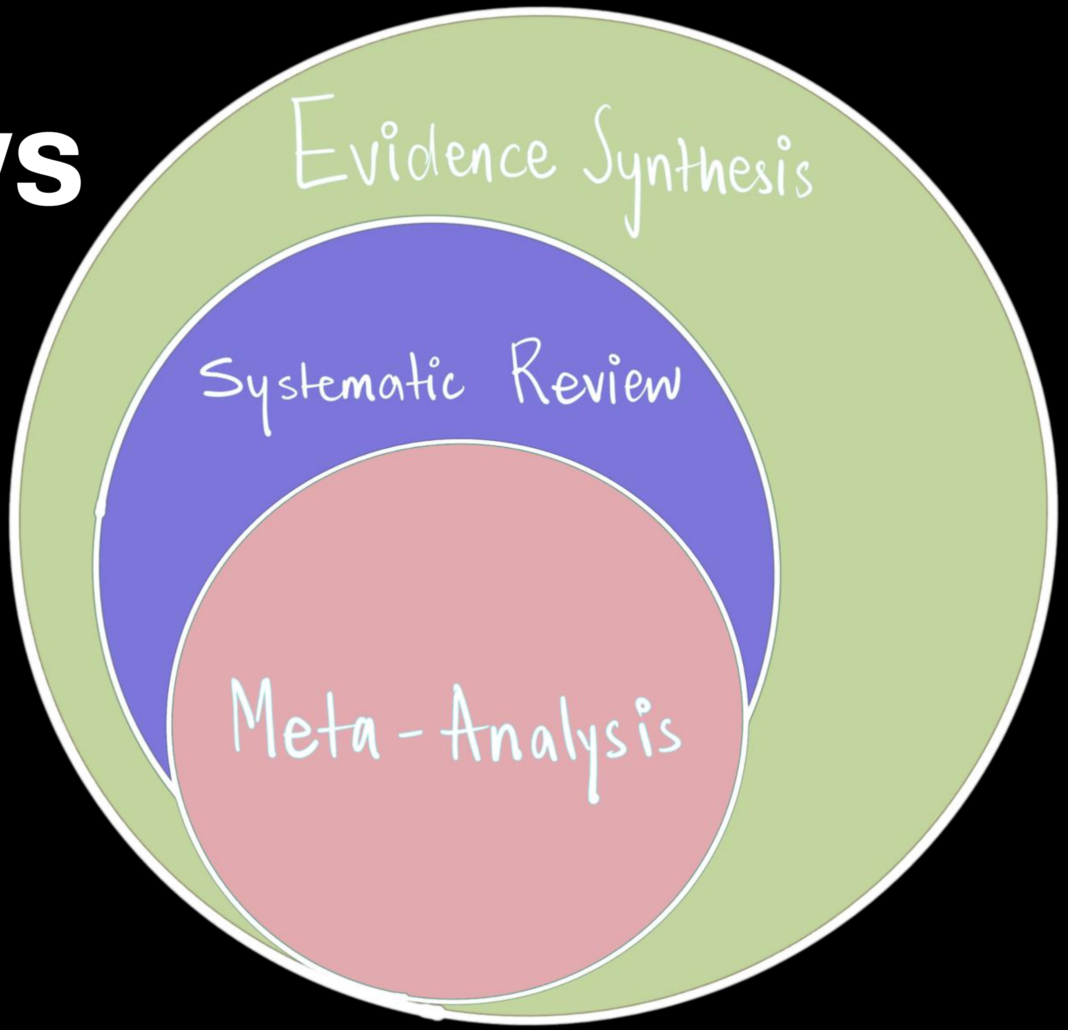
- Nakagawa, Lagisz, et al. (2020)
- Nakagawa, Lagisz, et al. (2023)
- Nakagawa et al. (2021)
- Pick, Nakagawa, and Noble (2018)
- Yang, Noble, et al. (2023)

And more..

- Cinar, Nakagawa, and Viechtbauer (2021)
- Shinichi Nakagawa et al. (2014)
- Senior, Viechtbauer, and Nakagawa (2020)
- Yang, Lagisz, and Nakagawa (2023)
- Yang, Williams, et al. (2024)

Any questions?

Introduction to Systematic Reviews



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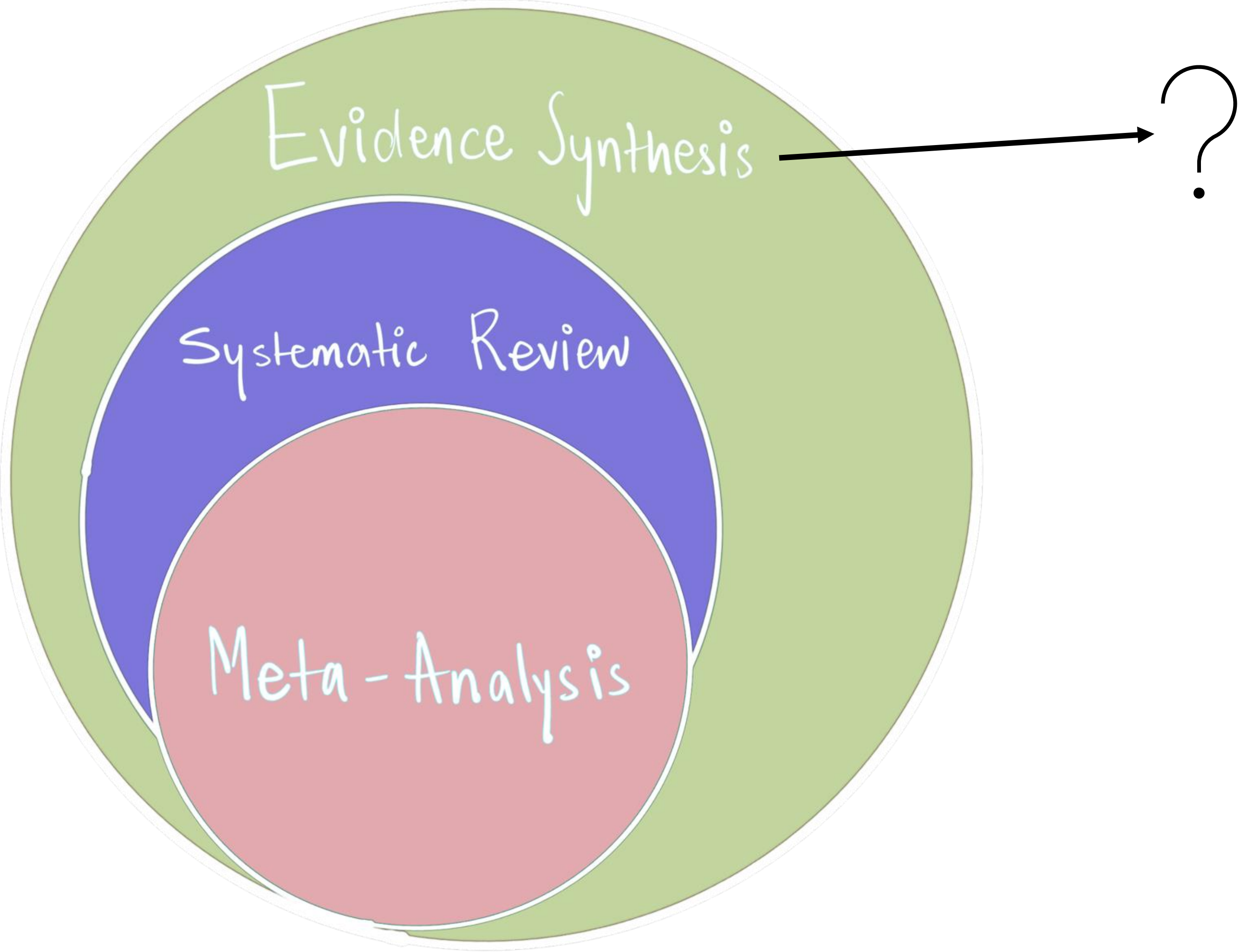
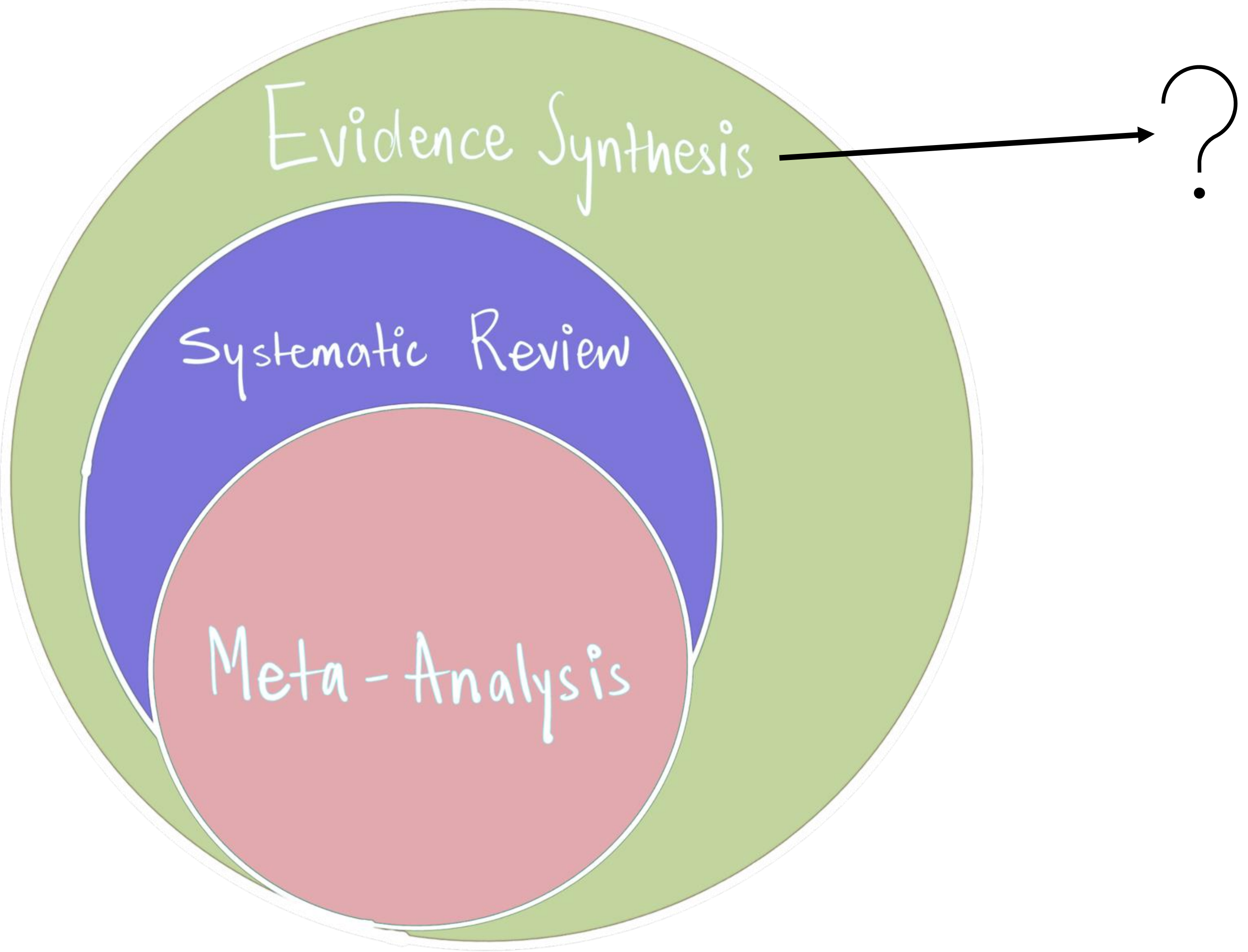
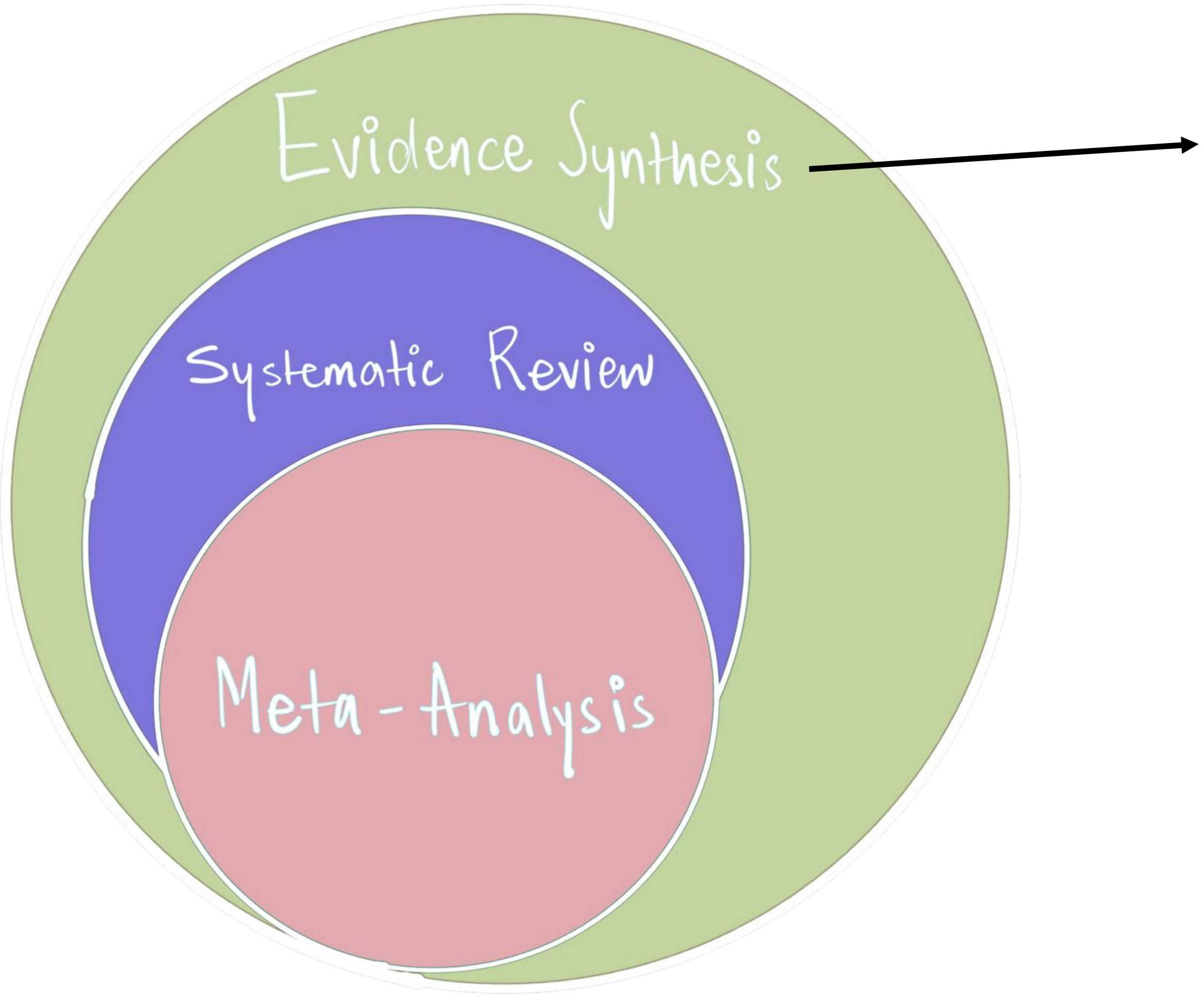




“The process of bringing together information from a range of sources and disciplines to inform debates and decisions on specific issues”

Process of:

- Gathering evidence
- Assessing evidence
- Combining evidence



- Narrative Review
- Rapid Review
- Scoping Review
- Systematic Map
- Umbrella Review
- Systematic Review
- Meta-Analysis

What is a literature review?

Literature review: “the process of conducting surveys of previously published material”

- Focus: empirical findings, methods, theory…
- Goals: integrate, critically analyse, identify issues
- Perspective: neutral representation vs. **espousal of a position**

“The reviewers accumulate and synthesise the literature in the service of demonstrating the value of **the particular point of view that they espouse**”.

Cooper, Hedges & Valentine 2019, 3rd edition

"a review maybe susceptible to a suite of biases and limitations if not conducted in a comprehensive, repeatable and systematic way"

Table 1 | A summary of the major limitations and biases that can affect a literature review across each stage of the review process

Review stage	Type of limitation/bias	Description	Possible mitigation measures
All stages	Lack of transparency	The methods used in the review are not reported in sufficient detail to allow replication or verification	Report all activities (including search terms/strings, languages and databases searched, screening process and test for consistency screening, number of included/excluded studies at each stage of the review process and so on) in detail in the Methods section and Supplementary Information
Protocol	Subject drift/question creep	The topic/scope or definitions of the review shift over the course of the review, for example driven by what reviewers find interesting in the evidence base	An a priori protocol sets out the planned methods for the review in detail and is published (and preferably peer-reviewed by subject and methodology experts) prior to commencing the review
Searching (general)	Selection bias/Cherry-picking	Studies included in the review are selected based on awareness of the reviewer	Systematic search across a suite of resources rather than selection of studies with which authors are already familiar
	Lack of comprehensiveness	The evidence base identified is incomplete and may not be representative	Comprehensive searching across multiple bibliographic databases and sources of grey literature, searches in multiple relevant languages
	Media attention bias	Reviewers may be influenced in their selection of articles based on what is discussed in the media	Systematic search across a suite of resources rather than selection of studies with which authors are already familiar with

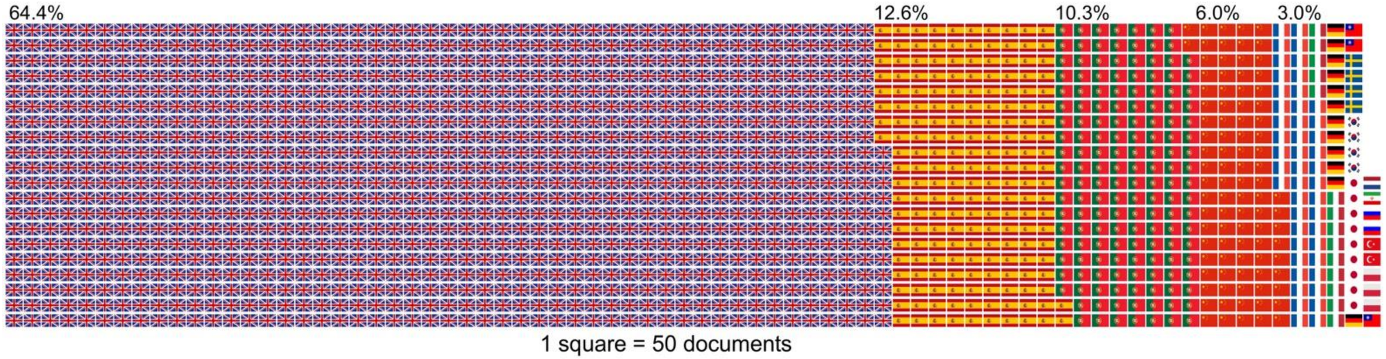
Haddaway & Macura 2018, Nature Climate Change

“a review maybe susceptible to a suite of biases and limitations if not conducted in a comprehensive, repeatable and systematic way”

Searching academic literature	Place of publication bias	Journals with a higher ‘impact’ and visibility are more likely to publish significant, positive/affirmative research	Comprehensive searching across multiple bibliographic databases
	Citation bias	Some studies, such as those with significant results, are more likely to be cited than others, resulting in a possible bias if citations are used as a basis for article selection	Searching using a variety of different methods, including bibliographic databases, screening bibliographies and citation tracking (forwards or backwards)
	Database bias	Bibliograhic databases select which journals to index and may omit certain bodies of evidence	Comprehensive searching across multiple bibliographic databases
Searching grey literature	Publication bias	Significant and/or positive results may be more likely to be published in traditional academic journals	Comprehensive searching across multiple resources
	Language bias	Two-fold: (1) non-English research may be less likely to be translated and published in ‘mainstream’ English journals; and (2) searches not performed in non-English languages may miss research published in other language journals	Searching for and inclusion of non-English language evidence

Haddaway & Macura 2018, Nature Climate Change

Language Bias



PLOS | BIOLOGY

PERSPECTIVE

Languages Are Still a Major Barrier to Global Science

Tatsuya Amano[1,2]*, **Juan P. González-Varo**[1], **William J. Sutherland**[1]

1 Conservation Science Group, Department of Zoology, University of Cambridge, Cambridge, United Kingdom, **2** Centre for the Study of Existential Risk, University of Cambridge, Cambridge, United Kingdom

* amatatsu830@gmail.com

OPEN ACCESS

Citation: Amano T, González-Varo JP, Sutherland WJ (2016) Languages Are Still a Major Barrier to Global Science. PLoS Biol 14(12): e2000933. doi:10.1371/journal.pbio.2000933

Published: December 29, 2016

Abstract

While it is recognized that language can pose a barrier to the transfer of scientific knowledge, the convergence on English as the global language of science may suggest that this problem has been resolved. However, our survey searching Google Scholar in 16 languages revealed that 35.6% of 75,513 scientific documents on biodiversity conservation published in 2014 were not in English. Ignoring such non-English knowledge can cause biases in our understanding of study systems. Furthermore, as publication in English has become prevalent, scientific knowledge is often unavailable in local languages. This hinders its use by field practitioners and policy makers for local environmental issues; 54% of protected area directors in Spain identified languages as a barrier. We urge scientific communities to make a more concerted effort to tackle this problem and propose potential approaches both for compiling non-English scientific knowledge effectively and for enhancing the multilingualization of new and existing knowledge available only in English for the users of such knowledge.

Language Bias

All systematic reviews and maps published in the journal Environmental Evidence (n=72) prior to January 2022 [...] 44% of the 72 assessed reviews did not search or include any non-English language literature.

Received: 8 June 2023 | Revised: 19 November 2023 | Accepted: 29 November 2023
DOI: 10.1002/jrsm.1699

RESEARCH ARTICLE

Research Synthesis Methods WILEY

Language inclusion in ecological systematic reviews and maps: Barriers and perspectives

Kelsey Hannah[1,2] | **Neal R. Haddaway**[3,4] | **Richard A. Fuller**[1,2] | **Tatsuya Amano**[1,2]

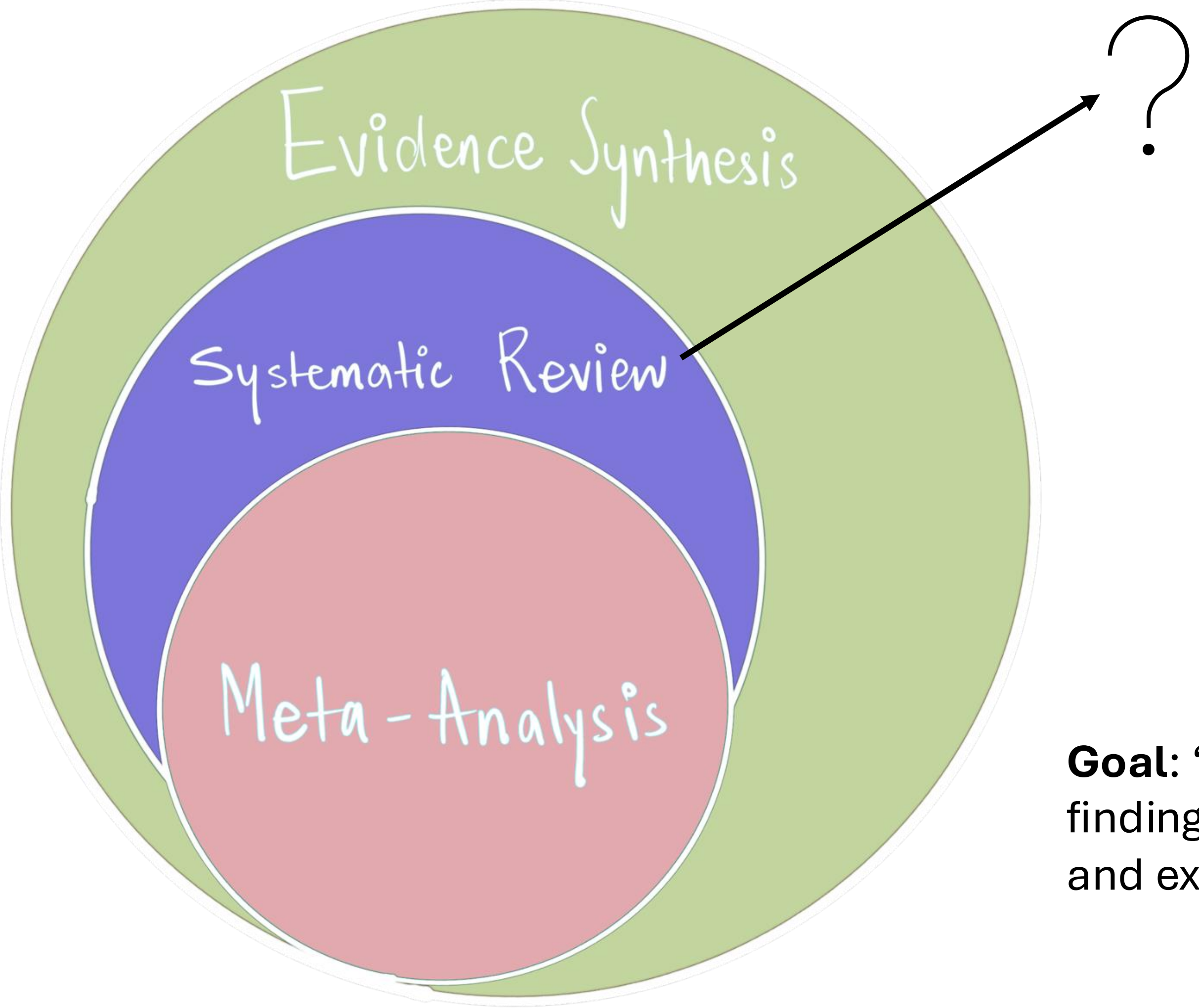
[1]School of the Environment, The University of Queensland, Brisbane, Queensland, Australia
[2]Centre for Biodiversity and Conservation Science, The University of Queensland, Brisbane, Queensland, Australia
[3]Leibniz-Centre for Agricultural Landscape Research (ZALF), Müncheberg, Germany

Abstract
Systematic reviews and maps are considered a reliable form of research evidence, but often neglect non-English-language literature, which can be a source of important evidence. To understand the barriers that might limit authors' ability or intent to find and include non-English-language literature, we assessed factors that may predict the inclusion of non-English-language literature in ecological systematic reviews and maps, as well as the review

“a review maybe susceptible to a suite of biases and limitations if not conducted in a comprehensive, repeatable and systematic way”

Eligibility screening	Lack of consistency	Different reviewers make different decisions regarding eligibility	Set out planned methods in detail in an a priori protocol developed through scoping of the literature. Test for consistency formally, discussing disagreements and revising criteria where necessary
Retrieval	Dissemination bias	Lack of accessibility (due to limitations of the reviewers' subscription, for example) meaning that some studies are omitted from the review	Comprehensive retrieval of studies
Data extraction	Lack of consistency	Different reviewers extract different data	Set out planned methods in detail in an a priori protocol developed through scoping of the literature. Test for consistency formally, discussing disagreements and revising criteria where necessary
Critical appraisal	Outcome reporting bias	Some study measured outcomes are omitted from a research article despite having been recorded by the authors, possibly due to a lack of significance	Critical appraisal
	Full publication bias	Researchers may only publish parts of their study, for example due to significance	Critical appraisal/author contact

Haddaway & Macura 2018, Nature Climate Change



"**Method** for synthesising evidence in a reliable manner that aims to maximise **transparency**, **comprehensiveness** and **objectivity***"

*Though comprehensiveness would be great, it should not be required as long as the literature review finds a representative sample of research findings

Goal: "provide a valid summary of primary research findings through [ideally] a pre-planned [e.g., pre-registered] and explicit procedure"

Haddaway & Macura 2018; Moher et al. 2015; Foo et al. 2021

Must read before you start!

Received: 14 January 2021 | Accepted: 10 May 2021
DOI: 10.1111/2041-210X.13654

REVIEW

Methods in Ecology and Evolution
BRITISH ECOLOGICAL SOCIETY

A practical guide to question formation, systematic searching and study screening for literature reviews in ecology and evolution

Yong Zhi Foo[1] | Rose E. O'Dea[1] | Julia Koricheva[2] | Shinichi Nakagawa[1] | Malgorzata Lagisz[1]

BIOLOGICAL REVIEWS
Cambridge Philosophical Society

Biol. Rev. (2021), **96**, pp. 1695–1722. 1695
doi: 10.1111/brv.12721

Preferred reporting items for systematic reviews and meta-analyses in ecology and evolutionary biology: a PRISMA extension

Rose E. O'Dea[1*], Malgorzata Lagisz[1], Michael D. Jennions[2], Julia Koricheva[3], Daniel W.A. Noble[1,2], Timothy H. Parker[4], Jessica Gurevitch[5], Matthew J. Page[6], Gavin Stewart[7], David Moher[8] and Shinichi Nakagawa[1*]

So how do we do this thing?

Steps:

(0) Formulating the question(s)

(1) Developing a protocol

(2) Searching across multiple sources of evidence

(3) Screening evidence for relevance

(4) Retrieving articles

(5) Critically appraising study validity

(6) Extracting data

(7) Performing synthesis (qualitative and/or quantitative)

(8) Documenting all activities in a detailed report

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Don’t be fooled!

The steps of a systematic search are far from straightforward; they are often difficult to execute.

- Many (subjective) decisions to take

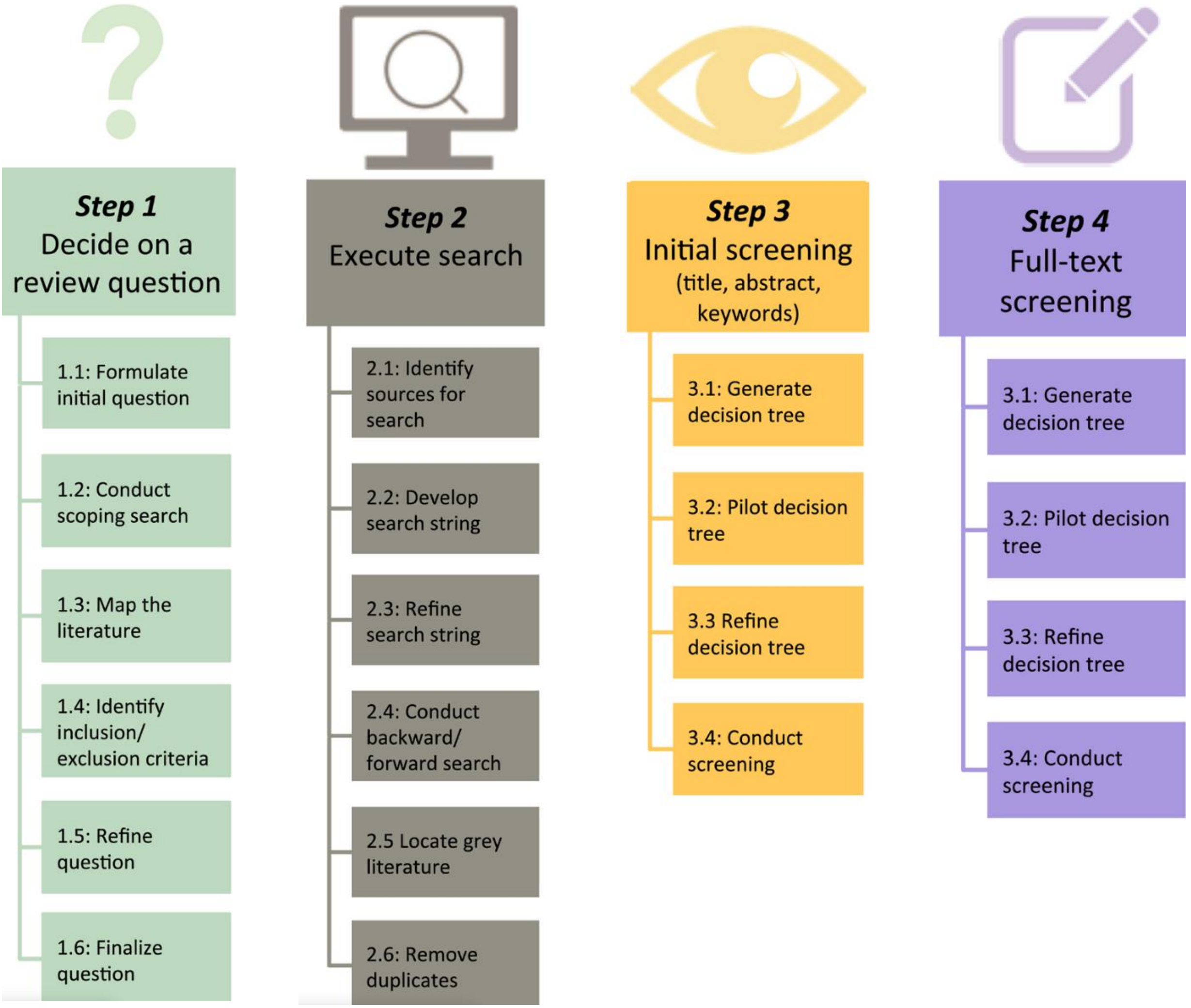
➢ Tools to minimise subjectivity and increase inter-observer agreement

➢ Transparent and detailed reporting

- Iterative process: you go back and forth

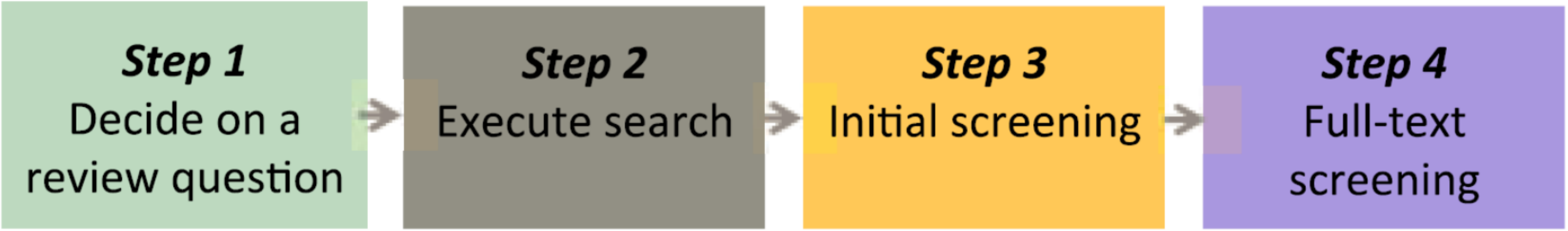
➢ Piloting before committing

Doing a systematic review



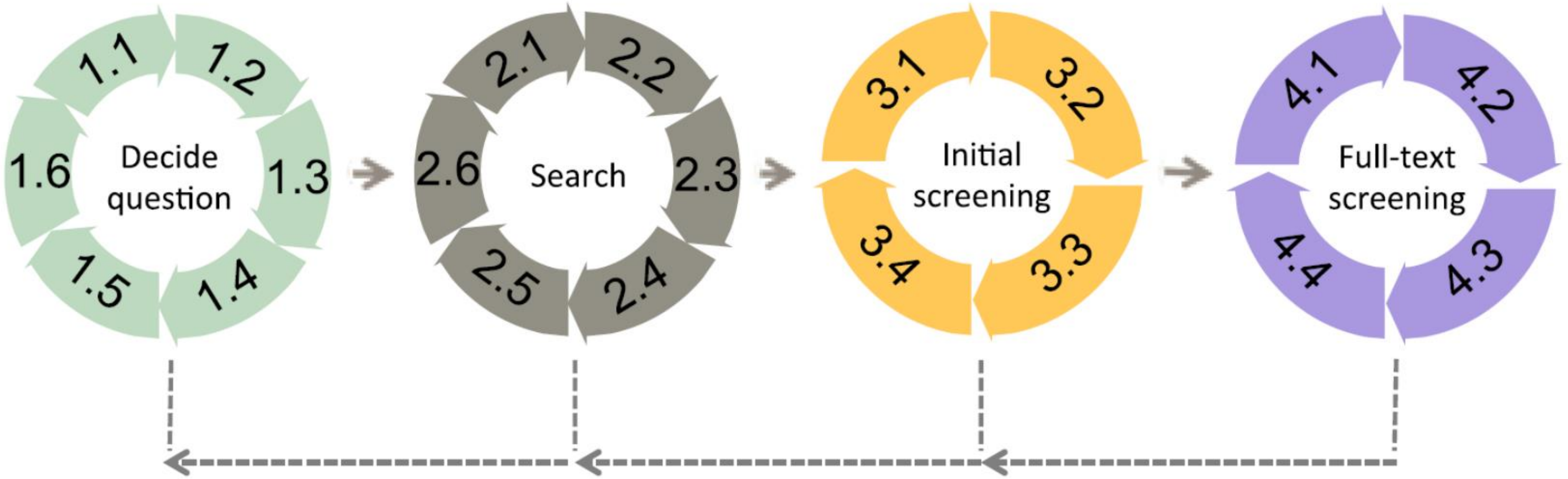
Foo et al. 2021

What most researchers think happens



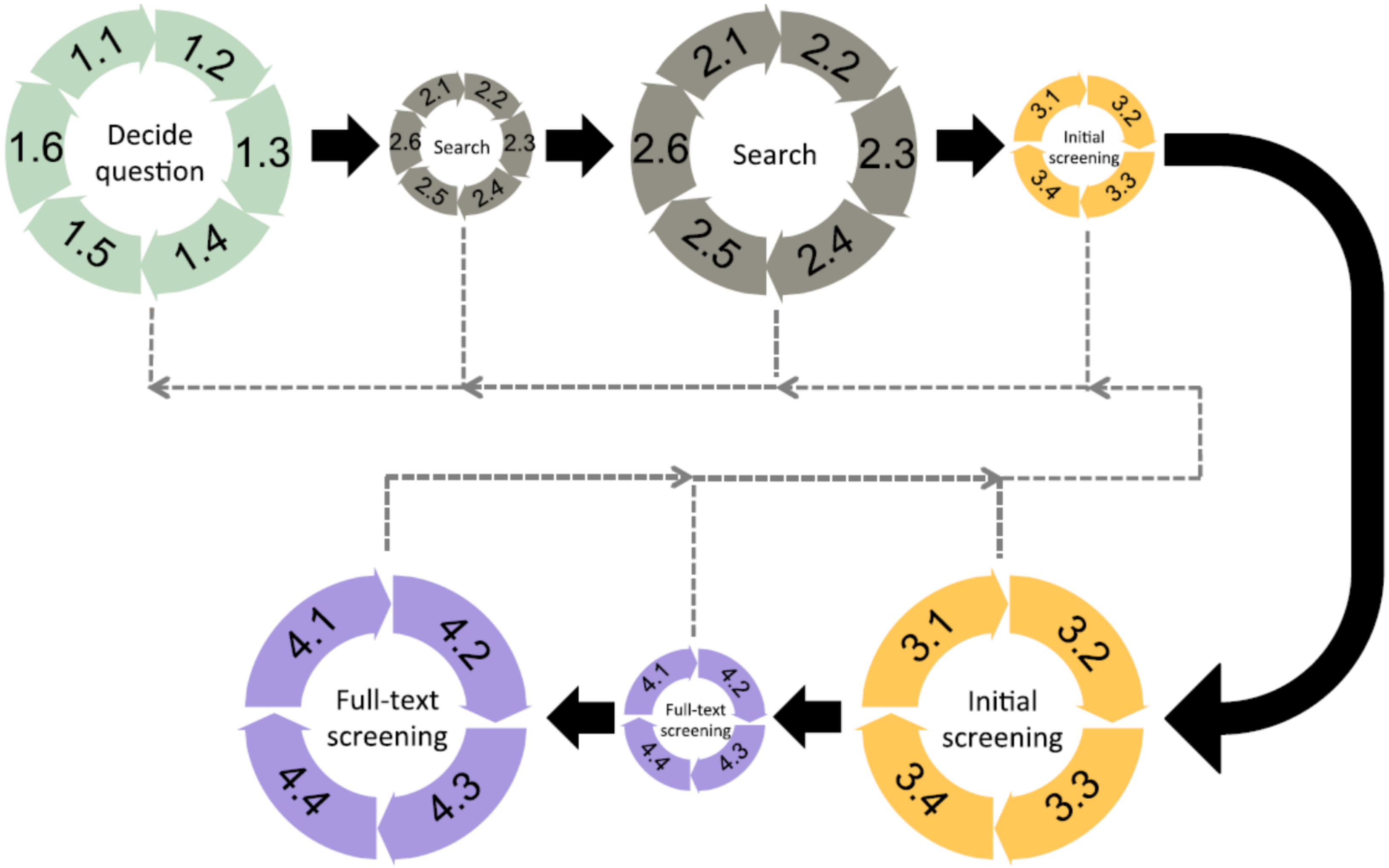
Foo et al. 2021

What likely happens in actual searches



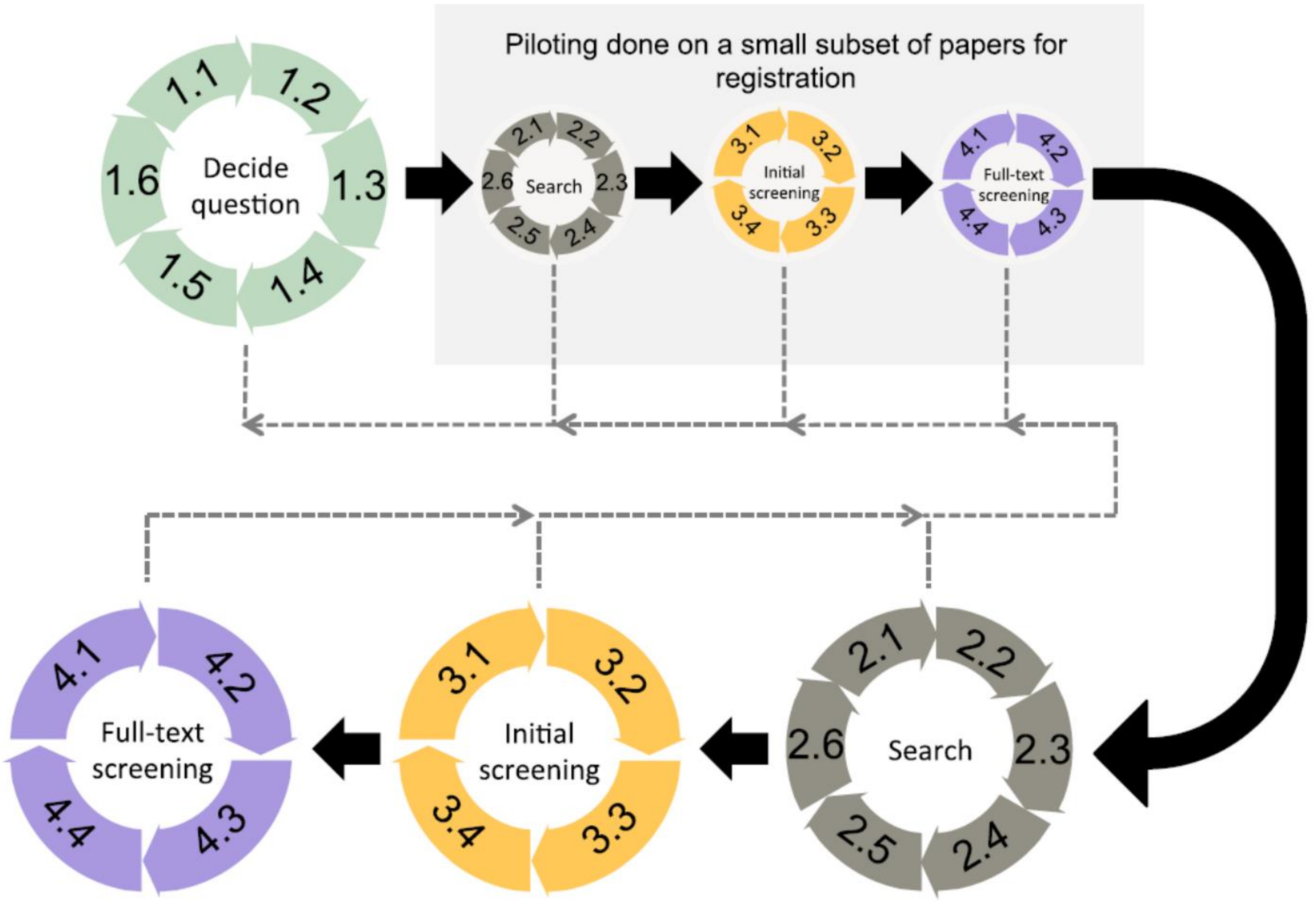
Foo et al. 2021

What should happen



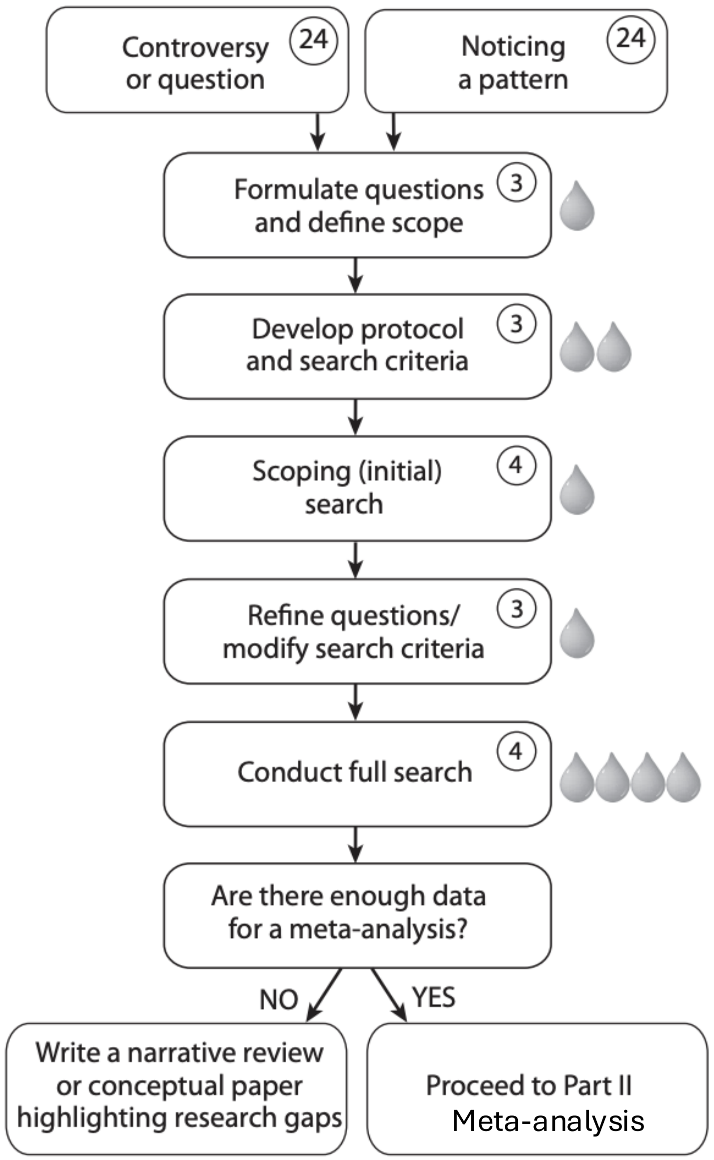
Foo et al. 2021

What should happen (for a pre-planned) search



Foo et al. 2021

Realistic expectations



Cote and Jennions (2013) in the Handbook of Meta-Analysis in Ecology and Evolution

Step 1: Decide on a review question

- Oftentimes, we have a previous question in mind/experience in a field
- The question should be sufficiently general, but not too broad to become impractical
- Use guiding framework to think/refine your questions

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PICO/PECO framework (origin: clinical experiments; Richardson et al., 1995)

PICO: Population, Intervention, Comparator, Outcome
Do vaccines cause autism in humans?

PECO: Population, Exposure, Comparator, Outcome
Does developmental stress affects fitness in animals?

PO: Population, Outcome
Is there a relationship between male plumage and dominance in birds?

PECO

CQUniversity AUSTRALIA

BE WHAT YOU WANT TO BE

cqu.edu.au

PECO AND ITS VARIATIONS

PECO is a variant of PICO which looks at Exposure instead of Intervention. It is also used in formulating clinical questions.

Population	• How is your population defined? (e.g. age, gender, ethnic group …)
Exposure	• What has your population been deliberately or inadvertently exposed to?
Comparison	• What are you comparing this with, e.g. a different level of exposure? (It's ok to leave this one blank if you are not doing a comparison)
Outcome(s)	• Which result(s) are you focusing on or measuring? • What are you hoping to improve?

Other variables that can be added to this PECO framework are:

Duration	• What is the follow up schedule?
Timeframe	• What is the duration of the exposure? • What is the follow up schedule? • Is this a longitudinal study?
Results	• Which reported results are you looking for in the literature on your topic?

Link here

[redacted]

Setting	• Where is your intervention of interest taking place?
Context	• Where is this happening? (e.g. geographical location or service location) • What is the social or cultural context for your population?

Variant frameworks:

- PICOC = **P**opulation, **I**ntervention, **C**omparison, **O**utcome, **C**ontext
- PICOS = **P**opulation, **I**ntervention, **C**omparison, **O**utcome, **S**tudy type
- PICOT = **P**opulation, **I**ntervention, **C**omparison, **O**utcome, **T**ime
- PICOTS = **P**opulation, **I**ntervention, **C**omparison, **O**utcome, **T**iming, **S**etting
- PICOTT = **P**opulation, **I**ntervention, **C**omparison, **O**utcome, **T**ype of question, **T**ype of study
- PIO = **P**opulation, **I**ntervention, **O**utcome

The **PICo** variation is useful for qualitative studies.

Population or Problem	• How is your population defined? (e.g. age, gender, ethnic group …) • Which of their problems, diseases or conditions are you looking at
Interest	• Which experience, activity, process or event are you interesting in focusing on?
Context	• Where is this happening? (Geographical location, e.g. Australia / Service location, e.g. hospital)

This variation of **PICO** is used for evaluating diagnostic tests.

Patient or Participants or Population	• How is your population defined? (e.g. age, gender, ethnic group …) • Which of their problems, diseases or conditions are you looking at
Index tests	• Which test are you evaluating?
Comparator or reference tests	• Which alternative are you comparing this with? (It's ok to leave this one blank if you are not doing a comparison)
Outcome	• Which result are you focusing on or measuring? • What are you hoping to improve?

What should happen (for a pre-planned) search

Broadening..

Narrowing..

Broadening..

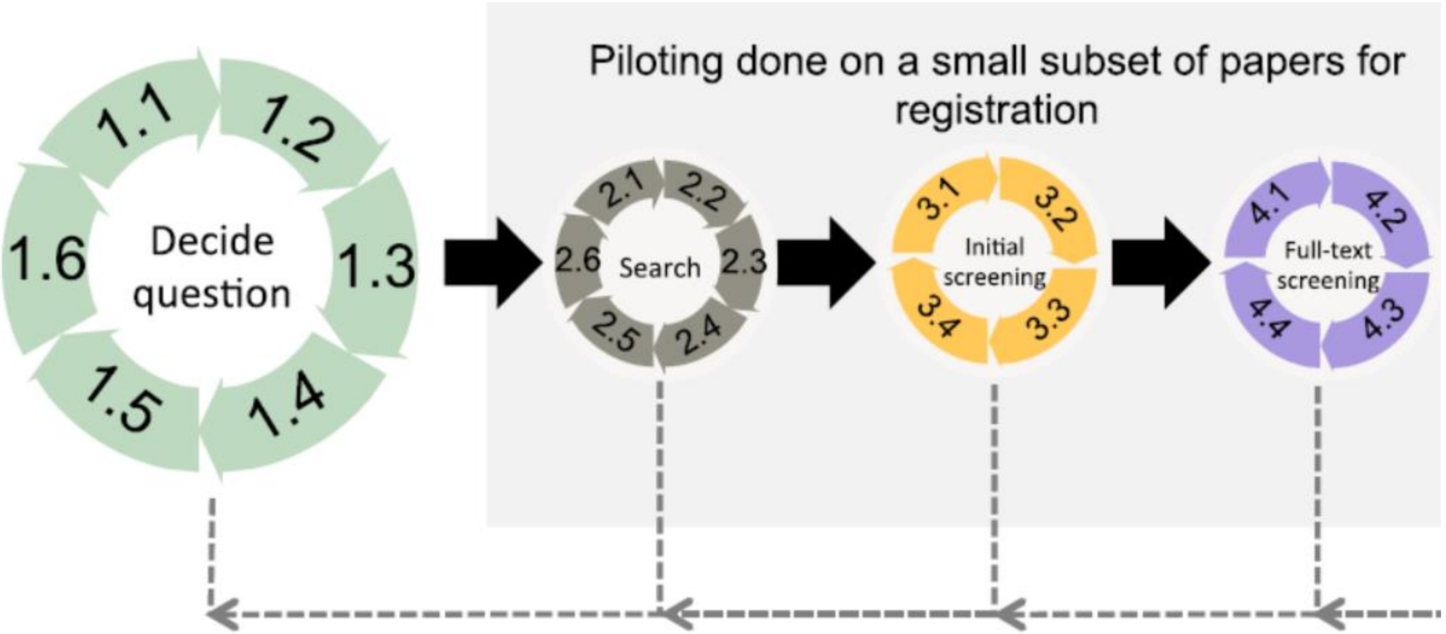
Narrowing..

…

…

…

.. Okay I think this is good



Step 2: Search for the literature

But first, you need to make some decisions and develop a search protocol

“What should I search?”

- Use a personal library (personal knowledge)
- Use Key Papers in the field (reviews/important empirical work)

- Start snowballing (Backwards and Forward search)

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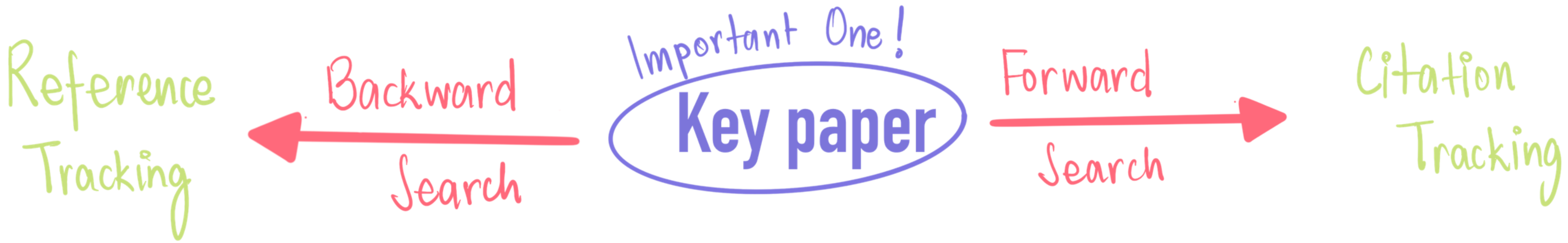


CONNECTED PAPERS   Green plant material in avian nests

Green Plant Material in Avian Nests   Prior works   Derivative works   List view   Filters   More

Origin paper
Green Plant Material in Avian Nests
A. Dubiec, Iga Góźdź, T. Mazgajski   2013

A review of the nest protection hypothesis: does inclusion of fresh green plant material in birds' nests...
James F Scott-Baumann, E. Morgan   2015

Interacting effects of aromatic plants and female age on nest-dwelling ectoparasites and blood-sucking flies in...
G. Tomás, Santiago Merino, J. Puente, Juan Moreno, J....   2012

Nest size and aromatic plants in the nest as sexually selected female traits in blue tits
G. Tomás, S. Merino, J. Puente, J. Moreno, J. Morales, J. R...   2013

Aromatic plants in nests of blue tits: positive effects on nestlings
Adèle Mennerat, P. Perret, Patrice Bourgault, J. Blondel, O....   2009

Aromatic plants in blue tit Cyanistes caeruleus nests: no negative effect on blood-sucking Protocalliphora blow...
Adèle Mennerat, P. Perret, S. Caro, P. Heeb, M. Lambrechts   2008

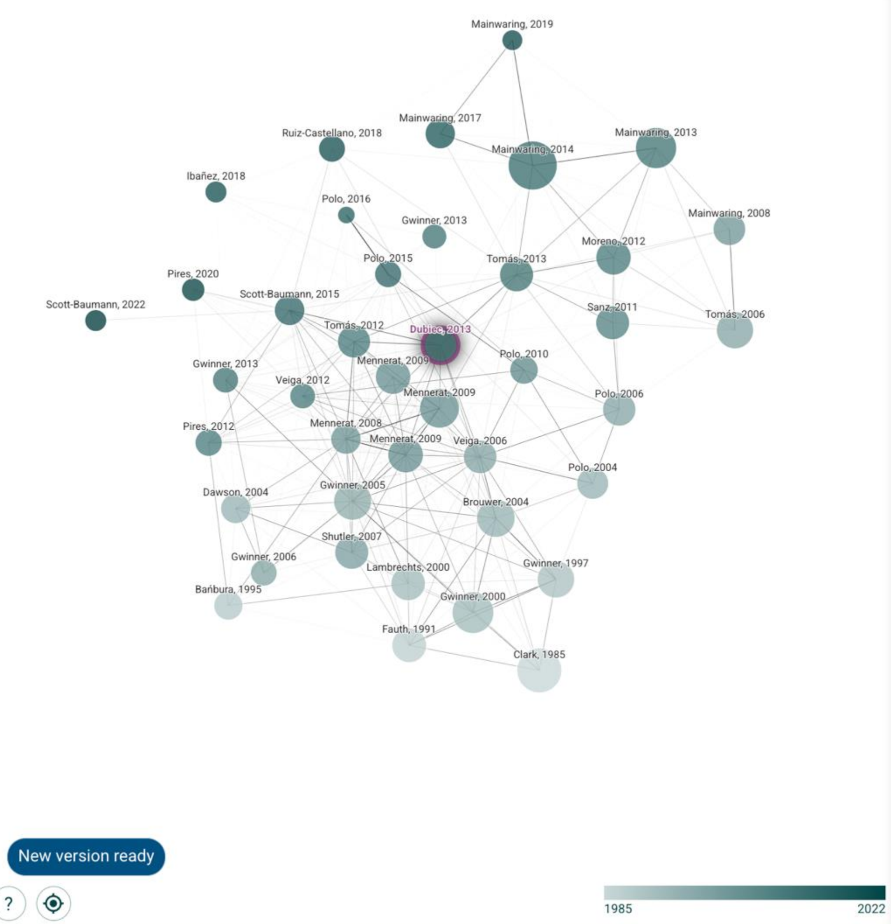
Nest Green Plants as a Male Status Signal and Courtship Display in the Spotless Starling
J. Veiga, V. Polo, J. Viñuela   2006

Aromatic plants in nests of the blue tit Cyanistes caeruleus protect chicks from bacteria
Adèle Mennerat, P. Mirleau, J. Blondel, P. Perret, M....   2009

Green nesting material has a function in mate attraction in the European starling
L. Brouwer, J. Komdeur   2004

European starlings: nestling condition, parasites and green nest material during the breeding season
H. Gwinner, S. Berger   2005

Nest material preferences by spotless starlings



Green Plant Material in Avian Nests
A. Dubiec, Iga Góźdź, T. Mazgajski
2013, Avian Biology Research
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Some bird species place in their nests fresh green plant materials, e.g. leaves, sprigs or branches of herbs, shrubs and trees, which are not a part of the basic nest structure. These additional materials are often characterised by a high content of volatile secondary metabolites and constitute a non-random, small fraction of plants available in the habitat. Several non-mutually exclusive hypotheses have been proposed to explain the function of green material in avian nests with three of them attracting the most attention. The courtship hypothesis proposes that green material is used by males in order to attract females to nesting sites. The nest protection hypothesis posits that birds place in the nest green material rich in volatile compounds to reduce the abundance of parasites and pathogens which, in turn, should mitigate their negative impact on the host. According to the drug hypothesis green material directly positively affects the health and development of nestlings, e.g. through stimulation of some components of the nestlings' immune system. Here, we present an overview of hypotheses explaining the phenomenon of green material in avian nests with a thorough description of three most frequently tested hypotheses and suggest the directions for future studies.

Step 2: Search for the literature

But first, you need to make some decisions and develop a search protocol

“What should I search?”

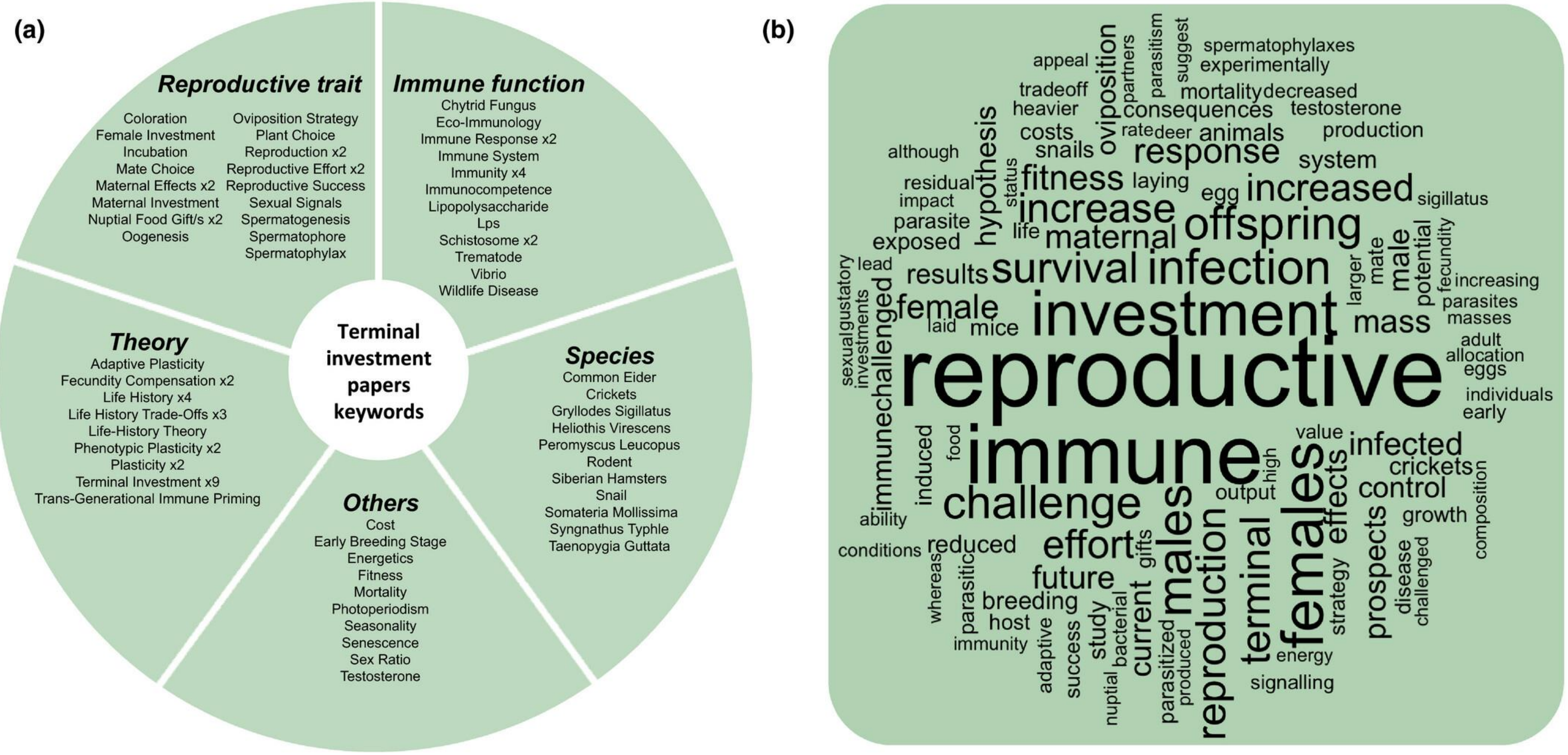
- Use a personal library (personal knowledge)

- Use Key Papers in the field (reviews/important empirical work)

- Start snowballing (Backward and Forward search)

- Use Scoping search and Literature maps

Step 2: Search for the literature



Foo et al. 2021

Step 2: Search for the literature

“Where do I search?”

Many literature databases:

- Web of Science (a platform of databases)
- Scopus
- PubMed
- Google Scholar
- … many more (Lens.org, Semantic Scholar: see Gusenbauer and Haddaway 2020)

Grey Literature (Unpublished data, Thesis, Reports, Pre-prints)

- EThOS [UK theses],
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(("status signal*" OR "badge*") AND "sparrow*")

Google Scholar

(("status signal*" OR "badge*") AND "sparrow*")

Articles

About 9,810 results (**0.07** sec)

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Since 2025
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[PDF] Evolution and maintenance of social status-signaling badges — [PDF] publicationslist.org

MJ Whiting, KA Nagy, PW Bateman - Lizard social behavior, 2003 - publicationslist.org

… In the absence of honest phenotypic lim itations, the Trojan **sparrow strategy** can successfully invade honest populations. Result: mixed fighting strategies, badges-of-status model …

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The cost of dominance and advantage of subordination in a **badge signaling** system — [PDF] jstor.org

S Rohwer, PW Ewald - Evolution, 1981 - JSTOR

… Past manipulations of the Harris' **sparrow badge** have provided two clues useful for this work… that the truthfulness of the Harris' **sparrow badge** was maintained as a consequence of …

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It is a little different from the rest, and maybe not the best choice

- High recall and low precision
- Low reproducibility
- Low transparency
- Potential relevance bias
- No Boolean operators
- No wildcards
- Download batch-by-batch

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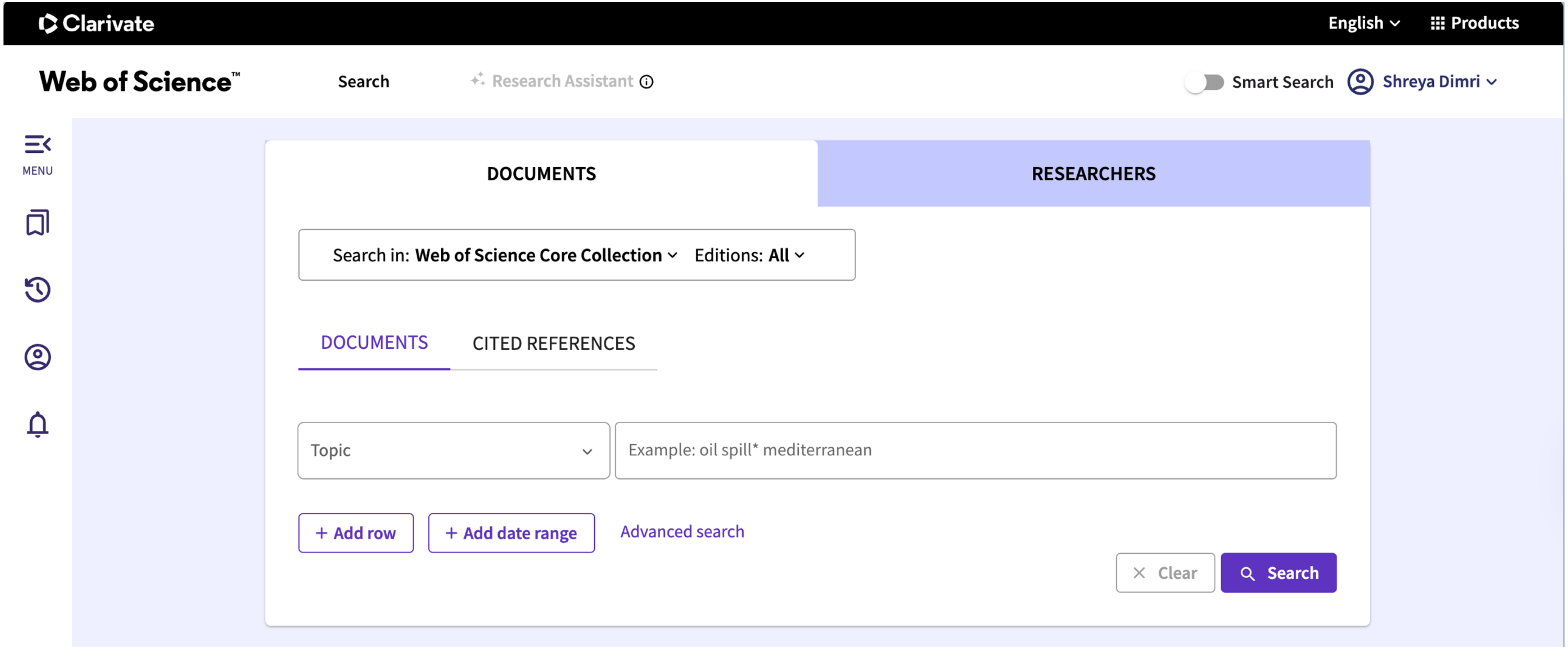
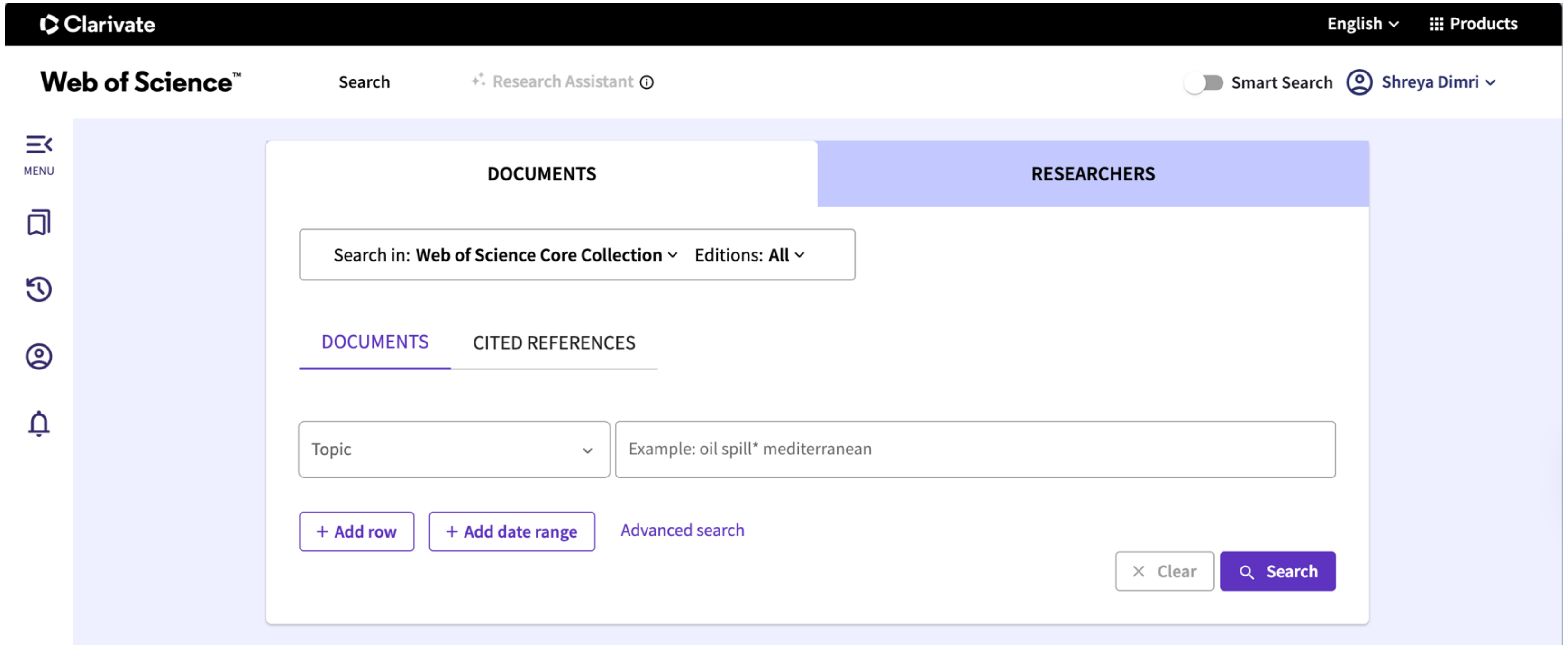
Step 2: Search for the literature

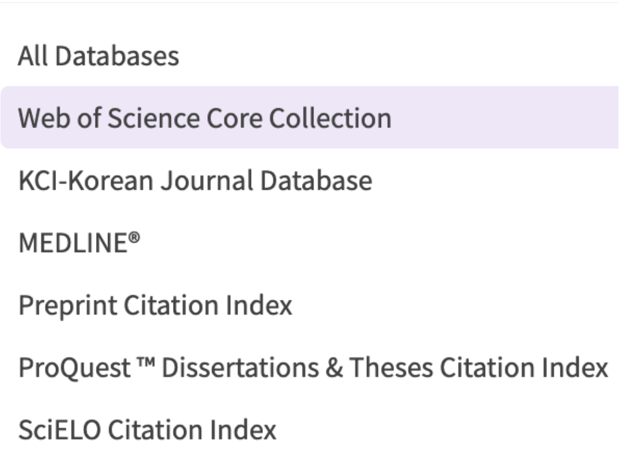
“Where do I search?”

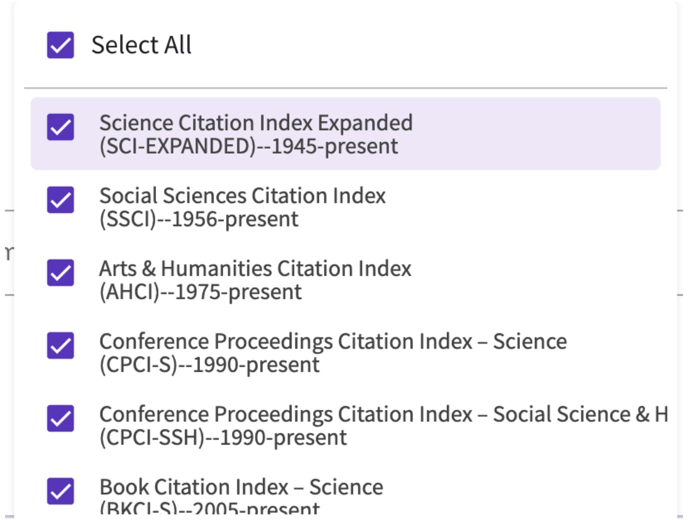


Step 2: Search for the literature

"Where do I search?"







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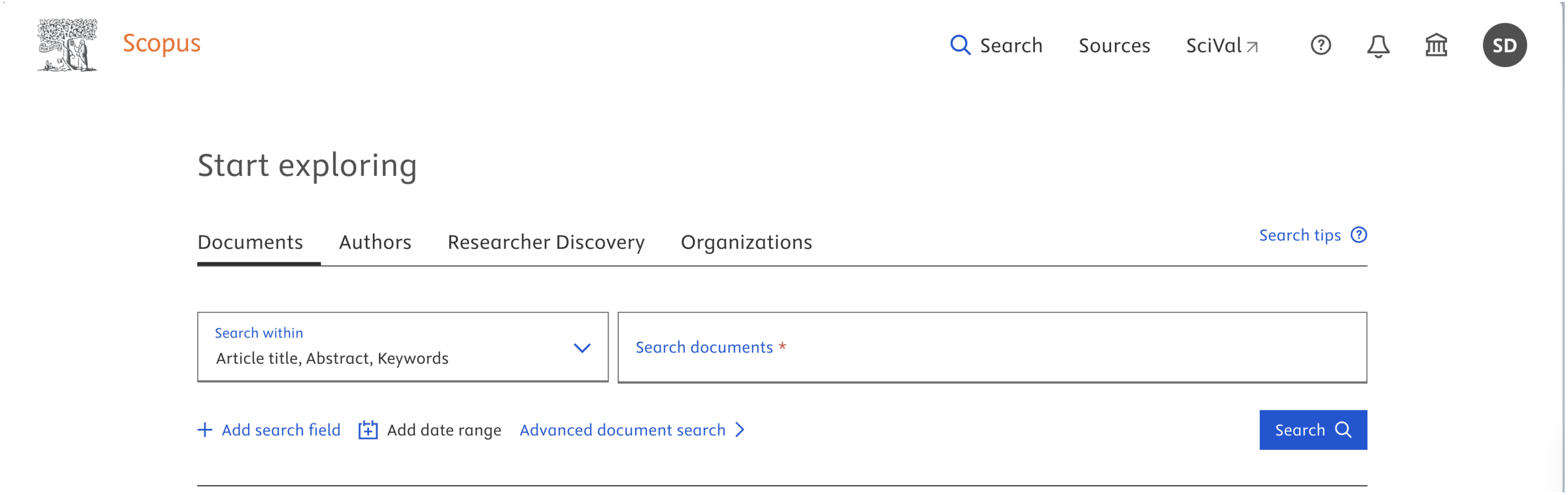
“Where do I search?”



Step 2: Search for the literature

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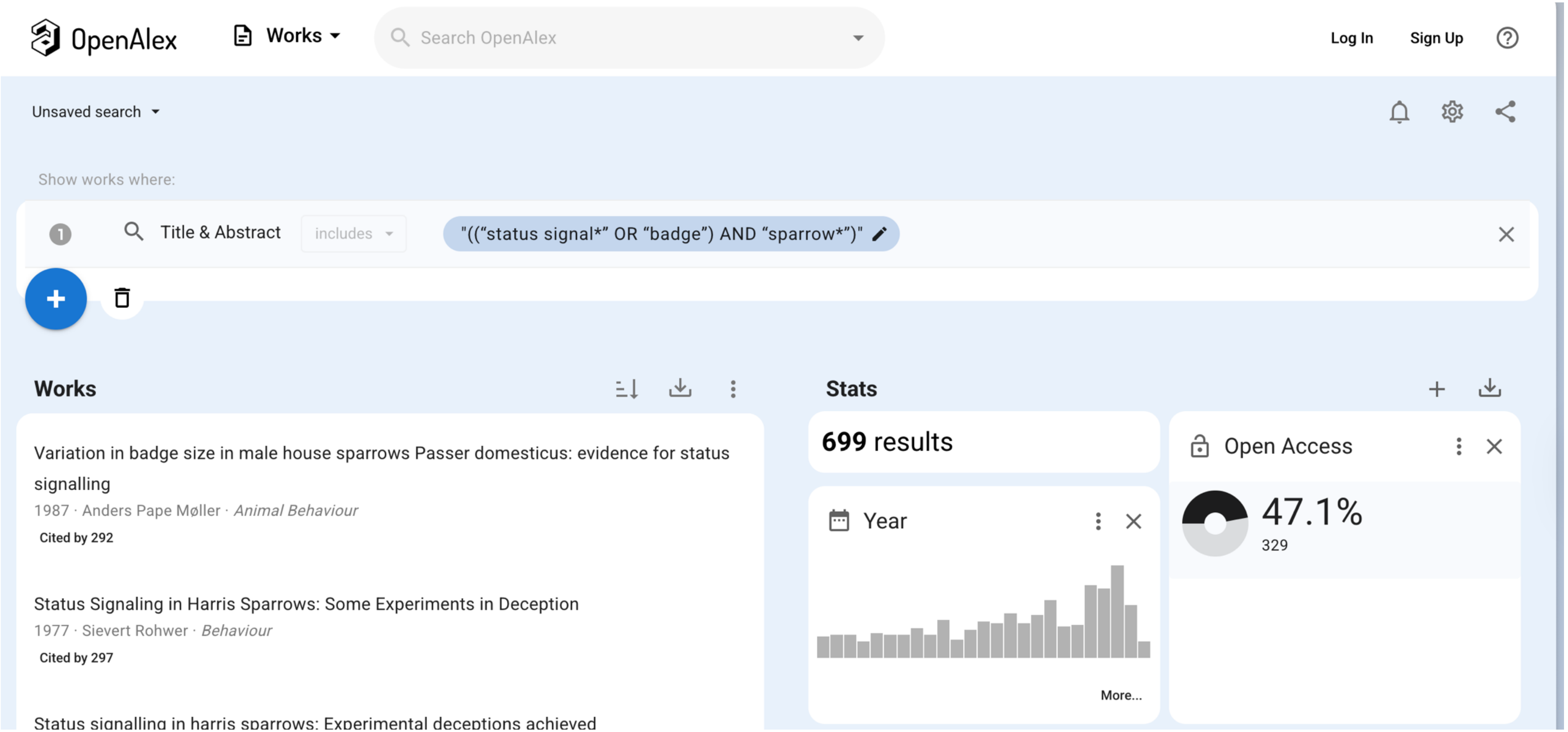


Step 2: Search for the literature

"How do I search?"

Boolean operators

AND, OR, NOT, NEAR

Wildcard symbols

* : any character(s) or no character

? : any single character

$: any single character or no character

An example of search

“What is the adaptive significance of fresh green material added by birds?”

Let’s say I want to look at the experimental evidence

An example of search

"What is the adaptive significance of fresh green material added by birds?"

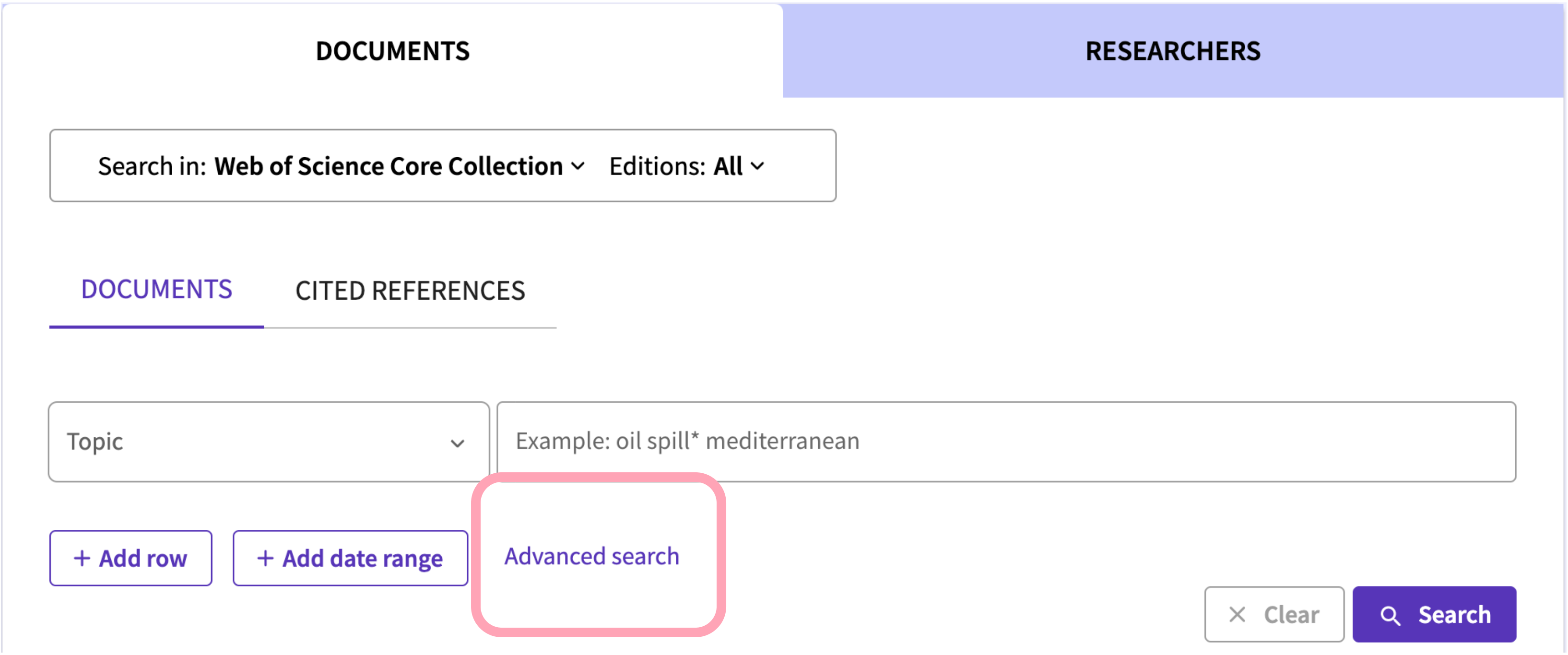
Let's say I want to look at the experimental evidence

An example of search

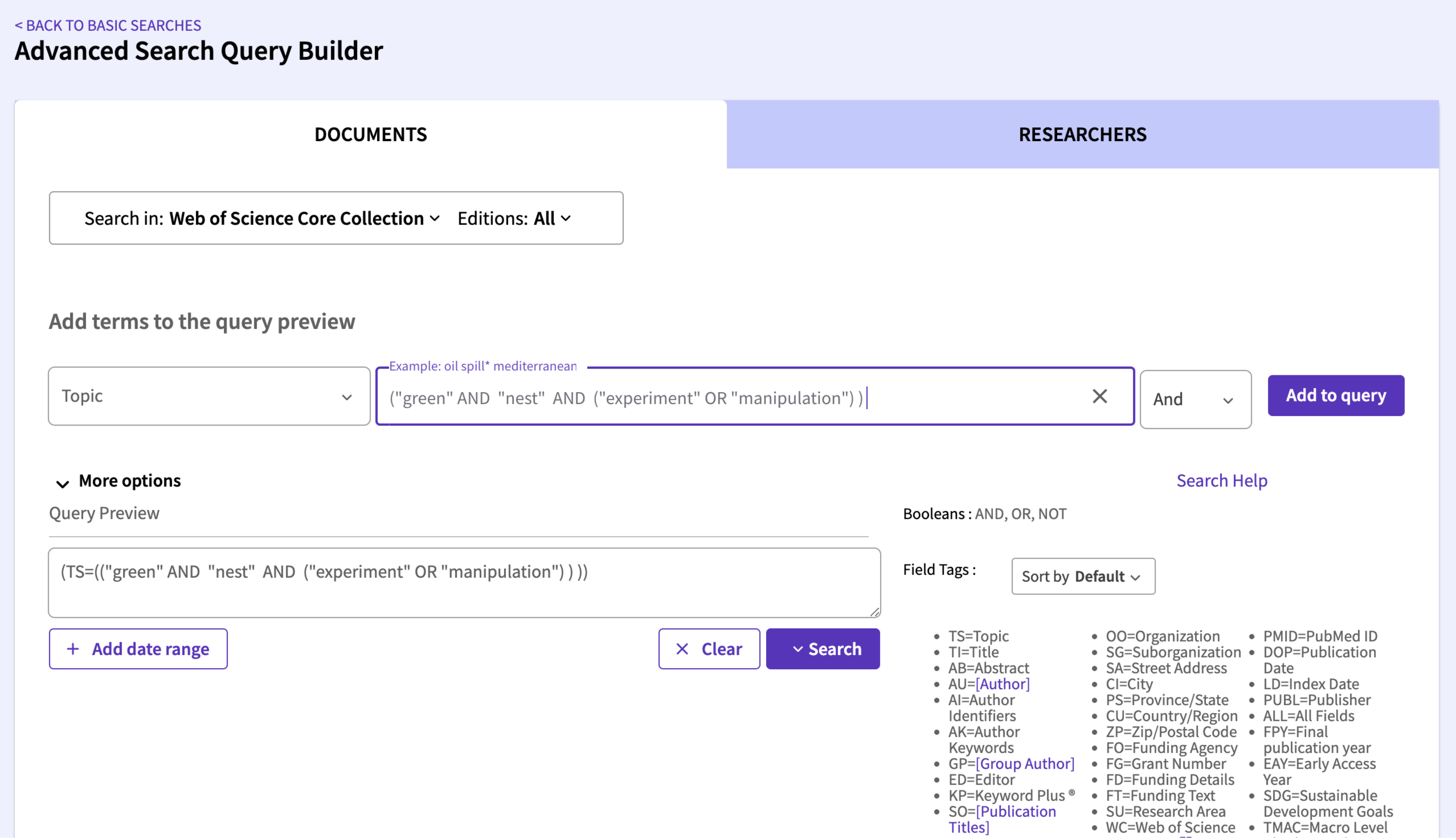
“What is the adaptive significance of fresh green material added by birds?”

Let’s say I want to look at the experimental evidence

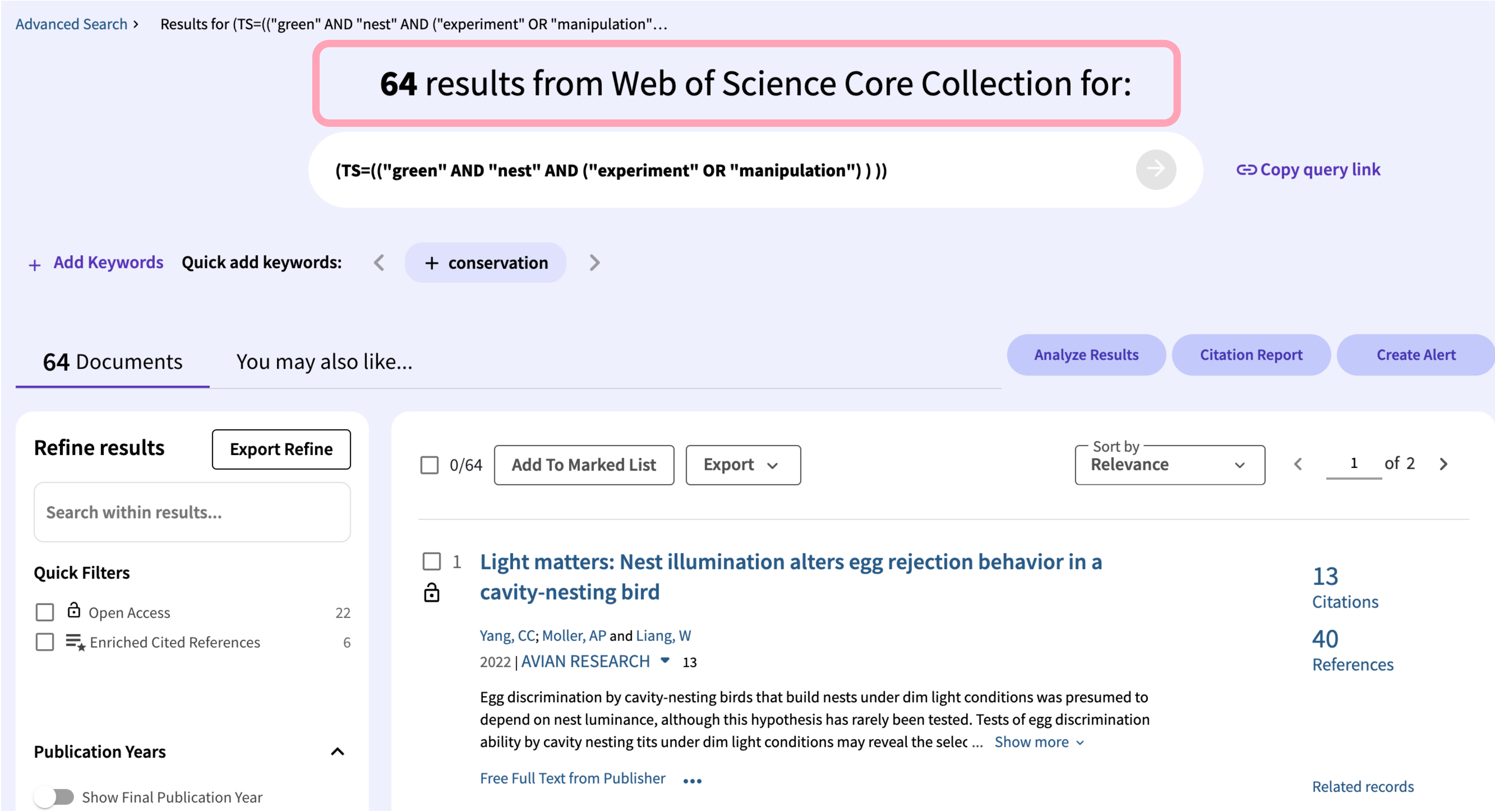
("green" AND "nest" AND ("experiment" OR "manipulation"))



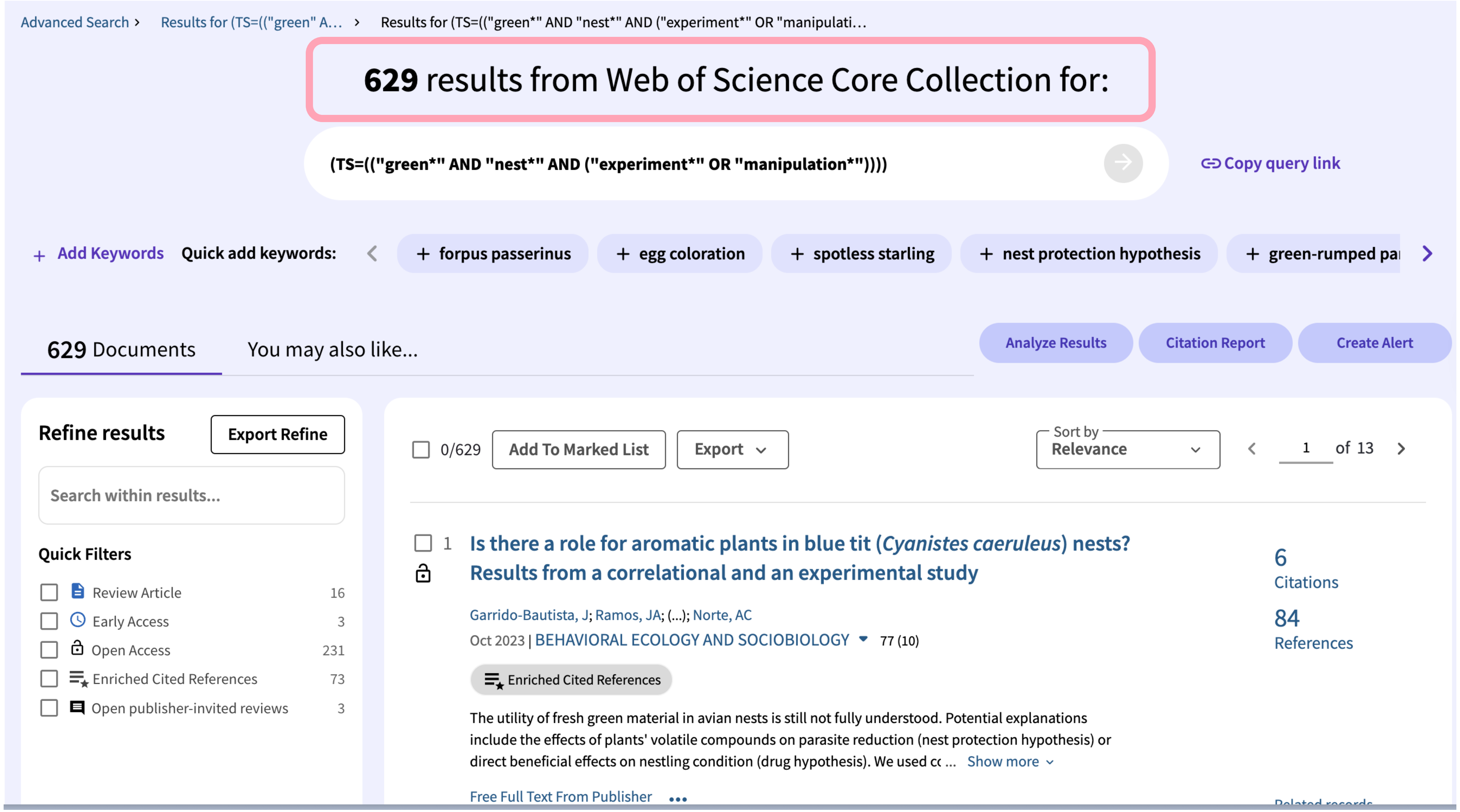
("green" AND "nest" AND ("experiment" OR "manipulation"))



("green" AND "nest" AND ("experiment" OR "manipulation"))

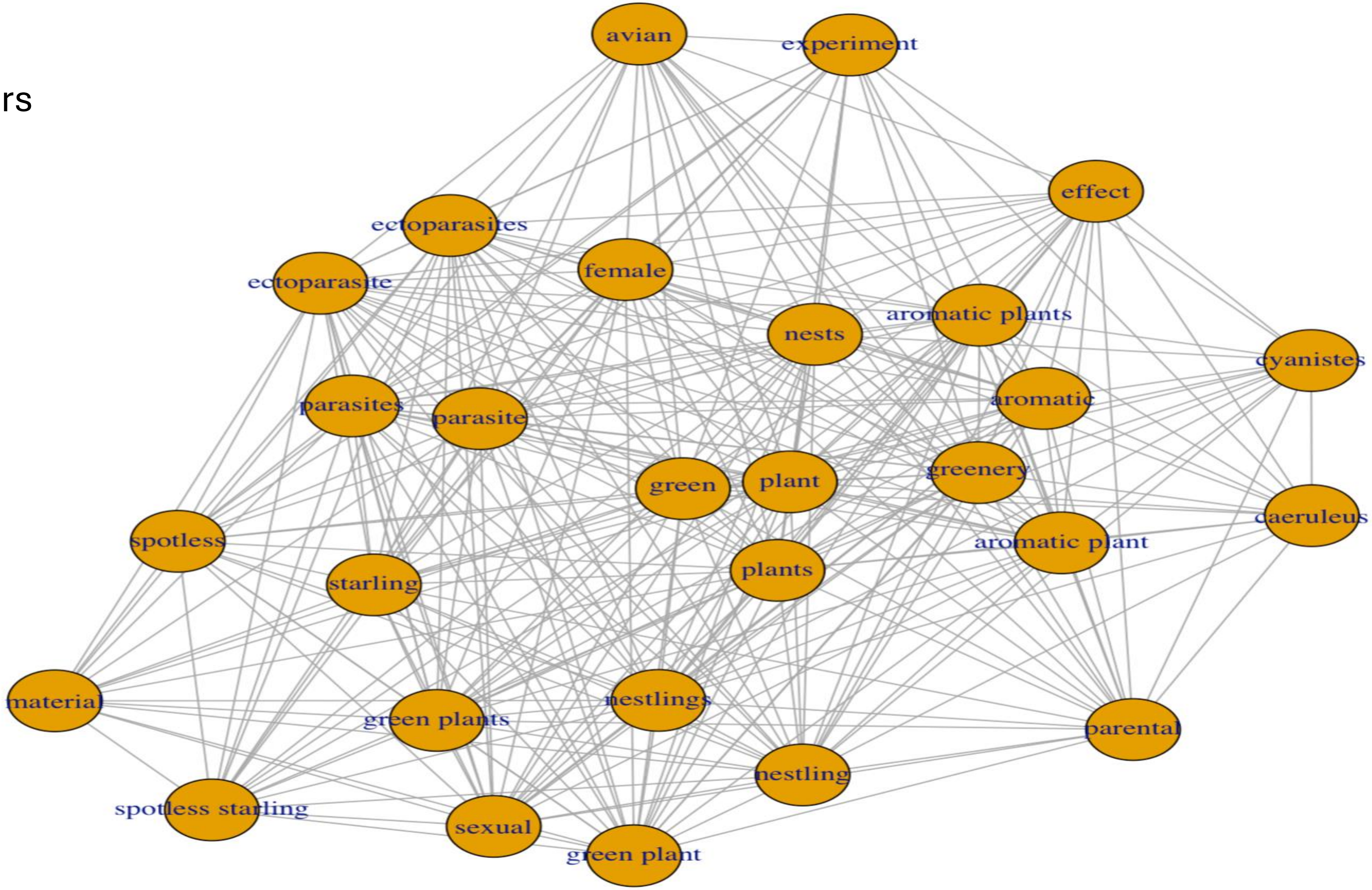


("green*" AND "nest*" AND ("experiment*" OR "manipulation*"))

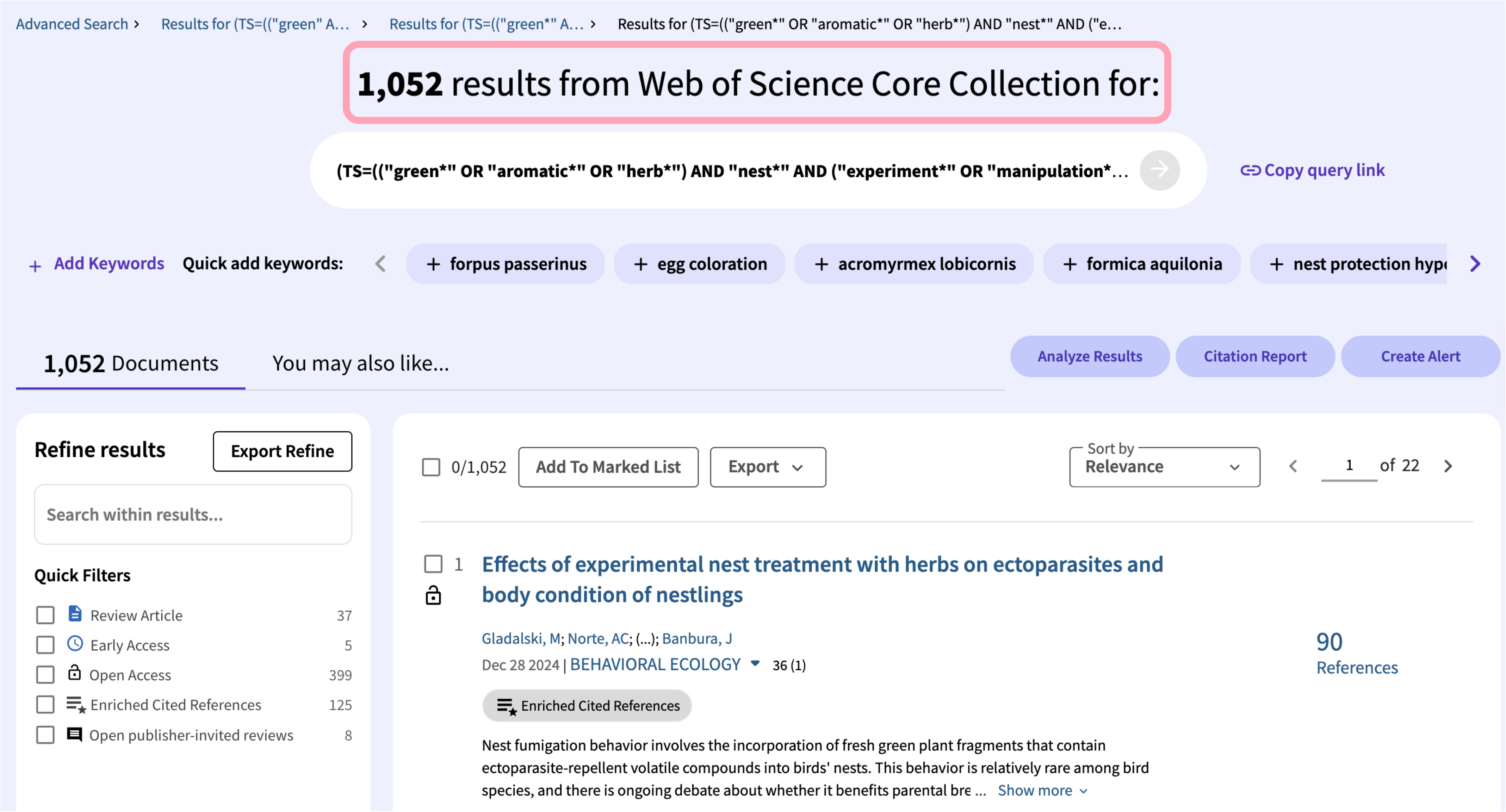


("green*" AND "nest*" AND ("experiment*" OR "manipulation*"))

litsearchr package
Network plot using 15 papers
from our own library



(("green*” OR “aromatic*” OR “herb*”) AND "nest*" AND ("experiment*" OR "manipulation*"))



[redacted]

TS=(("green*" OR "herb*" OR "aromatic*") AND "nest*" AND ("bird*" OR "aves" OR "avian" OR "ornithol*" OR "passerine*" OR "passeriform*" OR "songbird*" OR "Accipiter" OR "Aegypius" OR "Aquila" OR "Aviceda" OR "Busarellus" OR "Butastur" OR "Buteo" OR "Buteogallus" OR "Chondrohierax" OR "Circaetus" OR "Circus" OR "Clanga" OR "Elanoides" OR "Elanus" OR "Gampsonyx" OR "Geranoaetus" OR "Geranospiza" OR "Gypaetus" OR "Gypohierax" OR "Gyps" OR "Haliaeetus" OR "Haliastur" OR "Harpagus" OR "Harpia" OR "Hieraaetus" OR "Ictinaetus" OR "Ictinia" OR "Kaupifalco" OR "Leptodon" OR "Leucopternis" OR "Lophotriorchis" OR "Milvus" OR "Morphnarchus" OR "Morphnus" OR "Necrosyrtes" OR "Neophron" OR "Nisaetus" OR "Parabuteo" OR "Pernis" OR "Pithecophaga" OR "Polemaetus" OR "Pseudastur" OR "Rostrhamus" OR "Rupornis" OR "Sarcogyps" OR "Spilornis" OR "Spizaetus" OR "Stephanoaetus" OR "Terathopius" OR "Torgos" OR "Trigonoceps" OR "Cathartes" OR "Coragyps" OR "Gymnogyps" OR "Sarcoramphus" OR "Vultur" OR "Pandion" 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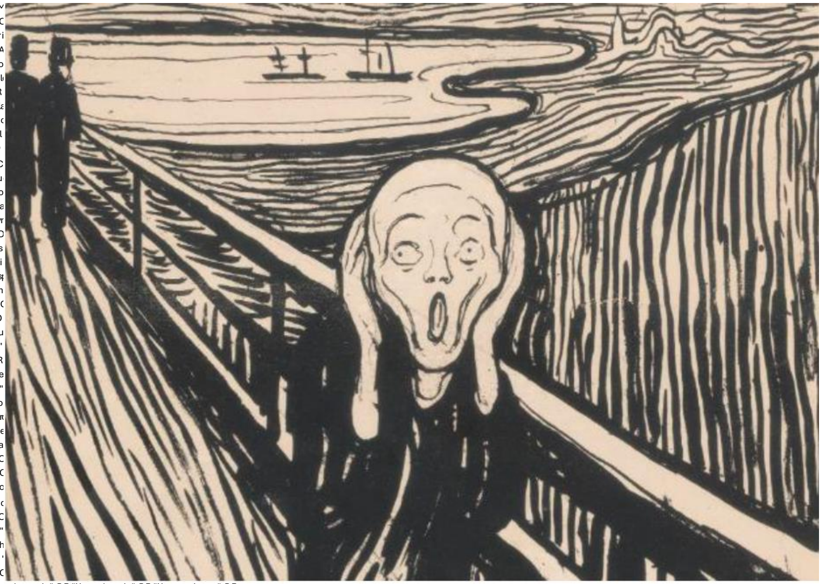
"Heteralocha" OR "Philesturnus" OR "Calyptophilus" OR "Campephaga" OR "Ceblepyris" OR "Coracina" OR "Edolisoma" OR "Lalage" OR "Malindangia" OR "Pericrocotus" OR "Amaurospiza" OR "Cardinalis" OR "Caryothraustes" OR "Chlorothraupis" OR "Cyanocompsa" OR "Cyanoloxia" OR "Granatellus" OR "Habia" OR "Passerina" OR
"Certhia" OR "Salpornis" OR "Abroscopus" OR "Cettia" OR "Horornis" OR "Phyllergates" OR "Tesia" OR "Urosphena" OR "Chloropsis" OR "Cinclus" OR "Apalis" OR "Artisornis" OR "Camaroptera" OR "Cisticola" OR "Eremomela" OR "Heliolais" OR "Hypergerus" OR "Neomixis" OR "Oreolais" OR "Orthotomus" OR "Phragmacia" OR "Poliola
OR "Climacteris" OR "Cormobates" OR "Conopophaga" OR "Struthidea" OR "Aphelocoma" OR "Calocitta" OR "Cissa" OR "Coloeus" OR "Corvus" OR "Cyanocitta" OR "Cyanocorax" OR "Cyanolyca" OR "Cyanopica" OR "Dendrocitta" OR "Garrulus" OR "Gymnorhinus" OR "Nucifraga" OR "Perisoreus" OR "Pica" OR "Platylophus" OR "Podo
"Urocissa" OR "Ampelioides" OR "Ampelion" OR "Carpornis" OR "Cephalopterus" OR "Conioptilon" OR "Cotinga" OR "Doliornis" OR "Gymnoderus" OR "Haematoderus" OR "Lipaugus" OR "Perissocephalus" OR "Phoenicircus" OR "Phytotoma" OR "Pipreola" OR "Porphyrolaema" OR "Procnias" OR "Pyroderus" OR "Querula" OR "Rupicol
"Dasyornis" OR "Dicaeum" OR "Prionochilus" OR "Dicrurus" OR "Donacobius" OR "Dulus" OR "Emberiza" OR "Amadina" OR "Amandava" OR "Cryptospiza" OR "Erythrura" OR "Estrilda" OR "Euodice" OR "Euschistospiza" OR "Lagonosticta" OR "Lonchura" OR "Mandingoa" OR "Neochmia" OR "Nesocharis" OR "Nigrita" OR "Ortygospiza"
"Spermophaga" OR "Stagonopleura" OR "Taeniopygia" OR "Uraeginthus" OR "Calyptomena" OR "Cymbirhynchus" OR "Eurylaimus" OR "Psarisomus" OR "Serilophus" OR "Smithornis" OR "Chamaeza" OR "Formicarius" OR "Acanthis" OR "Agraphospiza" OR "Akialoa" OR "Bucanetes" OR "Carduelis" OR "Carpodacus" OR "Chloris" OR "C
"Coccothraustes" OR "Crithagra" OR "Drepanis" OR "Eophona" OR "Euphonia" OR "Fringilla" OR "Haemorhous" OR "Hemignathus" OR "Himatione" OR "Leucosticte" OR "Linaria" OR "Linurgus" OR "Loxia" OR "Loxioides" OR "Loxops" OR "Magumma" OR "Melamprosops" OR "Oreomystis" OR "Paroreomyza" OR "Pinicola" OR "Procardu
OR "Pyrrhula" OR "Rhodopechys" OR "Rhodospiza" OR "Serinus" OR "Spinus" OR "Telespiza" OR "Anabacerthia" OR "Anabazenops" OR "Ancistrops" OR "Anumbius" OR "Aphrastura" OR "Asthenes" OR "Automolus" OR "Berlepschia" OR "Campylorhamphus" OR "Certhiasomus" OR "Certhiaxis" OR "Cinclodes" OR "Clibanornis" OR "Co
"Dendrexetastes" OR "Dendrocincla" OR "Dendrocolaptes" OR "Dendroplex" OR "Drymornis" OR "Drymotoxeres" OR "Furnarius" OR "Geocerthia" OR "Geositta" OR "Glyphorynchus" OR "Heliobletus" OR "Hellmayrea" OR "Hylexetastes" OR "Lepidocolaptes" OR "Leptasthenura" OR "Limnoctites" OR "Limnornis" OR "Lochmias" OR "Ma
OR "Microxenops" OR "Nasica" OR "Ochetorhynchus" OR "Phacellodomus" OR "Philydor" OR "Phleocryptes" OR "Premnoplex" OR "Premnornis" OR "Pseudasthenes" OR "Pseudocolaptes" OR "Pseudoseisura" OR "Pygarrhichas" OR "Roraimia" OR "Schoeniophylax" OR "Sclerurus" OR "Sittasomus" OR "Spartonoica" OR "Sylviorthorhyn
OR "Thripadectes" OR "Thripophaga" OR "Upucerthia" OR "Xenerpestes" OR "Xenops" OR "Xiphocolaptes" OR "Xiphorhynchus" OR "Grallaria" OR "Grallaricula" OR "Hylopezus" OR "Myrmothera" OR "Alopochelidon" OR "Atticora" OR "Cecropis" OR "Delichon" OR "Haplochelidon" OR "Hirundo" OR "Neochelidon" OR "Notiochelidon"
"Progne" OR "Psalidoprocne" OR "Pseudhirundo" OR "Pseudochelidon" OR "Ptyonoprogne" OR "Riparia" OR "Stelgidopteryx" OR "Tachycineta" OR "Hylioта" OR "Hypocolius" OR "Agelaioides" OR "Agelaius" OR "Agelasticus" OR "Amblycercus" OR "Amblyramphus" OR "Anumara" OR "Cacicus" OR "Chrysomus" OR "Curaeus" OR "Dives
"Gymnomystax" OR "Hypopyrrhus" OR "Icterus" OR "Lampropsar" OR "Leistes" OR "Macroagelaius" OR "Molothrus" OR "Nesopsar" OR "Oreopsar" OR "Psarocolius" OR "Pseudoleistes" OR "Quiscalus" OR "Sturnella" OR "Xanthocephalus" OR "Xanthopsar" OR "Icteria" OR "Ifrita" OR "Irena" OR "Corvinella" OR "Eurocephalus" OR "Lani
"Grammatoptila" OR "Heterophasia" OR "Ianthocincla" OR "Laniellus" OR "Leiothrix" OR "Liocichla" OR "Minla" OR "Pterorhinus" OR "Trochalopteron" OR "Turdoides" OR "Bradypterus" OR "Cincloramphus" OR "Helopsaltes" OR "Locustella" OR "Megalurus" OR "Poodytes" OR "Machaerirhynchus" OR "Grauena" OR "Hylia" OR "Macros
OR "Dryoscopus" OR "Laniarius" OR "Malaconotus" OR "Nilaus" OR "Rhodophoneus" OR "Tchagra" OR "Telophorus" OR "Amytornis" OR "Chenorhamphus" OR "Clytomyias" OR "Malurus" OR "Stipiturus" OR "Melampitta" OR "Melanocharis" OR "Oedistoma" OR "Toxorhamphus" OR "Melanopareia" OR "Acanthagenys" OR "Acanthorhyn
"Entomyzon" OR "Epthianura" OR "Glycifohia" OR "Gymnomyza" OR "Lichmera" OR "Manorina" OR "Melidectes" OR "Melilestes" OR "Meliphaga" OR "Melipotes" OR "Melithreptus" OR "Myzomela" OR "Nesoptilotis" OR "Philemon" OR "Phylidonyris" OR "Prosthemadera" OR "Ptiloprora" OR "Ptilotula" OR "Pycnopygius" OR "Timeliopsis"
OR "Dumetella" OR "Margarops" OR "Melanoptila" OR "Melanotis" OR "Mimus" OR "Oreoscoptes" OR "Ramphocinclus" OR "Toxostoma" OR "Lamprospiza" OR "Mitrospingus" OR "Moho" OR "Mohoua" OR "Arses" OR "Carterornis" OR "Clytorhynchus" OR "Grallina" OR "Hypothymis" OR "Monarcha" OR "Myiagra" OR "Symposiachrus" O
"Macronyx" OR "Motacilla" OR "Alethe" OR "Anthipes" OR "Brachypteryx" OR "Calliope" OR "Campicoloides" OR "Cercotrichas" OR "Chamaetylas" OR "Copsychus" OR "Cossypha" OR "Cyanoptila" OR "Cyornis" OR "Enicurus" OR "Erithacus" OR "Eumyias" OR "Ficedula" OR "Fraseria" OR "Irania" OR "Larvivora" OR "Leonardina" OR "Lu
"Myiomela" OR "Myioparus" OR "Myophonus" OR "Myrmecocichla" OR "Niltava" OR "Oenanthe" OR "Phoenicurus" OR "Pogonocichla" OR "Saxicola" OR "Sheppardia" OR "Sholicola" OR "Stiphrornis" OR "Tarsiger" OR "Thamnolaea" OR "Vauriella" OR "Aethopyga" OR "Anabathmis" OR "Anthreptes" OR "Arachnothera" OR "Chalcomitra"
"Hedydipna" OR "Kurochkinegramma" OR "Leptocoma" OR "Nectarinia" OR "Daphoenositta" OR "Nesospingus" OR "Nicator" OR "Notiomystis" OR "Aleadryas" OR "Oriolus" OR "Pitohui" OR "Sphecotheres" OR "Turnagra" OR "Colluricincla" OR "Falcunculus" OR "Melanorectes" OR "Pachycephala" OR "Pseudorectes" OR "Panurus" OR
"Paradisaea" OR "Ptiloris" OR "Paramythia" OR "Pardalotus" OR "Baeolophus" OR "Cephalopyrus" OR "Cyanistes" OR "Lophophanes" OR "Machlolophus" OR "Melaniparus" OR "Melanochlora" OR "Pardaliparus" OR "Parus" OR "Periparus" OR "Poecile" OR "Pseudopodoces" OR "Sittiparus" OR "Sylviparus" OR "Basileuterus" OR "Carde
"Helmitheros" OR "Leiothlypis" OR "Limnothlypis" OR "Mniotilta" OR "Myioborus" OR "Myiothlypis" OR "Oporornis" OR "Oreothlypis" OR "Parkesia" OR "Protonotaria" OR "Seiurus" OR "Setophaga" OR "Vermivora" OR "Aimophila" OR "Ammodramus" OR "Amphispiza" OR "Arremon" OR "Arremonops" OR "Artemisiospiza" OR "Atlapetes"
"Junco" OR "Melospiza" OR "Melozone" OR "Oreothraupis" OR "Oriturus" OR "Passerculus" OR "Passerella" OR "Peucaea" OR "Pezopetes" OR "Pipilo" OR "Pooecetes" OR "Pselliophorus" OR "Rhynchospiza" OR "Spizella" OR "Spizelloides" OR "Xenospiza" OR "Zonotrichia" OR "Carpospiza" OR "Gymnoris" OR "Hypocryptadius" OR "Mo
OR "Pyrgilauda" OR "Alcippe" OR "Gampsorhynchus" OR "Illadopsis" OR "Jabouilleia" OR "Kenopia" OR "Laticilla" OR "Malacocincla" OR "Malacopteron" OR "Napothera" OR "Pellorneum" OR "Rimator" OR "Trichastoma" OR "Amalocichla" OR "Drymodes" OR "Eopsaltria" OR "Eugerygone" OR "Heteromyias" OR "Melanodryas" OR "Micr
"Peneoenanthe" OR "Peneothello" OR "Petroica" OR "Poecilodryas" OR "Tregellasia" OR "Peucedramus" OR "Microligea" OR "Phaenicophilus" OR "Xenoligea" OR "Phylloscopus" OR "Picathartes" OR "Antilophia" OR "Ceratopipra" OR "Chiroxiphia" OR "Chloropipo" OR "Corapipo" OR "Cryptopipo" OR "Heterocercus" OR "Ilicura" OR "Le
OR "Neopelma" OR "Pipra" OR "Pseudopipra" OR "Tyranneutes" OR "Xenopipo" OR "Erythropitta" OR "Hydrornis" OR "Pitta" OR "Batis" OR "Platysteira" OR "Anaplectes" OR "Bubalornis" OR "Euplectes" OR "Malimbus" OR "Plocepasser" OR "Ploceus" OR "Quelea" OR "Sporopipes" OR "Pnoepyga" OR "Microbates" OR "Polioptila" OR "Ra
"Cinclosoma" OR "Ptilorrhoa" OR "Phainopepla" OR "Phainoptila" OR "Ptiliogonys" OR "Ailuroedus" OR "Amblyornis" OR "Ptilonorhynchus" OR "Sericulus" OR "Alophoixus" OR "Andropadus" OR "Arizelocichla" OR "Atimastillas" OR "Baeopogon" OR "Bleda" OR "Chlorocichla" OR "Criniger" OR "Eurillas" OR "Hemixos" OR "Hypsipetes" O
"Pycnonotus" OR "Spizixos" OR "Stelgidillas" OR "Tricholestes" OR "Regulus" OR "Anthoscopus" OR "Auriparus" OR "Remiz" OR "Rhagologus" OR "Eleoscytalopus" OR "Liosceles" OR "Psilorhamphus" OR "Pteroptochos" OR "Scelorchilus" OR "Scytalopus" OR "Teledromas" OR "Rhipidura" OR "Rhodinocichla" OR "Sapayoa" OR "Sitta" O
"Elminia" OR "Acridotheres" OR "Agropsar" OR "Ampeliceps" OR "Aplonis" OR "Basilornis" OR "Cinnyricinclus" OR "Creatophora" OR "Gracula" OR "Gracupica" OR "Hartlaubius" OR "Hylopsar" OR "Lamprotornis" OR "Leucopsar" OR "Mino" OR "Notopholia" OR "Onychognathus" OR "Pastor" OR "Poeoptera" OR "Rhabdornis" OR "Sarco
OR "Streptocitta" OR "Sturnia" OR "Sturnus" OR "Chamaea" OR "Chleuasicus" OR "Chloropeta" OR "Chrysomma" OR "Conostoma" OR "Fulvetta" OR "Lioparus" OR "Paradoxornis" OR "Pseudoalcippe" OR "Psittiparus" OR "Rhopophilus" OR "Sinosuthora" OR "Suthora" OR "Sylvia" OR "Teretistris" OR "Aleletos" OR "Ampelornis" OR "Apro
"Cercomacroides" OR "Cymbilaimus" OR "Dichrozona" OR "Drymophila" OR "Dysithamnus" OR "Epinecrophylla" OR "Euchrepomis" OR "Formicivora" OR "Frederickena" OR "Gymnocichla" OR "Gymnopithys" OR "Hafferia" OR "Herpsilochmus" OR "Hylophylax" OR "Hypocnemis" OR "Hypocnemoides" OR "Hypoedaleus" OR "Isleria" O
"Myrmeciza" OR "Myrmelastes" OR "Myrmoborus" OR "Myrmochanes" OR "Myrmoderus" OR "Myrmophylax" OR "Myrmorchilus" OR "Myrmornis" OR "Myrmotherula" OR "Neoctantes" OR "Oneillornis" OR "Percnostola" OR "Phaenostictus" OR "Phlegopsis" OR "Pithys" OR "Poliocrania" OR "Pygiptila" OR "Pyriglena" OR "Rhegmatorhina"
"Sciaphylax" OR "Sclateria" OR "Sipia" OR "Taraba" OR "Thamnistes" OR "Thamnomanes" OR "Thamnophilus" OR "Willisornis" OR "Xenornis" OR "Anisognathus" OR "Bangsia" OR "Buthraupis" OR "Calochaetes" OR "Catamblyrhynchus" OR "Catamenia" OR "Charitospiza" OR "Chlorochrysa" OR "Chlorophanes" OR "Chlorornis" OR "Ch
"Coereba" OR "Compsospiza" OR "Compsothraupis" OR "Conirostrum" OR "Conothraupis" OR "Coryphospingus" OR "Creurgops" OR "Cyanerpes" OR "Cyanicterus" OR "Cypsnagra" OR "Dacnis" OR "Delothraupis" OR "Diglossa" OR "Diuca" OR "Dolospingus" OR "Donacospiza" OR "Dubusia" OR "Emberizoides" OR "Embernagra" OR
OR "Haplospiza" OR "Hemispingus" OR "Hemithraupis" OR "Heterospingus" OR "Incaspiza" OR "Iridophanes" OR "Iridosornis" OR "Lanio" OR "Lophospingus" OR "Loxigilla" OR "Loxipasser" OR "Melanodera" OR "Melanospiza" OR "Melopyrrha" OR "Nemosia" OR "Nephelornis" OR "Oreomanes" OR "Oryzoborus" OR "Parkerthraustes" O
OR "Poospiza" OR "Pyrrhocoma" OR "Ramphocelus" OR "Rhodospingus" OR "Saltator" OR "Saltatricula" OR "Schistochlamys" OR "Sericossypha" OR "Sicalis" OR "Sporophila" OR "Stephanophorus" OR "Tachyphonus" OR "Tangara" OR "Tersina" OR "Thlypopsis" OR "Thraupis" OR "Tiaris" OR "Trichothraupis" OR "Volatinia" OR "Wetmorethraupis" OR "Xenodacnis" OR "Xenospingus" OR "Tichodroma" OR "Macronus" OR "Pomatorhinus" OR "Spelaeornis" OR "Stachyridopsis" OR "Stachyris" OR "Iodopleura" OR "Laniisoma" OR "Laniocera" OR "Myiobius" OR "Onychorhynchus" OR "Oxyruncus" OR "Pachyramphus" OR "Schiffornis" OR "Terenotriccus" OR "Tityra" OR "Xenopsaris" OR "Campylorhynchus" OR "Cantorchilus" OR "Catherpes" OR "Cinnycerthia" OR "Cistothorus" OR "Cyphorhinus" OR "Henicorhina" OR "Microcerculus" OR "Odontorchilus" OR "Pheugopedius" OR "Salpinctes" OR "Thryomanes" OR "Thryophilus" OR "Thryothorus" OR "Troglodytes" OR "Uropsila" OR "Catharus" OR "Cichlopsis" OR "Entomodestes" OR "Geokichla" OR "Hylocichla" OR "Ixoreus" OR "Myadestes" OR "Neocossyphus" OR "Ridgwayia" OR "Sialia" OR "Stizorhina" OR "Turdus" OR "Zoothera" OR "Agriornis" OR "Alectrurus" OR "Anairetes" OR "Aphanotriccus" OR "Arundinicola" OR "Atalotriccus" OR "Attila" OR "Camptostoma" OR "Capsiempis" OR "Casiornis" OR "Cnemarchus" OR "Cnemotriccus" OR "Cnipodectes" OR "Colonia" OR "Colorhamphus" OR "Conopias" OR "Contopus" OR "Corythopis" OR "Culicivora" OR "Deltarhynchus" OR "Elaenia" OR "Empidonax" OR "Empidonomus" OR "Euscarthmus" OR "Fluvicola" OR "Griseotyrannus" OR "Gubernetes" OR "Hemitriccus" OR "Heteroxolmis" OR "Hirundinea" OR "Hymenops" OR "Inezia" OR "Knipolegus" OR "Lathrotriccus" OR "Legatus" OR "Leptopogon" OR "Lessonia" OR "Lophotriccus" OR "Machetornis" OR "Mecocerculus" OR "Megarynchus" OR "Mionectes" OR "Mitrephanes" OR "Muscigralla" OR "Muscisaxicola" OR "Myiarchus" OR "Myiodynastes" OR "Myiopagis" OR "Myiophobus" OR "Myiornis" OR "Myiotheretes" OR "Myiotriccus" OR "Myiozetetes" OR "Neopipo" OR "Neoxolmis" OR "Nephelomyias" OR "Ochthoeca" OR "Ochthornis" OR "Oncostoma" OR "Ornithion" OR "Phaeomyias" OR "Philohydor" OR "Phyllomyias" OR "Phylloscartes" OR "Pipites" OR "Pitangus" OR "Platyrinchus" OR "Poecilotriccus" OR "Pogonotriccus" OR "Polioxolmis" OR "Polystictus" OR "Pseudelaenia" OR "Pseudocolopteryx" OR "Pseudotriccus" OR "Pyrocephalus" OR "Pyrrhomyias" OR "Ramphotrigon" OR "Rhynchocyclus" OR "Rhytipterna" OR "Satrapa" OR "Sayornis" OR "Serpophaga" OR "Silvicultrix" OR "Sirystes" OR "Stigmatura" OR "Sublegatus" OR "Suiriri" OR "Tachuris" OR "Taeniotriccus" OR "Todirostrum" OR "Tolmomyias" OR "Tumbezia" OR "Tyrannopsis" OR "Tyrannulus" OR "Tyrannus" OR "Uromyias" OR "Xenotriccus" OR "Xolmis" OR "Zimmerius" OR "Urocynchramus" OR "Artamella" OR "Calicalicus" OR "Cyanolanius" OR "Euryceros" OR "Falculea" OR "Hypositta" OR "Leptopterus" OR "Mystacornis" OR "Newtonia" OR "Oriolia" OR "Philentoma" OR "Prionops" OR "Pseudobias" OR "Schetba" OR "Tephrodornis" OR "Tylas" OR "Vanga" OR "Xenopirostris" OR "Anomalospiza" OR "Vidua" OR "Cyclarhis" OR "Erpornis" OR "Hylophilus" OR "Pteruthius" OR "Vireo" OR "Vireolanius" OR "Zeledonia" OR "Apalopteron" OR "Dasycrotapha" OR "Heleia" OR "Lophozosterops" OR "Sterrhoptilus" OR "Yuhina" OR "Zosterops" OR "Zosterornis" OR "Agamia" OR "Ardea" OR "Ardeola" OR "Botaurus" OR "Bubulcus" OR "Butorides" OR "Cochlearius" OR "Dupetor" OR "Egretta" OR "Gorsachius" OR "Ixobrychus" OR "Nyctanassa" OR "Nycticorax" OR "Pilherodius" OR "Syrigma" OR "Tigriornis" OR "Tigrisoma" OR "Zebrilus" OR "Balaeniceps" OR "Pelecanus" OR "Scopus" OR "Bostrychia" OR "Eudocimus" OR "Geronticus" OR "Lophotibis" OR "Mesembrinibis" OR "Nipponia" OR "Phimosus" OR "Platalea" OR "Plegadis" OR "Pseudibis" OR "Theristicus" OR "Threskiornis" OR "Phaethon" OR "Phoeniconaias" OR "Phoenicoparrus" OR "Phoenicopterus" OR "Bucco" OR "Chelidoptera" OR "Hapaloptila" OR "Hypnelus" OR "Malacoptila" OR "Micromonacha" OR "Monasa" OR "Nonnula" OR "Notharchus" OR "Nystalus" OR "Capito" OR "Eubucco" OR "Brachygalba" OR "Galbalcyrhynchus" OR "Galbula" OR "Jacamaralcyon" OR "Jacamerops" OR "Indicator" OR "Melichneutes" OR "Melignomon" OR "Prodotiscus" OR "Buccanodon" OR "Gymnobucco" OR "Lybius" OR "Pogoniulus" OR "Stactolaema" OR "Trachyphonus" OR "Psilopogon" OR "Blythipicus" OR "Campephilus" OR "Campethera" OR "Celeus" OR "Chloropicus" OR "Chrysocolaptes" OR "Chrysophlegma" OR "Colaptes" OR "Dendrocopos" OR "Dendrocoptes" OR "Dendropicos" OR "Dinopium" OR "Dryobates" OR "Dryocopus" OR "Gecinulus" OR "Geocolaptes" OR "Jynx" OR "Leiopicus" OR "Leuconotopicus" OR "Meiglyptes" OR "Melanerpes" OR "Micropternus" OR "Mulleripicus" OR "Nesoctites" OR "Picoides" OR "Piculus" OR "Picumnus" OR "Picus" OR "Reinwardtipicus" OR "Sasia" OR "Sphyrapicus" OR "Veniliornis" OR "Yungipicus" OR "Andigena" OR "Aulacorhynchus" OR "Pteroglossus" OR "Ramphastos" OR "Selenidera" OR "Semnornis" OR "Aechmophorus" OR "Podiceps" OR "Podilymbus" OR "Poliocephalus" OR "Rollandia" OR "Tachybaptus" OR "Diomedea" OR "Phoebastria" OR "Phoebetria" OR "Thalassarche" OR "Hydrobates" OR "Oceanodroma" OR "Fregetta" OR "Garrodia" OR "Nesofregetta" OR "Oceanites" OR "Pelagodroma" OR "Aphrodroma" OR "Ardenna" OR "Bulweria" OR "Calonectris" OR "Daption" OR "Fulmarus" OR "Halobaena" OR "Macronectes" OR "Pachyptila" OR "Pagodroma" OR "Pelecanoides" OR "Procellaria" OR "Pseudobulweria" OR "Pterodroma" OR "Puffinus" OR "Thalassoica" OR "Cacatua" OR "Callocephalon" OR "Calyptorhynchus" OR "Eolophus" OR "Lophochroa" OR "Nymphicus" OR "Probosciger" OR "Alipiopsitta" OR "Amazona" OR "Anodorhynchus" OR "Ara" OR "Aratinga" OR "Bolborhynchus" OR "Brotogeris" OR "Conuropsis" OR "Cyanoliseus" OR "Cyanopsitta" OR "Deroptyus" OR "Diopsittaca" OR "Enicognathus" OR "Eupsittula" OR "Forpus" OR "Graydidascalus" OR "Guarouba" OR "Guaruba" OR "Hapalopsittaca" OR "Leptosittaca" OR "Myiopsitta" OR "Nannopsittaca" OR "Orthopsittaca" OR "Pionites" OR "Pionopsitta" OR "Pionus" OR "Poicephalus" OR "Primolius" OR "Psilopsiagon" OR "Psittacara" OR "Psittacus" OR "Pyrilia" OR "Pyrrhura" OR "Rhynchopsitta" OR "Thectocercus" OR "Touit" OR "Triclaria" OR "Agapornis" OR "Alisterus" OR "Aprosmictus" OR "Barnardius" OR "Bolbopsittacus" OR "Chalcopsitta" OR "Charmosyna" OR "Coracopsis" OR "Cyanoramphus" OR "Cyclopsitta" OR "Eclectus" OR "Eos" OR "Eunymphicus" OR "Geoffroyus" OR "Glossopsitta" OR "Lathamus" OR "Loriculus" OR "Lorius" OR "Melopsittacus" OR "Micropsitta" OR "Neophema" OR "Neopsephotus" OR "Neopsittacus" OR "Northiella" OR "Oreopsittacus" OR "Parvipsitta" OR "Pezoporus" OR "Phigys" OR "Platycercus" OR "Polytelis" OR "Prioniturus" OR "Prosopeia" OR "Psephotellus" OR "Psephotus" OR "Pseudeos" OR "Psittacella" OR "Psittacula" OR "Psittaculirostris" OR "Psitteuteles" OR "Psittinus" OR "Psittrichas" OR "Purpureicephalus" OR "Tanygnathus" OR "Trichoglossus" OR "Vini" OR "Nestor" OR "Strigops" OR "Pterocles" OR "Syrrhaptes" OR "Rhea" OR "Aptenodytes" OR "Eudyptes" OR "Eudyptula" OR "Megadyptes" OR "Pygoscelis" OR "Spheniscus" OR "Aegolius" OR "Asio" OR "Athene" OR "Bubo" OR "Glaucidium" OR "Ketupa" OR "Lophostrix" OR "Megascops" OR "Micrathene" OR "Ninox" OR "Otus" OR "Pseudoscops" OR "Psiloscops" OR "Pulsatrix" OR "Sceloglaux" OR "Scotopelia" OR "Strix" OR "Surnia" OR "Xenoglaux" OR "Phodilus" OR "Tyto" OR "Aepyornis" OR "Mullerornis" OR "Struthio" OR "Anhinga" OR "Fregata" OR "Leucocarbo" OR "Microcarbo" OR "Nannopterum" OR "Phalacrocorax" OR "Morus" OR "Sula" OR "Crypturellus" OR "Eudromia" OR "Nothocercus" OR "Nothoprocta" OR "Nothura" OR "Rhynchotus" OR "Tinamotis" OR "Tinamus" OR "Apalharpactes" OR "Apaloderma" OR "Harpactes" OR "Pharomachrus" OR "Trogon" OR "Tetrastes" OR "Hesperiphona") AND ("experiment*" OR "manipulat*"))



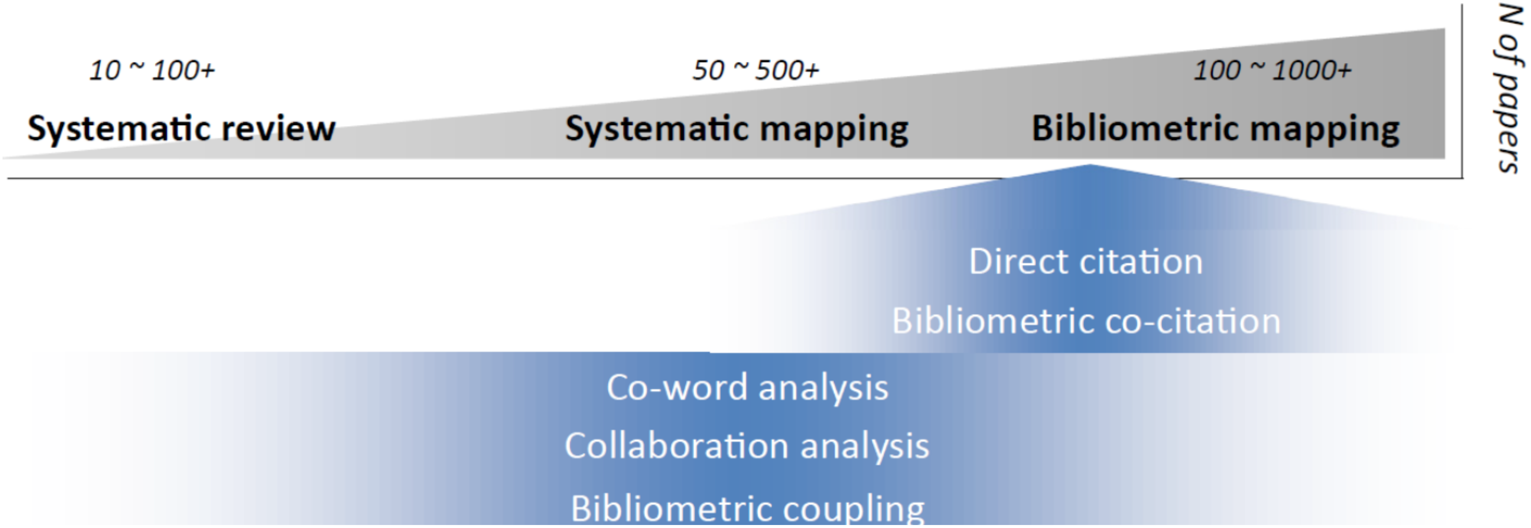
What is the minimum number of studies needed?

Depends:

- Scope of the question
- Aim of the study
- Type of synthesis
- Practical considerations
- Heterogeneity of studies

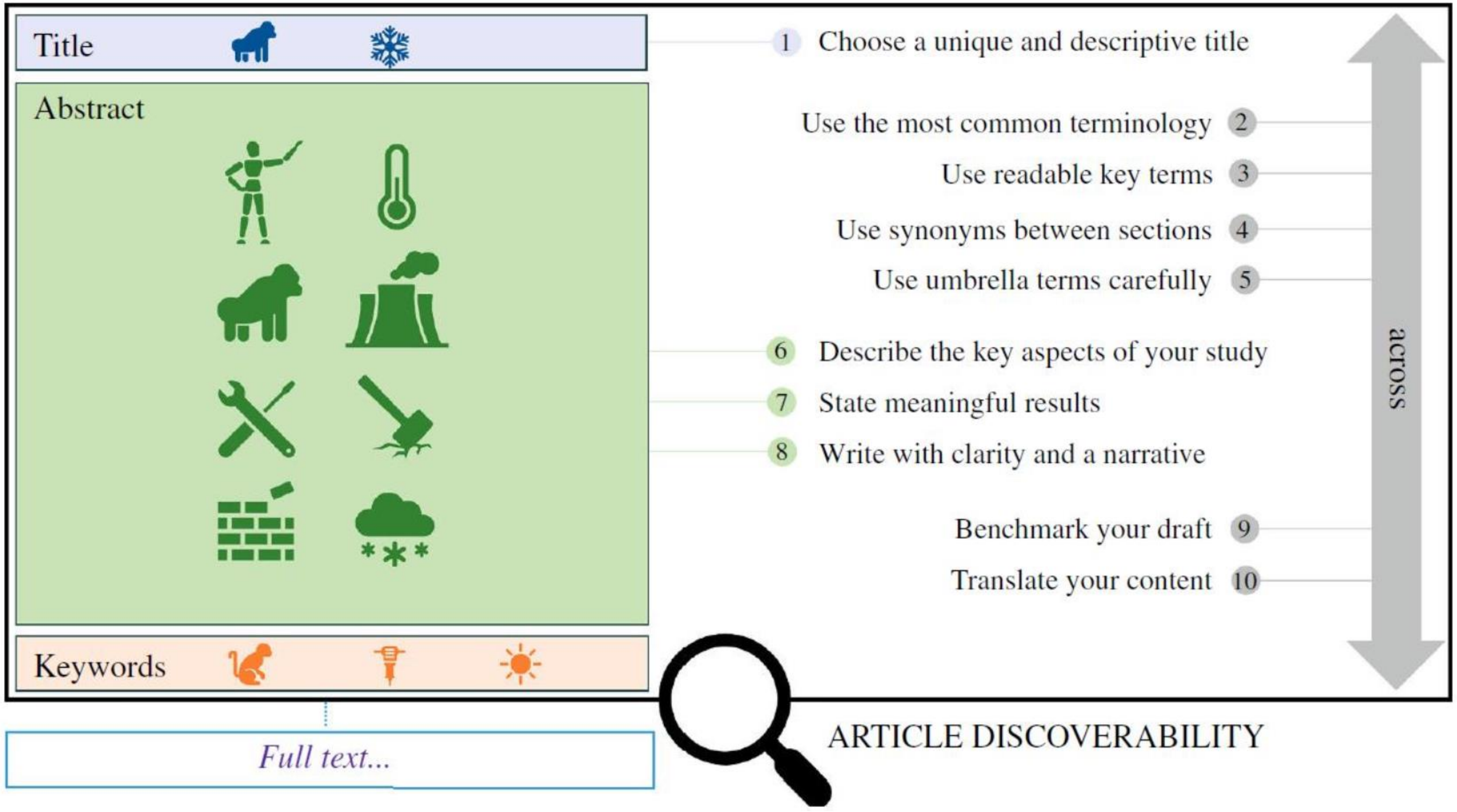
But also…

"Of fully eligible articles for the meta-analysis, systematic review, found by your search…?"



Nakagawa et al. 2018, TREE

Ideally every researcher would follow guidelines such as...

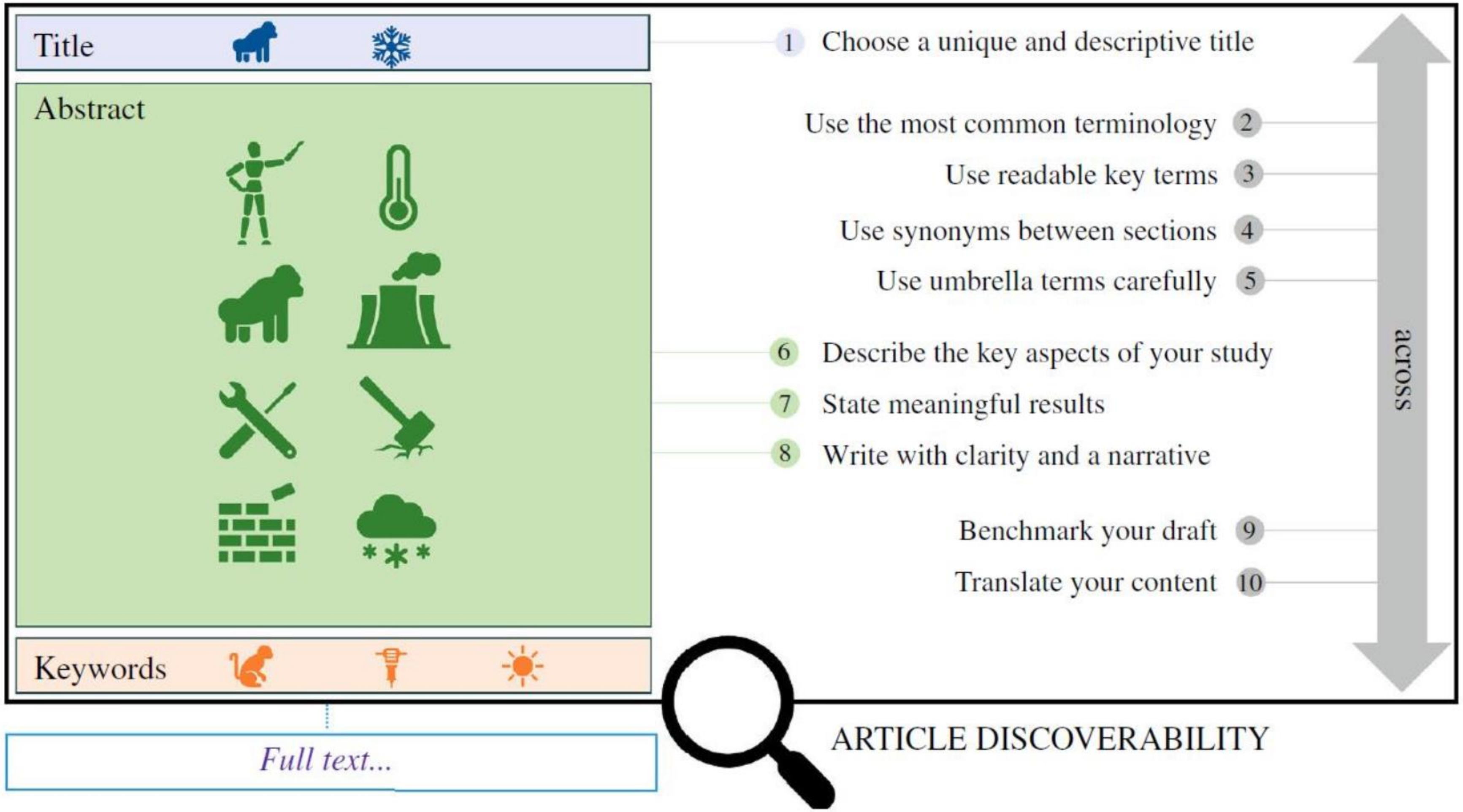


- Avoid using redundant key terms between the title, abstract, and keyword sections

- Include taxonomic group, species name, location, study type, and variables investigated in the title or abstract

Pottier et al. 2024, Proc B

Ideally every researcher would follow guidelines such as...

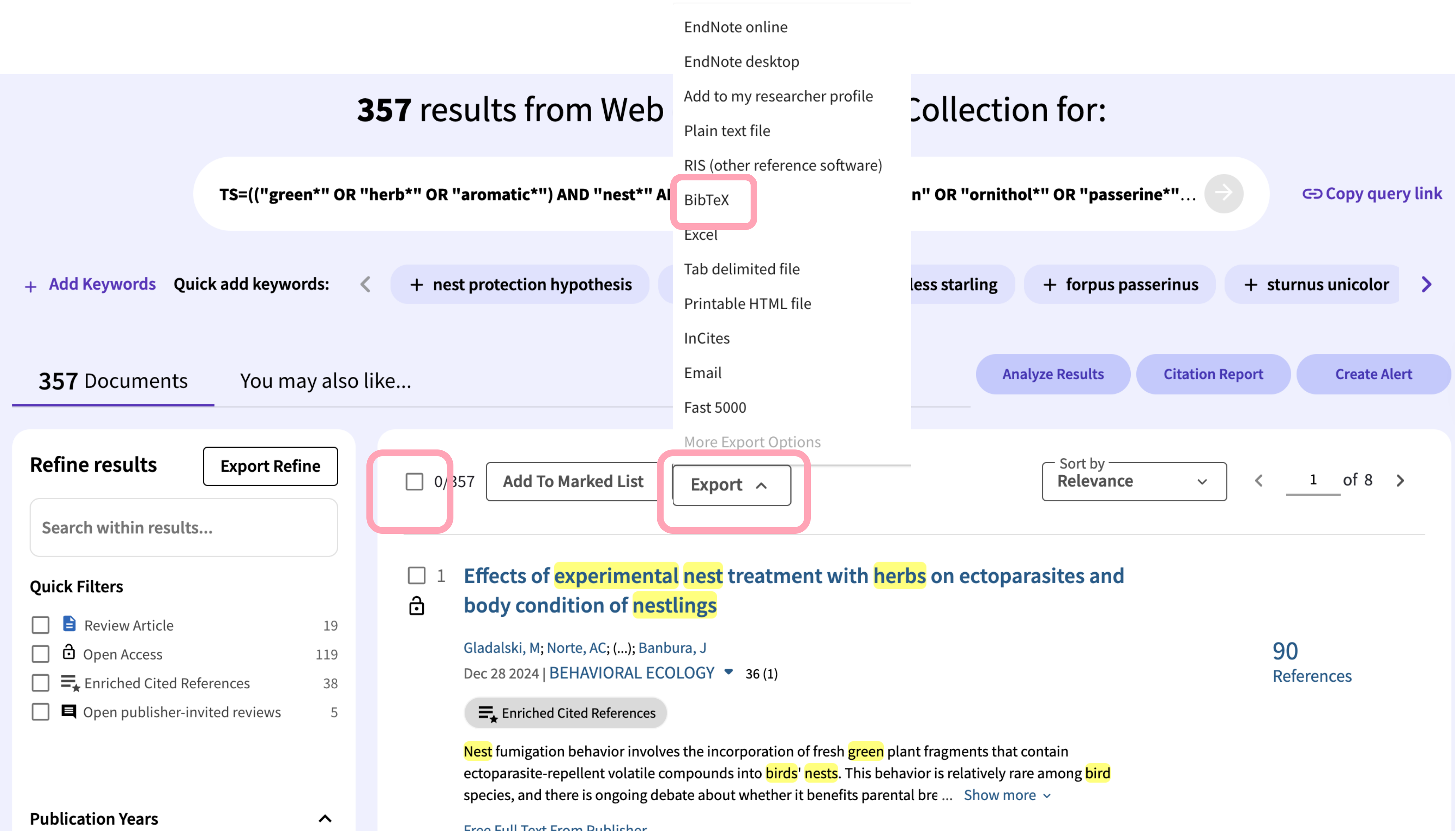


- Avoid using redundant key terms between the title, abstract, and keyword sections

- Include taxonomic group, species name, location, study type, and variables investigated in the title or abstract

But we are far from it, so plan in accordance

Pottier et al. 2024, Proc B


EndNote online
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Add to my researcher profile
Plain text file
RIS (other reference software)
BibTeX
Excel
Tab delimited file
Printable HTML file
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Fast 5000
More Export Options
357 results from Web
Collection for:
TS=(("green*" OR "herb*" OR "aromatic*") AND "nest*" AI
n" OR "ornithol*" OR "passerine*"...
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sturnus unicolor
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Relevance
1 of 8
1
Effects of experimental nest treatment with herbs on ectoparasites and body condition of nestlings
Gladalski, M; Norte, AC; (...); Banbura, J
Dec 28 2024 | BEHAVIORAL ECOLOGY 36 (1)
90
References
Enriched Cited References
Nest fumigation behavior involves the incorporation of fresh green plant fragments that contain ectoparasite-repellent volatile compounds into birds' nests. This behavior is relatively rare among bird species, and there is ongoing debate about whether it benefits parental bre ... Show more
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Author, Title, Source

- [x] Author(s)
- [x] Title
- [x] Source
- [x] Conf.Info/Sponsors
- [x] Times Cited Count
- [x] Accession Number
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Abstract, Keyword, Addresses

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- [x] Addresses
- [x] Affiliations
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Step 2: Search for the literature

"Where do I search?"

Why should we search across multiple databases?

- Different coverage
- Around 50% overlap, but as low as 11%

Bar-Ilan, 2018; Mongeon & Paul-Hus, 2016

Step 2: Search for the literature

“Where do I search?”

Why should we search across multiple databases?

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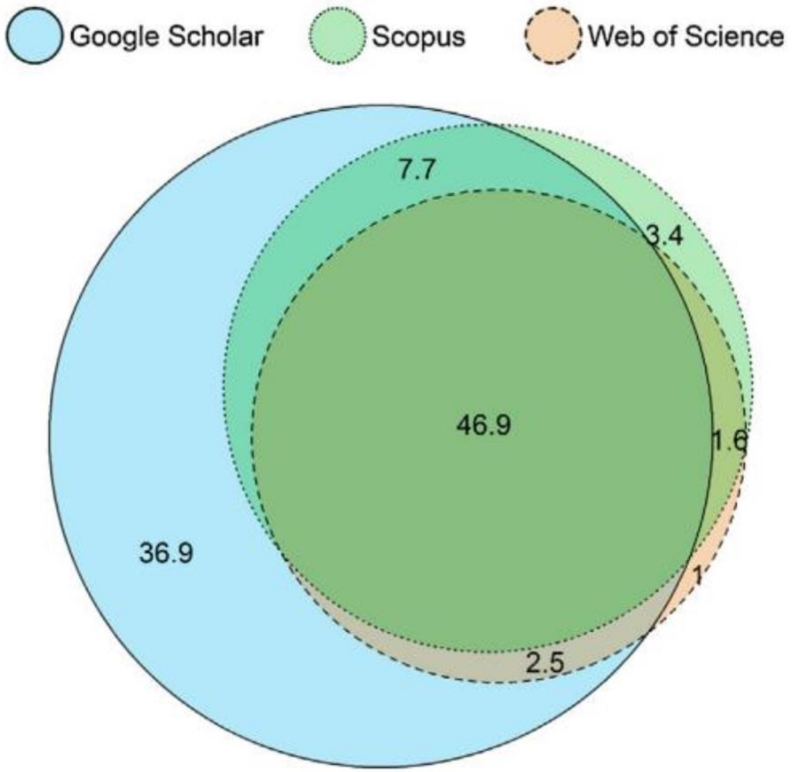


. Percentage of unique and overlapping citations in google scholar, Scopus, and Web of Science. n = 2,448,055 citations from all subject areas.

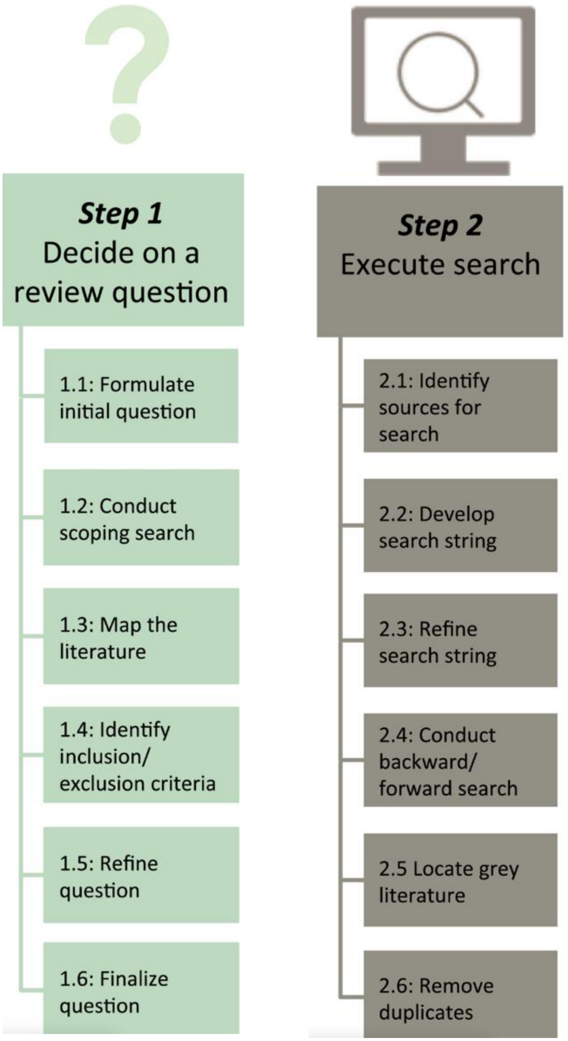
Martin-Martin et al. 2018

Working with search .bib files in R

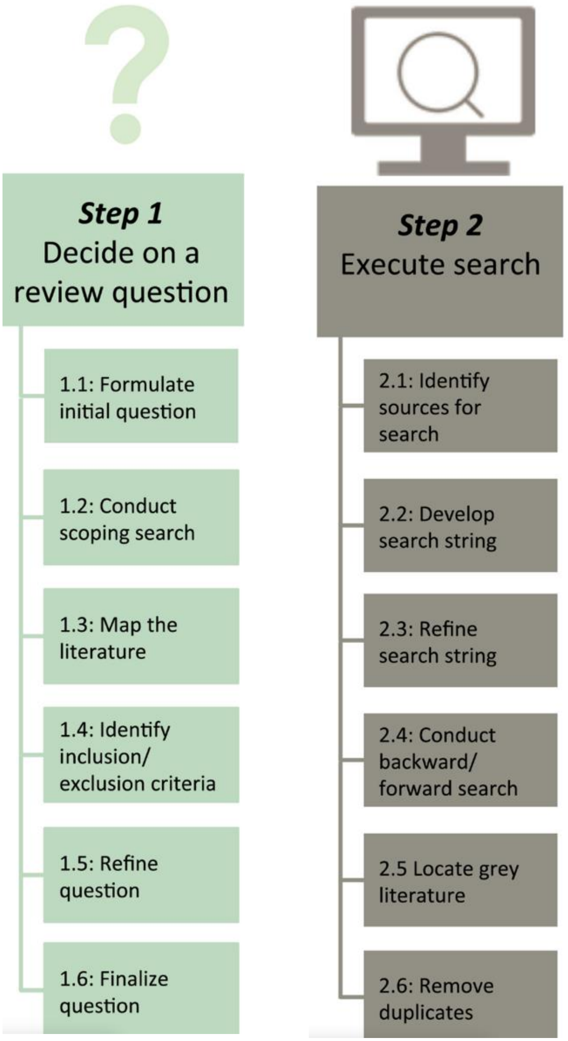
- Import into R using ‘synthesisr’
- Deduplicate across the database (automatic [exact]-automatic [fuzzy matching]-manual[-automatic]).

Exercise Time!

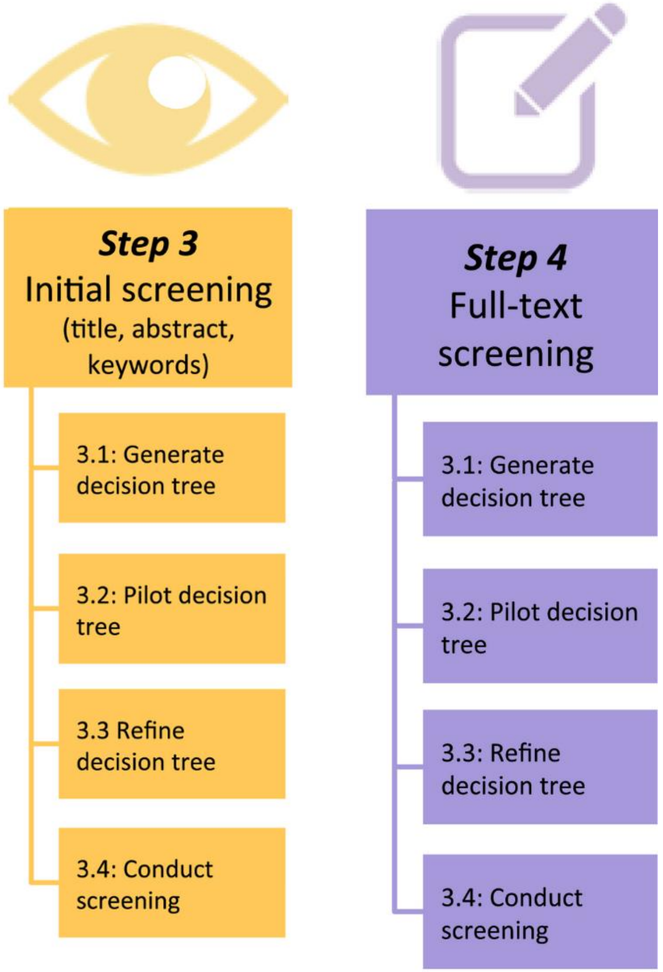
What we have done so far!



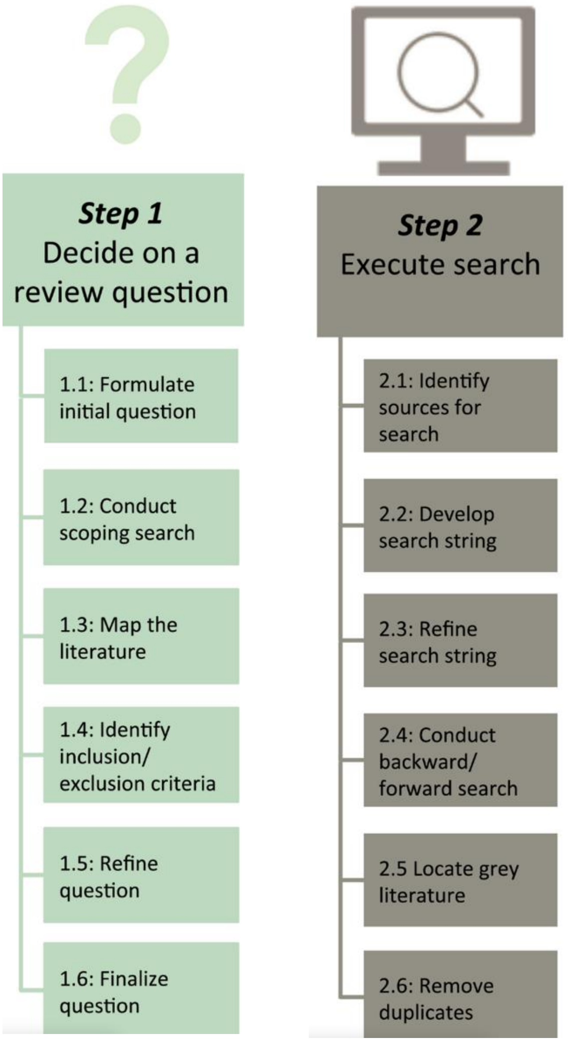
What we have done so far!



Now we need to screen evidence for relevance



How to



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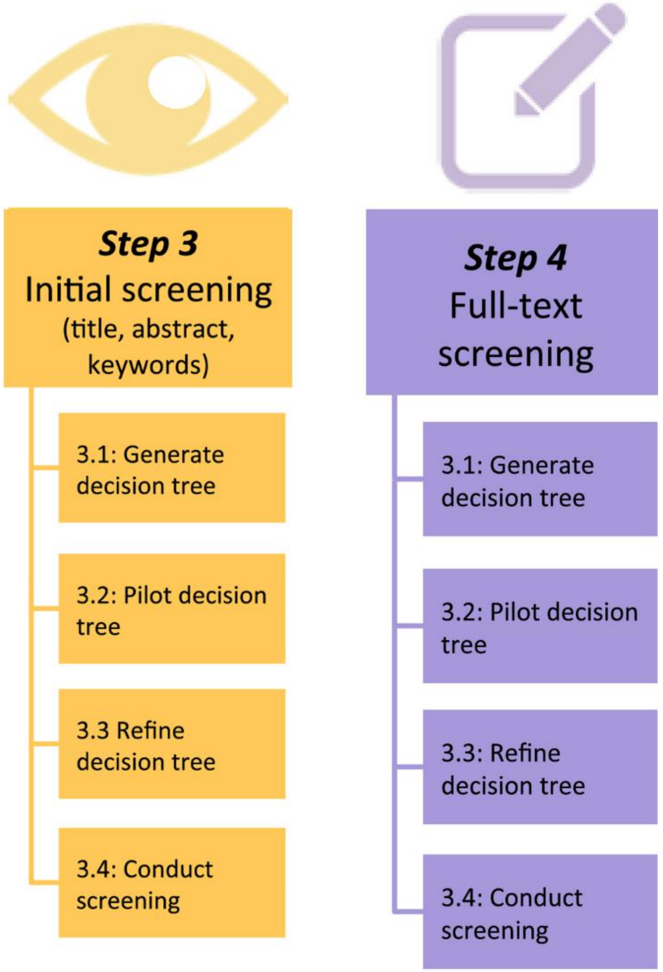


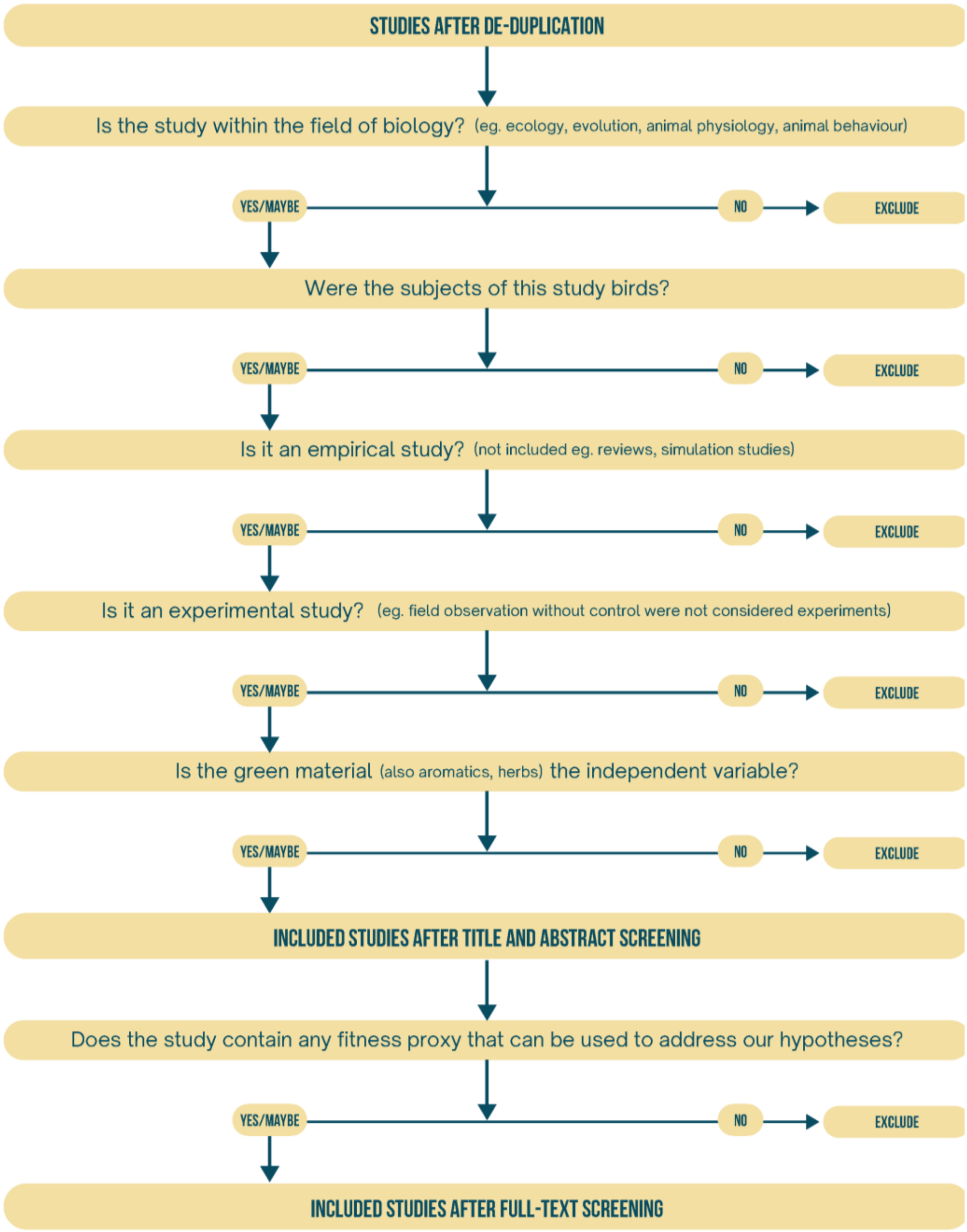
Step 3: Screening for relevant articles

“Where do I begin the screening process?”

- With a decision tree:
  clear, well-thought-out Inclusion/Exclusion criteria

“What is the adaptive significance of fresh green material added by birds?”

Let’s say I want to look at the experimental evidence



Pilot the screening

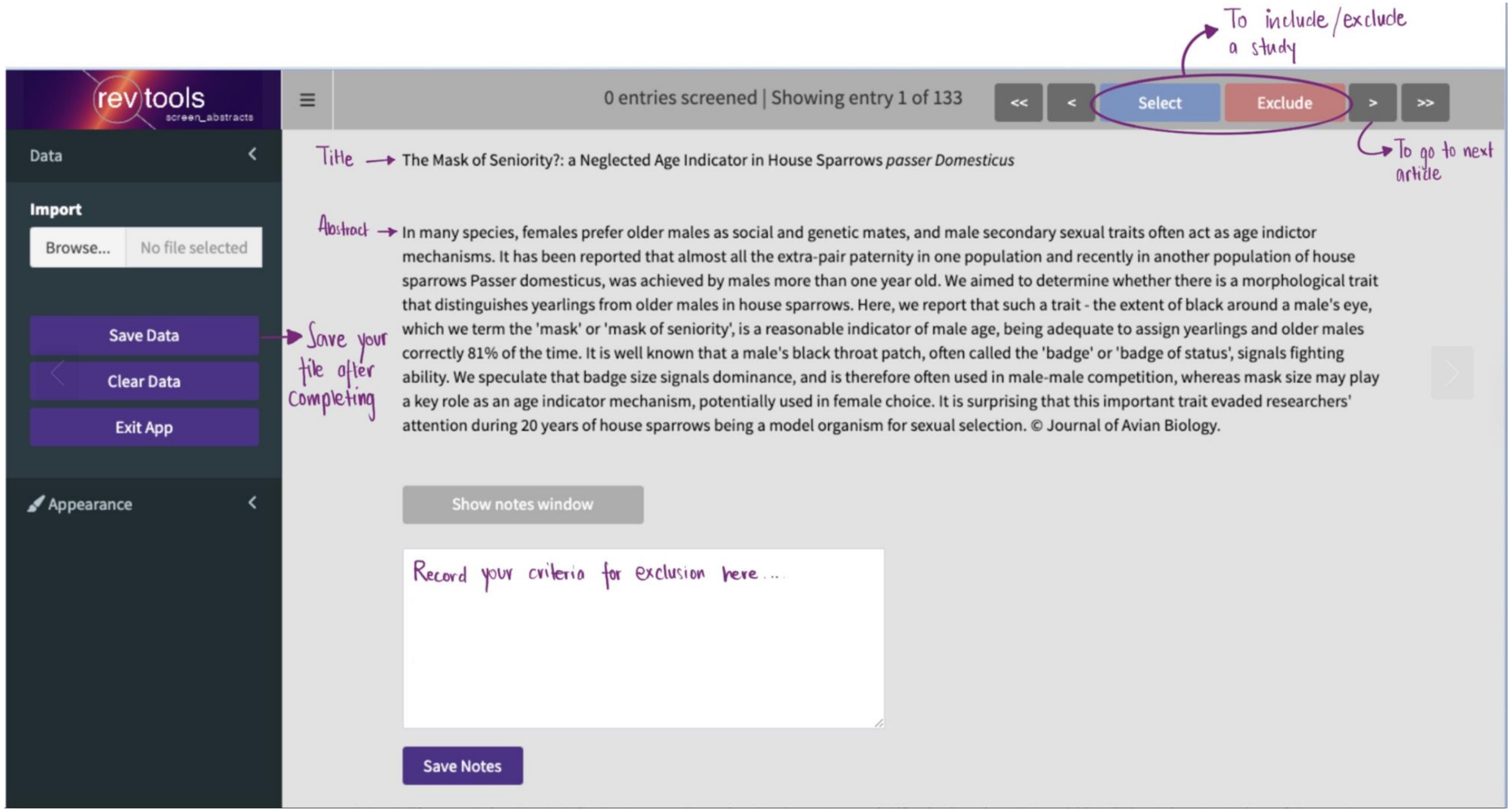
“How do I begin screening?”

- Revtools Package GUI
- MetScreen ShinyApp
- Rayyan Software

Pilot the screening

"How do I begin screening?"

- Revtools Package GUI
- MetScreen ShinyApp
- Rayyan Software

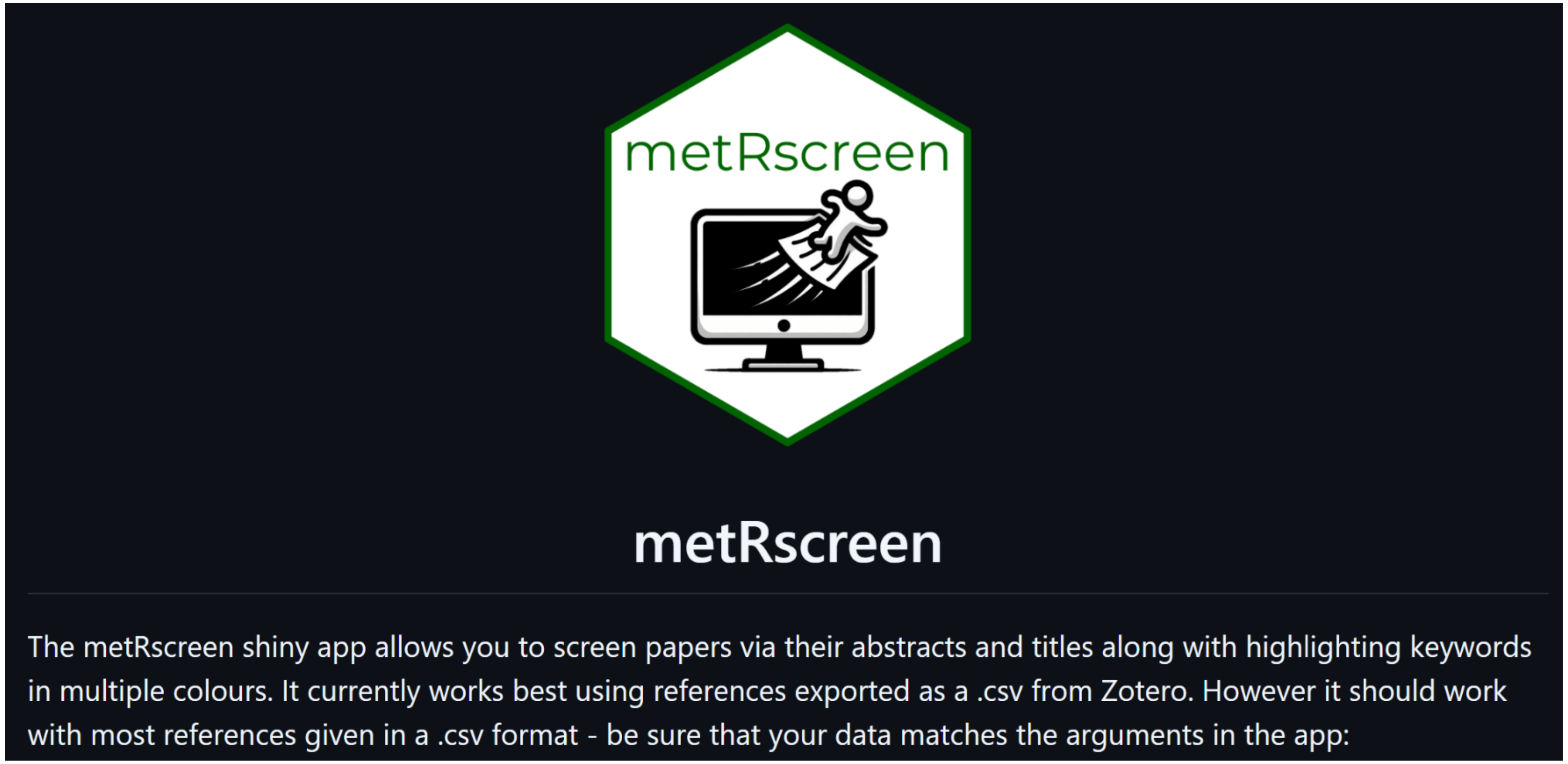


Westgate 2019

Pilot the screening

"How do I begin screening?"

- Revtools Package GUI
- MetScreen ShinyApp
- Rayyan Software

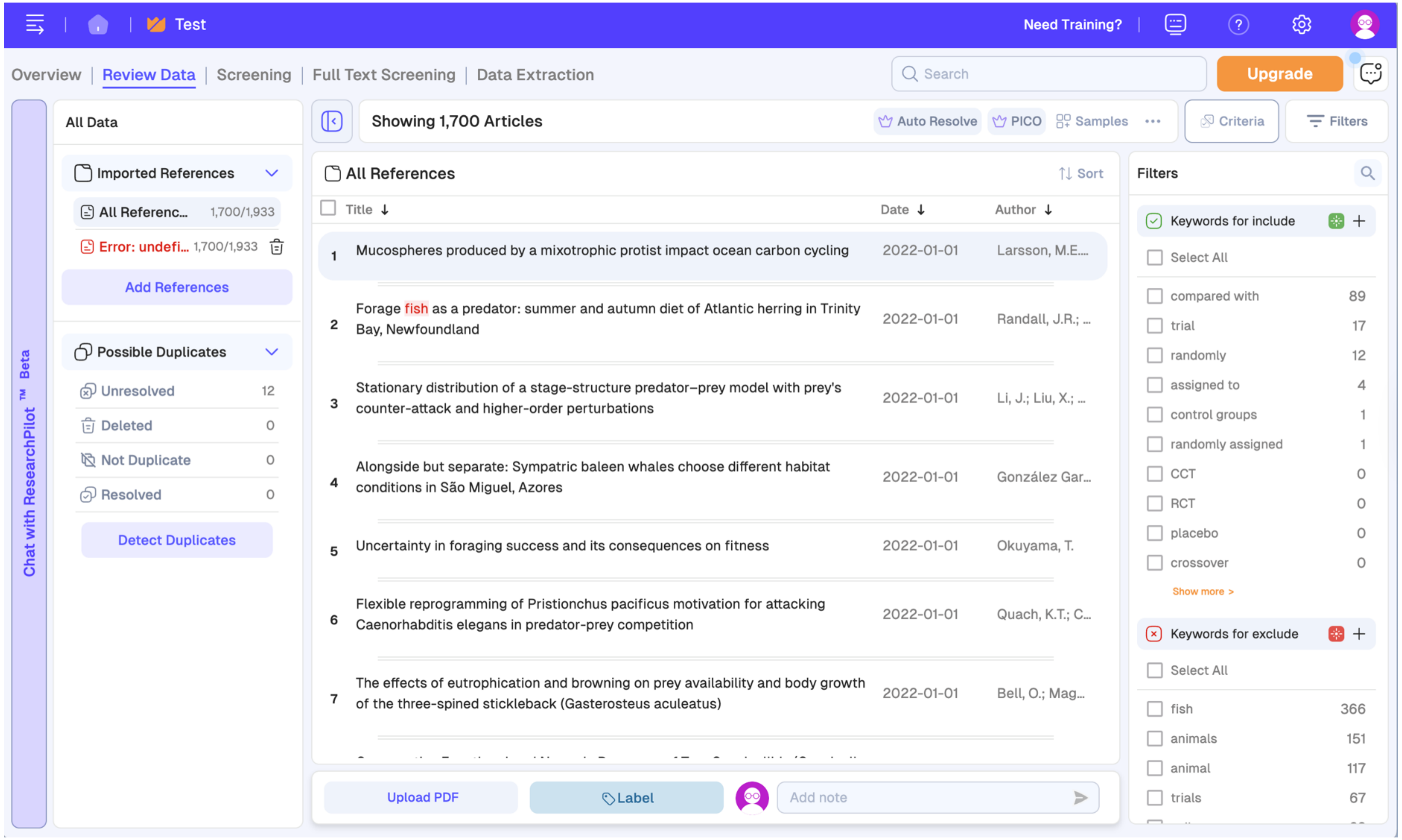


Ivimey-Cook 2024

Pilot the screening

"How do I begin screening?"

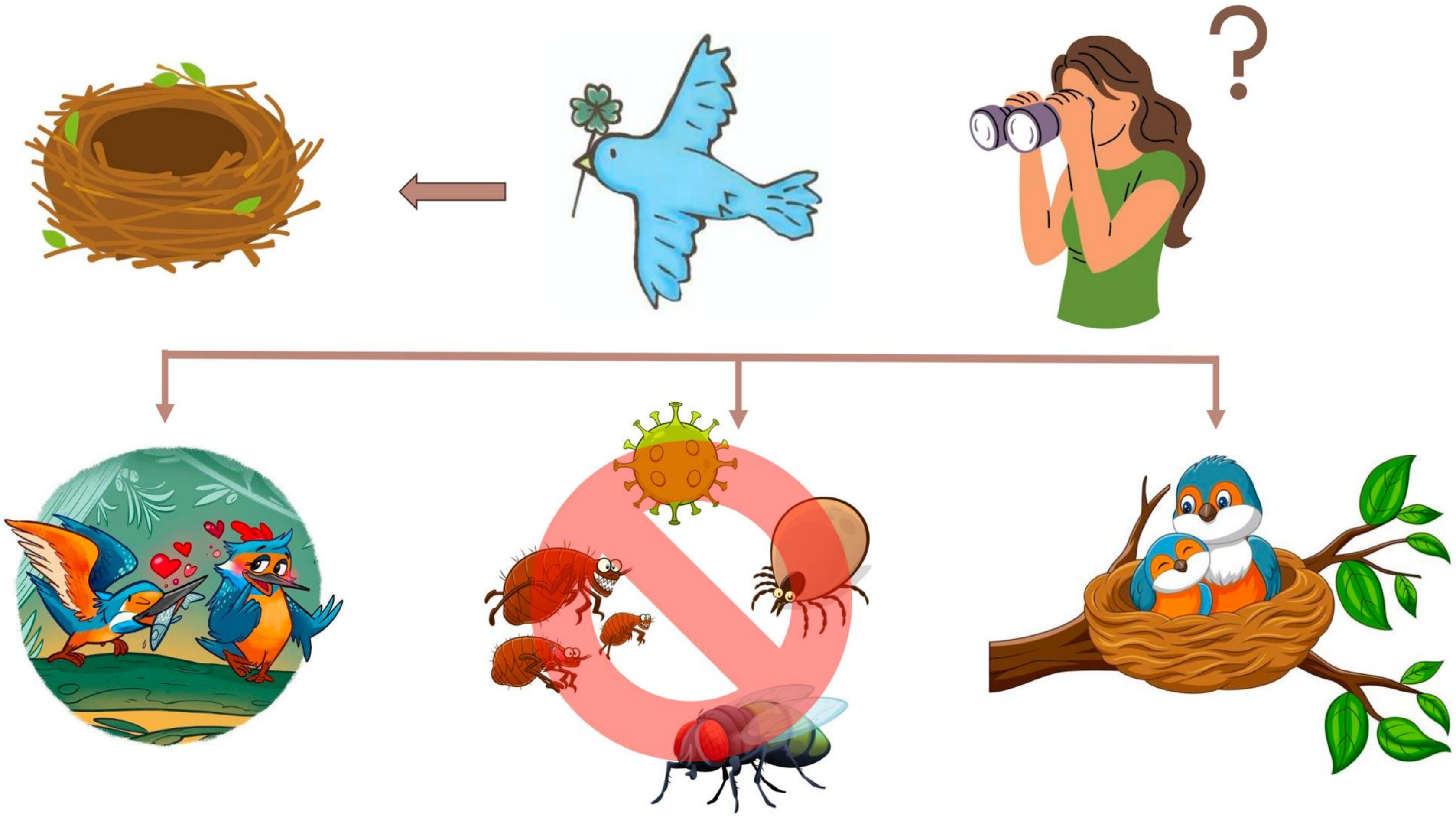
- Revtools Package GUI
- MetScreen ShinyApp
- Rayyan Software



Exercise Time?

Adaptive value of green nest material in bird species

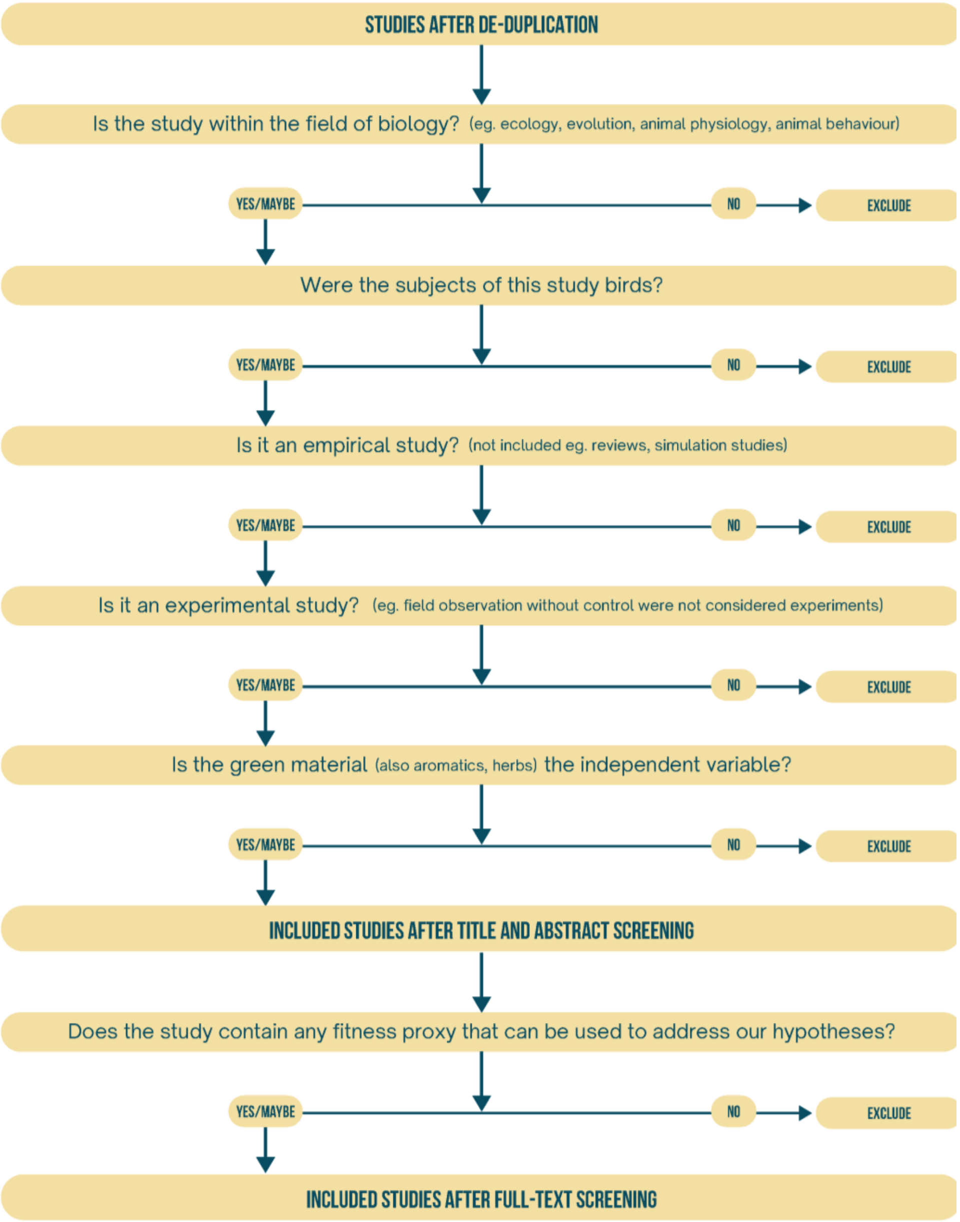
Adaptive value of green nest material in bird species



Distinct Determinants of Sparse Activation during Granule Cell Maturation

Abstract

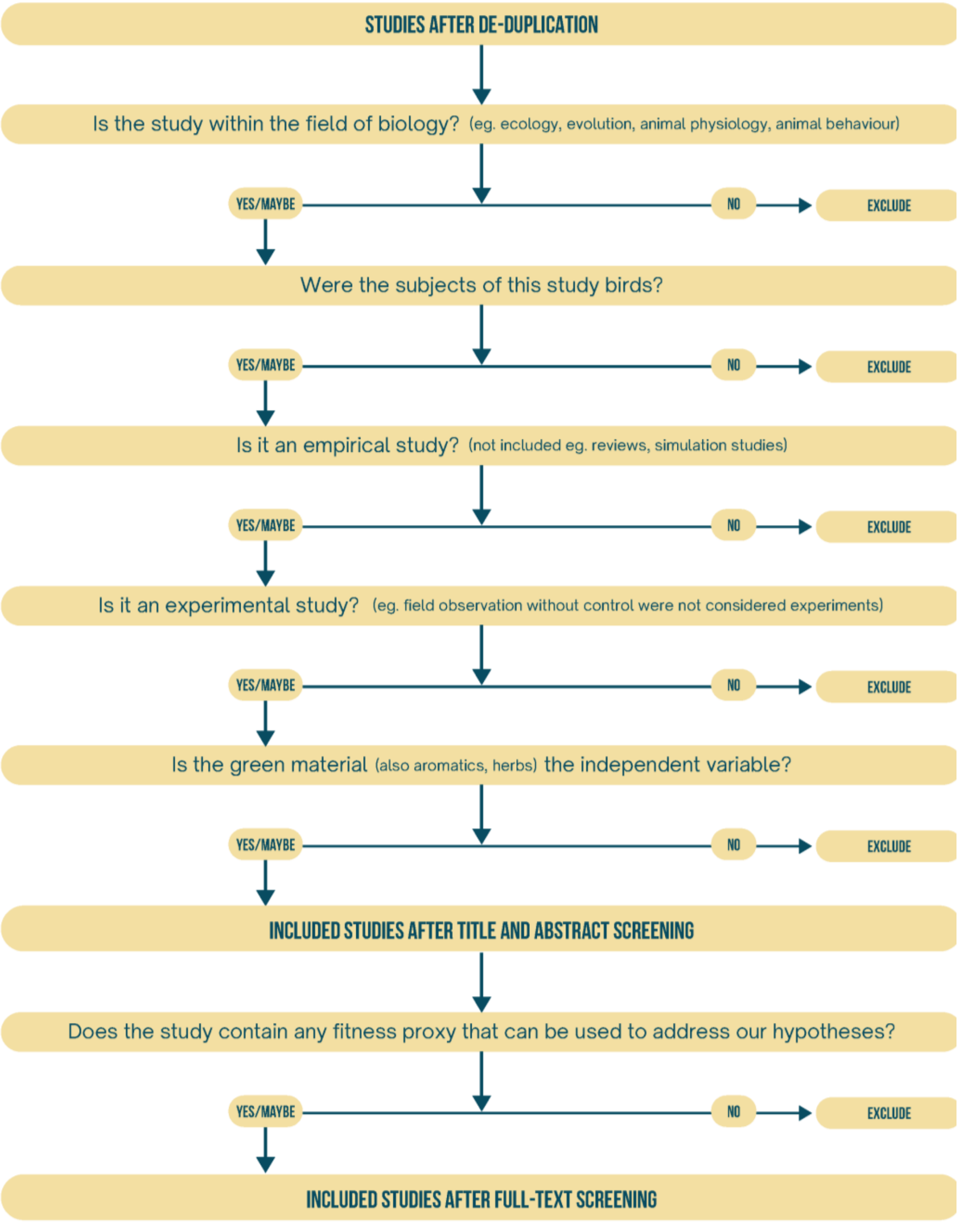
Adult neurogenesis continually produces a small population of immature granule cells (GCs) within the dentate gyrus. The physiological properties of immature GCs distinguish them from the more numerous mature GCs and potentially enables distinct network functions. To test how the changing properties of developing GCs affect spiking behavior, we examined synaptic responses of mature and immature GCs in hippocampal slices from adult mice. Whereas synaptic inhibition restricted GC spiking at most stages of maturation, the relative influence of inhibition, excitatory synaptic drive, and intrinsic excitability shifted over the course of maturation. Mature GCs received profuse afferent innervation such that spiking was suppressed primarily by inhibition, whereas immature GC spiking was also limited by the strength of excitatory drive. Although the input resistance was a reliable indicator of maturation, it did not determine spiking probability at immature stages. Our results confirm the existence of a transient period during GC maturation when perforant path stimulation can generate a high probability of spiking, but also reveal that immature GC excitability is tempered by functional synaptic inhibition and reduced excitatory innervation, likely maintaining the sparse population activity observed *in vivo*.



Internal seed dispersal by parrots: an overview of a neglected mutualism

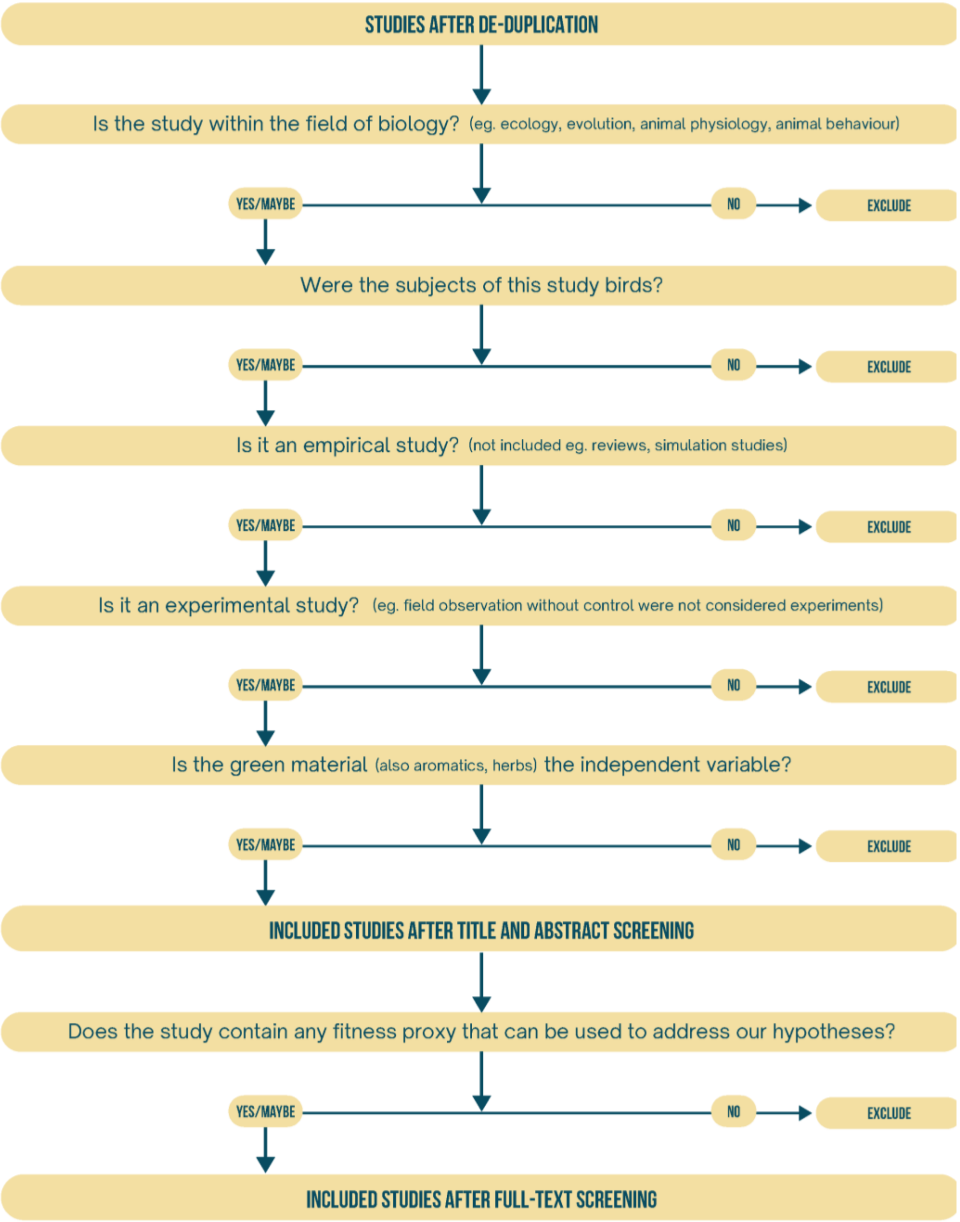
ABSTRACT

Despite the fact that parrots (Psitacifformes) are generalist apex frugivores, they have largely been considered plant antagonists and thus neglected as seed dispersers of their food plants. Internal dispersal was investigated by searching for seeds in faeces opportunistically collected at communal roosts, foraging sites and nests of eleven parrot species in different habitats and biomes in the Neotropics. Multiple intact seeds of seven plant species of five families were found in a variable proportion of faeces from four parrot species. The mean number of seeds of each plant species per dropping ranged between one and about sixty, with a maximum of almost five hundred seeds from the cacti *Pilosocereus pachycladus* in a single dropping of Lear's Macaw (*Anodorhynchus leari*). All seeds retrieved were small (<3 mm) and corresponded to herbs and relatively large, multiple-seeded fleshy berries and infrutescences from shrubs, trees and columnar cacti, often also dispersed by stomatochory. An overview of the potential constraints driving seed dispersal suggest that, despite the obvious size difference between seeds dispersed by endozoochory and stomatochory, there is no clear difference in fruit size depending on the dispersal mode. Regardless of the enhanced or limited germination capability after gut transit, a relatively large proportion of cacti seeds frequently found in the faeces of two parrot species were viable according to the tetrazolium test and germination experiments. The conservative results of our exploratory sampling and a literature review clearly indicate that the importance of parrots as endozoochorous dispersers has been largely under-appreciated due to the lack of research systematically searching for seeds in their faeces. We encourage the evaluation of seed dispersal and other mutualistic interactions mediated by parrots before their generalized population declines contribute to the collapse of key ecosystem processes.



Experimental evidence of a testosterone-induced shift from paternal to mating behaviour in a facultatively polygynous songbird

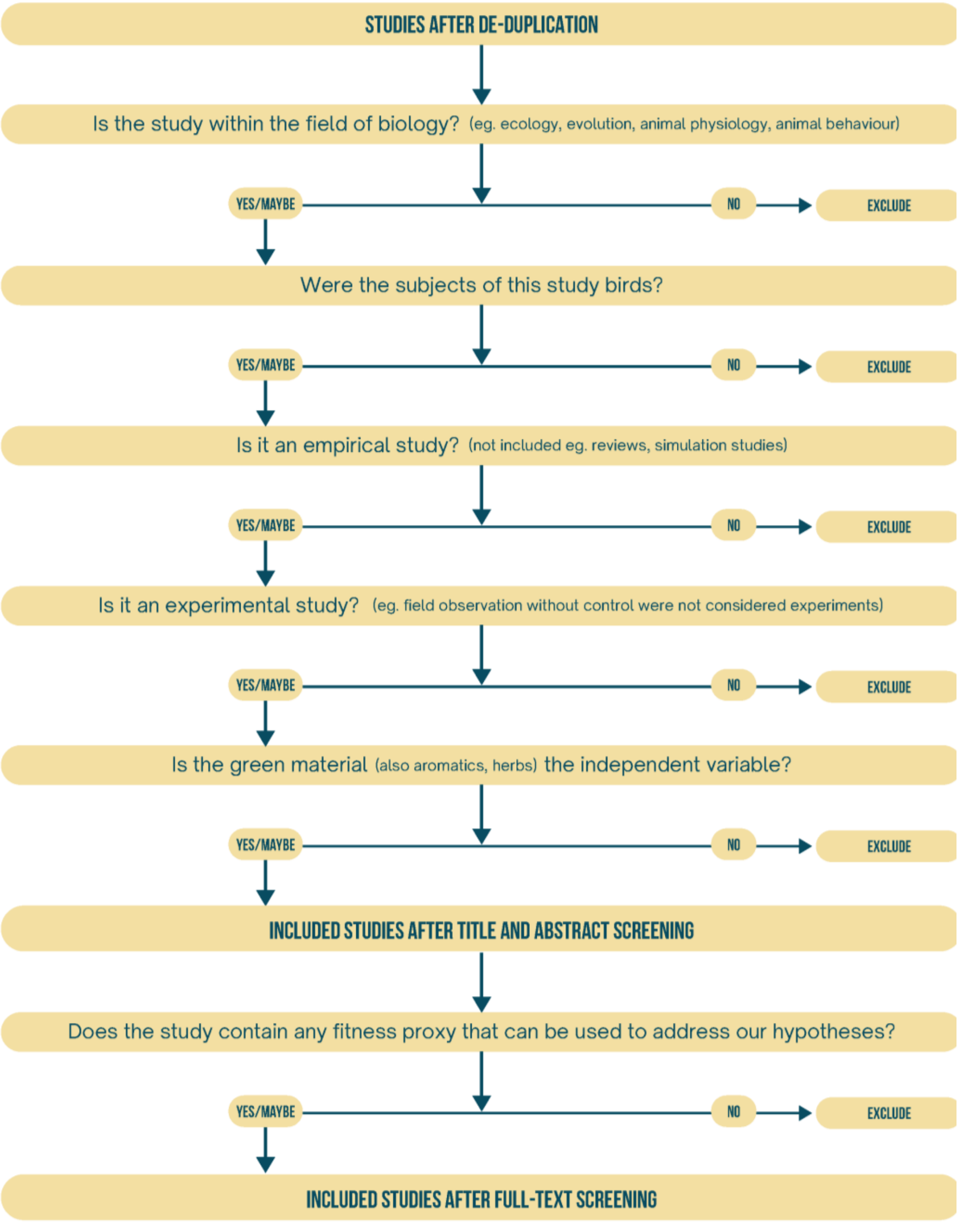
Abstract Previous studies have suggested that testosterone (T) profiles of male birds reflect a trade-off between mate attraction behaviours (requiring high T levels) and parental care activities (requiring low T levels). In this study, we experimentally elevated T levels of monogamous males in the facultatively polygynous European starling (*Sturnus vulgaris*), and compared mate attraction and paternal behaviour of T-treated males with those of controls (C-males). T-males significantly reduced their participation in incubation and fed nestlings significantly less often than C-males. Females paired to T-treated males did not compensate for their mate's lower paternal effort. The observed reduction in a male's investment in incubating the eggs was accompanied by an increased investment in typical female-attracting behaviours: T-males spent a significantly higher proportion of their time singing to attract additional females. They also occupied more additional nestboxes than C-males, although the differences just failed to be significant, and carried significantly more green nesting materials into an additional nestbox (a behaviour previously shown to serve a courtship function). T-males also behaved significantly more aggressively than C-males. During the nestling period, the frequency of mate-attracting behaviours by T-treated and control males no longer differed significantly. Despite the reduced paternal effort by T-males and the lack of compensation behaviour by females, hatching and breeding success did not differ significantly between T- and C-pairs.



Aromatic plants in nests of the blue tit *Cyanistes caeruleus* protect chicks from bacteria

Abstract Several bird species add fresh fragments of plants which are rich in volatile secondary compounds to their nests. It has been suggested, although never tested, that birds use fresh plants to limit the growth of nest micro-organisms. On Corsica, blue tits (*Cyanistes caeruleus*) incorporate fresh fragments of aromatic plants into their nests. These plants do not reduce infestation by nest ecto-parasites, but have been shown to improve growth and condition of chicks at fledging. To understand the mechanisms underlying such benefits, we experimentally tested the effects of these plants on the bacteria living on blue tits.

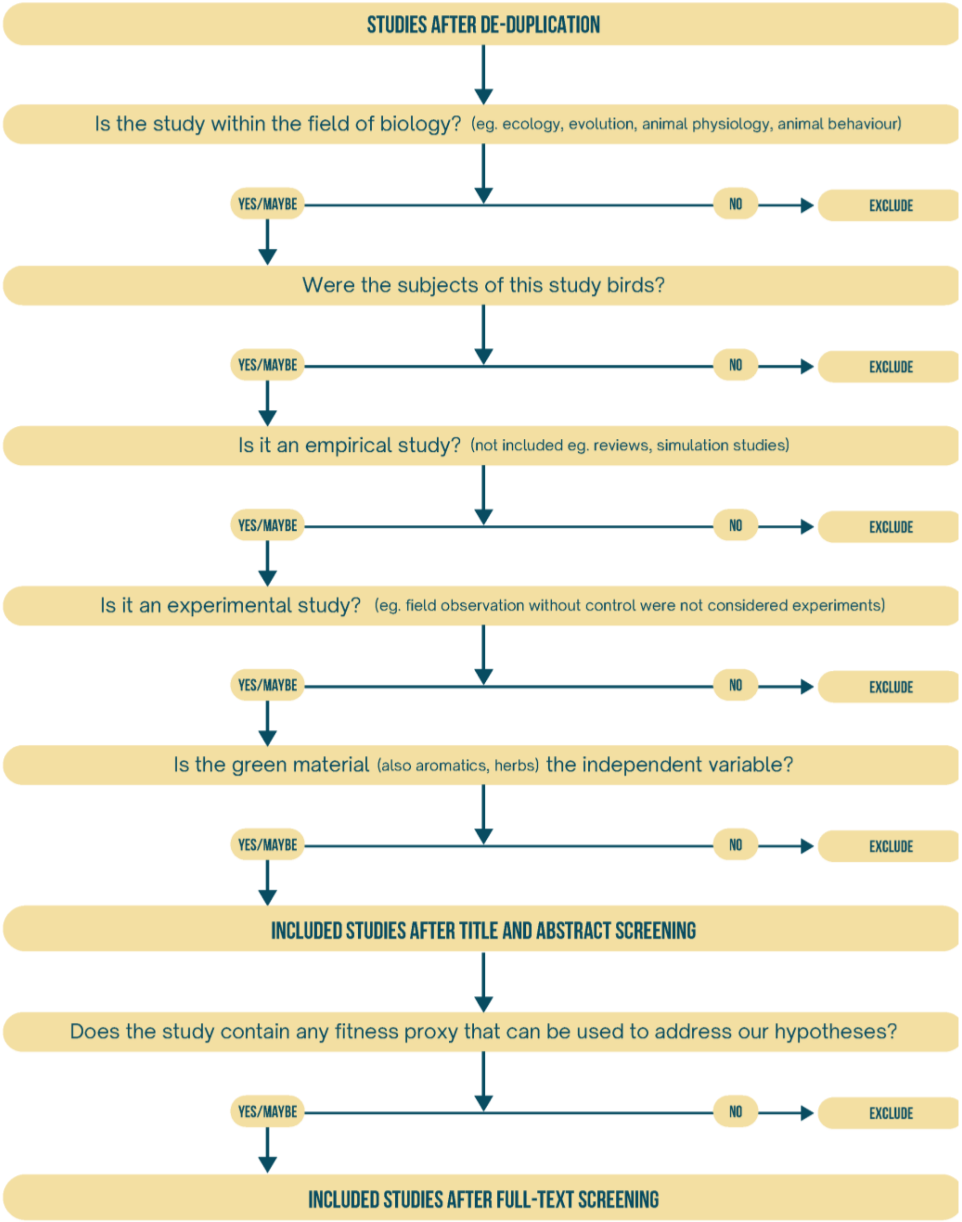
Aromatic plants significantly affected the structure of bacterial communities, in particular reducing bacterial richness on nestlings. In addition, in this population where there is a strong association between bacterial density and infestation by blood-sucking *Protocalliphora* blow fly larvae, these plants reduced bacterial density on the most infested chicks. Aromatic plants had no significant effect on the bacteria living on adult blue tits. This study provides the first evidence that fresh plants brought to the nests by adult birds limit bacterial richness and density on their chicks.



Nest composition and yolk hormones: do female European starlings adjust yolk androgens to nest quality?

ABSTRACT

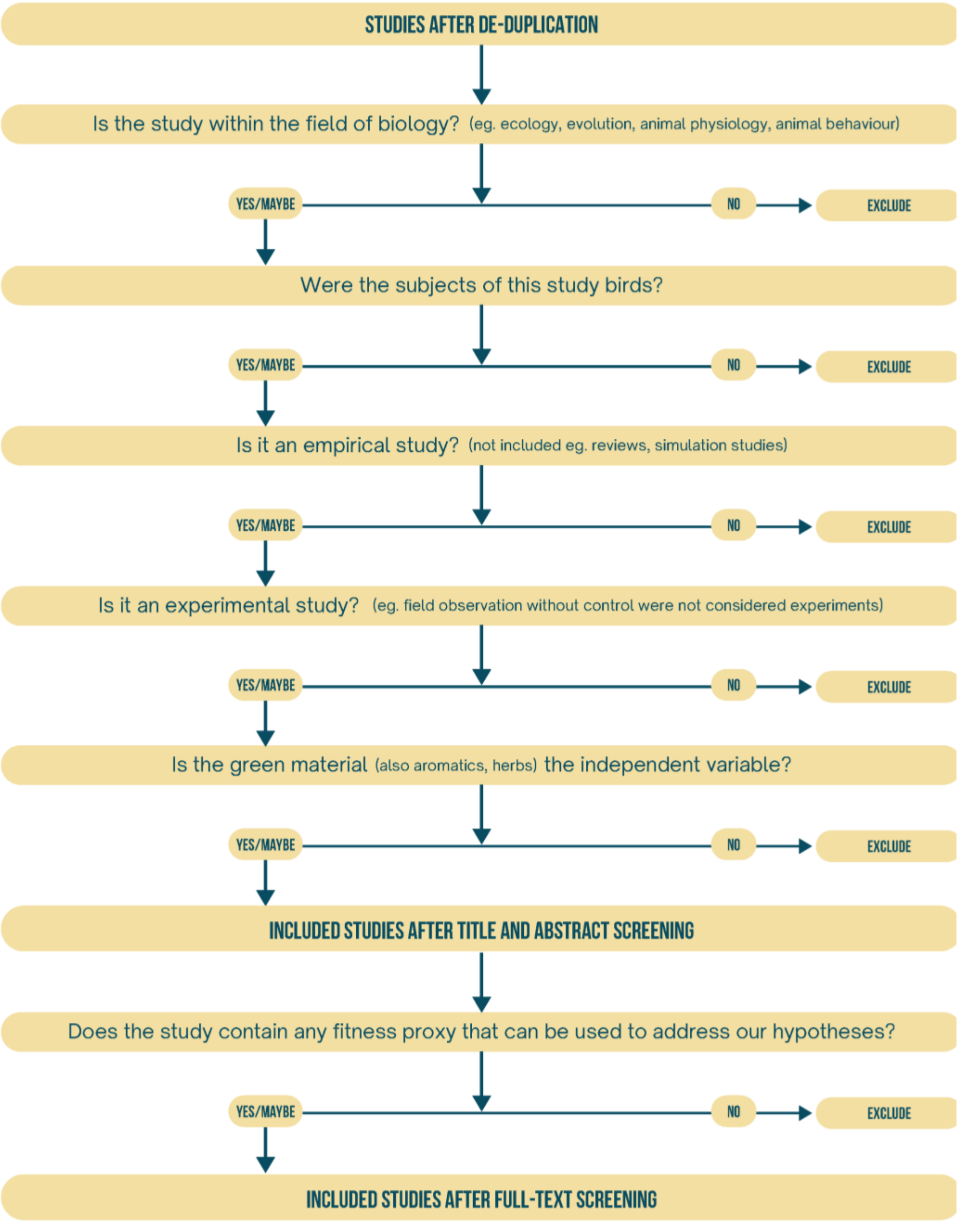
The nest is a key element of avian reproductive fitness. It provides the developmental environment for the embryo and nestling thereby affecting their quality and survival. Nests are often constructed by the male and nest quality, elaboration, and ornamentation vary among males in a species. Therefore, male nest-building behaviour is likely under sexual selection and females may use male nest-building behaviour and nest features as cues in mate choice and their own reproductive investment. We tested this hypothesis in European starlings (*Sturnus vulgaris*). Male starlings present green plants to females during their courtship display and furbish their nests with variable quantities of fresh green plants. The highest frequency of plant incorporation into the nest occurs during pair formation and oogenesis. We determined nest composition and measured yolk testosterone concentrations in eggs. Yolk testosterone concentrations were positively correlated with the amount of green plant material in nests. Because males use green nest material in courtship and green nest material positively affects nestlings' development, we suggest that female European starlings use male nest-building effort, ritualised courtship display of green plants, and nest characteristics in mate choice, and adjust yolk testosterone levels in the eggs to the quality of the nest to optimise offspring growth.



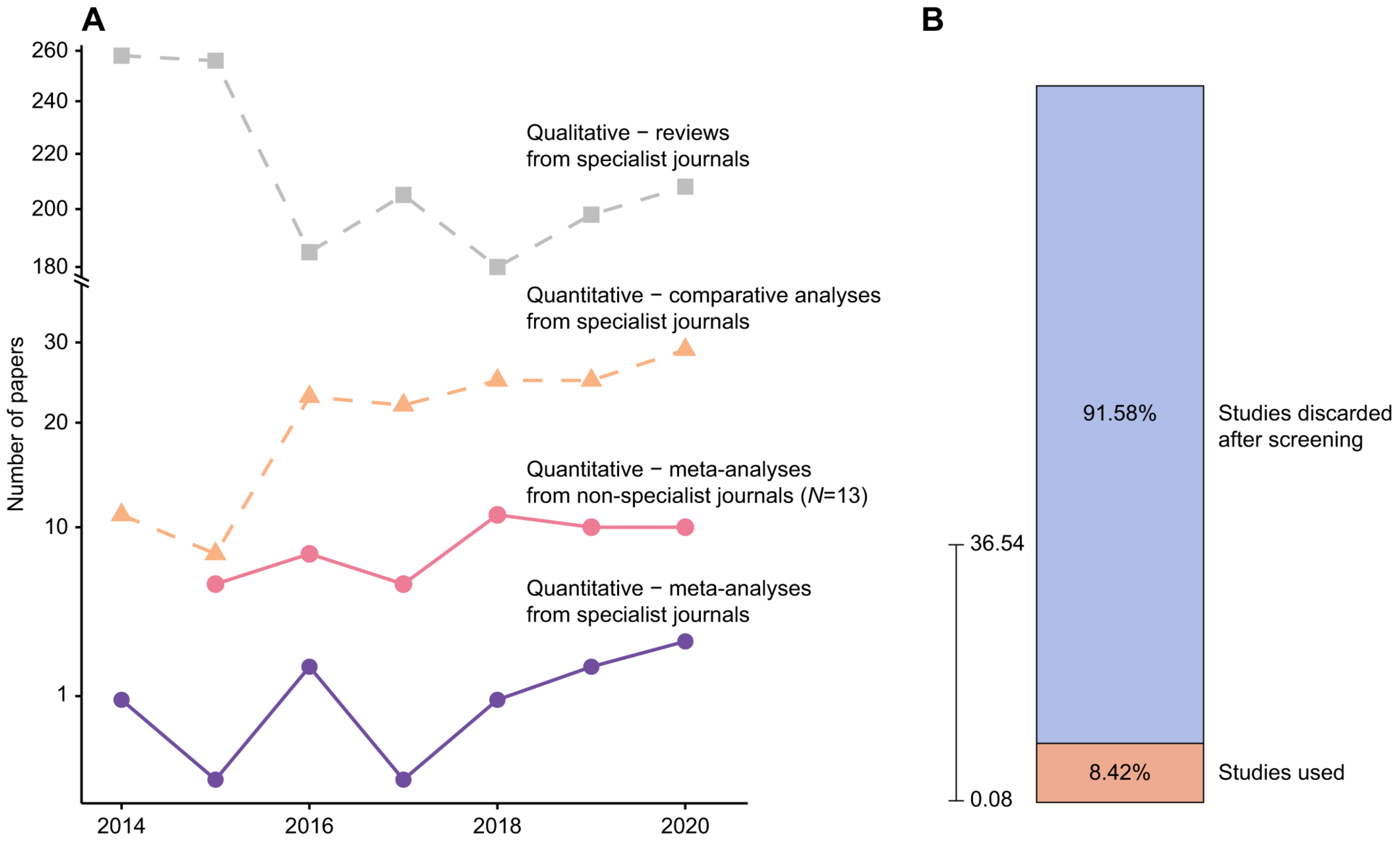
Environmental cues and dietary antioxidants affect breeding behavior and testosterone of male European starlings (*Sturnus vulgaris*)

A B S T R A C T

Environmental cues, such as photoperiod, regulate the timing of major life-history events like breeding through direct neuroendocrine control. Less known is how supplementary environmental cues (e.g., nest sites, food availability) interact to influence key hormones and behaviors involved in reproduction, specifically in migratory species with gonadal recrudescence largely occurring at breeding sites. We investigated the behavioral and physiological responses of male European starlings to the sequential addition of nest boxes and nesting material, green herbs, and female conspecifics and how these responses depend on the availability of certain antioxidants (anthocyanins) in the diet. As expected, cloacal protuberance volume and plasma testosterone of males generally increased with photoperiod. More notably, testosterone levels peaked in males fed the high antioxidant diet when both nest box and herbal cues were present, while males fed the low antioxidant diet showed no or only a muted testosterone response to the sequential addition of these environmental cues; thus our results are in agreement with the oxidation handicap hypothesis. Males fed the high antioxidant diet maintained a constant frequency of breeding behaviors over time, whereas those fed the low antioxidant diet decreased breeding behaviors as environmental cues were sequentially added. Overall, sequential addition of the environmental cues modulated physiological and behavioral measures of reproductive condition, and dietary antioxidants were shown to be a key factor in affecting the degree of response to each of these cues. Our results highlight the importance of supplementary environmental cues and key resources such as dietary antioxidants in enhancing breeding condition of males, which conceivably aid in attraction of high quality females and reproductive success.



How many studies do we usually exclude?



Schwanz et al. 2022

Step 4: Full-text screening

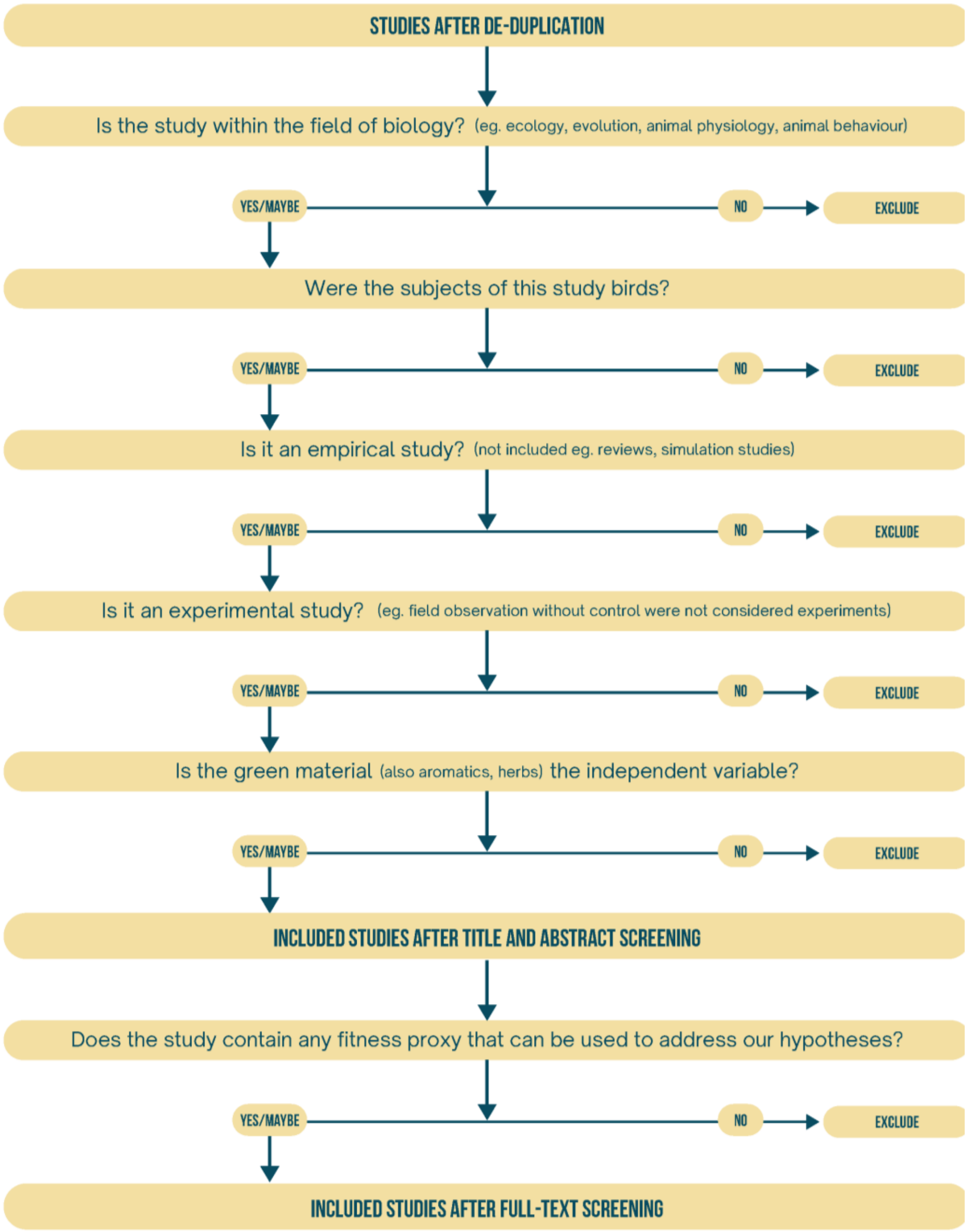
How to do it:

- Reference managers:
- Endnote $$$ (free trial)
- Zotero + Zotero-SciHub plugin

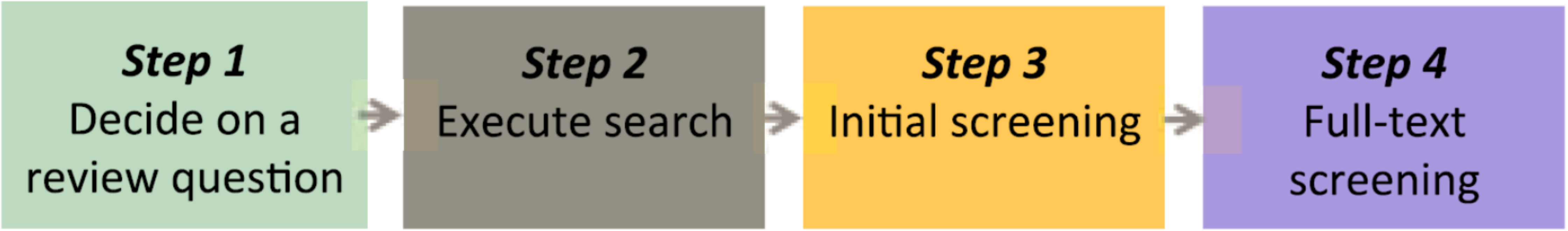
…

Manual downloading

Use an excel to make the decision and reason



We would have completed all the steps to get our evidence base



Foo et al. 2021

“a review maybe susceptible to a suite of biases and limitations if not conducted in a comprehensive, repeatable and systematic way”

Eligibility screening	Lack of consistency	Different reviewers make different decisions regarding eligibility	Set out planned methods in detail in an a priori protocol developed through scoping of the literature. Test for consistency formally, discussing disagreements and revising criteria where necessary
Retrieval	Dissemination bias	Lack of accessibility (due to limitations of the reviewers’ subscription, for example) meaning that some studies are omitted from the review	Comprehensive retrieval of studies
Data extraction	Lack of consistency	Different reviewers extract different data	Set out planned methods in detail in an a priori protocol developed through scoping of the literature. Test for consistency formally, discussing disagreements and revising criteria where necessary
Critical appraisal	Outcome reporting bias	Some study measured outcomes are omitted from a research article despite having been recorded by the authors, possibly due to a lack of significance	Critical appraisal
	Full publication bias	Researchers may only publish parts of their study, for example due to significance	Critical appraisal/author contact

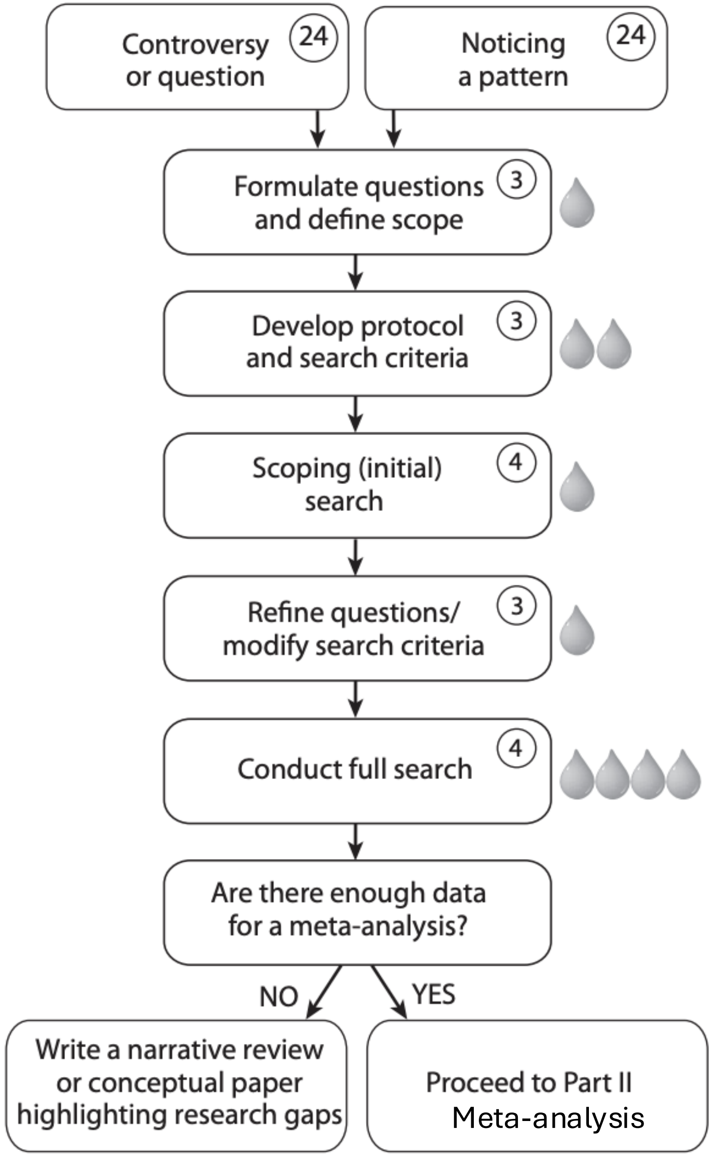
Haddaway & Macura 2018, Nature Climate Change

Is there enough quantitative data to synthesise?

If yes: WELCOME TO META-ANALYSIS TERRITORY!

If No?

You have choices and wise decisions to make…



Cote and Jennions (2013) in the Handbook of Meta-Analysis in Ecology and Evolution

Is there enough quantitative data to synthesise?

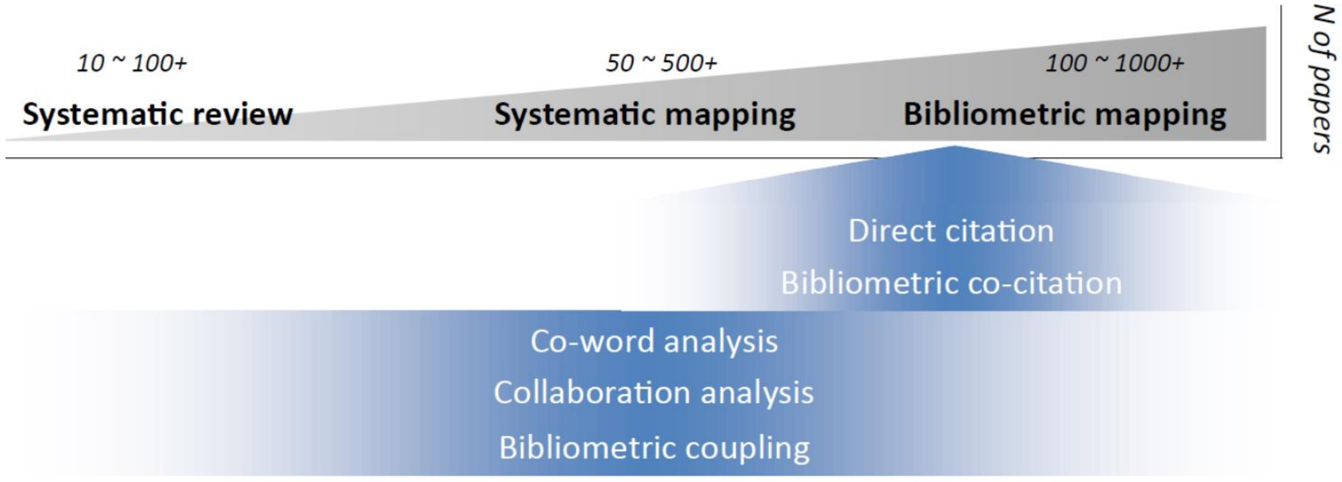
If yes: WELCOME TO META-ANALYSIS TERRITORY!

If No?

Narrative Review or a concept paper if you like

Qualitative Review

- Use Systematic Maps
- Bibliometric Analysis



Nakagawa et al. 2018, TREE

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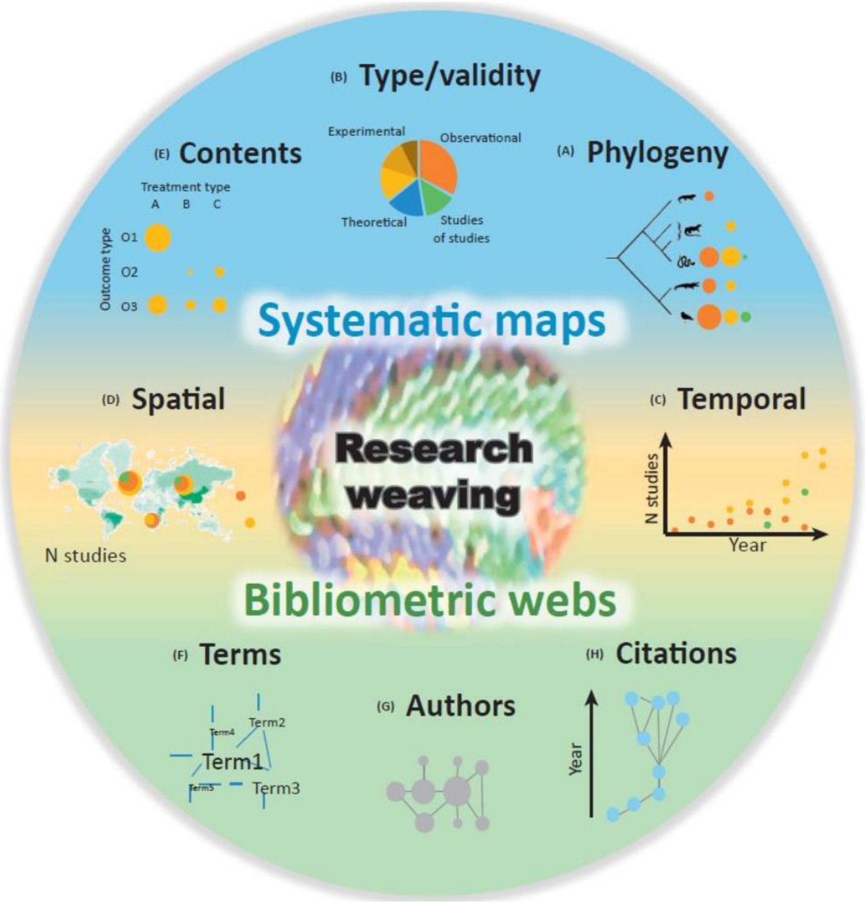

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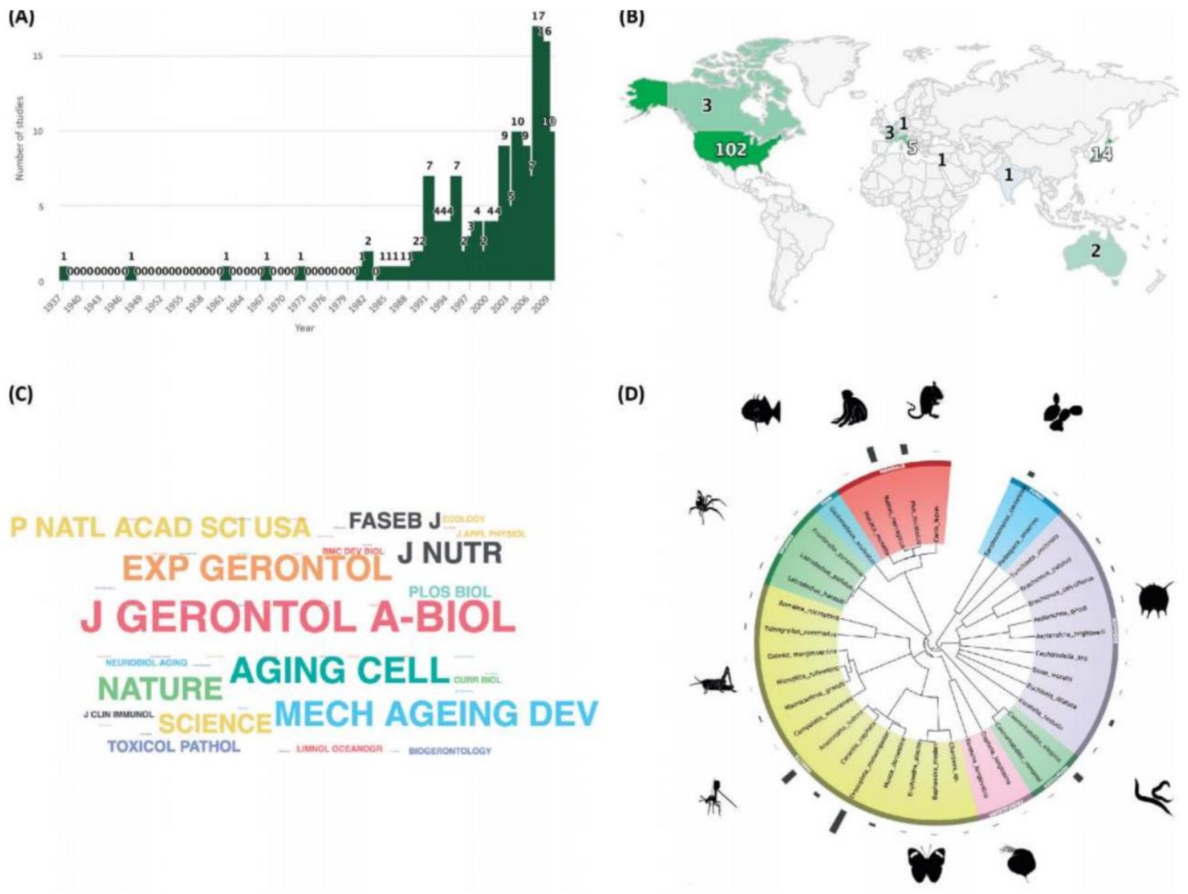
- Use Systematic Maps
- Bibliometric Analysis



Nakagawa et al. 2018, TREE

Is there enough quantitative data to synthesise?

If yes: WELCOME TO META-ANALYSIS TERRITORY!

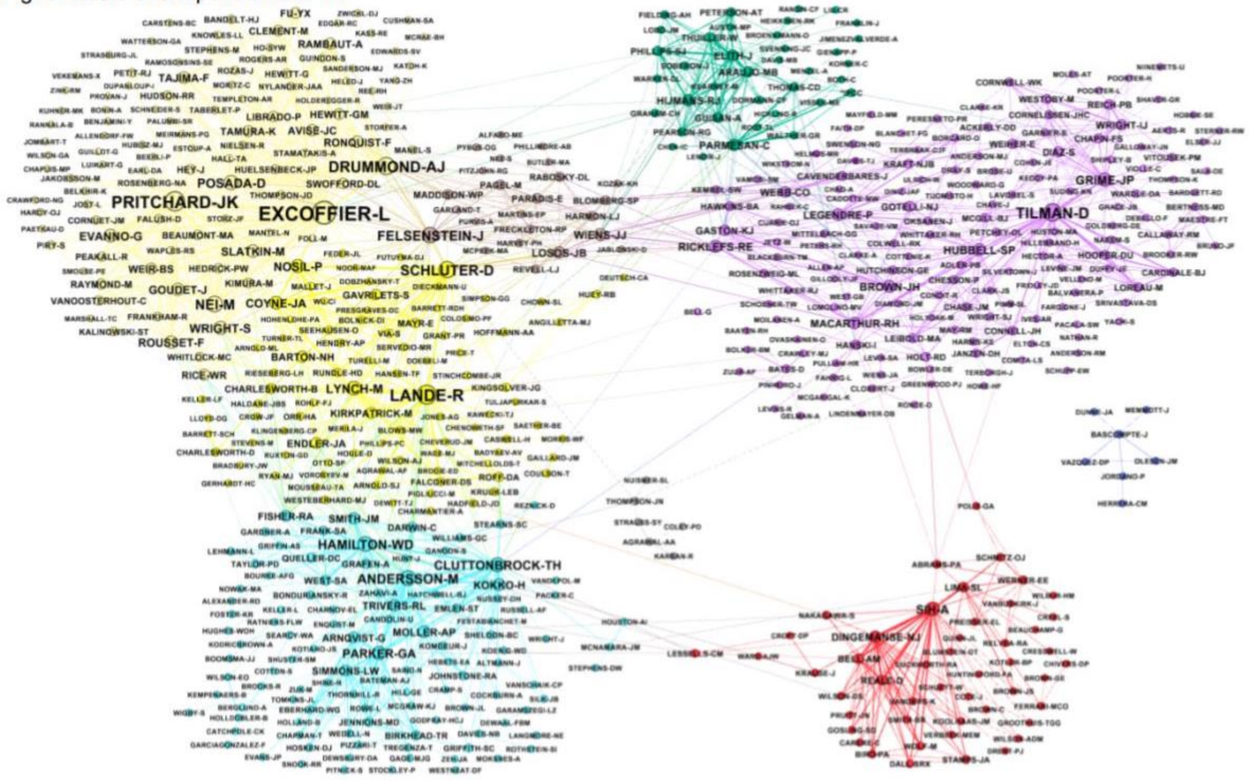
If No?

Narrative Review or a concept paper if you like

Qualitative Review

- Use Systematic Maps
- Bibliometric Analysis

Fig. 8. Réale et al.: period 2010-14



Nakagawa et al. 2018, TREE

Is there enough quantitative data to synthesise?

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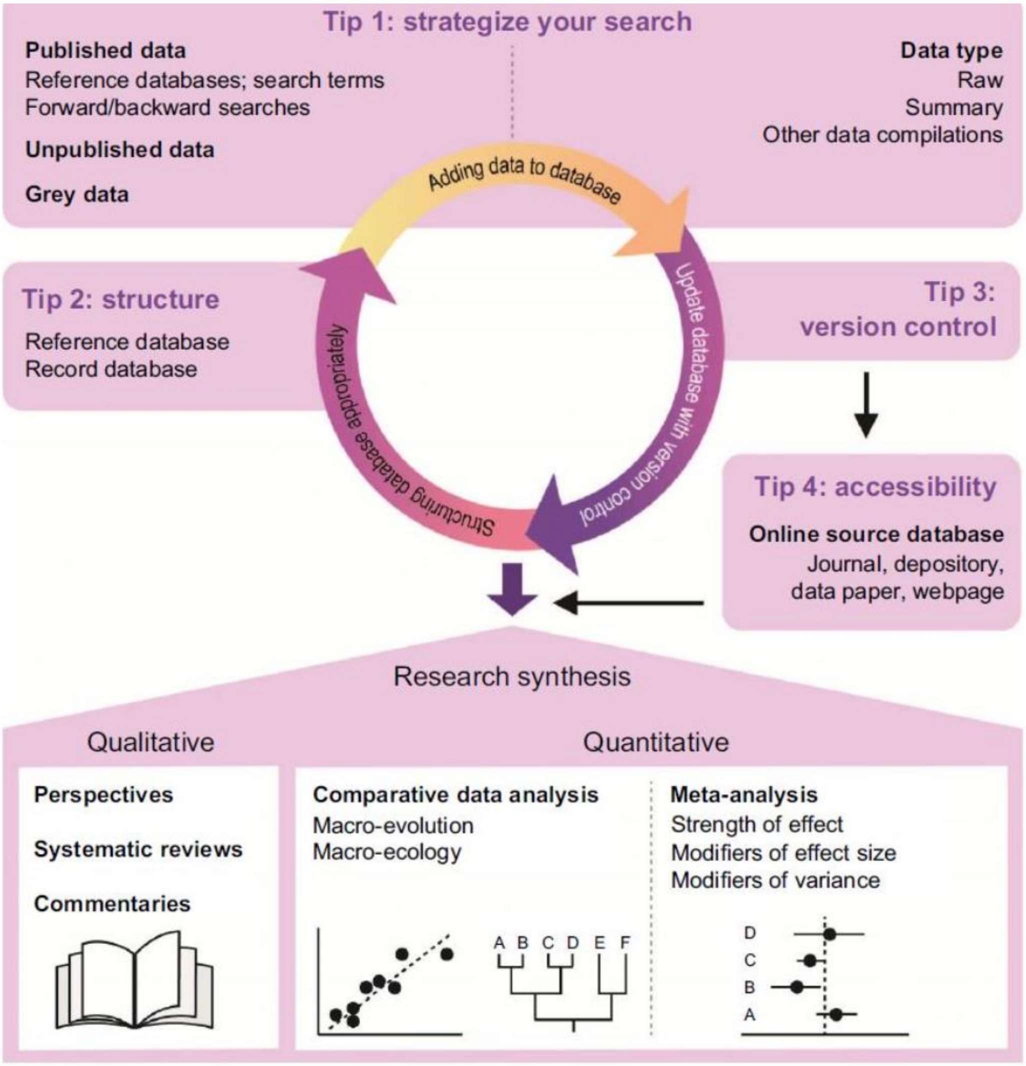
If No?

Narrative Review or a concept paper if you like

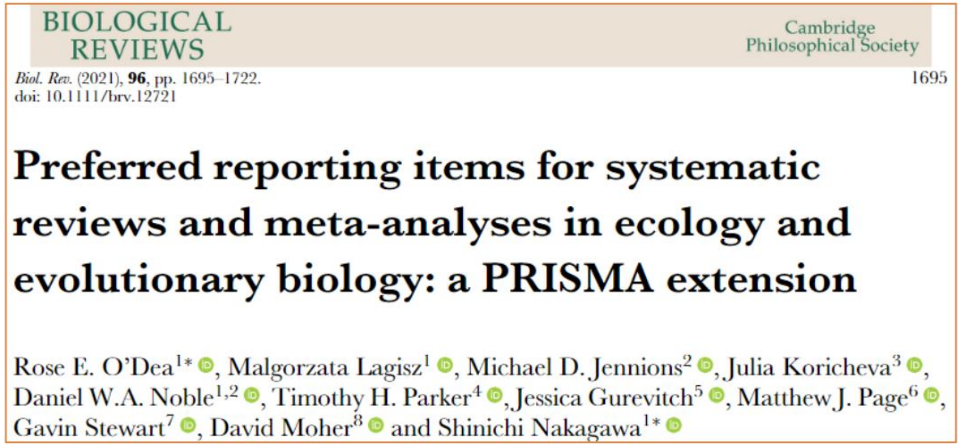
Qualitative Review

- Use Systematic Maps
- Bibliometric Analysis



Schwanz et al. 2022

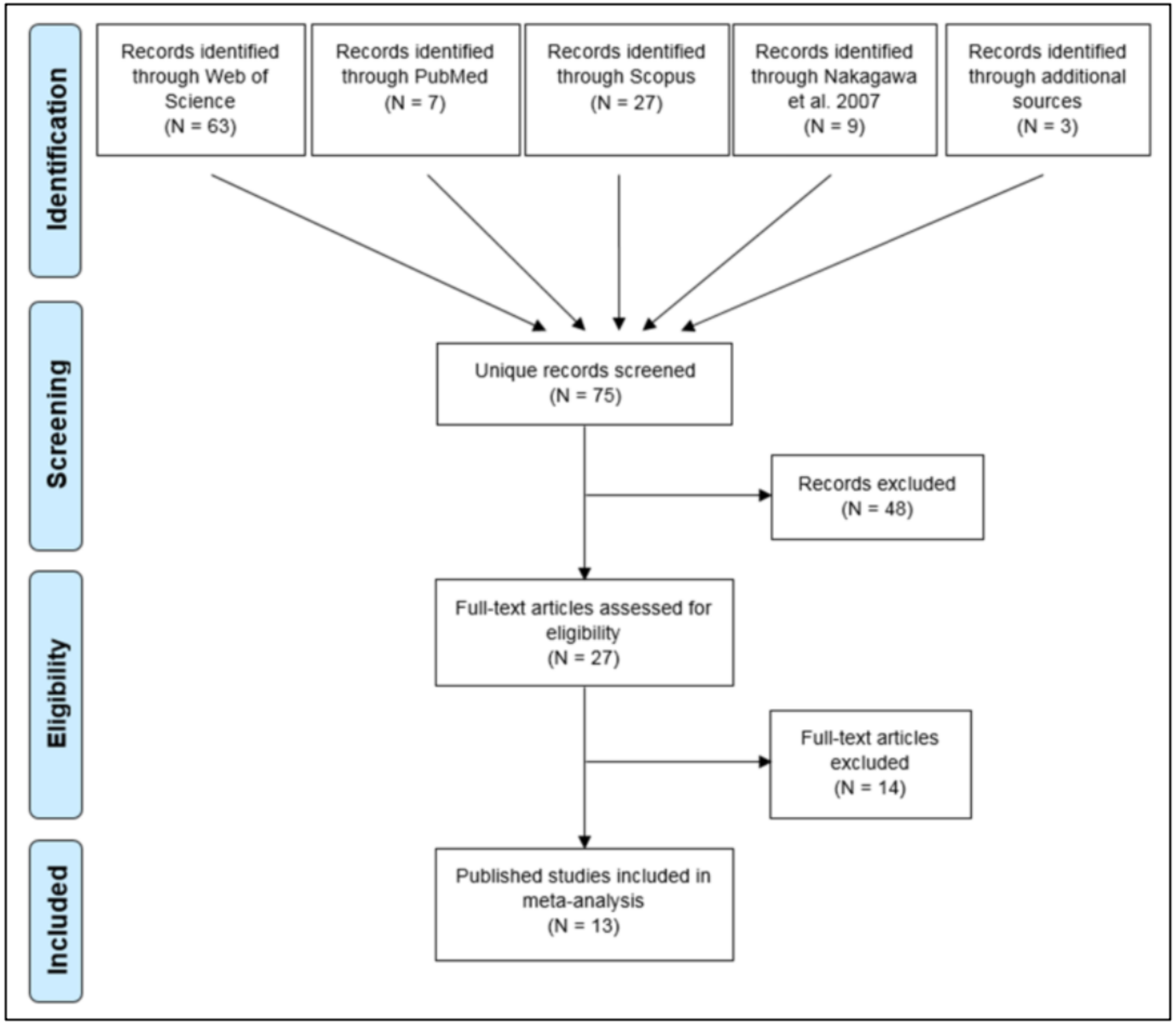
Whatever you decide, document and report well!



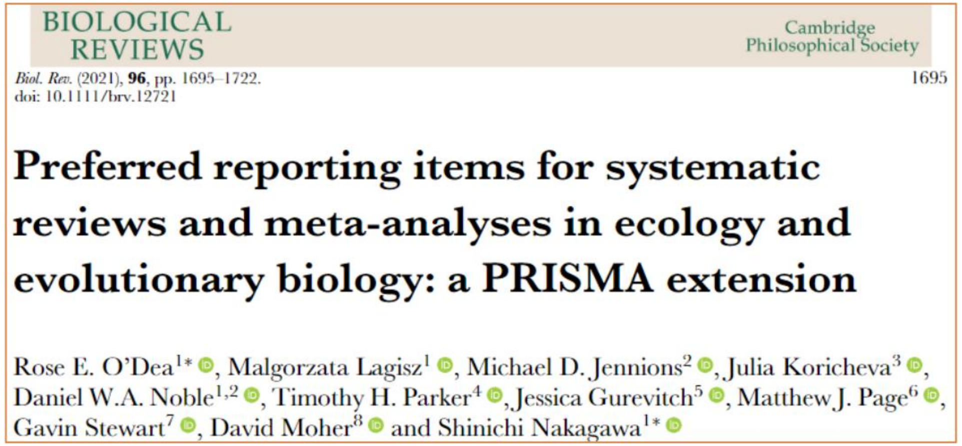
BIOLOGICAL REVIEWS
Cambridge Philosophical Society
Biol. Rev. (2021), **96**, pp. 1695–1722. 1695
doi: 10.1111/brv.12721

Preferred reporting items for systematic reviews and meta-analyses in ecology and evolutionary biology: a PRISMA extension

Rose E. O'Dea[1]*, Malgorzata Lagisz[1], Michael D. Jennions[2], Julia Koricheva[3], Daniel W.A. Noble[1,2], Timothy H. Parker[4], Jessica Gurevitch[5], Matthew J. Page[6], Gavin Stewart[7], David Moher[8] and Shinichi Nakagawa[1]*



Don’t let your study be part of low score group!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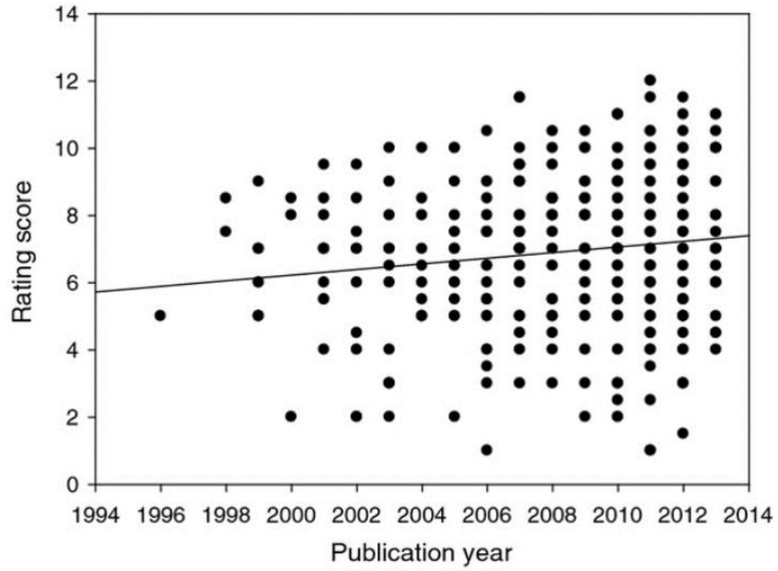
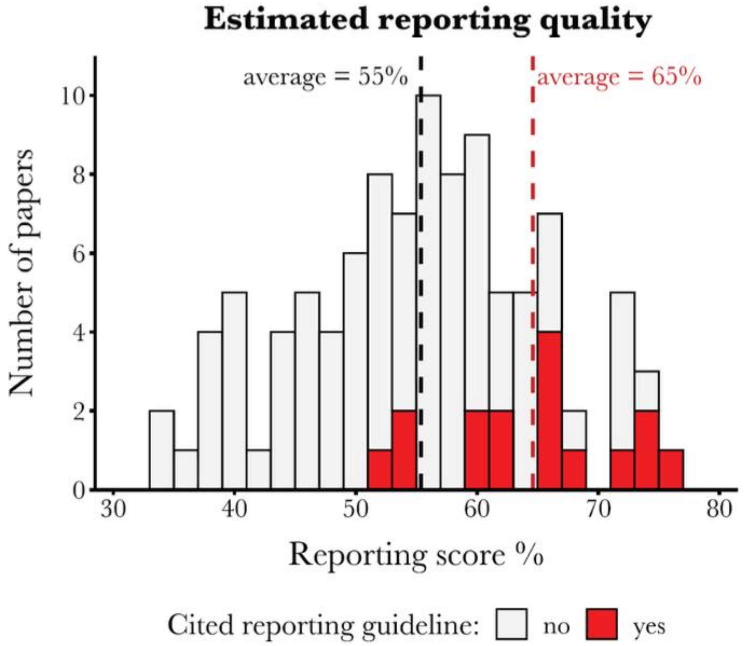
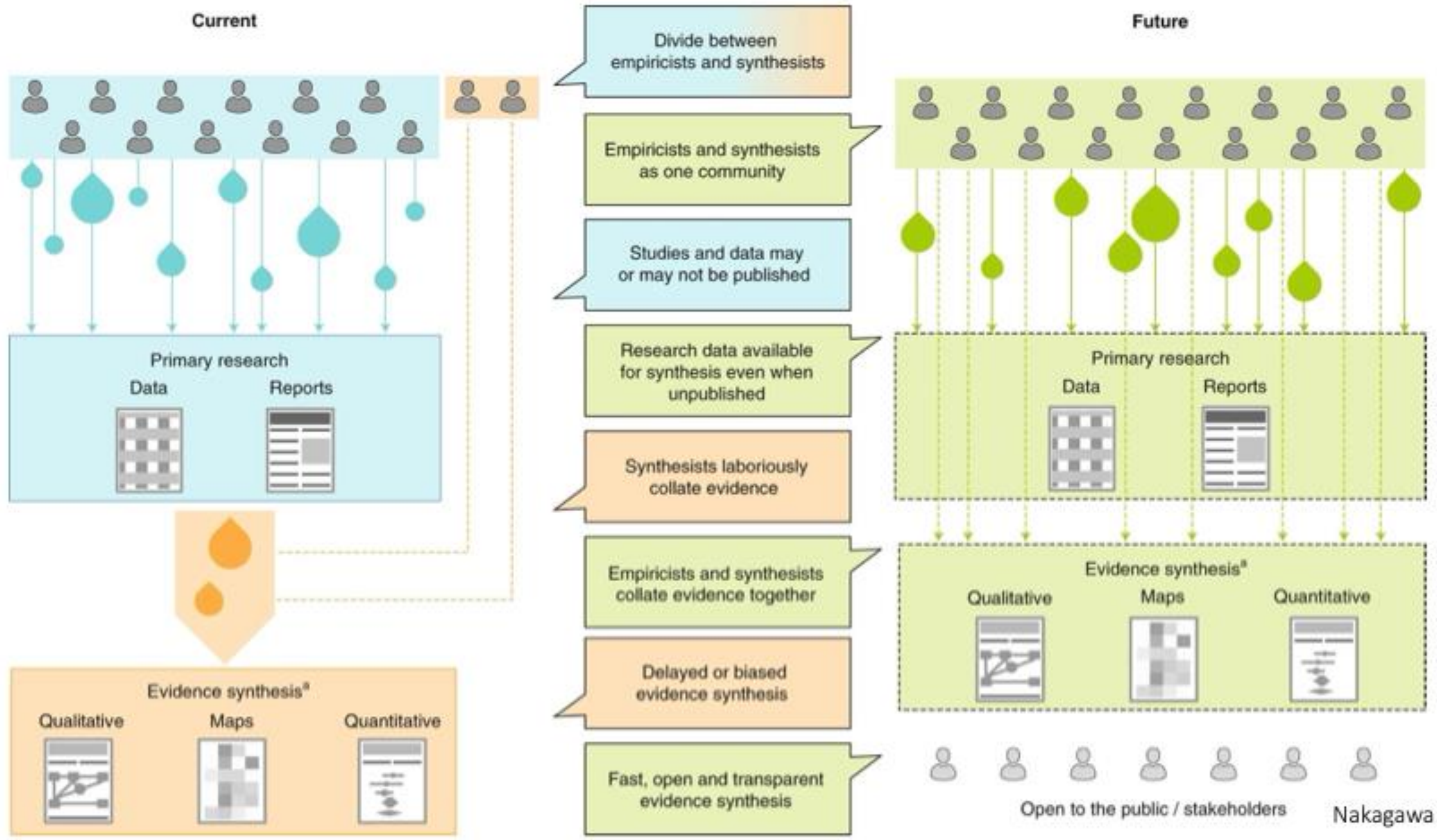




Koricheva & Gurevitch 2014,
O’Dea et al. 2021

Future?



Nakagawa et al. 2020, NEE

Now you can conduct a cool and thorough Systematic Review!

Introduction to Meta-Analysis

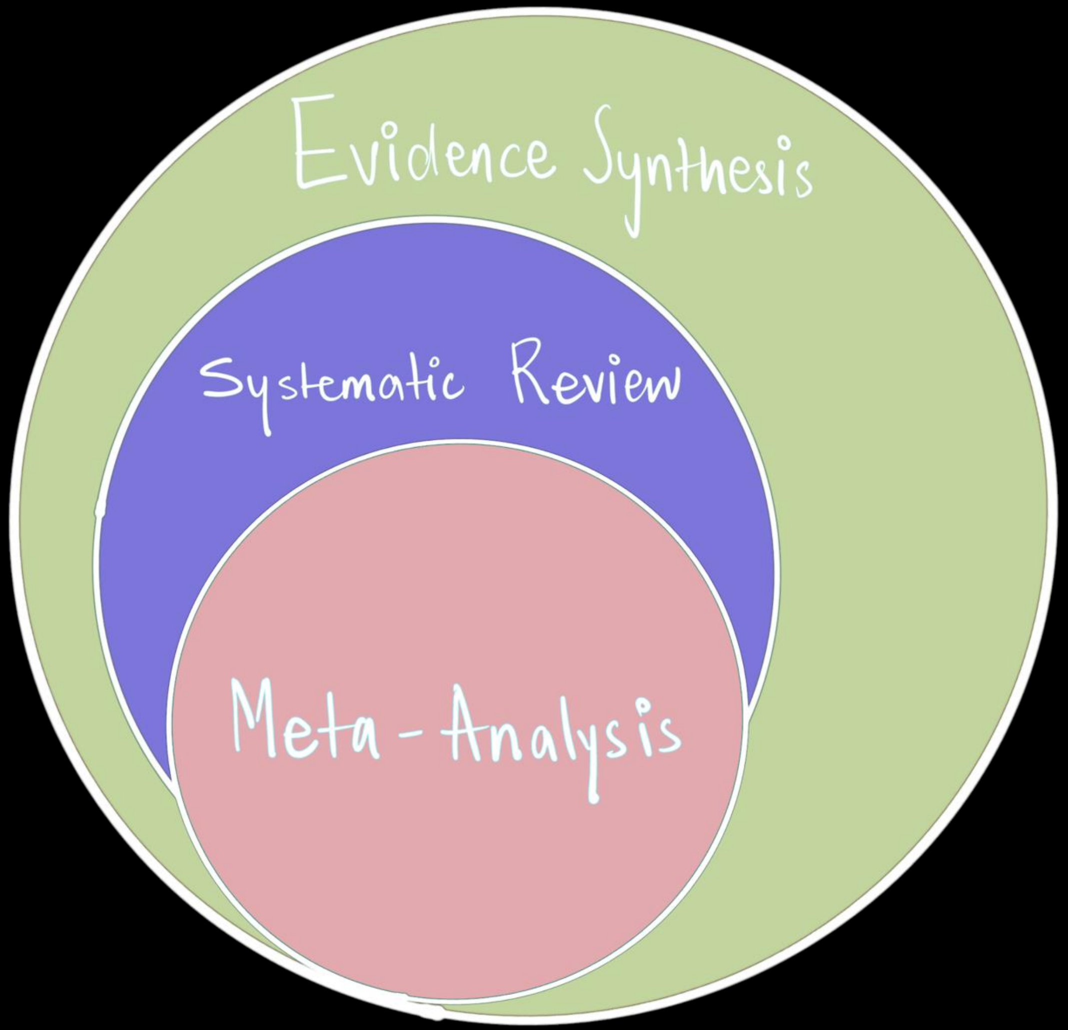
Pre-conference Workshop: Behaviour 2025

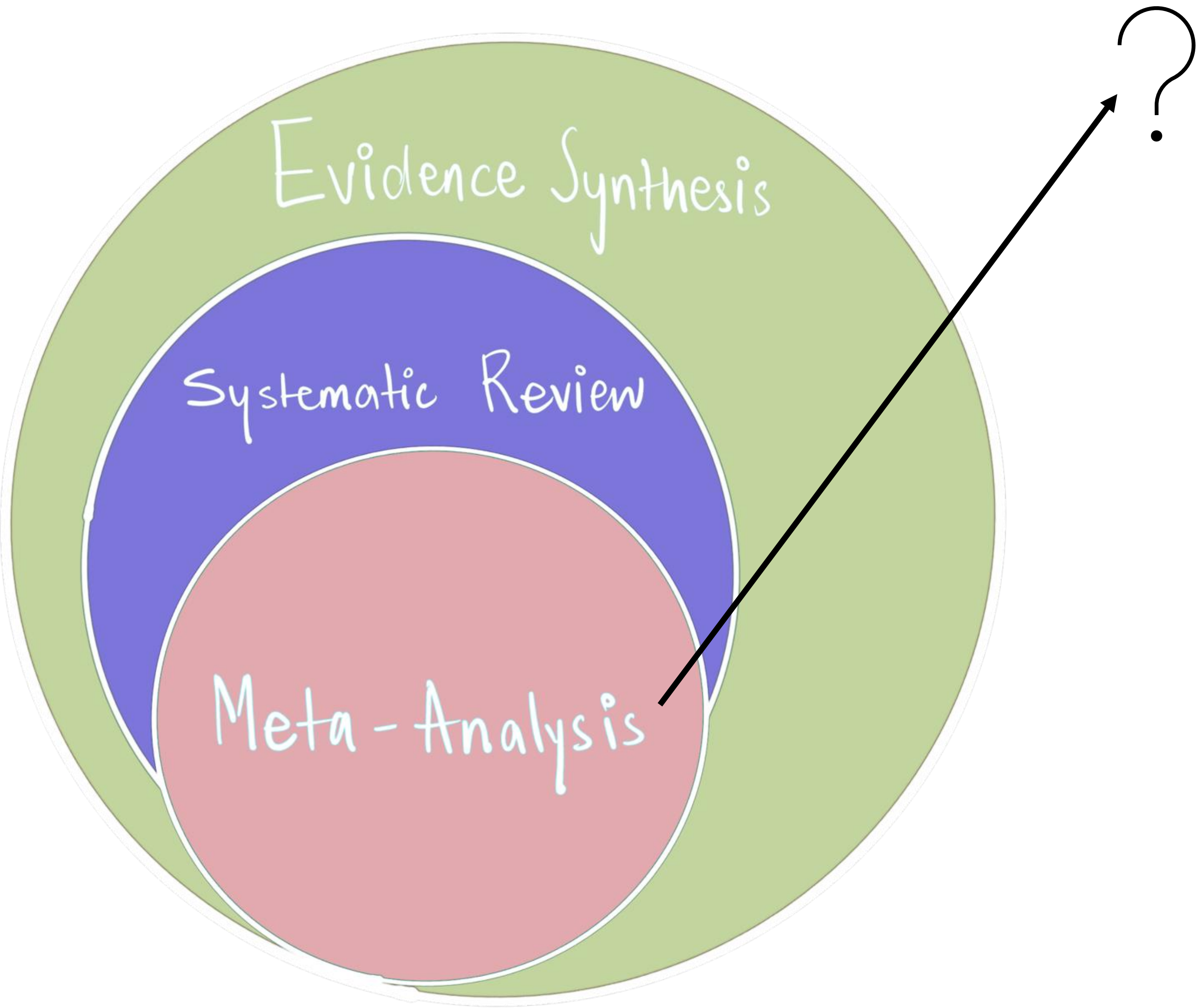
Instructor:

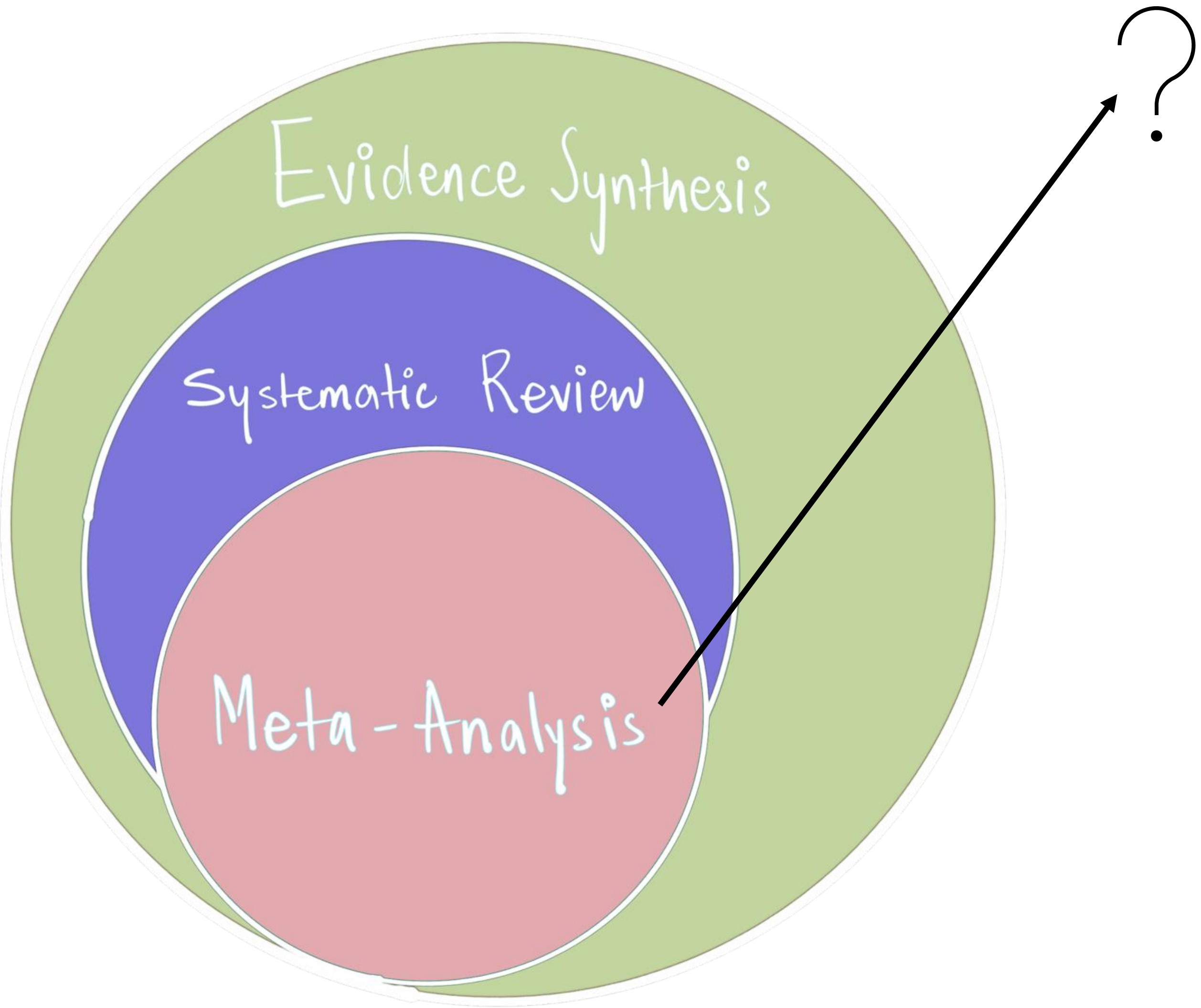
Shreya Dimri

Department of Evolutionary Biology

Bielefeld University (Germany)




Evidence Synthesis
Systematic Review
Meta-Analysis



"A meta-analysis combines evidence, most often by weighting evidence according to how much we trust it. This consequently allows us to increase power (and precision) when compared to the primary studies."

Meta-analysis **has 3 main goals**:

- Test overall effects and generality
Meta-analytic models (i.e., intercept-only models) estimate and explore heterogeneity (i.e., variation in effects not explained by differences in precision).

- Understand pattern of effects (i.e., try to explain heterogeneity)
Meta-regressions (i.e., including moderators): Biological, Methodological, including (and very importantly) publication biases

- Generate hypotheses

What is the average height of humans?

Study A

Study E

Study B

Study D

Study C

What is the average height of humans?

Study A

Study E

Study B

Study D

Study C

Put them all together

Voila! Meta-analysis

What is the average height of humans?

Study A

Study E

Study B

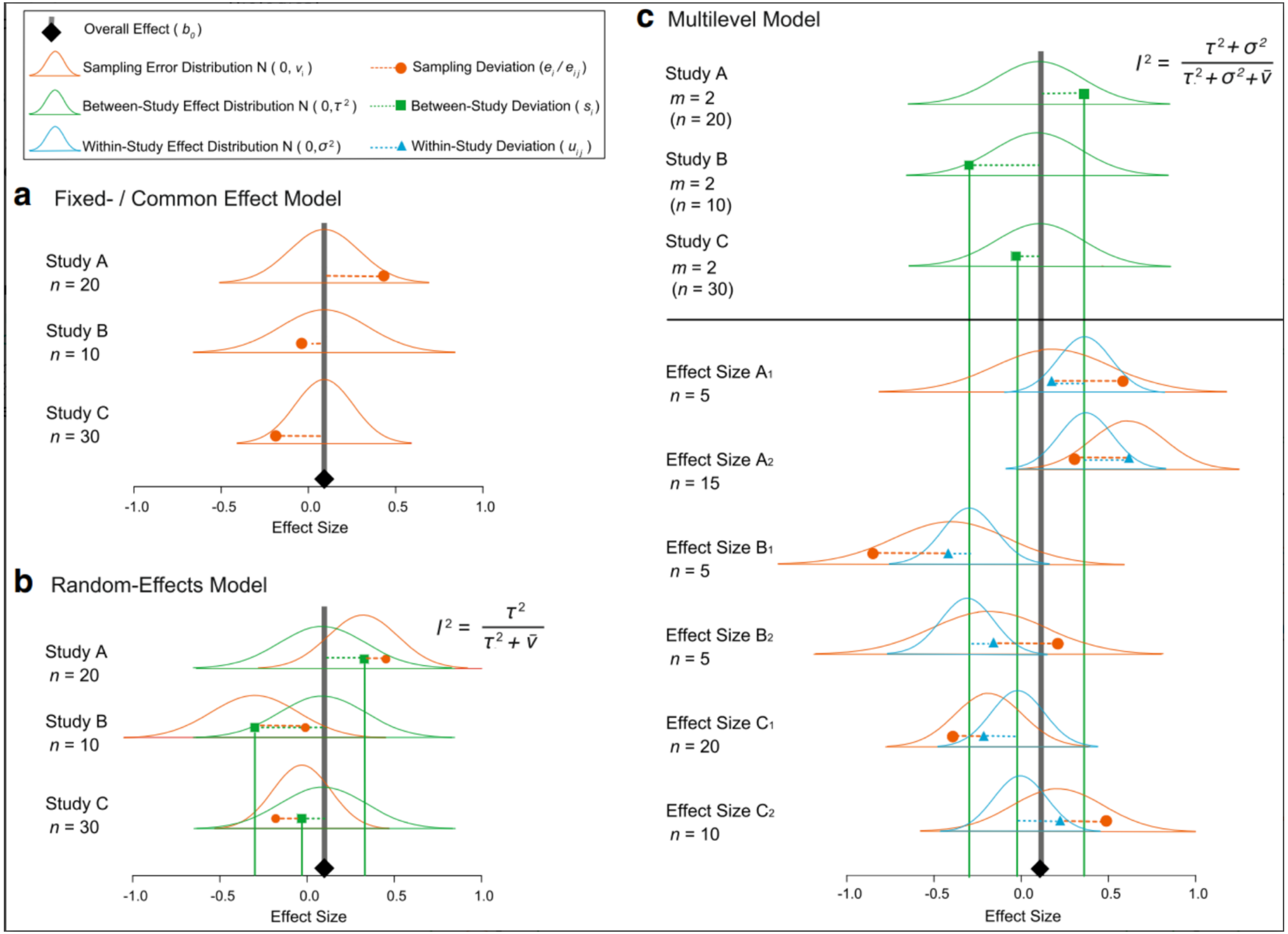
Study D

Study C

Put them all together

Voila! Meta-analysis

Mean?
Weighted mean?
Are the units the same?



Overwhelming, but we will get to it!

Nakagawa et al. 2017

Effect Sizes: The common currency of meta-analysis

A measure of the magnitude and direction of a difference, association or central tendency.

Effect size value (primary study): corresponds to one individual effect size

- The mean difference between the treatment and control groups is 0.66.
- The mean of the treatment group is 1.33 times larger than the mean of the control group.
- The mean of the treatment group is 33% higher than the mean of the control group.
- Any of the inferential statistics explored above (e.g., t-value, F-value, etc, but beware of the sign)

Effect Sizes: The common currency of meta-analysis

A measure of the magnitude and direction of a difference, association or central tendency.

Effect size statistic (meta-analysis): refers to the meta-analytic effect sizes that we use to integrate and compare results from different sources,
e.g. *Cohen's d*, *Hedges' g*, *lnRR*, Zr etc.

Most effect size statistics focus on the magnitude and direction of a **difference** or an **association**, but we can also meta-analyse variation.

Something important about effect size statistics is that they have an associated **sampling variance**, which is essential for meta-analysis.

Effect Sizes: The common currency of meta-analysis

A measure of the magnitude and direction of a difference, association or central tendency.

According to Cooper, Hedges, and Valentine 2019,
Four major considerations should drive the choice of an effect size statistic:

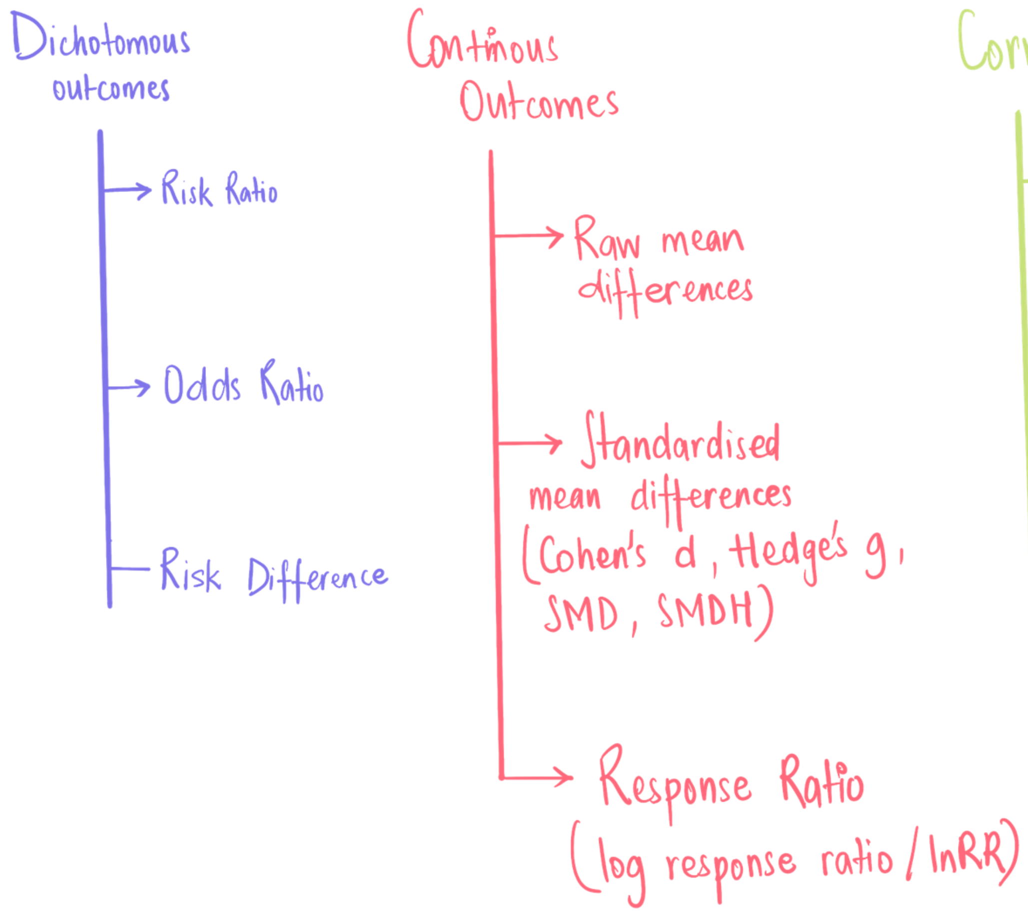
1. Comparable to one another in the sense that they measure (at least approximately) the same thing.

2. Substantively interpretable (AKA meaningful for the question at hand).

3. Computable from the information that is likely to be reported in published research reports. That is, it should *generally* not require the reanalysis of the raw data.

4. Have good technical properties (e.g., its sampling distribution should be known so that variances and confidence intervals can be computed).

What effect sizes are out there?

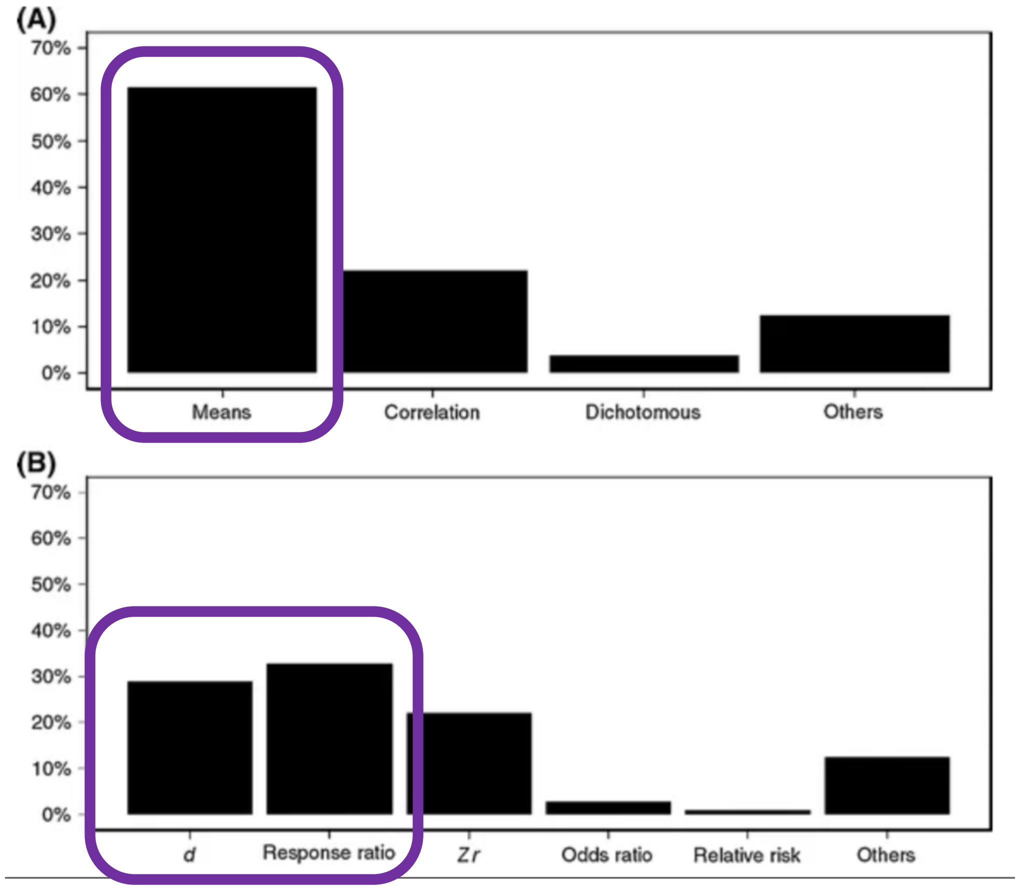


For variance:

- Coefficient of variation (CV)
- lnCVR
- Variance standardised variance components

What effect sizes are out there?

Let's focus on means today!



Nakagawa and Santos 2021

Whenever the focus is on comparing two groups!

We can meta-analyse standardised mean differences or mean ratios (or both!)

Standardised Mean Differences (SMD):

including Cohen's *d* or its *betterself*, Hedges' *g, SMDH*

$$\mathrm{SMD}_i = \frac{\overline{\mathrm{X}}_{2i} - \overline{\mathrm{X}}_{1i}}{\sqrt{\frac{(n_{1i}-1)SD_{1i}^2 + (n_{2i}-1)SD_{2i}^2}{n_{1i}+n_{2i}-2}}},$$

$$\mathrm{Var}\left(\mathrm{SMD}_i\right) = \frac{n_{1i}+n_{2i}}{n_{1i}n_{2i}} + \frac{\mathrm{SMD}_i^2}{2\left(n_{1i}+n_{2i}\right)},$$

$$J = 1 - \frac{3}{4(N_1+N_2-2)-1};$$

Whenever the focus is on comparing two groups!

We can meta-analyse standardised mean differences or mean ratios (or both!)

Log Reponse Ratio (lnRR):

Sometimes referred to as the Ratio of the Means (ROM)

$$\ln RR_i = \ln\left(\frac{\overline{X}_{2i}}{\overline{X}_{1i}}\right), \quad \mathrm{Var}\left(\ln RR_i\right) = \frac{SD_{1i}^2}{n_{1i}\overline{X}_{1i}^2} + \frac{SD_{2i}^2}{n_{2i}\overline{X}_{2i}^2},$$

You can easily transform them into % change, making them easy to interpret

```
percent_change <- 100*(exp(lnRR)-1)
```

Experimental studies only

Trait measured no. of fledgelings, parasites in nest

Treatment (green nest material)

$Mean_{treatment}$

$Variance_{treatment}$

$N_{treatment}$

Control (no green nest material)

$Mean_{control}$

$Variance_{control}$

$N_{control}$

Standardized Mean Difference

$$\mathrm{SMD}_i = \frac{\boxed{\bar{X}_{2i} - \bar{X}_{1i}}}{\sqrt{\frac{(n_{1i}-1)SD_{1i}^2 + (n_{2i}-1)SD_{2i}^2}{n_{1i}+n_{2i}-2}}},$$

$$\mathrm{Var}\left(\mathrm{SMD}_i\right) = \frac{n_{1i}+n_{2i}}{n_{1i}n_{2i}} + \frac{\mathrm{SMD}_i^2}{2\left(n_{1i}+n_{2i}\right)},$$

The raw mean difference is simply $(\mathrm{m1i} - \mathrm{m2i})$, wh

measure="SMD", $\mathrm{sdi} = \sqrt{\frac{(\mathrm{n1i}-1)\mathrm{sd1i}^2 + (\mathrm{n2i}-1)\mathrm{sd2i}^2}{\mathrm{n1i}+\mathrm{n2i}-2}}$ i

```
dataset_SMDH<- escalc(measure = "SMD",
           vtype= "LS",
                  n1i = effective_n_experiment,
                  n2i = effective_n_control,
                  m1i = measure_central_tendency_experiment,
                  m2i = measure_central_tendency_control,
                  sd1i = sd_experiment,
                  sd2i = sd_control,
                  data = dataset_SMDH,
                  var.names = c('SMDH',
                                'SMDH_variance'),
                  add.measure = FALSE,
                  append = TRUE)
```

Log Response Ratio

$$\ln\mathrm{RR}_i = \ln\boxed{\left(\frac{\bar{X}_{2i}}{\bar{X}_{1i}}\right)},$$

$$\mathrm{Var}\left(\ln\mathrm{RR}_i\right) = \frac{SD_{1i}^2}{n_{1i}\bar{X}_{1i}^2} + \frac{SD_{2i}^2}{n_{2i}\bar{X}_{2i}^2},$$

```
dataset_lnRR<- escalc(measure = "ROM",
           vtype= "LS",
                  n1i = effective_n_experiment,
                  n2i = effective_n_control,
                  m1i = measure_central_tendency_experiment,
                  m2i = measure_central_tendency_control,
                  sd1i = sd_experiment,
                  sd2i = sd_control,
                  data = dataset_lnRR,
                  var.names = c('lnRR',
                                'lnRR_variance'),
                  add.measure = FALSE,
                  append = TRUE)
```

Package: metafor

But do you think this is possible?

We need:

Raw means of the two groups

Variance of the two groups

Sample sizes of the two groups

But do you think this is possible?

We need:

Raw means of the two groups

Variance of the two groups

Sample sizes of the two groups

RARELY POSSIBLE!!

Table 11.1 Computing *d*, Independent Groups

Reported	Computation of Needed Quantities
$\bar{Y}_1, \bar{Y}_2\, S_{Pooled}, n_1, n_2$	$d = \frac{Y_1 - Y_2}{S_{Pooled}}, v = \frac{n_1+n_2}{n_1 n_2} + \frac{d^2}{2(n_1+n_2)}$
t, n_1, n_2	$d = t\sqrt{\frac{n_1+n_2}{n_1 n_2}}, v = \frac{n_1+n_2}{n_1 n_2} + \frac{d^2}{2(n_1+n_2)}$
F, n_1, n_2	$d = \pm\sqrt{\frac{F(n_1+n_2)}{n_1 n_2}}, v = \frac{n_1+n_2}{n_1 n_2} + \frac{d^2}{2(n_1+n_2)}$
p(one-tailed), n_1, n_2	$d = \pm t^{-1}(p)\sqrt{\frac{n_1+n_2}{n_1 n_2}}, v = \frac{n_1+n_2}{n_1 n_2} + \frac{d^2}{2(n_1+n_2)}$
p(two-tailed), n_1, n_2	$d = \pm t^{-1}\left(\frac{p}{2}\right)\sqrt{\frac{n_1+n_2}{n_1 n_2}}, v = \frac{n_1+n_2}{n_1 n_2} + \frac{d^2}{2(n_1+n_2)}$

SOURCE: Authors' tabulation.
NOTE: The function $t^{-1}(p)$ is the inverse of the cumulative distribution function o Student's t with $n_1 + n_2 - 2$ degrees of freedom. Many computer programs anc spreadsheets provide functions that can be used to compute $t - 1$. Assume $n_1 = n_2 =$ 10, so that $df = 18$. Then, in Excel, for example, if the reported p-value is 0.0 (two-tailed) TINV(p,df) = TINV(0.05,18) will return the required value (2.1009). I the reported p-value is 0.05 (one-tailed), TINV(2p,df) = TINV(0.10,18) will retur the required value 1.7341. The F in row 3 is the F-statistic from a one-way analysi of variance. In rows 3 through 5, the sign of d must reflect the direction of the mear difference.

Cooper, Hedges, and Valentine 2019

$\bar{Y}_1, \bar{Y}_2\, S_{Difference}$, r, n (number of pairs)	$d = \left(\frac{\ldots}{S_{Difference}}\right)\ldots, v = \left(\frac{1}{n} + \frac{\ldots}{2n}\right)\ldots$
t (from paired t-test), r, n	$d = t\sqrt{\frac{2(1-r)}{n}}, v = \left(\frac{1}{n} + \frac{d^2}{2n}\right)2(1-r)$
F (from repeated measures ANOVA), r, n	$d = \pm\sqrt{\frac{2F(1-r)}{n}}, v = \left(\frac{1}{n} + \frac{d^2}{2n}\right)2(1-r)$
p (one-tailed), r, n	$d = \pm t^{-1}(p)\sqrt{\frac{2(1-r)}{n}}, v = \left(\frac{1}{n} + \frac{d^2}{2n}\right)2(1-r)$
p (two-tailed), r, n	$d = \pm t^{-1}\left(\frac{p}{2}\right)\sqrt{\frac{2(1-r)}{n}}, v = \left(\frac{1}{n} + \frac{d^2}{2n}\right)2(1-r)$

SOURCE: Authors' tabulation.
NOTE: The function $t^{-1}(p)$ is the inverse of the cumulative distribution function of Student's t with $n - 1$ degrees of freedom. Many computer programs and spreadsheets provide functions that can be used to compute t^{-1}. Assume $n = 19$, so that $df = 18$. Then, in Excel, for example, if the reported p-value is 0.05 (2-tailed), TINV(p,df) = TINV(0.05,18) will return the required value (2.1009). If the reported p-value is 0.05 (1-tailed), TINV(2p,df) = TINV(0.10,18) will return the required value 1.7341. The F in row 3 of the table is the F-statistic from a one-way repeated measures analysis of variance. In rows 3 through 5, the sign of d must reflect the direction of

t (from ANCOVA), n_1, n_2, R, q	$d = t\sqrt{\frac{n_1+n_2}{n_1 n_2}}\sqrt{1-R^2}, v = \frac{(n_1+n_2)(1-R^2)}{n_1 n_2} + \frac{d^2}{2(n_1+n_2)}$
F (from ANCOVA), n_1, n_2, R, q	$d = \pm\sqrt{\frac{F(n_1+n_2)}{n_1 n_2}}\sqrt{1-R^2}, v = \frac{(n_1+n_2)(1-R^2)}{n_1 n_2} + \frac{d^2}{2(n_1+n_2)}$
p (1-tailed, from ANCOVA), n_1, n_2, R, q	$d = \pm t^{-1}(p)\sqrt{\frac{n_1+n_2}{n_1 n_2}}\sqrt{1-R^2}, v = \frac{(n_1+n_2)(1-R^2)}{n_1 n_2} + \frac{d^2}{2(n_1+n_2)}$
p (2-tailed, from ANCOVA), n_1, n_2, R, q	$d = \pm t^{-1}\left(\frac{p}{2}\right)\sqrt{\frac{n_1+n_2}{n_1 n_2}}\sqrt{1-R^2}, v = \frac{(n_1+n_2)(1-R^2)}{n_1 n_2} + \frac{d^2}{2(n_1+n_2)}$

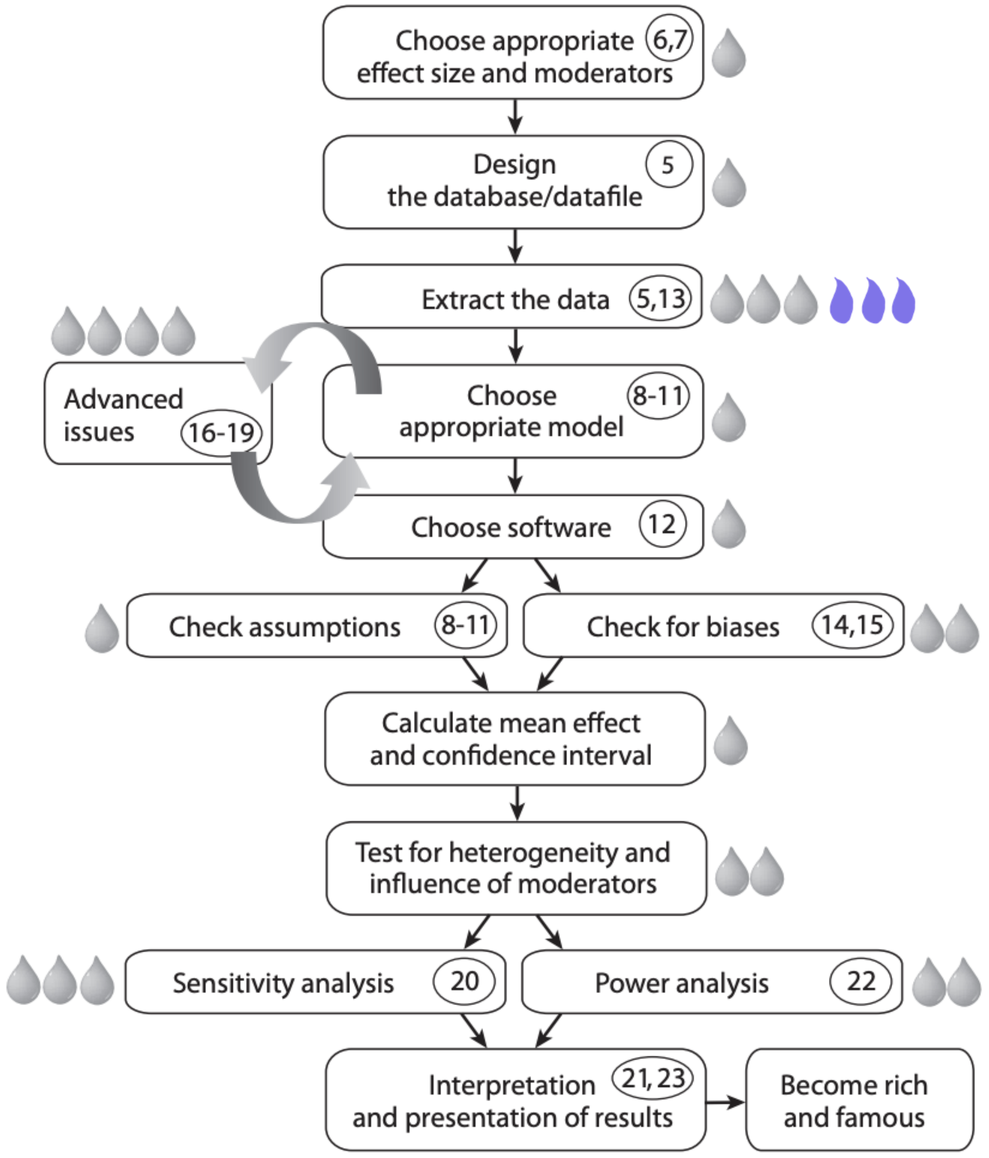
SOURCE: Authors' tabulation.
NOTE: The function $t^{-1}(p)$ is the inverse of the cumulative distribution function of Student's t with $n_1 + n_2 - 2 - q$ degrees of freedom, q is the number of covariates, and R is the covariate outcome correlation or multiple correlation. Many computer programs and spreadsheets provide functions that can be used to compute t^{-1}. Assume $n_1 = n_2 = 11$, and $q = 2$, so that $df = 18$. Then, in Excel, for example, if the reported p-value is 0.05 (2-tailed), TINV(p,df) = TINV(0.05,18) will return the required value (2.1009). If the reported p-value is 0.05 (1-tailed), TINV(2p,df) = TINV(0.10,18) will return the required value 1.7341. The F in row 3 of the table is the F-statistic from a one-way analysis of covariance. In rows 3 through 5, the sign of d must reflect the direction of the mean difference.

So now we need data

It gets tough here!

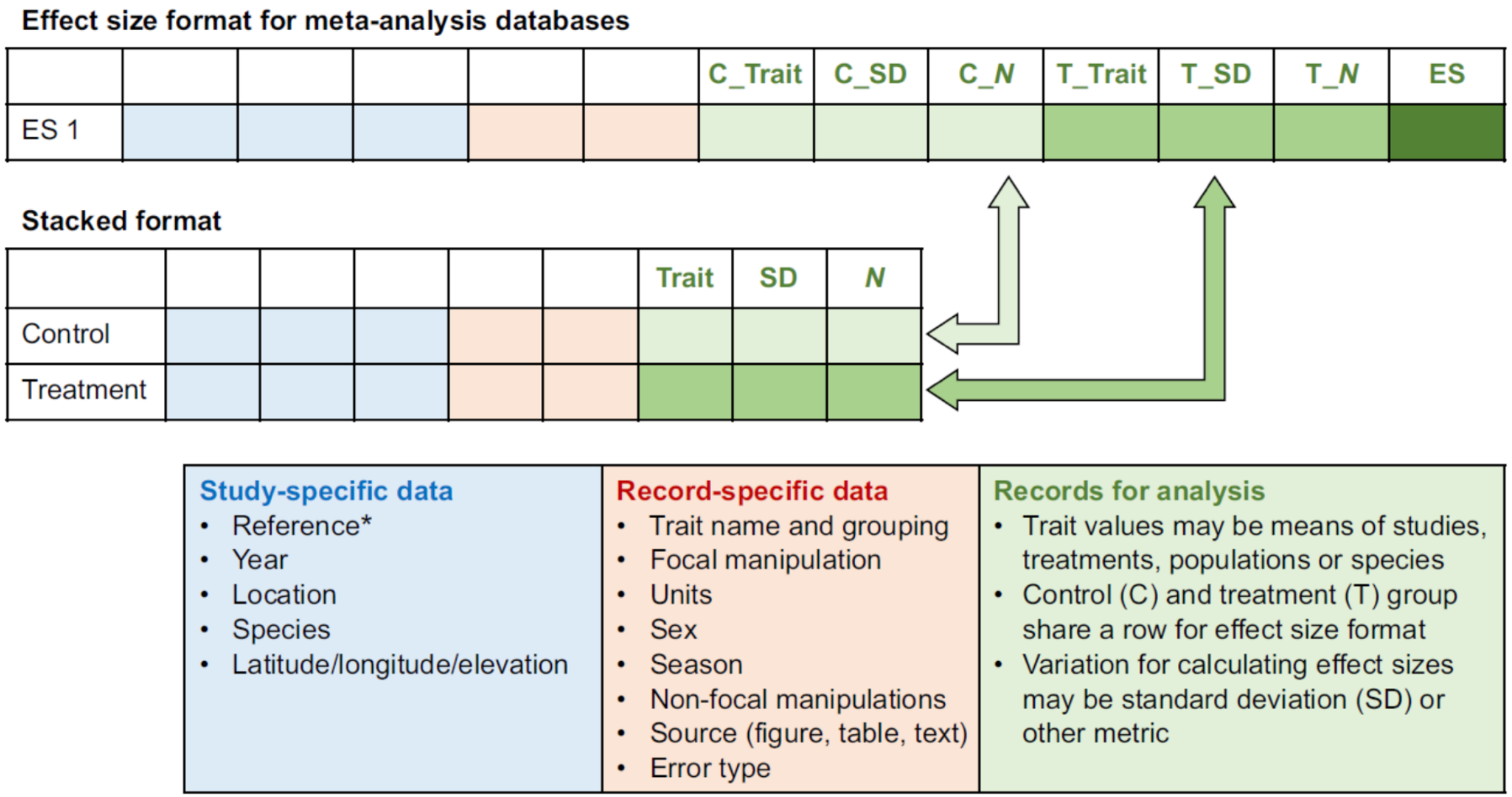
"We often spend too much time thinking about the models and little time thinking about the data and how to collect it. Remember that "the point of statistics is not to do myriad rigorous mathematical calculations; the point is to gain insight into meaningful social phenomena. Statistical inference is really just the marriage of [...] data and probability (with a little help from the central limit theorem)"

(Charles Wheelan in "Naked Statistics")



Cote and Jennions (2013) in the Handbook of Meta-Analysis in Ecology and Evolution

So now we need data, let’s try!



Schwanz et al. 2022

So now we need data, let’s try!

Experimental studies only

Trait measured no. of fledgelings, parasites in nest

Treatment (green nest material)

Control (no green nest material)

$\text{Mean}_{treatment}$

$\text{Variance}_{treatment}$

$N_{treatment}$

$\text{Mean}_{control}$

$\text{Variance}_{control}$

$N_{treatment}$

paper_ID	measure_central_tendency_experiment	type_measure_central_tendency_experiment	measure_dispersion_experiment	type_measure_dispersion_experiment	n_experiment	measure_central_tendency_control	type_measure_central_tendency_control	measure_dispersion_control	type_measure_dispersion_control	n_control	fitness_proxy
GNM_001	123.1	mean	2.3	SE	17	115.2	mean	2.4	SE	16	hemoglobin cor
GNM_001	8.77	mean	1.68	SD	17	8.25	mean	1.88	SD	16	number of nest
GNM_001	235.75	mean	44.58	SD	17	252.27	mean	57.35	SD	16	glucose concen
GNM_002	1.8359024	mean	0.01990244	SD	2	1.7449756	mean	0.02458537	SD	3	Nestling tarsus
GNM_002	1.8515122	mean	0.01717073	SD	6	1.8148293	mean	0.01404878	SD	4	Nestling tarsus
GNM_002	1.8300488	mean	0.01326829	SD	4	1.8226341	mean	0.01717073	SD	7	Nestling tarsus
GNM_002	1.8065548	mean	0.01653784	SD	4	1.8316795	mean	0.02003623	SD	3	Nestling tarsus
GNM_002	1.8383583	mean	0.01621981	SD	4	1.8434469	mean	0.01303945	SD	6	Nestling tarsus
GNM_002	1.845037	mean	0.02417069	SD	2	1.8230926	mean	0.01590177	SD	4	Nestling tarsus
GNM_016	82.2	mean	0.52	SE	27	84.1	mean	0.45	SE	27	fledgling body r
GNM_016	79.5	mean	0.49	SE	27	79.3	mean	0.44	SE	27	fledgling body r
GNM_016	30.4	mean	0.05	SE	27	30.4	mean	0.05	SE	27	fledling tarsus-
GNM_016	30	mean	0.06	SE	27	29.84	mean	0.05	SE	27	fledling tarsus-
GNM_016	2.5	mean	0.15	SE	27	2.58	mean	0.16	SE	27	number of fledl
GNM_016	0.04035088	mean	0.01754386	SE	27	0.10964912	mean	0.02719298	SE	27	probability of lc
GNM_016	0.13596491	mean	0.03947368	SE	27	0.15964912	mean	0.04035088	SE	27	probability of lc
GNM_016	0.06666667	mean	0.0245614	SE	27	0.11754386	mean	0.02894737	SE	27	probability of lc
GNM_016	0.04561404	mean	0.02017544	SE	27	0.0877193	mean	0.02631579	SE	27	probability of lc
GNM_024	1.828125	mean	0.04640388	SE	8	1.475	mean	0.02261335	SE	11	females' testos
GNM_024	4.937931	mean	0.04310345	SE	16	4.103448	mean	0.05221186	SE	19	Clutch size
GNM_024	3.78	mean	0.04	SD	16	3.79	mean	0.03	SD	19	egg size (mm3)
GNM_024	0.88	mean	0.15	SD	16	0.93	mean	0.18	SD	19	Yolk Log T (test
GNM_024	0.99	mean	0.09	SD	16	0.96	mean	0.14	SD	19	Yolk Log A4(anc

[redacted]

paper_ID	fulltext_screening	authors	year_publication	population_location	bird_species	treatment_plant_species	control_plant_species
GNM_001	included	Gladalski M.; Ba	2020	Lodz_Poland	Cyanistes caeruleus	Lavandula angusti	grass
GNM_001	included	Gladalski M.; Ba	2020	Lodz_Poland	Cyanistes caeruleus	Lavandula angusti	grass
GNM_001	included	Gladalski M.; Ba	2020	Lodz_Poland	Cyanistes caeruleus	Lavandula angusti	grass
GNM_002	included	Pires B. A.; Belo	2020	Portuguese Air Force	Cyanistes caeruleus	Lavandula dentata	grass
GNM_002	included	Pires B. A.; Belo	2020	Portuguese Air Force	Cyanistes caeruleus	Lavandula dentata	grass
GNM_002	included	Pires B. A.; Belo	2020	Portuguese Air Force	Cyanistes caeruleus	Lavandula dentata	grass
GNM_002	included	Pires B. A.; Belo	2020	Portuguese Air Force	Cyanistes caeruleus	Lavandula dentata	grass
GNM_002	included	Pires B. A.; Belo	2020	Portuguese Air Force	Cyanistes caeruleus	Lavandula dentata	grass
GNM_002	included	Pires B. A.; Belo	2020	Portuguese Air Force	Cyanistes caeruleus	Lavandula dentata	grass
GNM_016	included	Polo V.; Rubalca	2015	Manzanares el Real	Sturnus unicolor	Lavandula stoecha	blank
GNM_016	included	Polo V.; Rubalca	2015	Manzanares el Real	Sturnus unicolor	Lavandula stoecha	blank
GNM_016	included	Polo V.; Rubalca	2015	Manzanares el Real	Sturnus unicolor	Lavandula stoecha	blank
GNM_016	included	Polo V.; Rubalca	2015	Manzanares el Real	Sturnus unicolor	Lavandula stoecha	blank
GNM_016	included	Polo V.; Rubalca	2015	Manzanares el Real	Sturnus unicolor	Lavandula stoecha	blank
GNM_016	included	Polo V.; Rubalca	2015	Manzanares el Real	Sturnus unicolor	Lavandula stoecha	blank
GNM_016	included	Polo V.; Rubalca	2015	Manzanares el Real	Sturnus unicolor	Lavandula stoecha	blank
GNM_016	included	Polo V.; Rubalca	2015	Manzanares el Real	Sturnus unicolor	Lavandula stoecha	blank
GNM_016	included	Polo V.; Rubalca	2015	Manzanares el Real	Sturnus unicolor	Lavandula stoecha	blank
GNM_024	included	Lopez-Rull I.; Gil	2009	Soto del Real, Madri	Sturnus unicolor	Lavandula stoecha	blank
GNM_024	included	Lopez-Rull I.; Gil	2009	Soto del Real, Madri	Sturnus unicolor	Lavandula stoecha	blank
GNM_024	included	Lopez-Rull I.; Gil	2009	Soto del Real, Madri	Sturnus unicolor	Lavandula stoecha	blank
GNM_024	included	Lopez-Rull I.; Gil	2009	Soto del Real, Madri	Sturnus unicolor	Lavandula stoecha	blank
GNM_024	included	Lopez-Rull I.; Gil	2009	Soto del Real, Madri	Sturnus unicolor	Lavandula stoecha	blank

paper_ID	Observation_ID	experiment_ID	experiment_ID_coded	group_ID	group_ID_coded	repeated_trait_ID	repeated_trait_ID_coded	comparision_type	CH	PCH
GNM_001	1	1	GNM_001_1	1	GNM_001_1	1	GNM_001_1_1	1	0	1
GNM_001	2	1	GNM_001_1	1	GNM_001_1	2	GNM_001_1_2	1	1	1
GNM_001	3	1	GNM_001_1	1	GNM_001_1	3	GNM_001_1_3	1	0	1
GNM_002	7	2	GNM_002_2	2	GNM_002_2	4	GNM_002_2_4	1	1	1
GNM_002	8	3	GNM_002_3	2	GNM_002_2	4	GNM_002_2_4	1	1	1
GNM_002	9	4	GNM_002_4	2	GNM_002_2	4	GNM_002_2_4	1	1	1
GNM_002	10	5	GNM_002_5	3	GNM_002_3	4	GNM_002_3_4	1	1	1
GNM_002	11	2	GNM_002_2	3	GNM_002_3	4	GNM_002_3_4	1	1	1
GNM_002	12	3	GNM_002_3	3	GNM_002_3	4	GNM_002_3_4	1	1	1
GNM_016	13	1	GNM_016_1	1	GNM_016_1	1	GNM_016_1_1	2	1	1
GNM_016	14	1	GNM_016_1	2	GNM_016_2	2	GNM_016_2_2	2	1	1
GNM_016	15	1	GNM_016_1	1	GNM_016_1	3	GNM_016_1_3	2	1	1
GNM_016	16	1	GNM_016_1	2	GNM_016_2	4	GNM_016_2_4	2	1	1
GNM_016	17	1	GNM_016_1	3	GNM_016_3	5	GNM_016_3_5	2	1	1
GNM_016	18	1	GNM_016_1	4	GNM_016_4	5	GNM_016_4_5	2	1	1
GNM_016	19	1	GNM_016_1	5	GNM_016_5	5	GNM_016_5_5	2	1	1
GNM_016	20	1	GNM_016_1	6	GNM_016_6	5	GNM_016_6_5	2	1	1
GNM_016	21	1	GNM_016_1	7	GNM_016_7	5	GNM_016_7_5	2	1	1
GNM_024	22	1	GNM_024_1	1	GNM_024_1	1	GNM_024_1_1	2	1	0
GNM_024	23	1	GNM_024_1	1	GNM_024_1	2	GNM_024_1_2	2	1	1
GNM_024	24	1	GNM_024_1	1	GNM_024_1	3	GNM_024_1_3	2	1	1
GNM_024	25	1	GNM_024_1	2	GNM_024_2	4	GNM_024_2_4	2	1	0
GNM_024	26	1	GNM_024_1	2	GNM_024_2	5	GNM_024_2_5	2	1	0

fitness_proxy	trait_type	proxies_sign	statistics_type	test_statistics_type	statistics_value	p_value	sign_relationship	total_sample_size	DF	data_location	parasite_type	time_of_gnm_addition	extractor_ID	extractor_comments	data_checker_ID	data_checker_comments	blinding	random_assignment	missing_data
hemoglobin concentr	Physiology	1	NA	NA	NA	NA	NA	NA	NA	First paragraph	NA	c	JMGS	I considered that	SD	control_plant_s	no	yes	yes
number of nestlings	Reproduction	1	NA	NA	NA	NA	NA	NA	NA	First paragraph	NA	c	JMGS	NA	SD	I am not sure if	no	yes	yes
glucose concentratio	Physiology	-1	NA	NA	NA	NA	NA	NA	NA	Provided by autl	NA	c	JMGS	Missing primary	SD	NA	no	yes	yes
Nestling tarsus lengt	Morphology	1	NA	NA	NA	NA	NA	NA	NA	Fig 1	NA	c	SD	I have extracted t	SD	NA	no	no	no
Nestling tarsus lengt	Morphology	1	NA	NA	NA	NA	NA	NA	NA	Fig 1	NA	c	SD	I have extracted t	SD	NA	no	no	no
Nestling tarsus lengt	Morphology	1	NA	NA	NA	NA	NA	NA	NA	Fig 1	NA	c	SD	I have extracted t	SD	NA	no	no	no
Nestling tarsus lengt	Morphology	1	NA	NA	NA	NA	NA	NA	NA	Fig 1	NA	c	SD	I have extracted t	SD	NA	no	no	no
Nestling tarsus lengt	Morphology	1	NA	NA	NA	NA	NA	NA	NA	Fig 1	NA	c	SD	I have extracted t	SD	NA	no	no	no
Nestling tarsus lengt	Morphology	1	NA	NA	NA	NA	NA	NA	NA	Fig 1	NA	c	SD	I have extracted t	SD	NA	no	no	no
fledgling body mass i	Morphology	1	NA	NA	NA	NA	NA	NA	NA	Sample sizes fr	NA	b	JMGS	***I considere th	SD	Authors also sa	no	yes	no
fledgling body mass i	Morphology	1	NA	NA	NA	NA	NA	NA	NA	Sample sizes fr	NA	b	JMGS	***I considere th	SD	NA	no	yes	no
fledling tarsus-metat	Morphology	1	NA	NA	NA	NA	NA	NA	NA	Sample sizes fr	NA	b	JMGS	It seemed strang	SD	NA	no	yes	no
fledling tarsus-metat	Morphology	1	NA	NA	NA	NA	NA	NA	NA	Sample sizes fr	NA	b	JMGS	***I considere th	SD	NA	no	yes	no
number of fledlings ;	Reproduction	1	NA	NA	NA	NA	NA	NA	NA	First paragraph	NA	b	JMGS	write to authors !	SD	NA	no	yes	no
probability of local re	Reproduction	1	NA	NA	NA	NA	NA	NA	NA	Fig 2	NA	b	JMGS	***I considere th	SD	Authors also sa	no	yes	no
probability of local re	Reproduction	1	NA	NA	NA	NA	NA	NA	NA	Fig 2	NA	b	JMGS	***I considere th	SD	NA	no	yes	no
probability of local re	Reproduction	1	NA	NA	NA	NA	NA	NA	NA	Fig 2	NA	b	JMGS	***I considere th	SD	NA	no	yes	no
probability of local re	Reproduction	1	NA	NA	NA	NA	NA	NA	NA	Fig 2	NA	b	JMGS	***I considere th	SD	NA	no	yes	no
females' testosteron	Physiology	1	NA	NA	NA	NA	NA	NA	NA	Fig.2	NA	b	JMGS	The data was full	SD	NA	no	yes	no
Clutch size	Reproduction	1	NA	NA	NA	NA	NA	NA	NA	Fig.3	NA	b	JMGS	The data was full	SD	NA	no	yes	no
egg size (mm3)	Reproduction	1	NA	NA	NA	NA	NA	NA	NA	Last paragraph (	NA	b	JMGS	*notice that the t	SD	I am more likel	no	yes	no
Yolk Log T (testosten	Physiology	1	NA	NA	NA	NA	NA	NA	NA	Text: Results: Yc	NA	b	JMGS	***notice that th	SD	I am more likel	no	yes	no
Yolk Log A4(androste	Physiology	1	NA	NA	NA	NA	NA	NA	NA	Text: Results: Yc	NA	b	JMGS	***notice that th	SD	I am more likel	no	yes	no

Here are some tips and tricks

- Each meta-analysis is unique and will encounter different challenges. Embrace them and learn from them.
- Meta-analysis is teamwork; find people with expertise in meta-analysis, experimental design, etc.
- Extracting data is much more objective than you imagine; **multiple observers** are needed!
- Build a **data sheet template with its associated metadata** (Think well in advance!)
- Perform **pilot extractions** (discuss and adjust to reduce disagreements)
- Column for **Notes and explanation on extraction** (Trust me! You will thank me later!)
- **Prefer raw statistics** (mean, variance and sample size) and correlations (r, tau etc.) over inferential statistics (t, F) whenever possible
- **Don't confuse SE with SD**; remember $SE = \frac{SD}{\sqrt{N}}$

Here are some tips and tricks

- Studies **can have a shared control group** that they compare to two treatments.
  To deal with this non-independence there are a few ways.

- Whenever **extracting inferential statistic** (t-values, F-values, Chi-square values and, in some occasions, p-values), **it needs be clear where they come from**!

  For example, a t-value from a independent t-test, dependent t-test, linear regression or a mixed-effects model **DON'T mean the same**.

Here are some tips and tricks

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$\bar{Y}_1, \bar{Y}_2, S_{Difference}, r, n$ (number of pairs)	$d = \left(\frac{\bar{Y}_1 - \bar{Y}_2}{S_{Difference}}\right)\sqrt{2(1-r)}, v = \left(\frac{1}{n}+\frac{d^2}{2n}\right)2(1-r)$
t (from paired t-test), r, n	$d = t\sqrt{\frac{2(1-r)}{n}}, v = \left(\frac{1}{n}+\frac{d^2}{2n}\right)2(1-r)$
F (from repeated measures ANOVA), r, n	$d = \pm\sqrt{\frac{2F(1-r)}{n}}, v = \left(\frac{1}{n}+\frac{d^2}{2n}\right)2(1-r)$
p (one-tailed), r, n	$d = \pm t^{-1}(p)\sqrt{\frac{2(1-r)}{n}}, v = \left(\frac{1}{n}+\frac{d^2}{2n}\right)2(1-r)$
p (two-tailed), r, n	$d = \pm t^{-1}\left(\frac{p}{2}\right)\sqrt{\frac{2(1-r)}{n}}, v = \left(\frac{1}{n}+\frac{d^2}{2n}\right)2(1-r)$

SOURCE: Authors' tabulation.
NOTE: The function $t^{-1}(p)$ is the inverse of the cumulative distribution function of Student's t with $n - 1$ degrees of freedom. Many computer programs and spreadsheets provide functions that can be used to compute t^{-1}. Assume $n = 19$, so that $df = 18$. Then, in Excel, for example, if the reported p-value is 0.05 (2-tailed), TINV(p,df) = TINV(0.05,18) will return the required value (2.1009). If the reported p-value is 0.05 (1-tailed), TINV($2p$,df) = TINV(0.10,18) will return the required value 1.7341. The F in row 3 of the table is the F-statistic from a one-way repeated measures analysis of variance. In rows 3 through 5, the sign of d must reflect the direction of

t (from ANCOVA), n_1, n_2, R, q	$d = t\sqrt{\frac{n_1+n_2}{n_1n_2}}\sqrt{1-R^2}, v = \frac{(n_1+n_2)(1-R^2)}{n_1n_2}+\frac{d^2}{2(n_1+n_2)}$
F (from ANCOVA), n_1, n_2, R, q	$d = \pm\sqrt{\frac{F(n_1+n_2)}{n_1n_2}}\sqrt{1-R^2}, v = \frac{(n_1+n_2)(1-R^2)}{n_1n_2}+\frac{d^2}{2(n_1+n_2)}$
p (1-tailed, from ANCOVA), n_1, n_2, R, q	$d = \pm t^{-1}(p)\sqrt{\frac{n_1+n_2}{n_1n_2}}\sqrt{1-R^2}, v = \frac{(n_1+n_2)(1-R^2)}{n_1n_2}+\frac{d^2}{2(n_1+n_2)}$
p (2-tailed, from ANCOVA), n_1, n_2, R, q	$d = \pm t^{-1}\left(\frac{p}{2}\right)\sqrt{\frac{n_1+n_2}{n_1n_2}}\sqrt{1-R^2}, v = \frac{(n_1+n_2)(1-R^2)}{n_1n_2}+\frac{d^2}{2(n_1+n_2)}$

SOURCE: Authors' tabulation.
NOTE: The function $t^{-1}(p)$ is the inverse of the cumulative distribution function of Student's t with $n_1 + n_2 - 2 - q$ degrees of freedom, q is the number of covariates, and R is the covariate outcome correlation or multiple correlation. Many computer programs and spreadsheets provide functions that can be used to compute t^{-1}. Assume $n_1 = n_2 = 11$, and $q = 2$, so that $df = 18$. Then, in Excel, for example, if the reported p-value is 0.05 (2-tailed), TINV(p,df) = TINV(0.05,18) will return the required value (2.1009). If the reported p-value is 0.05 (1-tailed), TINV($2p$,df) = TINV(0.10,18) will return the required value 1.7341. The F in row 3 of the table is the F-statistic from a one-way analysis of covariance. In rows 3 through 5, the sign of d must reflect the direction of the mean difference.

Here are some tips and tricks

- Studies **can have a shared control group** that they compare to two treatments.
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- Whenever **extracting inferential statistic** (t-values, F-values, Chi-square values and, in some occasions, p-values), **it needs be clear where they come from**!

  For example, a t-value from a independent t-test, dependent t-test, linear regression or a mixed-effects model **DON'T mean the same**.

- **Signs matter A LOT!** All effect sizes need to be comparable
  In inferential statistics make sure to know the direction of the effect (know the reference of each model)
  What do the traits mean (survival vs. mortality)

- Sometimes you need to **contact the authors** to know things!

- **Open data** (or data obtained directly from the authors) can be used to calculate missing results

Here are some tips and tricks

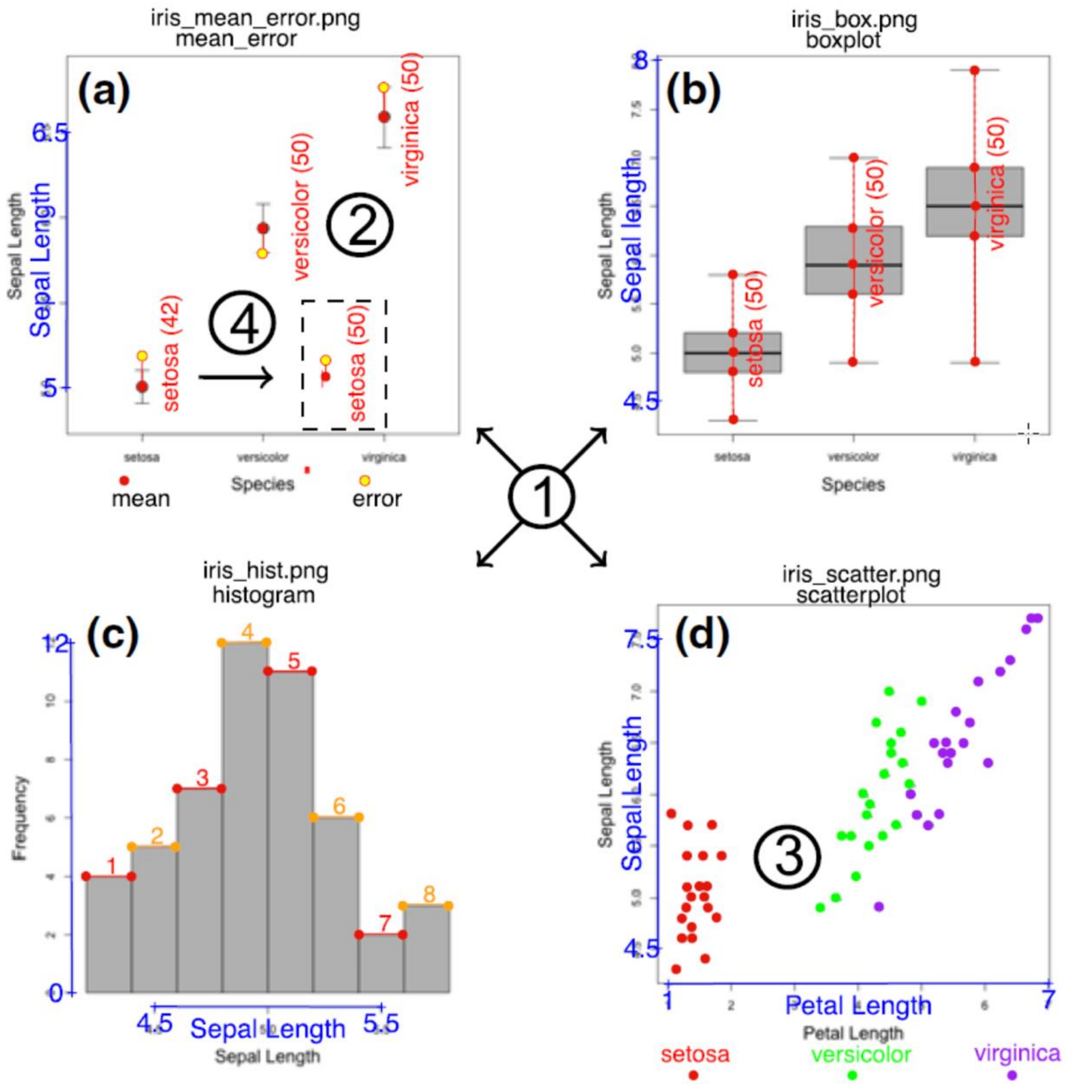
- You can extract data from the plots

  metaDigitise or shinyDigitise

Data output

⑥ ⑤

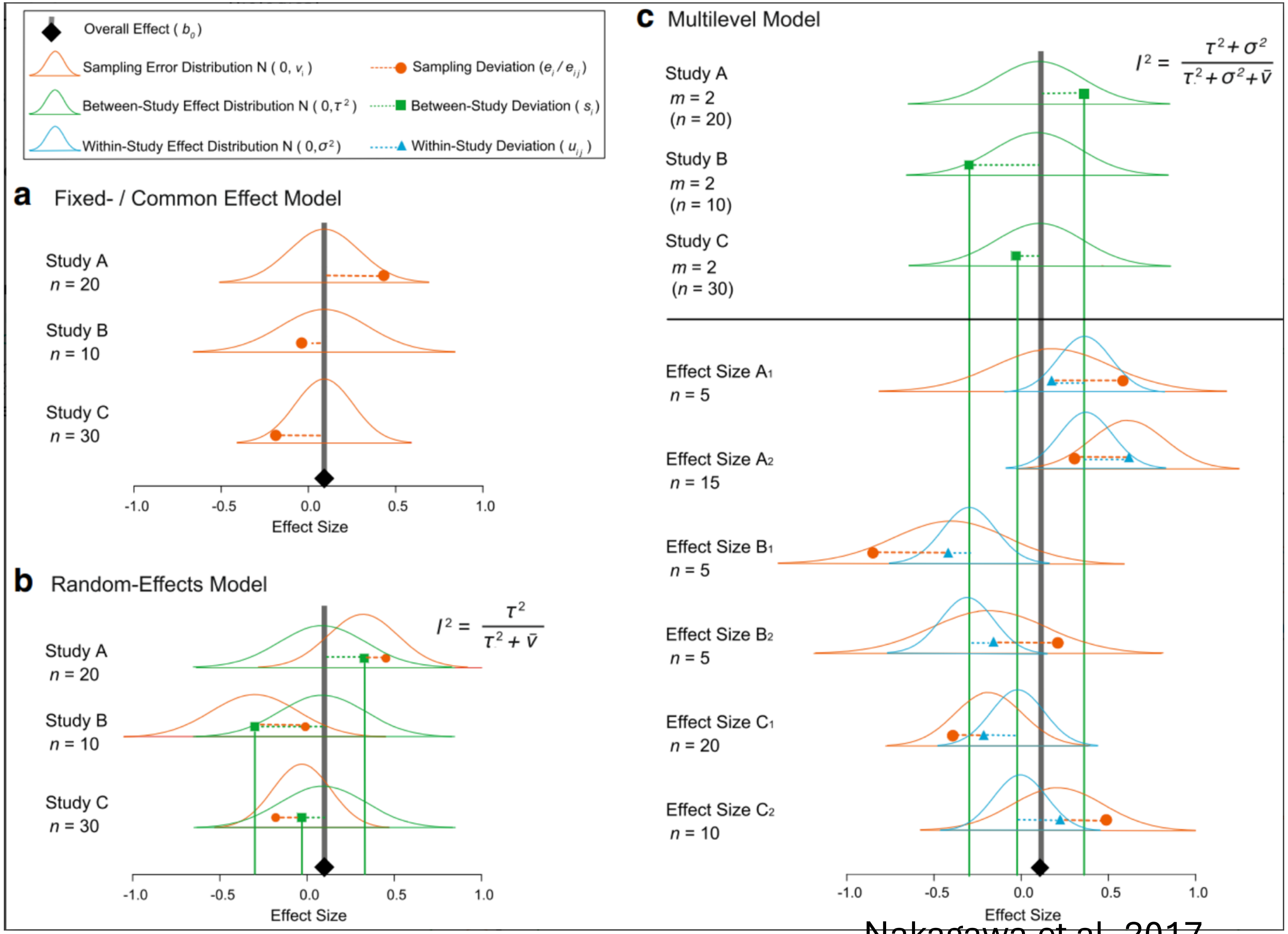
filename	variable	group_id	mean	sd	n	r	plot_type
iris_box.png	Sepal length	setosa	5.01	0.317	50	NA	boxplot
iris_box.png	Sepal length	versicolor	5.93	0.497	50	NA	boxplot
iris_box.png	Sepal length	virginica	6.49	0.603	50	NA	boxplot
iris_hist.png	Sepal Length	setosa	4.95	0.364	50	NA	histogram
iris_mean_error.png	Sepal Length	setosa	5.01	0.680	50	NA	mean_error
iris_mean_error.png	Sepal Length	versicolor	5.94	1.025	50	NA	mean_error
iris_mean_error.png	Sepal Length	virginica	6.59	1.251	50	NA	mean_error
iris_scatter.png	Petal Length	setosa	1.44	0.215	20	0.109	scatterplot
iris_scatter.png	Sepal Length	setosa	5.03	0.427	20	0.109	scatterplot
iris_scatter.png	Petal Length	versicolor	4.29	0.415	20	0.786	scatterplot
iris_scatter.png	Sepal Length	versicolor	5.97	0.603	20	0.786	scatterplot
iris_scatter.png	Petal Length	virginica	5.66	0.668	20	0.932	scatterplot



Pick, Nakagawa, and Noble (2018)

Now time to do the meta-analysis!!

What model do I use?



Nakagawa et al. 2017

What model do I use?

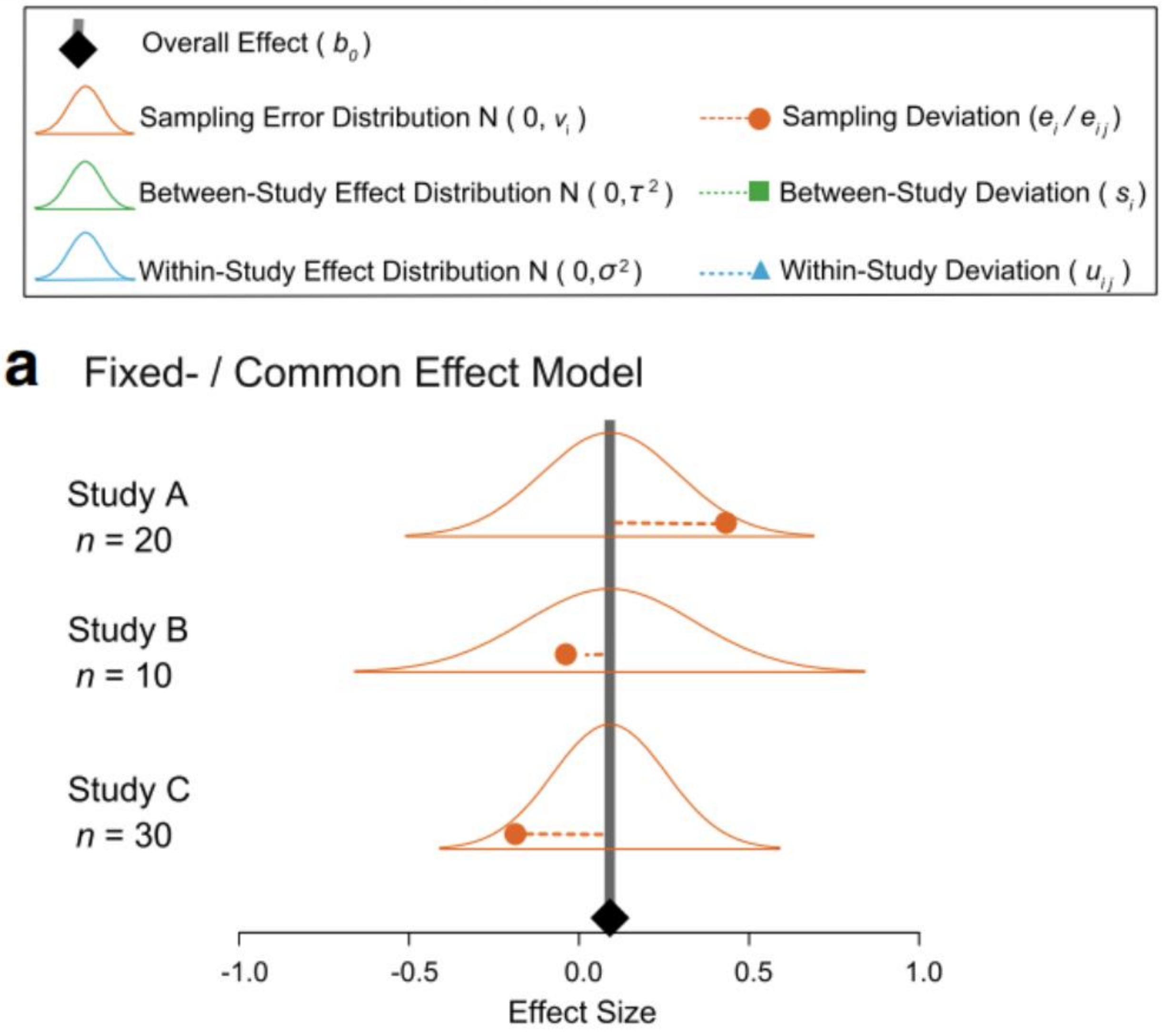
Understand the assumptions:

- All effect sizes are independent
  (one from each study,
  no other source of non-independence)

- All effect sizes share a common mean,
  and thus that variation among data is
  solely attributable to sampling error

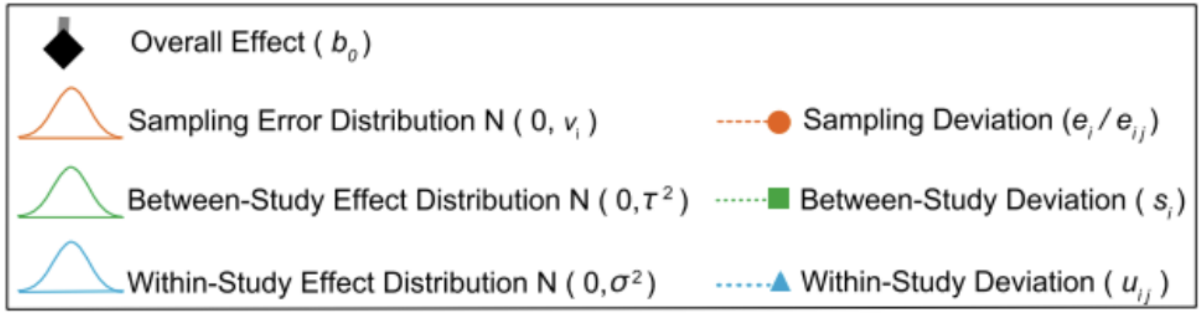
$$\hat{\mu} = \frac{\sum_{i=1}^{k} w_i \, y_i}{\sum_{i=1}^{k} w_i}$$

$$w_i = \frac{1}{v_i}$$



Nakagawa et al. 2017

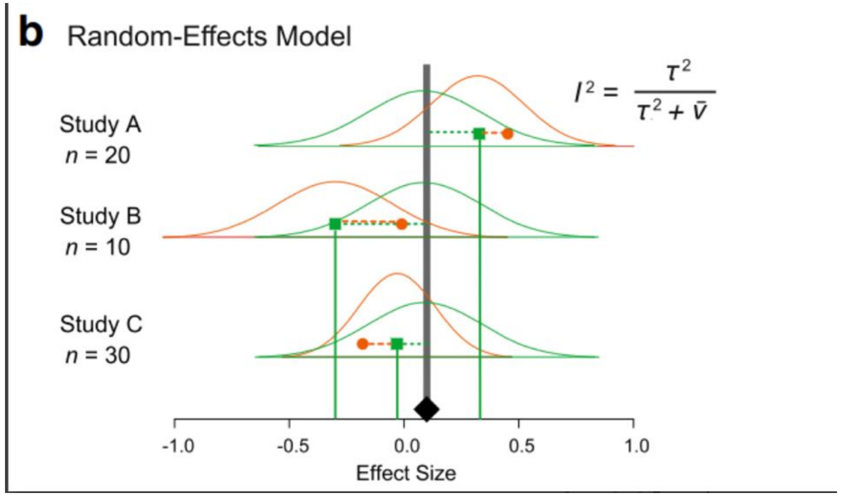
What model do I use?



Understand the assumptions:

- All effect sizes are independent
  (one from each study,
  no other source of non-independence)

- Relax the assumption of a common population,
  needs to quantify between-study variance
  (for this sample size over 10)



$$\hat{\mu} = \frac{\sum_{i=1}^{k} w_i \, y_i}{\sum_{i=1}^{k} w_i}$$

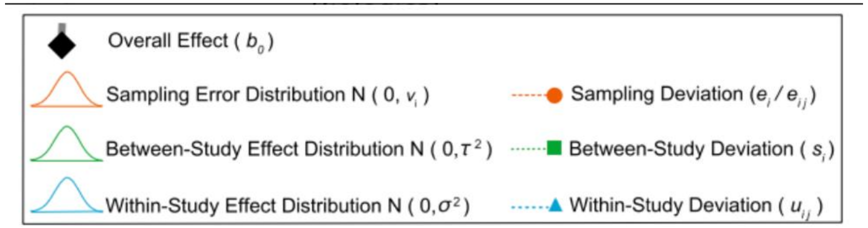
$$w_i = \frac{1}{\hat{\tau}^2 + v_i}$$

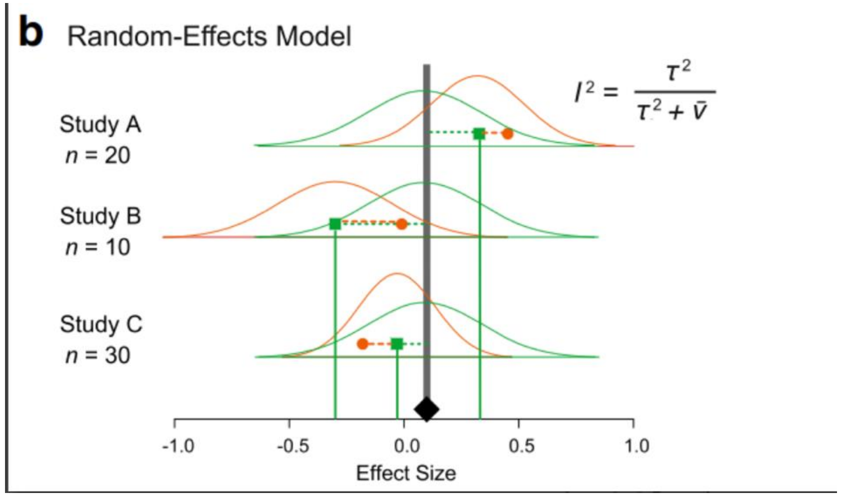
Nakagawa et al. 2017

What model do I use?

```
RE.model <- metafor::rma(yi = es.RE, vi = V, method = "DL", data = Fixed.and.Random.Effect)
summary(RE.model)
```

```
Random-Effects Model (k = 12; tau^2 estimator: DL)

  logLik  deviance       AIC       BIC      AICc
 -1.9498   16.8195    7.8996    8.8695    9.2330

tau^2 (estimated amount of total heterogeneity): 0.0335 (SE = 0.0352)
tau (square root of estimated tau^2 value):      0.1829
I^2 (total heterogeneity / total variability):   41.32%
H^2 (total variability / sampling variability):  1.70

Test for Heterogeneity:
Q(df = 11) = 18.7466, p-val = 0.0658

Model Results:

estimate      se     zval    pval   ci.lb   ci.ub
  0.5754  0.0842  6.8298  <.0001  0.4103  0.7405  ***

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```





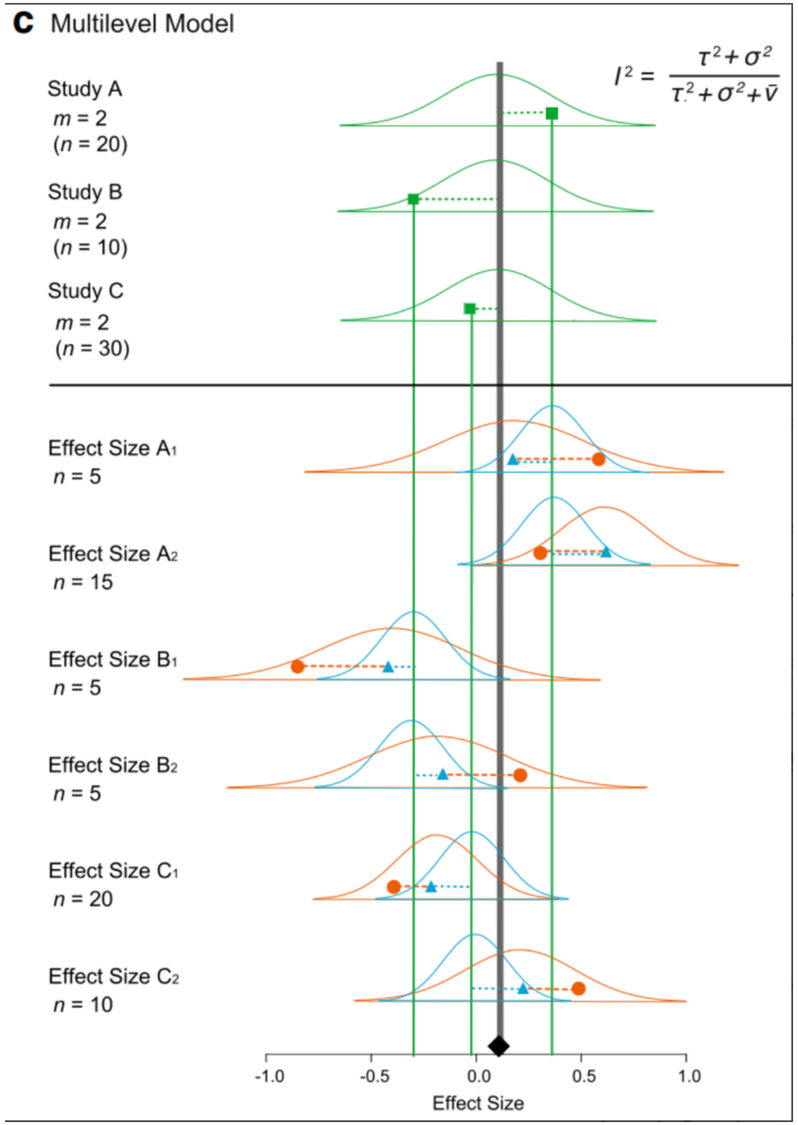
Nakagawa et al. 2017

What model do I use?

Understand the assumptions:

- Relaxes the assumption of independence
- Relax the assumption of a common population, needs to quantify between-study
  (for this sample size over 10)

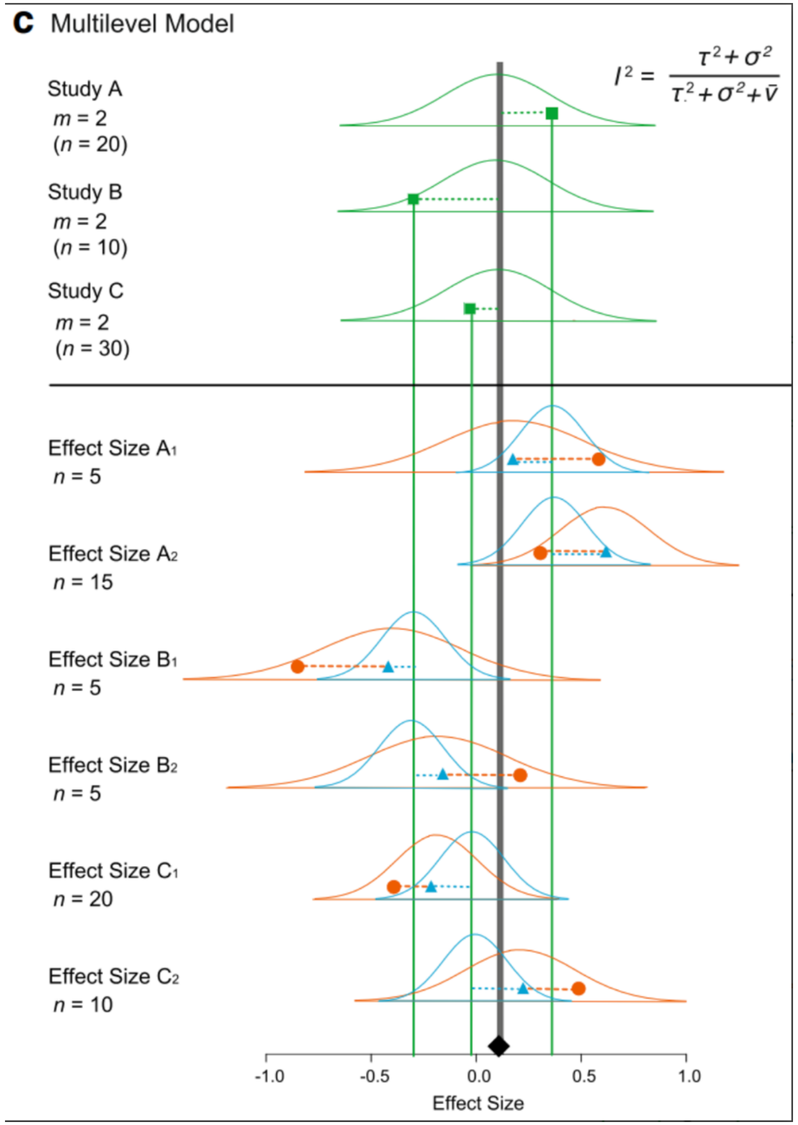
$$w_i = \frac{1}{\hat{\sigma}_1^2 + \hat{\sigma}_2^2 + v_i}$$



Nakagawa et al. 2017

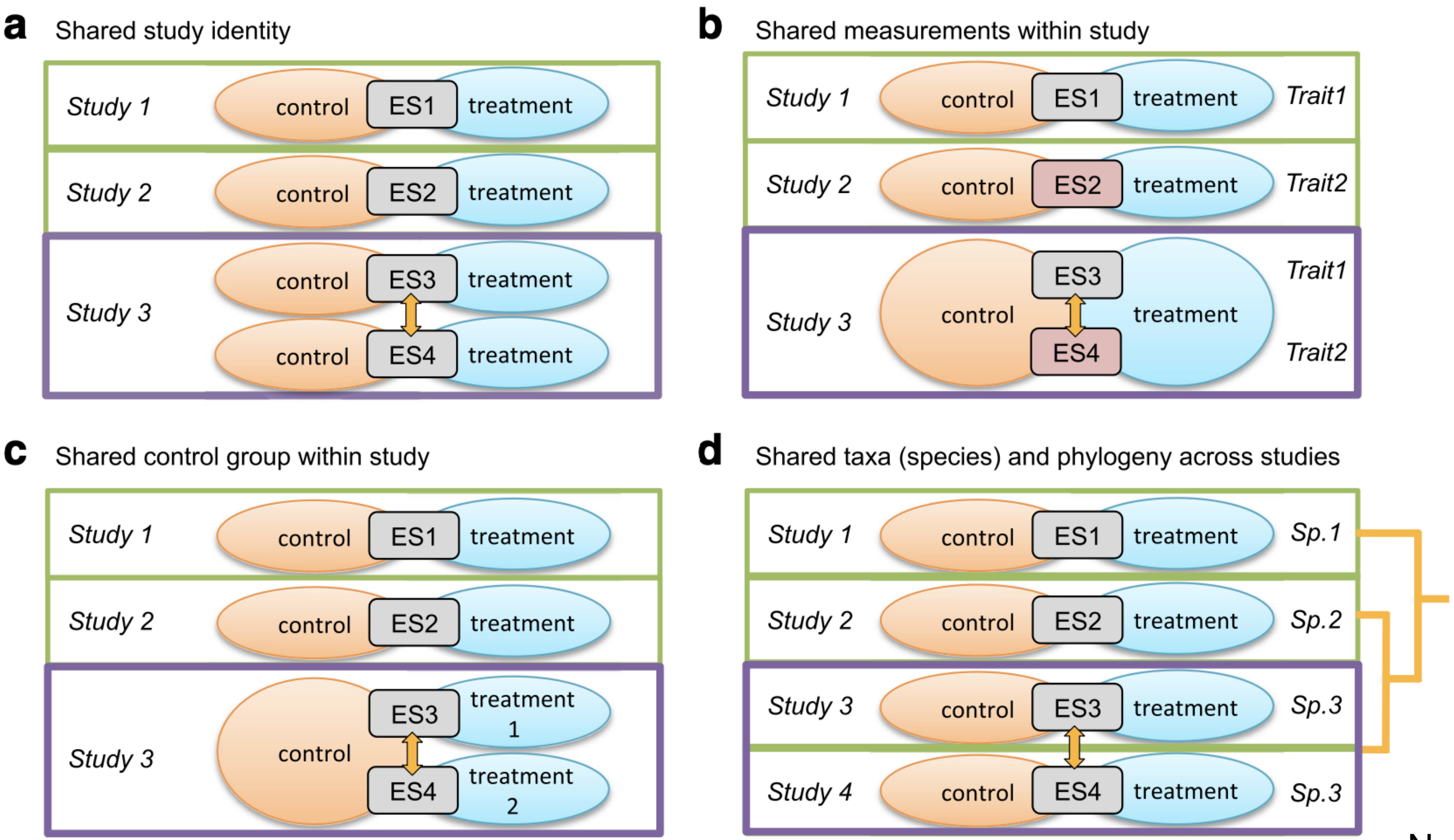
What model do I use?

```
main_model_vector <- rma.mv(yi = Zr,
                            V = VZr,
                            mods = ~ 1,
                            random = list(~ 1 | paperID,
                                          ~ 1 | popID2,
                                          ~ 1 | EffectID),
                            method = "REML",
                            test = "t", # "t" is more conservative than the default "z",
                            data = meta_data,
                            slab = paperID) # for the forestplot
```



Nakagawa et al. 2017

Why do we need multilevel model?



Nakagawa et al. 2017

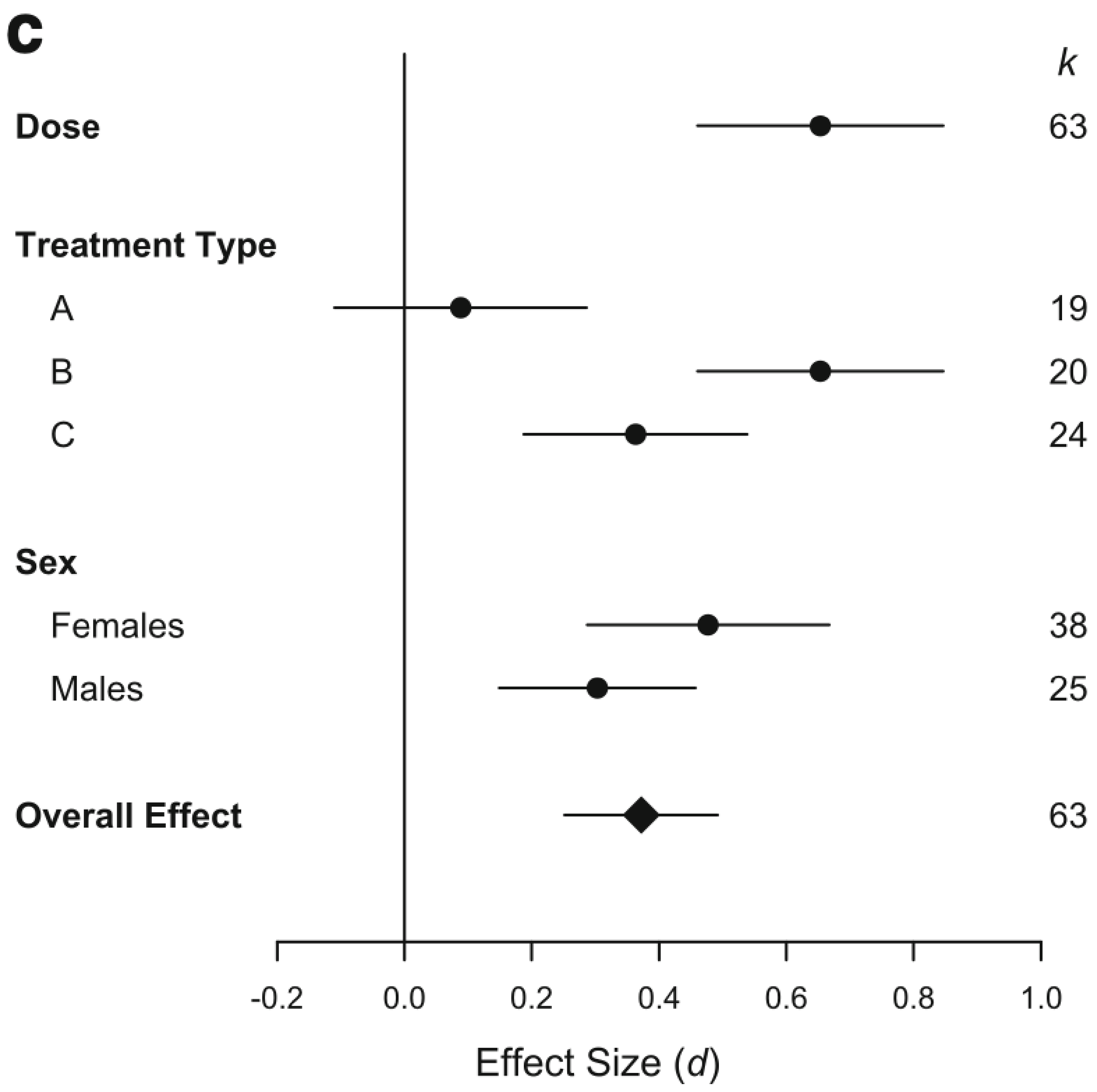
What model do I use?

```
main_model_vector <- rma.mv(yi = Zr,
                            V = VZr,
                            mods = ~ 1,
                            random = list(~ 1 | paperID,
                                          ~ 1 | popID2,
                                          ~ 1 | EffectID),
                            method = "REML",
                            test = "t", # "t" is more conservative than the default "z",
                            data = meta_data,
                            slab = paperID) # for the forestplot
```

```
Multivariate Meta-Analysis Model (k = 87; method: REML)

Variance Components:

            estim    sqrt  nlvls  fixed    factor
sigma^2.1  0.091  0.301     19     no   paperID
sigma^2.2  0.000  0.000     13     no    popID2
sigma^2.3  0.057  0.239     87     no  EffectID

Test for Heterogeneity:
Q(df = 86) = 169.352, p-val < .001

Model Results:

estimate      se    tval  df    pval   ci.lb   ci.ub
   0.193  0.092  2.112  86  0.038  0.011  0.376  *

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.
```

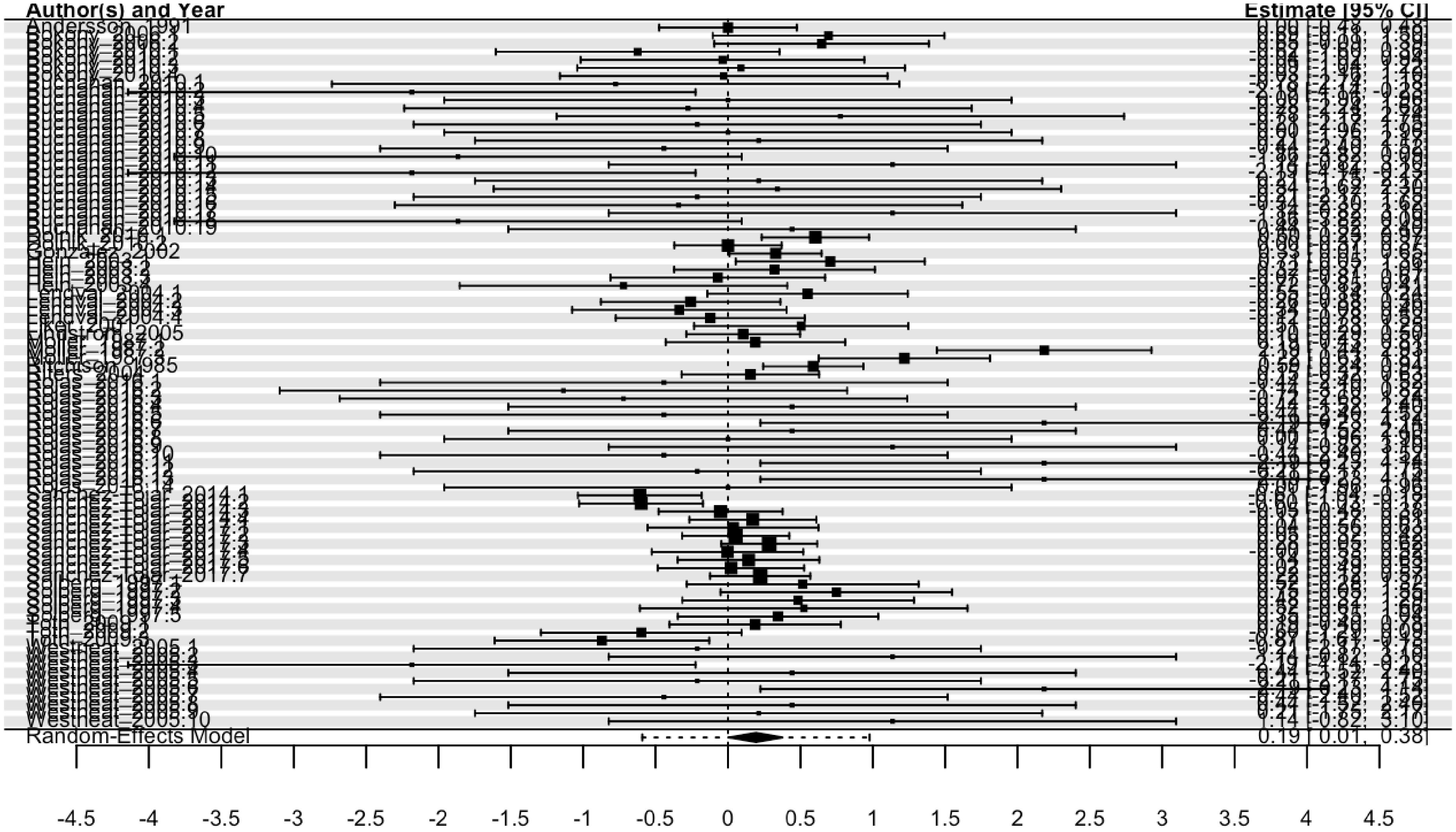
What are all these plots?

Forest plot



What are all these plots?

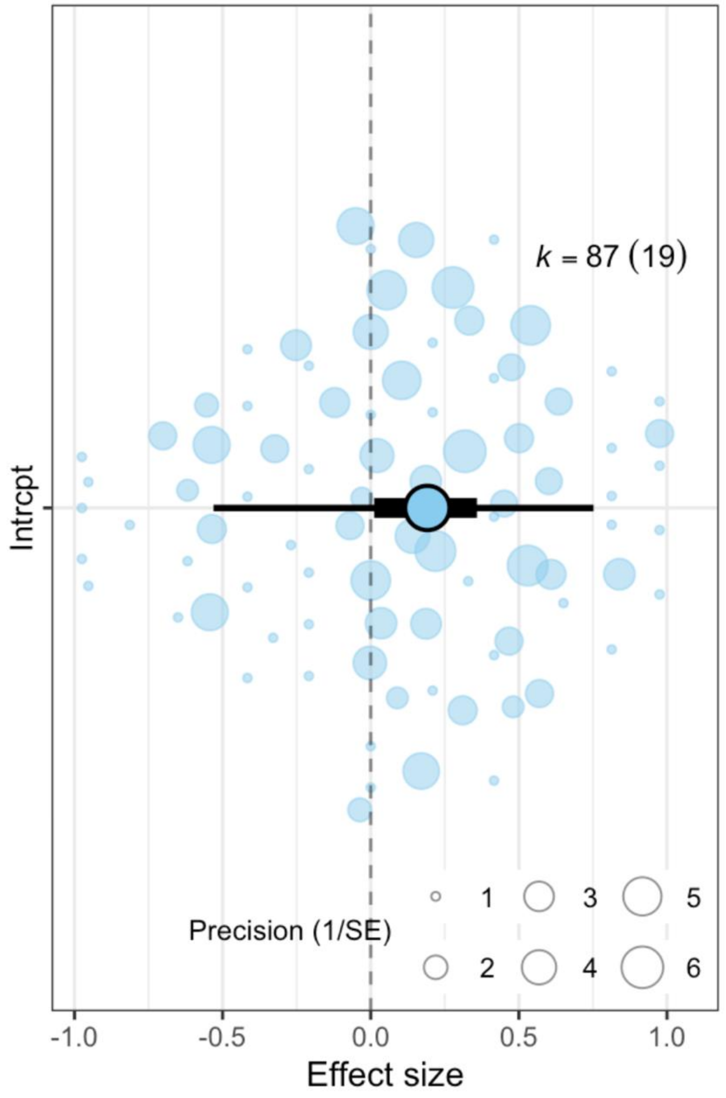
Forest plot



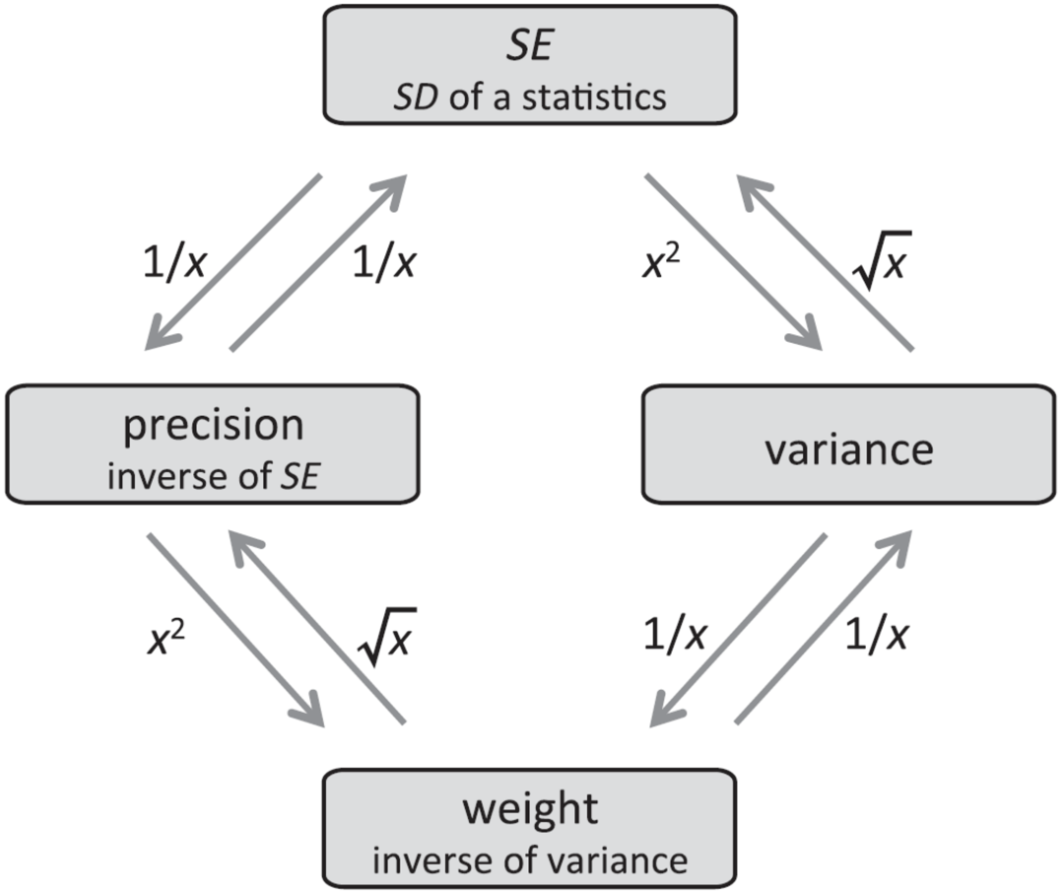
What are all these plots?

Orchard Plot

Package: OrchaRd



What are all these plots?



What is this heterogeneity you speak of?

Heterogeneity is the variance in true effects or said differently, the variability in study outcomes.
Thus, heterogeneity can also be thought of as the inconsistency among effect sizes.

σ^2, I^2, *CVH2, M2*

Heterogeneity indicates the degree of *context dependency* of the study findings:
The larger the heterogeneity, the lower the generality of the meta-analytic mean.

The very cool thing about finding heterogeneity is that we get to wear our detective hat and ask ourselves:

- Where does the heterogeneity come from? (decomposing heterogeneity)

- Can we explain some of it? (the power of meta-regressions)

Where does the heterogeneity come from?

σ^2, I^2, *CVH2, M2*

σ^2 Absolute variance in the effect sizes

$$I^2 = \frac{\sigma^2}{\sigma^2 + \bar{v}}$$

Quantifies the proportion of variance due to differences among effect sizes rather than by statistical noise (i.e., sampling variance)

$$CV = \frac{\sigma}{|\mu|}$$

$$M = \frac{\sigma}{\sigma + |\mu|}$$

Pluralistic approach : Yang et al. 2024

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Pluralistic approach : Yang et al. 2024

Where does the heterogeneity come from?

```
# # Total unstandardized raw variance (i.e. total heterogeneity)
sigma2_main_model_vector <- sum(main_model_vector$sigma2)
I2_main_model_vector <- orchaRd::i2_ml(main_model_vector)
CV_main_model_vector <- orchaRd::cvh2_ml(main_model_vector)
M_main_model_vector <- orchaRd::m2_ml(main_model_vector)
```

```
round(I2_main_model_vector,1)
```

```
   I2_Total   I2_paperID    I2_popID2 I2_EffectID 
       51.6         31.7          0.0        19.9 
```

```
round(CV_main_model_vector,3)
```

```
   CVH2_Total   CVH2_paperID    CVH2_popID2 CVH2_EffectID 
        3.940          2.419          0.000         1.521 
```

```
round(M_main_model_vector,3)
```

```
   M2_Total   M2_paperID    M2_popID2 M2_EffectID 
      0.798        0.490        0.000       0.308 
```

I^2_{total} was 51.6%, showing that heterogeneity in our dataset is, on average, only 1.06 times larger than statistical noise (or 6.5% larger than statistical noise). I^2_{total} was below the 25th (73%) percentile for Zr (from Yang et al. 2024), indicating a small value.

$CVH2_{total}$ was 3.94, showing that heterogeneity in our dataset is, on average, 3.94 times larger than the magnitude of our meta-analytic mean (meta-analytic mean, μ, 0.19). $CVH2_{total}$ was between the 50th (2.8) and 75th (8.5) percentile for Zr, indicating a moderate-to-high value.

$M2_{total}$ was 0.798, showing that heterogeneity in our dataset is, on average, indeed 3.94 times larger than the magnitude of our meta-analytic mean, a value that is between the 50th (0.73) and 75th (0.9) percentile for Zr, indicating a moderate value.

Can we explore this heterogeneity?

By running meta-regressions

```
main_model_vector <- rma.mv(yi = Zr,
                            V = VZr,
                            mods = ~ 1,
                            random = list(~ 1 | paperID,
                                          ~ 1 | popID2,
                                          ~ 1 | EffectID),
                            method = "REML",
                            test = "t", # "t" is more conservative than the default "z",
                            data = meta_data,
                            slab = paperID) # for the forestplot
```

Can we explore this heterogeneity?

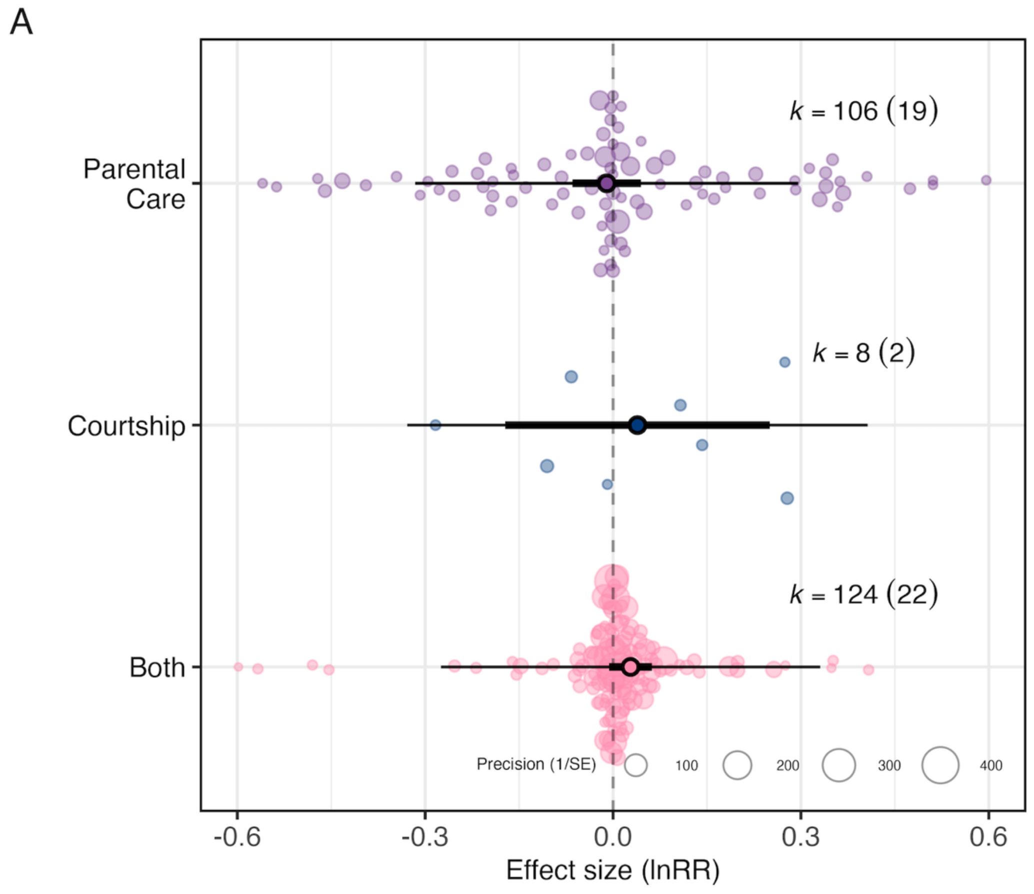
By running meta-regressions

```
lnRR_hypothesis <-
  metafor::rma.mv(yi = lnRR_sign, # specify lnRR as the effect size measure;
                  V = VCV_lnRR,# specify lnRR's sampling variance;
                  mods = ~ - 1 + Hypothesis,
                  random = list(~ 1 | paper_ID,
                                ~ 1 | Observation_ID,
                                ~ 1 | group_ID_coded,
                                ~ 1 | population_ID,
                                ~ 1 | bird_species),
                  method = "REML",
                test = "t",
                control = list(optimizer="optim"),
                data = dataset_lnRR)
```

```
> summary(lnRR_hypothesis)

Multivariate Meta-Analysis Model (k = 238; method: REML)

  logLik  Deviance       AIC       BIC      AICc
-26.0690   52.1380   68.1380   95.8146   68.7751

Variance Components:

             estim    sqrt  nlvls  fixed          factor
sigma^2.1  0.0000  0.0001     26     no        paper_ID
sigma^2.2  0.0233  0.1527    238     no  Observation_ID
sigma^2.3  0.0000  0.0005     46     no  group_ID_coded
sigma^2.4  0.0000  0.0000     17     no   population_ID
sigma^2.5  0.0000  0.0000      7     no    bird_species

Test for Residual Heterogeneity:
QE(df = 235) = 2275.7451, p-val < .0001

Test of Moderators (coefficients 1:3):
F(df1 = 3, df2 = 235) = 1.0183, p-val = 0.3853

Model Results:

                estimate      se     tval   df    pval    ci.lb   ci.ub
Hypothesisboth    0.0279  0.0174   1.6061  235  0.1096  -0.0063  0.0622
HypothesisCH      0.0390  0.1072   0.3636  235  0.7165  -0.1722  0.2502
HypothesisPCH    -0.0102  0.0278  -0.3647  235  0.7157  -0.0650  0.0447

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Can we explore this heterogeneity?

```
# Plot lnRR meta-regression on hypothesis

lnRR_hypothesis_plot <-
  orchard_plot(mod_results(lnRR_hypothesis,
                           mod = "Hypothesis",
                           group = "paper_ID"),
               trunk.size = 0.25,
               branch.size = 1,
               twig.size = 0.5,
               alpha = 0.4,
               xlab ="") +
  scale_colour_manual(values = c("#FF8CAE", "#003A7D", "#7E4794")) +
  scale_fill_manual(values = c("#FF8CAE", "#003A7D", "#7E4794")) +
  labs(x = "",
       y = "Effect size (lnRR)")+
  theme(
    panel.border = element_rect(colour = "black", fill = NA, linewidth = 0.8),
    legend.position.inside = c(0.99, 0.01),
    legend.background = element_blank(),
    legend.key = element_blank(),
    legend.justification = c(1, 0),
    legend.direction = "horizontal",
    legend.title = element_text(size = 6),
    legend.text = element_text(size = 6),
    axis.title = element_text(size = 10),
    axis.text.x = element_text(size = 10),
    axis.text.y = element_text(angle = 0, vjust = 0.5, hjust = 1))+
  scale_x_discrete(labels = c("Both","Courtship","Parental Care"))
```

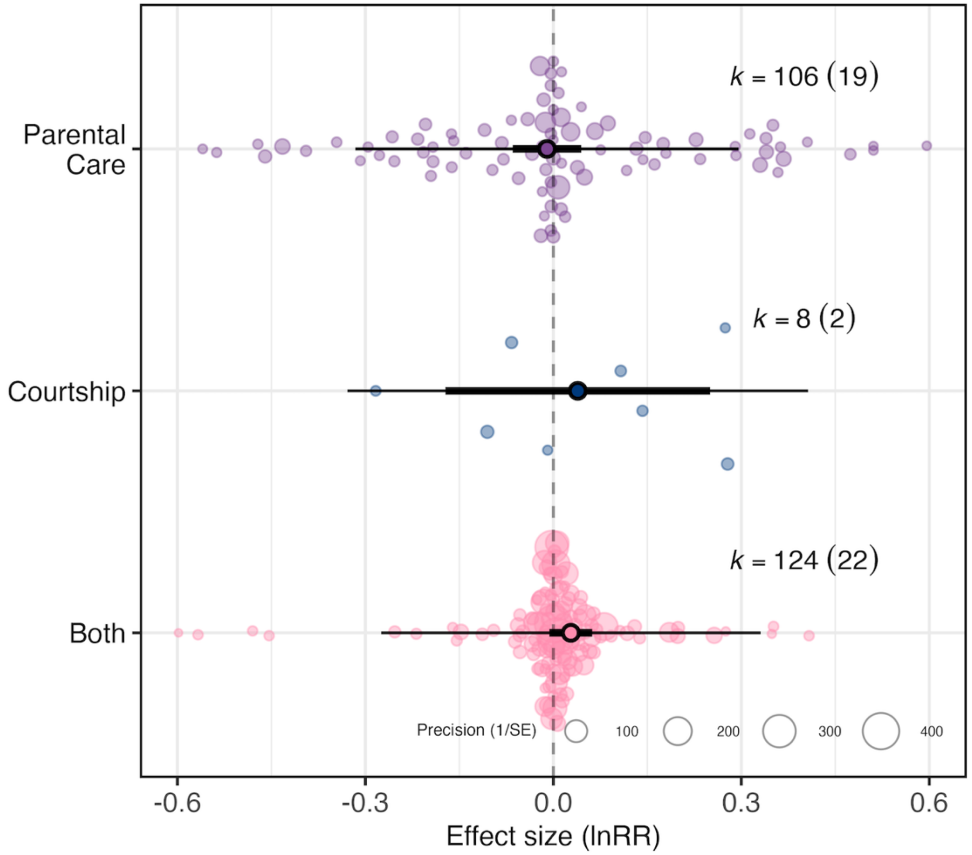


Can we explore this heterogeneity?

orchaRd::r2_ml(lnRR_hypothesis)

$R^2_{Marginal} = 0.0158$

A



Publication Bias

Publication bias commonly occurs when studies with statistically significant results are published more frequently than those with statistically non-significant findings (or are published more quickly)

File drawer problem

Problem?

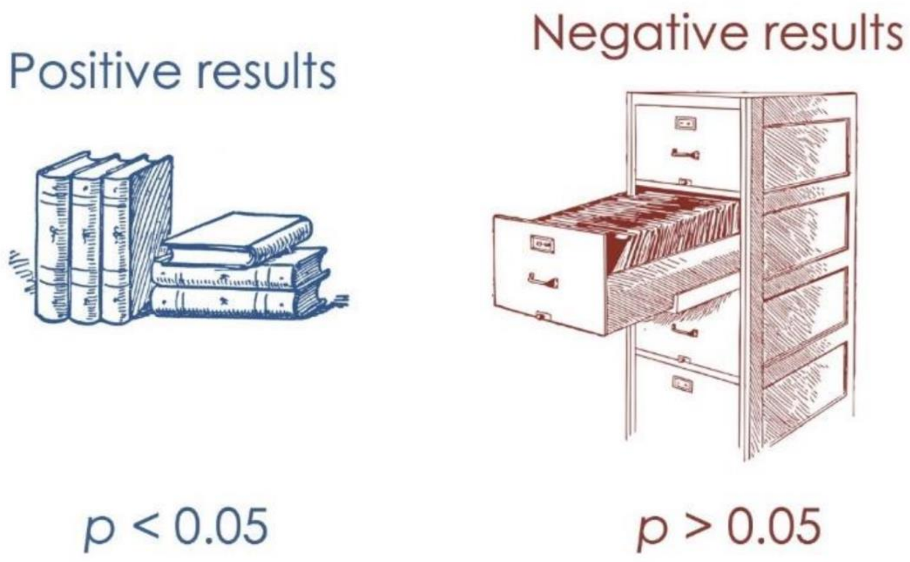
Distorted (biased) view of how much support there is for a hypothesis

Publication Bias

Publication bias commonly occurs when studies with statistically significant results are published more frequently than those with statistically non-significant findings (or are published more quickly)

Not all studies are 'published'
- No resources
- Lost interest

❖ $p > 0.05$
❖ methodological flaws
❖ topic ain't hot no more



Publication Bias

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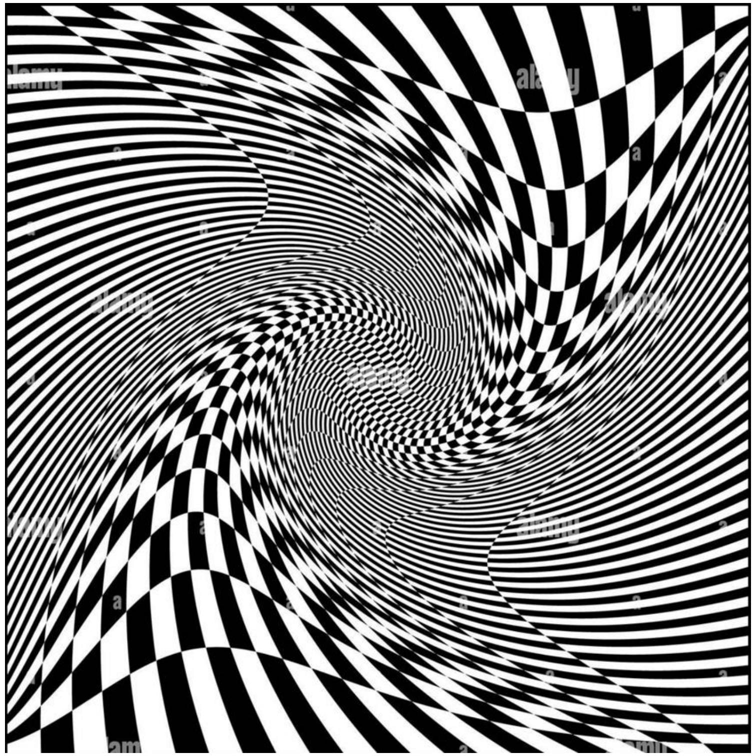
But also, some are

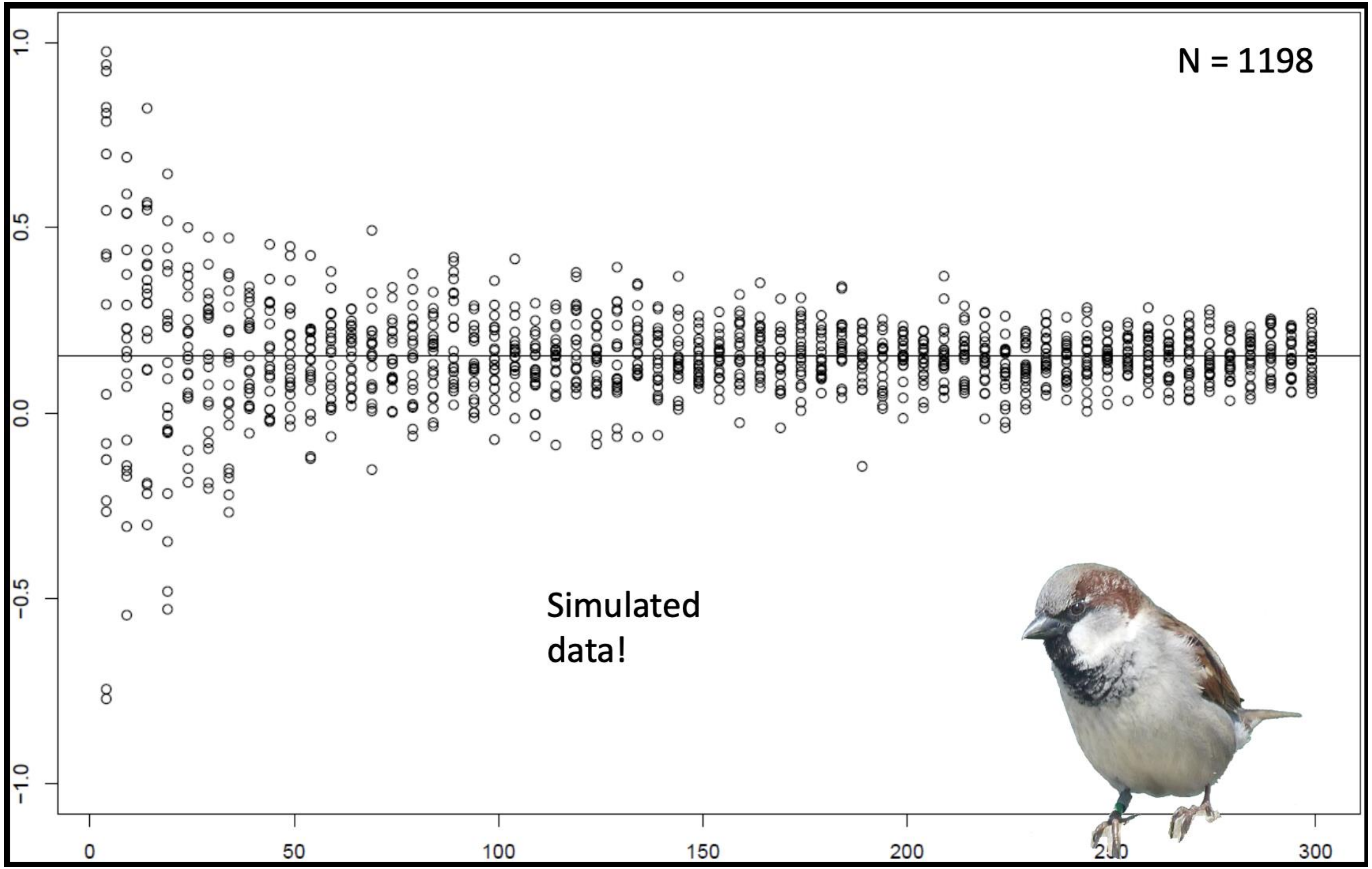
- disseminated less (e.g. citation biases)
- published later
- not accessible or difficult to find and access

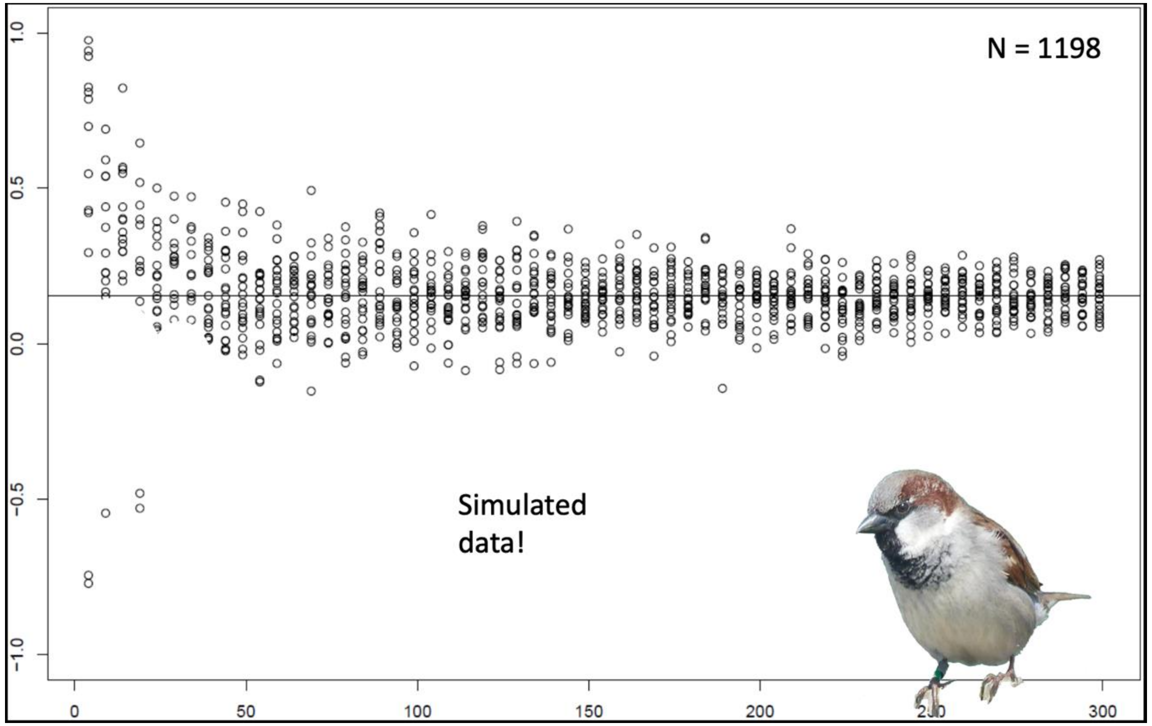


What can we do then for our meta-analysis?

We can test for publication bias and WE SHOULD!

But we cannot know what we do not know!


N = 1198
Simulated
data!
1.0
0.5
0.0
−0.5
−1.0
0
50
100
150
200
300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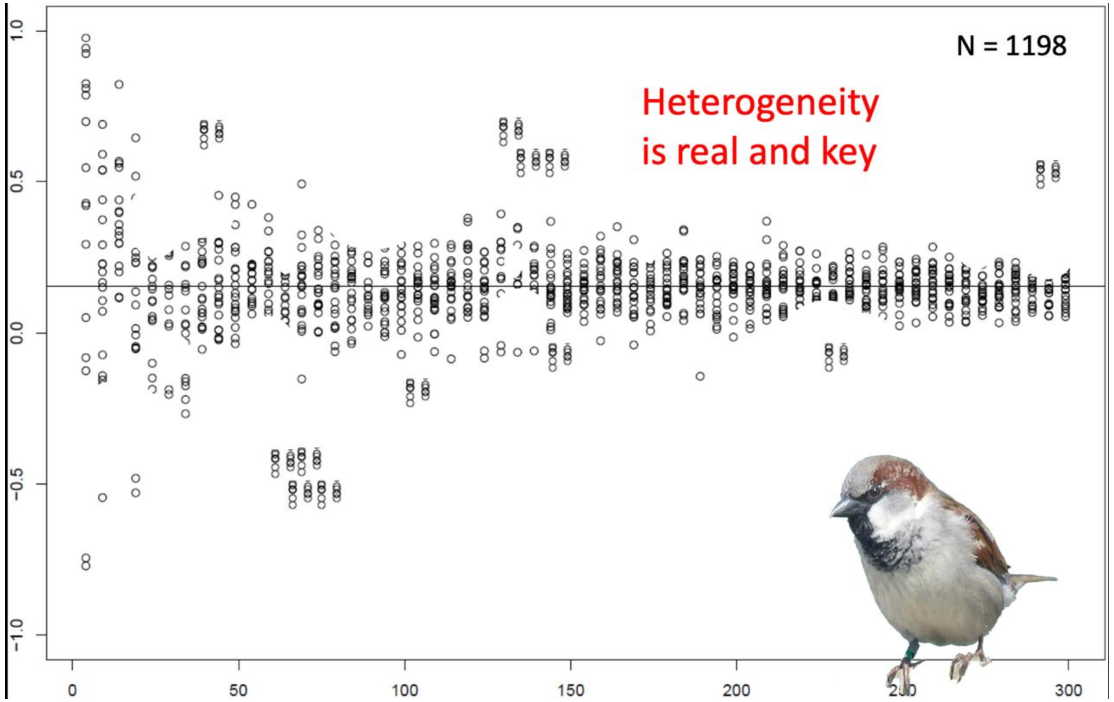
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Heterogeneity
is real and key
1.0
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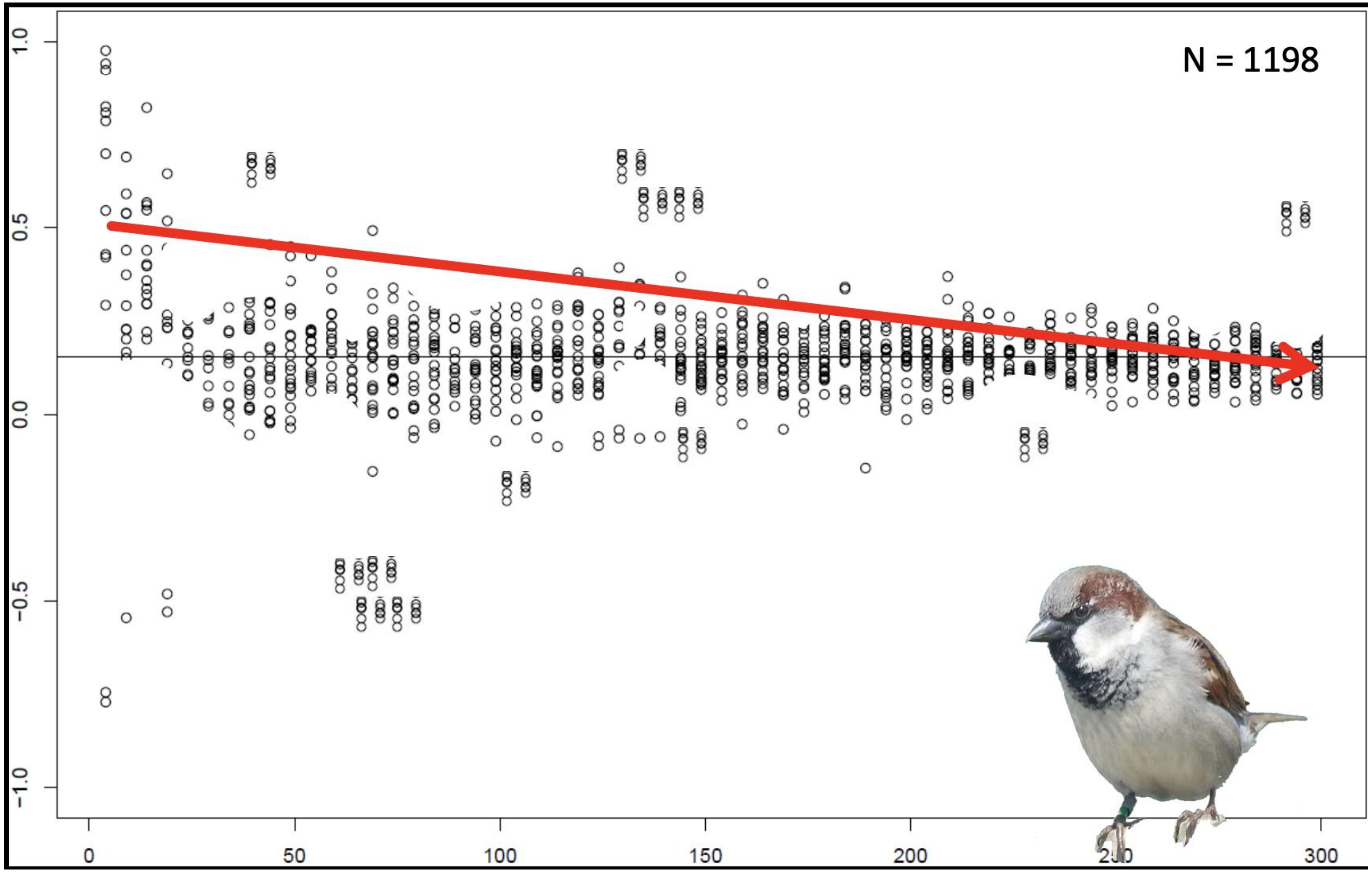
0.0
−0.5
−1.0
0
50
100
150
200
300

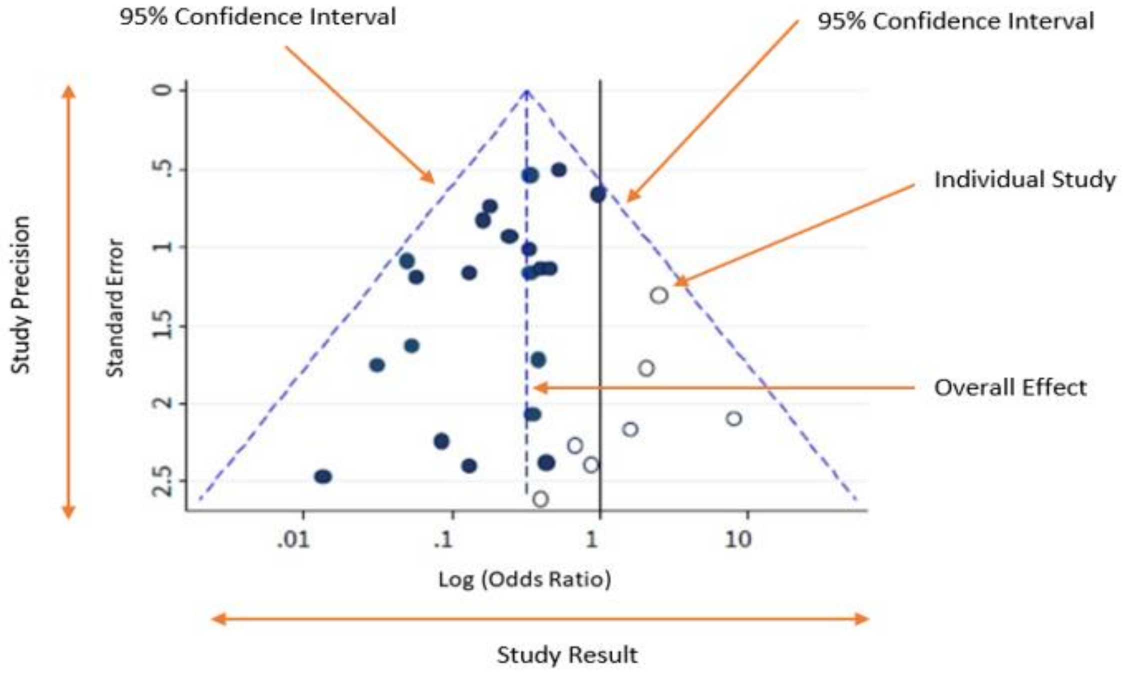


VOILA WE HAVE AN EFFECT

Small-study effects

miss small effect sizes (i.e., close to 0, often also potentially large, but against common knowledge) based on small sample sizes (i.e., of low precision).

Refers to asymmetry in the funnel plot



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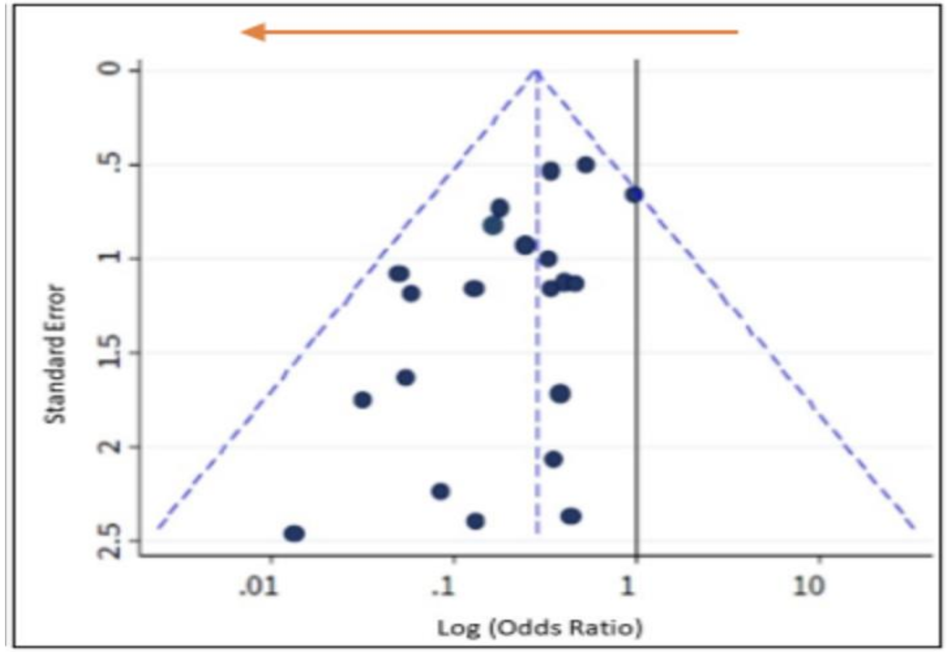

Small-study effects

You need a variable (square root of the inverse of the effective sample size)

$$\sqrt{\frac{1}{\tilde{n}_i}} = \sqrt{\frac{n_1 + n_2}{n_1 n_2}}$$

Small-study effects

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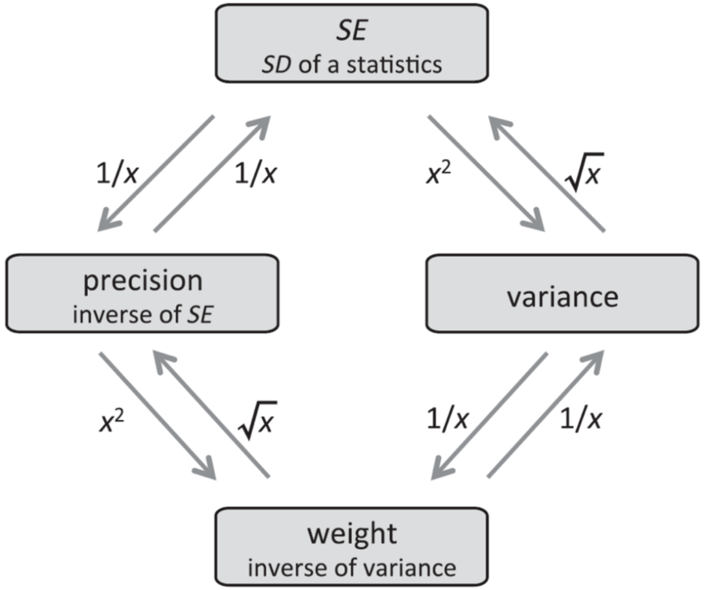
OR Standard Error

```
meta_data$sei <- sqrt(meta_data$VZr) # SE = sqrt(SV)
```

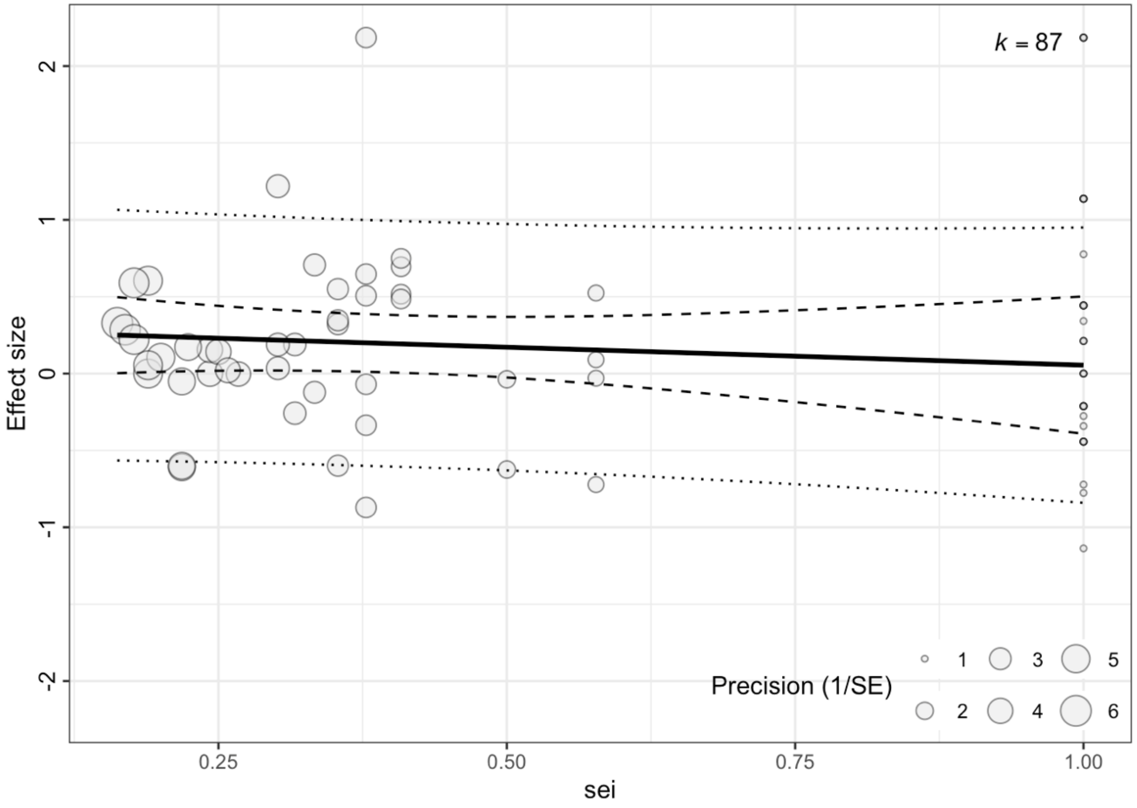


Small-study effects

```
small_study_effects <- rma.mv(Zr,
                              VZr,
                              mods = ~ 1 + sei,
                              random = list(~ 1 | paperID,
                                            ~ 1 | popID2,
                                            ~ 1 | EffectID),
                              method = "REML",
                              test = "t",
                              data = meta_data)
```

Small-study effects

```
figure_small_study_effects <- orchaRd::bubble_plot(small_study_effects,
                                                   mod = "sei",
                                                   group = "paperID",
                                                   xlab = "sei",
                                                   legend.pos = "bottom.right")
figure_small_study_effects
```




```
round(r2_ml(small_study_effects)*100, 2)[1]
```

```
R2_marginal
       4.13
```

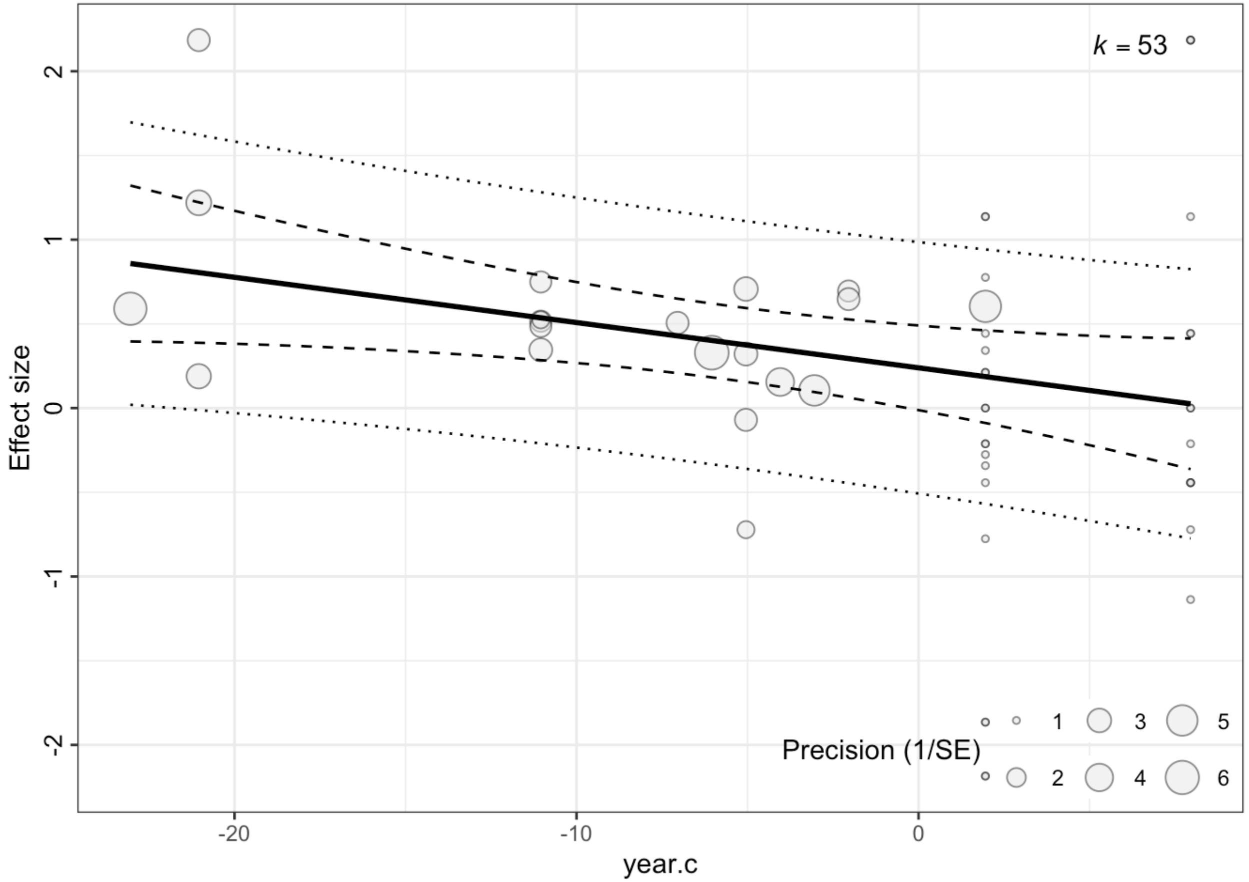
Decline effects

Overall effect becomes smaller over time

Easy-to-test

Less evidence, but important to test

Nothing magical about this effect,
just another way of exploring
publication bias.



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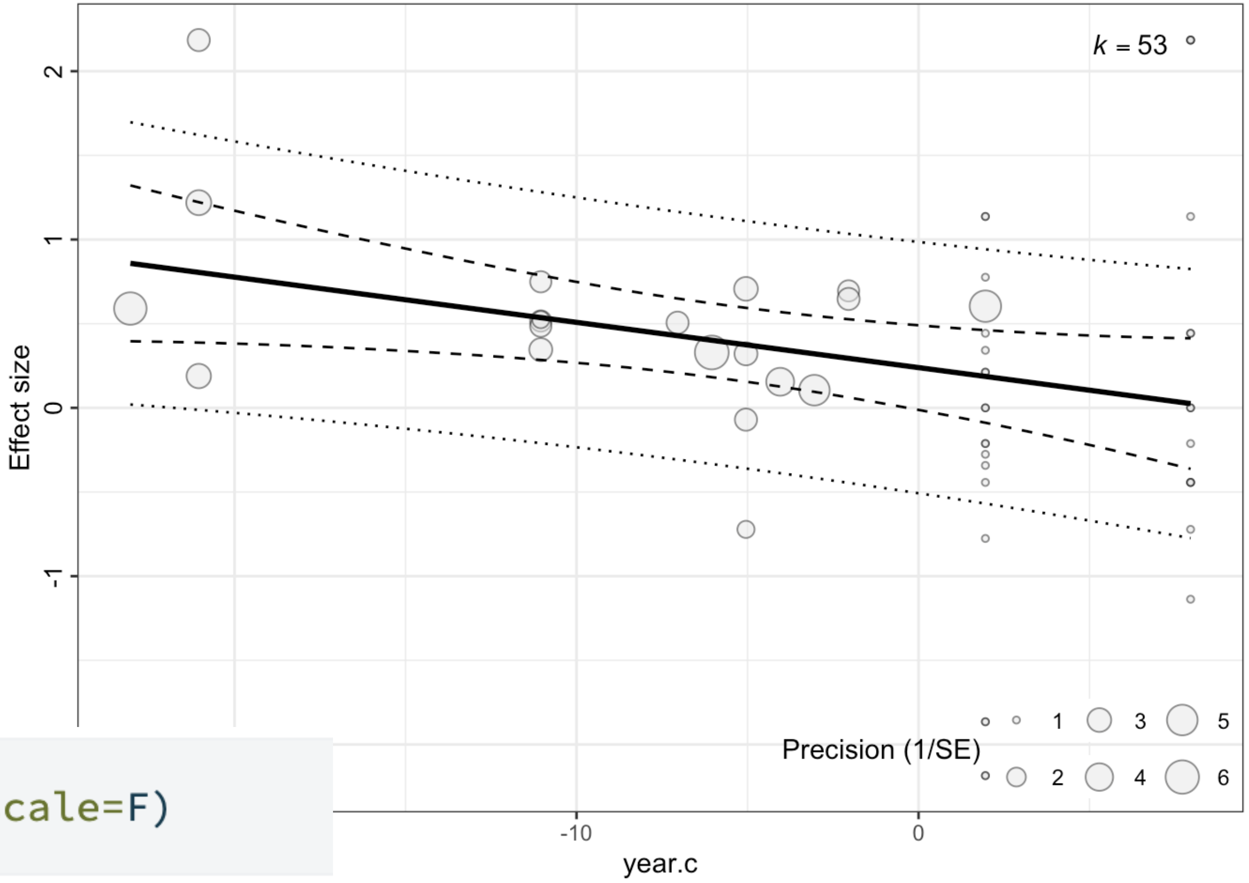


```r
meta_data$year.c <- scale(meta_data$year,scale=F)
```

Decline effects

```
# Select only the published studies..
meta_data_published<-meta_data%>%filter(published=="yes")

decline_effects <- rma.mv(Zr,
                          VZr,
                          mods = ~ 1 + year.c,
                          random = list(~ 1 | paperID,
                                        ~ 1 | popID2,
                                        ~ 1 | EffectID),
                          method = "REML",
                          test = "t",
                          data = meta_data_published)

# Printing the summary results of the model
print(decline_effects, digits = 3)
```

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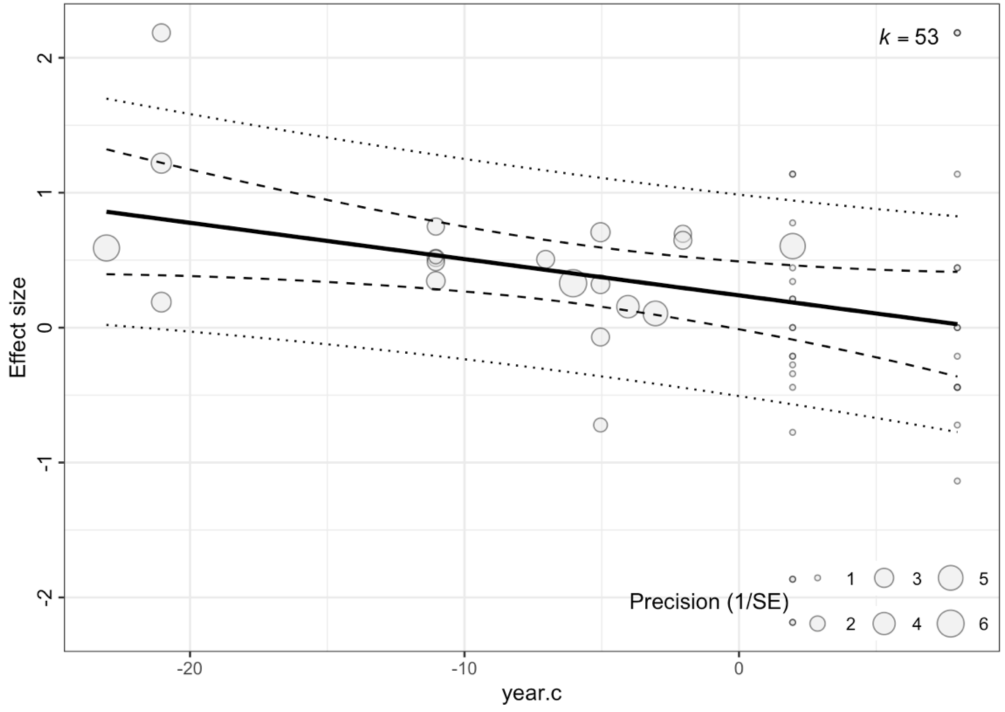
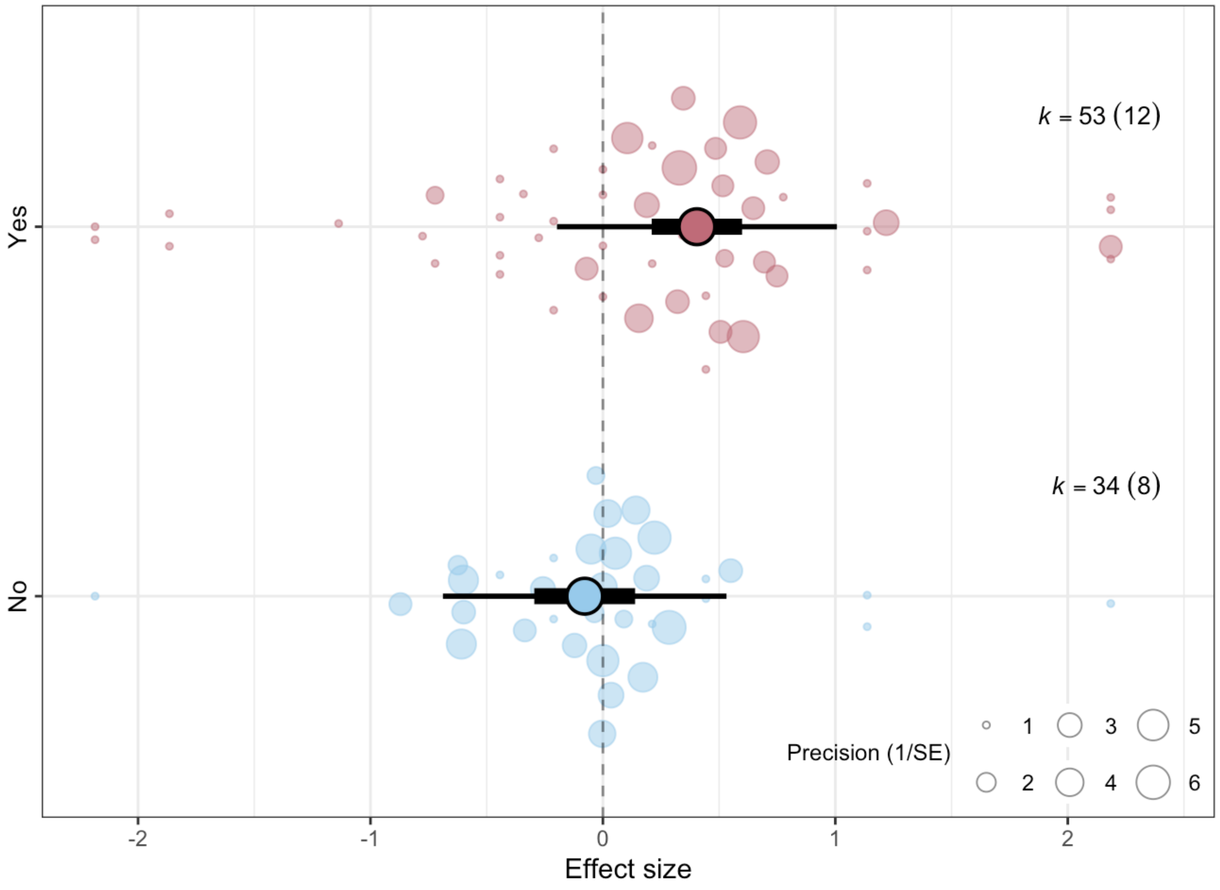
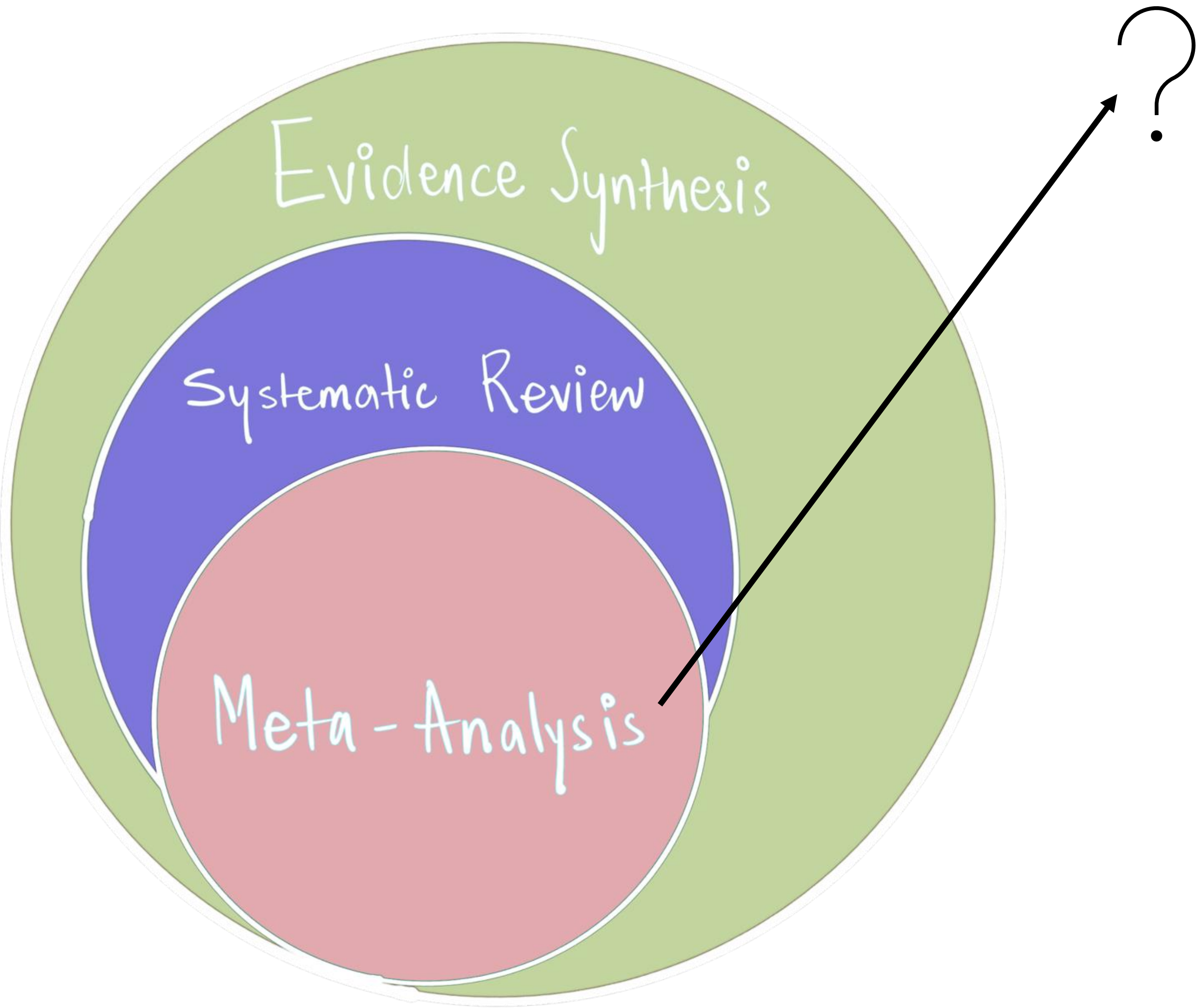


Decline effects



Published Vs. Unpublished studies



“A meta-analysis combines evidence, most often by weighting evidence according to how much we trust it. This consequently allows us to increase power (and precision) when compared to the primary studies.”

Meta-analysis **has 3 main goals**:

- Test overall effects and generality
Meta-analytic models (i.e., intercept-only models) estimate and explore heterogeneity (i.e., variation in effects not explained by differences in precision).

- Understand pattern of effects (i.e., try to explain heterogeneity)
Meta-regressions (i.e., including moderators): Biological, Methodological, including (and very importantly) publication biases

- Generate hypotheses

Take home messages!

1. Meta-analysis is a statistical tool to synthesise outcomes from different studies
2. All meta-analysis need effect sizes as a common currency
3. You can be interested in the mean effect but ...
4. Exploring heterogeneity is very meaningful in a meta-analysis
5. Test for publication bias and do sensitivity analysis
6. Document well! Report transparently and ensure your work is reproducible!